



# Full text of "Nuclear war survival skills"

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Nuclear War  
Survival  
Skills

Updated and Expanded  
1987 Edition

by

Cresson H. Kearny

With Foreword by Dr. Edward Teller

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LIFESAVING NUCLEAR FACTS AND SELF-HELP INSTRUCTIONS

Introduction to: Nuclear War Survival Skills

by Edward Teller  
January 14, 1994

With the collapse of the Soviet Union, the unimaginable catastrophe of all out nuclear war probable. At the same time this unexpected event taught a lesson: being prepared for trouble the source of trouble. Perhaps, after all, the atomic age might become a happy age.

Possible but not yet probable. Proliferation of nuclear weapons is more of a danger than a danger is now different. What may happen is still horrible but it is no longer a catastrophe. Describing it or preventing it.

As long as the superpowers faced each other with tens of thousands of megaton-class weapons seemed insufficient. It was a palliative of nuclear war. It played more of a role as part of us it seemed to be a necessity. But in efforts to convince the general public we made little could not be resolved by reason alone.

The problem of ideological conflict is disappearing. The problem of a violent dictator is one of mass destruction he could do enormous damage. Furthermore, the proliferation of ballistic missiles is a frightening and growing reality. But we are now no longer facing tens of thousands of missiles need to worry about at most hundreds. Defense, therefore, has become a rational possibility.

But if defense is possible, it is also most important for four connected but, at the same is that in the case of war defense may save many thousands, maybe even millions of lives. those who say that the main problem is to prevent war itself. I do disagree when prevention is the only problem.

The second reason is that defense helps to prevent proliferation of weapons of aggression. neglected these weapons of attack become effective. They become available and desirable in the hands of a dictator, even if his means are limited. Weapons of mass destruction could become equalized in size and small, highly developed and primitive, if defense is neglected. If defense is developed for general prevention of war, weapons of aggression will become less desirable. Thus defense is probable.

The third reason is of a most general character. One psychological defense mechanism again forget about it. This attitude is as common as it is disastrous. It may turn a limited danger into a world war.

The last and most important reason is that the world has become thoroughly interdependent and must come for the positive use of this interdependence. International cooperation is obviously necessary. It is best started by modest activities that are obviously in everyone's interest. War seems to be a good candidate for such cooperation. This would be particularly true if the cooperation is and effective.

This book is an excellent example of an international initiative that with a minimal effort has had a beneficial effect. It describes simple procedures of individual defensive measures which show the areas of danger including those where it is wrongly believed that defense is impossible. It is for all countries and in countries at an early stage of development. Electronics makes the book available to the world.

This book will not satisfy the demands of those who are interested only in final solutions and think that final solutions exist. The more important and difficult a problem is the more it becomes clear that it lies in a careful development consisting of small steps. This book prepares us, throughout the world, for the small steps that must be taken if the twenty-first century is to escape the curse of war.

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Updated and Expanded 1987 Edition

The purpose of this book is to provide Americans and other unprepared people with instructions that will significantly increase their chances of surviving a nuclear attack. Instructions that, if followed by a large fraction of Americans during a crisis that precedes loss of lives. The author is convinced that the vulnerability especially of Americans to nuclear attack has been reduced and that the wide dissemination of the information contained in this book will help preserve freedom.

Underlying the advocacy of Americans learning these down-to-earth survival skills is the hope that, even if the worst, the worst is less likely to happen. Effective American civil defense is hampered by the probability of nuclear blackmail and war. Yet in our world of increasing dangers, it is surprising that the United States spends much less per capita on civil defense than many other countries. The United States spends about 10 cents per capita, and only a few cents of this is spent on war-related civil defense. Unless these conditions are improved, you are unlikely to receive from official sources much of the survival information you need.

Over 400,000 copies of the Oak Ridge National Laboratory original 1979 edition of Nuclear War Survival Skills have been sold by various private publishers. A few additions and modifications, some helpful, were made in several of these private printings. This updated and expanded edition is needed because of changes in nuclear weapons and strategies between 1979 and 1987, and because of improvements in self-defense and instructions.

The 1987 edition provides current information on how the Soviet Union's continuing deployment of nuclear weapons, more numerous warheads should affect your shelter-building and evacuation plans.

In the first chapter the myths and facts about the consequences of a massive nuclear attack are presented. Post-1979 myths have been added: the myth of blinding post-attack increased ultra-violet radiation, the survivable "nuclear winter" - along with refuting facts.

A new chapter, "Permanent Family Fallout Shelters for Dual Use", has been added, because there have been received many requests for instructions for building permanent small shelters better and less expensive than described in official civil defense hand-outs. Another new chapter, "Trans-Pacific Fallout", discusses the dangers that you will face if one or more nations use nuclear weapons, but none are exploded.

Improved instructions are given for making and using a KFM, based on the findings of numerous tests since 1979. (The KFM still is the only accurate and dependable fallout radiation meter that millions of people can make for themselves in a few hours, using only common household materials - if they have the tools and instructions with patterns.) Field-tested instructions for easily made Directional Fans, the simple and effective "Ventilation and Cooling of Shelters" chapter. Also included in this edition are new facts and datings likely to help save lives if nuclear war strikes.

A new appendix gives instructions for a home makeable Plywood Double-Action Piston Pump, and a description of a wooden air pump the author saw being used in China in 1982.

This first-of-its-kind book is primarily a compilation and summary of civil defense measures developed at Oak Ridge National Laboratory and elsewhere over the past 24 years, and field tested by typical Americans in many states, from Florida to Washington. The reader is urged to make at least some of these preparations before a crisis arises. The main emphasis, however, is on survival preparations that could be made in the first few days of a worsening crisis. The author wrote the original, uncopyrighted Nuclear War Survival Skills as a research engineer at Oak Ridge National Laboratory. As a result, he has no proprietary interest in the book, nothing but satisfaction from past sales. Nor will he gain materially from future sales, and

copyright notice covering this edition. Civil defense professionals and others concerned with survival information can reproduce parts or all of this 1987 edition without getting permission, as long as they comply with the terms of the copyright notice.

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## Foreword

There are two diametrically opposite views on civil defense. Russian official policy holds that civil defense is feasible even in a nuclear war. American official policy, or at any rate the implementation of it, is based on the assumption that civil defense is useless.

The Russians, having learned a bitter lesson in the second world war, have bent every effort to prepare for a nuclear war under all circumstances. They are spending several billion dollars per year on this activity. They have plans to evacuate their cities before they let loose a nuclear strike. They have strong shelters for people to remain in the cities. They are building up protected food reserves to tide them over a crisis.

All this may mean that in a nuclear exchange, which we must try to avoid or to deter, the Russian population will probably not exceed ten million. Tragic as such a figure is, the Russian nation would survive. In the event of a nuclear war, the United States they can commandeer food, machinery and manpower from the rest of the world and recover rapidly. They would have attained their goal: world domination.

In the American view the Russian plan is unfeasible. Those who argue on this side point out that the Russians have no nuclear weapons. In this they are right. Their argument is particularly impressive in its simplicity.

But this argument has never been backed up by a careful quantitative analysis which takes into account the planned dispersal and sheltering of the Russian population and the other measures which they are committed to.

That evacuation of our own citizens can be extremely useful if we see that the Russians are doing it is a matter of common sense. With the use of American automobiles an evacuation could be faster and more efficient than it is in Russia. To carry it out we need not resort to the totalitarian methods of the iron curtain. We can warn our people and advise them where to go, how to protect themselves. The Federal Emergency Administration contains the beginnings on which such a policy might be built.

The present book does not, and indeed cannot, make the assumption that such minimal yet effective government guidance will be available. Instead it outlines the skills that individuals or groups can learn to apply in order to improve their chances of survival.

This book is not a description of civil defense. It is a guide to "Stop-gap" civil defense measures that individuals can carry out for themselves, if need be, with no expenditures by our government. It fills the gap between the civil defense that we have today and the highly effective survival preparations that we could have in a few years from now. However, if we go no further than what we can do on the basis of this book, we cannot survive a major nuclear war.

Yet this book, besides being realistic and objectively correct, serves two extremely important purposes. The first will help to save lives. The second purpose is to show that with relatively inexpensive government action and popular participation, an educated American public could, indeed, defend itself. We could survive a nuclear war.

This is an all-important goal. Its most practical aspect lies in the fact that the men in



they cannot count on destroying us they probably will never launch their nuclear arsenal and once the most peaceful and the most effective deterrent of nuclear war.

Some may argue that the Russians could evacuate again and again and thus, by forcing us in a corner, exhaust us. I believe that in reality they would anger us sufficiently so that we would reach the point the Russians want to accomplish.

Others may say that the Russians could strike without previous evacuation. This could result in their part which, I hope, they will not risk.

Civil defense as here described will not eliminate the danger of nuclear war. It will only reduce the probability.

This book takes a long overdue step in educating the American people. It does not suggest that it does not prove that national survival is possible. But it can save lives and it will stimulate action. It will be crucial in our two main purposes: to preserve freedom and to avoid war.

Edward Teller

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#### Nuclear War Survival Skills Video Tapes

The Oregon Institute of Science and Medicine, which distributes extensive written, audio, and video information on expedient and permanent civil defense procedures and preparations, has produced a series of video tapes in which the field tested instructions in Nuclear War Survival Skills and facts about nuclear war are demonstrated by civil defense volunteers including demonstrations and explanations by Civil Defense Volunteers.

Shelter construction and ventilation, water purification, food preparation, radiation monitoring, and first aid saving procedures - these essential survival skills are performed just as they would be in a nuclear emergency. This is six hours of video viewing that should be experienced by every American.

Part 1 : Expedient Blast and Radiation Shelters (102 minutes)

Part 2: Shelter Ventilation and Various Other Survival Skills (78 minutes)

Part 3 : Home-makeable and Commercial Fallout Radiation Meters (117 minutes)

Part 4: Nuclear War Facts as Told to Teenagers (74 minutes)

Complete Set - Four parts - Four tapes: \$95.00 VHS \$ 105.00 Beta

Each Tape Alone: \$29.50 VHS \$32.00 Beta

#### Nuclear War Survival Skills Quantity Book Discounts

This book should be in every American home and place of business. It should be a part of every family's defense preparations. In this nuclear age, prior preparation and knowledge are the primary defense against nuclear war. This book provides that essential knowledge.

It is published on a non-profit, non-royalty basis by the Oregon Institute of Science and public foundation). These low prices also are made possible by continuing donations to the science and Medicine given specifically to help meet the cost of publication and wide distribution enlarged edition.

Nuclear War Survival Skills is available postage paid within the United States at the following

1 copy \$12.50  
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#### About the Author

When the U.S. Atomic Energy Commission authorized me in 1964 to initiate the Civil Defense Ridge National Laboratory, one of the first researchers I recruited was Cresson H. Kearny. preparation, unplanned and planned, for writing this guide to help people unfamiliar with ons improve their chances of surviving a nuclear attack. During the past 15 years he has of practical field work on basic survival problems, without always conforming to the chang

After I returned to my professional duties at Princeton in 1966, the civil defense effort Laboratory was first headed by James C. Bresee, and is now headed by Conrad V. Chester. Bo edly supported Kearny's down-to- earth research, and Chester was not only a codeveloper of items described in this book, but also participated in the planning of the experiments tes

Kearny's concern with nuclear war dangers began while he was studying for his degree in ci Princeton he graduated summa cum laude in 1937. His Princeton studies had already acquaint nitude of an explosion in which nuclear energy is liberated, then only a theoretical possi Scholarship, Kearny earned two degrees in geology at Oxford. Still before the outbreak of the effective preparations made in England to reduce the effects of aerial attacks. He had ships, whether from the right or left, and during the Munich crisis he acted as a courier helping anti-Nazis escape from Czechoslovakia.

Lollowing graduation from Oxford, Kearny did geological exploration work in the Andes of E of Venezuela. He has traveled also in Mexico, China, and the Philippines.

A year before Pearl Harbor, realizing that the United States would soon be at war and that should have at least as good personal equipment, food, and individual medical supplies as quit his job with the Standard Oil Company of Venezuela, returned to the United States, an infantry reserve lieutenant. Kearny was soon assigned to Panama as the Jungle Experiment C Mobile Force. In that capacity he was able to improve or invent, and then thoroughly jungl ized equipment and rations used by our jungle infantrymen in World War II. For this work h and awarded the Legion of Merit.

To take his chances in combat, in 1944 the author volunteered for duty with the Office of demolition specialist helping to limit the Japanese invasion then driving into the wintry saw mass starvation and death first hand. The experiences gained in this capacity also res standing of both the physical and emotional problems of people whose country is under atta

Worry about the increasing dangers of nuclear war and America's lack of civil defense caused him to consult Herman Kahn, a leading nuclear strategist. Kahn, who was at that time forming an organization, the Hudson Institute, offered him work as a research analyst. Two years of work at the "think tank" made the author much more knowledgeable of survival problems.

In 1964 he joined the Oak Ridge civil defense project and since then Oak Ridge has been his home, except for two years during the height of the Vietnam war. For his Vietnam work on civil defense, and also for his contributions to preparations for improving survivability in the event of a nuclear war, he received the Army's Decoration for Distinguished Civilian Service in 1972.

This book draws extensively on Kearny's understanding of the problems of civil defense acquired from his own field testing of shelters and other survival needs, and also from an intensive study of civil defense preparations undertaken by other countries, including Switzerland, Sweden, the USSR, and China. He also edited the Oak Ridge National Laboratory translations of Soviet civil defense handbooks and gained additional knowledge from these new sources. Trips to England, Europe, and Israel provided information on survival measures, which contributed to the Nuclear War Survival Skills. However, principally those do-it-yourself instructions that field tests have proved to be practical.

(Signature)

Eugene P. Wigner. Physicist, Nobel Laureate, and the only surviving initiator of the Nuclear War Survival Skills.  
May, 1979

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#### Acknowledgments

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Robert H. Kupperman, Ph.D., physicist, in 1979 the Chief Scientist, U.S. Arms Control and Disarmament Agency, review of the final draft of the 1979 edition.

David B. Nelson, Ph.D., electrical engineer and mathematician, for years a civil defense authority on electromagnetic pulse (EMP) problems-for manuscript review and contributions to fallout monitoring instruments, and communications.

Lewis V. Spencer, Ph.D., for many years a physicist with the Radiation Physics Division, Oak Ridge National Laboratory-for his calculations and advice regarding needed improvements in the design of protective pants against excessive exposure to initial nuclear radiation.

Edward Teller, Ph.D., nuclear physicist, leading inventor of offensive and defensive weapons, Oak Ridge National Laboratory and worldwide-for contributing the Foreword, originally written for the 1979 edition, and his urging which motivated the author to work on this 1987 edition.

Eugene P. Wigner, Ph.D., physicist and mathematician, Nobel laureate. Professor Emeritus of Physics, Princeton University, principal initiator of the Nuclear Age and a prominent leader of the civil defense movement. He contributed to this book, contributing the About the Author section, and improving drafts, especially of the Introduction.

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Introduction

## SELF-HELP CIVIL DEFENSE

Your best hope of surviving a nuclear war in this century is self-help civil defense - knowing nuclear weapon effects and what you, your family, and small groups can do to protect yourselves. It continues to downgrade war-related survival preparations and spends only a few cents a year against possible war dangers. During the 10 years or more before the Strategic Defense weapons can be invented, developed and deployed, self-help civil defense will continue to be the only way of surviving if we suffer a nuclear attack.

Most Americans hope that Star Wars will lead to the deployment of new weapons capable of destroying missiles and warheads in flight. However, no defensive system can be made leak-proof. If the Strategic Defense research project, leads to a deployed defensive system, then self-help civil defense will still be the only way for, truly defensive system to prevent aggressions and to reduce losses if deterrence fails.

## PURPOSE AND SCOPE OF THIS BOOK

This book is written for the majority of Americans who want to improve their chances of surviving a nuclear war. It brings together field-tested instructions that have enabled untrained Americans to make air pumps to ventilate and cool shelters, fallout meters, and other expedient life-support equipment. It describes equipment that can be made by untrained citizens in the home, and is guided solely by field-tested, written instructions and using only widely available materials. It shows expedient ways to remove even dissolved radioiodine from water, and to process and cook beans, our main food reserves. Successive versions of these instructions have been used successfully in working under simulated crisis conditions, and have been improved repeatedly by Oak Ridge National Laboratory defense researchers and others over a period of 14 years. These improved instructions are the 1987 edition of the original Oak Ridge National Laboratory survival book first published in 1973.

The average American has far too little information that would help him and his family and community survive a nuclear attack, and many of his beliefs about nuclear war are both false and dangerous. Since Hiroshima and Nagasaki hurled mankind into the Nuclear Age, only during a recognized crisis threatening the United States have Americans been seriously interested in improving their chances of surviving a nuclear attack. During the Cuban Missile Crisis in 1962, millions of Americans built fallout shelters or tried to improve their shelters. At that time most of the available survival information was inadequate, and dangerously false. This is still the case in 1987. Widespread recognition of these civil defense shortcomings has contributed to the false beliefs of Americans of one or both of two false beliefs:

One of these false beliefs is that nuclear war would be such a terrible catastrophe that it would be impossible to survive. If this were true, there would be no logical reason to worry about nuclear war or to try to survive a nuclear attack.

The second false belief is that, if a nuclear war were to break out, it would be the end of the world. If true, a rational person would not try to improve his chances of surviving the unsurvivable.

This book gives facts that show these beliefs are false. History shows that once a weapon is developed and ready for use in the arsenals of some nations and in time will be used. Researchers who have made great effort learning the facts about effects of nuclear weapons now know that all-out nuclear war would destroy mankind or of civilization. Even if our country remained unprepared and were to be subject to a nuclear attack, many millions of Americans would survive and could live through the difficult post-attack period.

## WHY YOU AND YOUR FAMILY AND ALMOST ALL OTHER AMERICANS ARE LEFT UNPROTECTED HOSTAGES TO THE SOVIET UNION

Unknown to most Americans, our Government lacks the defense capabilities that would enable to stop being dependent on a uniquely American strategic policy called Mutual Assured Destruction. It maintains that if both the United States and Russia do not or can not adequately protect their industries, then neither will attack the other.

An influential minority of Americans still believe that protecting our citizens and our vital interests will require us to continue the arms race and increase the risk of war. No wonder that President Reagan's defense initiative, derisively called Star Wars, is subjected to impassioned opposition by those who are threatened even by research to develop new weapons designed to destroy weapons launched against them. No wonder that even a proposed small increase in funding for civil defense to save lives is met with stronger opposition from MAD supporters than do most much larger expenditures for weapons.

## RUSSIAN, SWISS, AND AMERICAN CIVIL DEFENSE

No nation other than the United States has advocated or adopted a strategy that purposely protected hostages to its enemies. The rulers of the Soviet Union never have adopted a MAD strategy to prepare the Russians to fight, survive, and win all types of wars. Almost all Russians are taught about the effects of nuclear and other mass-destruction weapons, and what their chances of surviving. Comprehensive preparations have been made for the crisis evacuation of urban areas, where they and rural Russians would make high-protection-factor expedient fall to protect millions have been built in the cities and near factories where essential workers would be during a crisis. Wheat reserves and other foods for war survivors have been stored outside the cities. 100,000 civil defense troops are maintained for control, rescue, and post-attack recovery. The estimated cost of Russian civil defense preparations, if made at costs equivalent to those in the United States, is estimated to be between \$8 and \$20.

Switzerland has the best civil defense system, one that already includes blast shelters for its citizens. Swiss investment in this most effective kind of war-risk insurance has continued to grow. According to Dr. Fritz Sager, the Vice Director of Switzerland's civil defense, in 1984 the cost was \$12.60 per capita.

In contrast, our Federal Emergency Management Agency, that includes nuclear attack preparedness as one of its many responsibilities, will receive only about \$126 million in fiscal 1987. This will amount to only a few dollars per American. And only a small fraction of this pittance will be available for nuclear attack preparedness. The cost of nuclear attack self-help survival instructions is about all that FEMA could afford to do to improve American survival in the event of a nuclear war, unless FEMA's funding for war-related civil defense is greatly increased.

## PRACTICALITY OF MAKING SURVIVAL PREPARATIONS DURING A CRISIS

The emphasis in this book is on survival preparations that can be made in the last few days before a crisis. However, the measures put into effect during such a crisis can be very much more effective if the preparations are completed well in advance. It is hoped that persons who read this book will be able to make the preparations outlined in Chapter 16, Minimum Pre-Crisis Preparations.

Well-informed persons realize that a nuclear attack by the Soviet Union is unlikely to be a surprise attack, launched without warning. Strategists agree that a nuclear war most likely would be preceded by months of worsening crisis. The most realistic of the extensive Russian plans and preparations for a nuclear war is the plan to survive a nuclear war by surviving a long and worsening crisis.

war are based on using at least several days during an escalating crisis to get most urban and other high risk areas, to build or improve shelters in all parts of the Soviet Union, and the like. The Russians know that if they are able to complete evacuation and sheltering of nuclear war, the number of their people killed would be a small fraction of those who c

satellites and other sources of intelligence would reveal such massive movements within a most likely circumstances Americans would have several days in which to make life-saving p

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The Russians have learned from the devastating wars they have survived that people are the to be saved. Russian civil defense publications emphasize Lenin's justly famous statement: factor of all humanity is the laboring man, the worker. If he survives, we can save everyt . but we shall perish if we are not able to save him." Strategists conclude that those in very unlikely to launch a nuclear attack until they have protected most of their people.

The reassurance of having at least a few days of pre-attack warning, however, is lessening bers of Soviet blast shelters and of first-strike offensive weapons capable of destroying weapons will reduce the importance of pre-attack city evacuation as a means of saving Russ developments will make it less likely that Americans will have a few days' warning before fore should motivate our Government both to deploy truly defensive Star Wars weapons and t protect urban Americans.

Nuclear weapons that could strike the United States continue to increase in accuracy as we most modern warheads usually can hit within a few hundred feet of their precise targets. T has enough warheads to target all militarily important fixed site objectives. These includ mand and control centers, military installations, oil refineries and other industrial plan long runways, and major electric generating plants. Many of these are either in or near ci cans live in cities that contain strategically important targets, urban Americans' best ch clear attack is to get out of cities during a worsening crisis and into fallout shelters a

Most American civil defense advocates believe that it would be desirable for our Governmen permanent blast shelters. However, such permanent shelters would cost many tens of billion likely to be undertaken as a national objective. Therefore, field-tested instructions and urban evacuees and rural Americans to build expedient shelters and life-support equipment

#### SMALLER NUCLEAR ATTACKS ON AMERICA

Many strategists believe that the United States is more likely to suffer a relatively small out Soviet onslaught. These possible smaller nuclear attacks include:

- ° A limited Soviet attack that might result if Russia's rulers were to conclude that an Am be likely to capitulate rather than retaliate if a partially disarming first strike knocke and retaliatory weapons, but spared the great majority of our cities. Then tens of million away from missile silos and Strategic Air Force bases would need only fallout protection. who live in large metropolitan areas and doubt that they could successfully evacuate durin should realize that in the event of such a limited attack they would have great need for r s kil ls.

- ° An accidental or unauthorized launching of one or several nuclear weapons that would exp



Complex computerized weapon systems and/or their human operators are capable of making let

- ° A small attack on the United States by the fanatical ruler of an unstable country that m nuclear weapons and a primitive delivery system.

- ° A terrorist attack, that will be a more likely possibility once nuclear weapons become a nations. Fallout dangers could extend clear across America. For example, a single small nu exploded in a West Coast city would cause lethal fallout hazards to unsheltered persons fo downwind from the part of the city devastated by blast and fire. It also would result in c downwind localities up to hundreds of miles away, with radiation dose rates hundreds of ti normal background. Fallout would be especially heavy in areas of rain- out; pregnant women children in those areas, following peacetime standards for radiation protection, might nee for weeks. Furthermore, in localities spotted across the United States, milk would be cont radioiodine.

Surely in future years nuclear survival know-how will become an increasingly important par person's education.

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WHY THIS 1987 EDITION?

This updated and augmented edition is needed to give you:

- ° Information on how changes since 1979 in the Soviet nuclear arsenal - especially the gre sizes of Russian warheads and increases in their accuracy and number - both decrease and i dangers we all face. You need this information to make logical decisions regarding essenti planning, including whether you should evacuate during a worsening crisis or build or impr near your home.

- ° Instructions for making and using self-help survival items that have been rediscovered, since 1979. These do-it- yourself items include: (1) Directional Fanning, the simplest way through large openings; (2) the Plywood Double-Action Piston Pump, to ventilate shelters t (3) the improved KFM, the best homemadeable fallout meter.

- ° Facts that refute two demoralizing anti- defense myths that have been conceived and prop 1979: the myth of blinding post-attack ultra-violet radiation and the myth of unsurvivable

- ° Current information on advantages and disadvantages, prices, and sources of some manufac items for which there is greatest need.

- ° Updated facts on low cost survival foods and on expedient means for processing and cooki grains, soybeans, and other over- produced basic foods. Our Government stores no food as a has not given even civil defense workers the instructions needed to enable survivors to ma America's unplanned, poorly distributed, large stocks of unprocessed foods.

- ° Updated information on how to obtain and use prophylactic potassium iodide to protect yo injury both from war fallout, and also from peacetime fallout if the United States suffers nuclear power reactor accident releasing life endangering radiation.

° Instructions for building, furnishing, and stocking economical, permanent home fallout shelters for dual use-in a new chapter.

° Information on what you can do to prevent sickness if fallout from an overseas nuclear war with the United States is not a belligerent is blown across the Pacific and deposited on America -

#### EXOTIC WEAPONS

Chemical and biological weapons and neutron warheads are called "exotic weapons". Protection against these weapons are not emphasized in this book, because its purpose is to help Americans in surviving what is by far the most likely type of attack on the United States: a nuclear attack on related strategic targets.

Chemical Weapons are inefficient killing agents compared to typical nuclear warheads and bombs. Terminating the unprepared population of a specified large area were an enemy's objective, the required payload of deadly chemical weapons many hundreds of times heavier than if large nuclear weapons were employed.

Biological Weapons are more effective but less reliable than chemical weapons. They are more dependent on favorable meteorological conditions, and could destroy neither our retaliatory weapons nor our cities. They could not kill or incapacitate well protected military personnel manning our radar and missile defenses. A biological attack could not prevent, but would invite, U.S. nuclear retaliatory strikes.

Neutron Warheads are small, yet extremely expensive. A 1-kiloton neutron warhead costs about the same as a 100-kiloton ordinary warhead, but the ordinary warhead not only has 1000 times the explosive power, but also a surface airburst to cover a very large area with deadly fallout.

#### REWARDS

My greatest reward for writing Nuclear War Survival Skills is the realization that the hundreds of copies of the original edition which have been sold since 1979 already have provided many people with survival information that may save their lives. Especially rewarding have been the thanks from mothers with small children - for having given them hope of surviving a nuclear war. Rekindled hope has caused some readers to work to improve their and their families' chances of surviving, rationed their food, to evacuate high risk areas during an all too possible worsening crisis, to building fallout shelters.

Because I wrote the original Nuclear War Survival Skills while working at Oak Ridge National Laboratory, for the American taxpayers

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expense, I have no proprietary interest either in the original 1979 Government edition or in its reproductions. I have gotten nothing but satisfaction from the reported sales of over 400,000 copies, and sold between 1979 and 1987. Nor will I receive any monetary reward in the future from the sale of survival instructions to people who want to improve their chances of surviving a nuclear attack.

#### AVAILABILITY

None of the material that appeared in the original Oak Ridge National Laboratory unclassified report is available elsewhere.

can be covered by a legitimate copyright; it can be reproduced by anyone, without receiving permission, of the material, which I have written since my retirement in 1979 from Oak Ridge National Laboratory. This new material is printed in a different type. To assure that this new material also can be made widely available at low cost, without getting permission from or paying anyone, I have copyrighted my new material by this 1987 edition's copyright notice.

#### RECOMMENDED ACTIONS

Work to persuade the President, your Congressmen, your Senators, and other leaders to support war survival preparations, starting with increased funding for war-related civil defense. Fund the early deployment of truly defensive weapons that tests already have proven capable of stopping enemy heads in flight. (Attempts to develop perfect defenses postpone or prevent the attainment of a secure world.)

Obtain and study the best survival instructions available long before a crisis occurs. Bet on the best instructions, such as the ones described in this book, to increase your and your family's chances of survival.

During a crisis threatening nuclear attack, present uncertainties regarding the distribution of survival instructions seem likely to continue. Thoroughly field-tested survival instructions are not likely to be widely available to all Americans. Furthermore, even a highly intelligent citizen, if given excellent instructions and ample time to learn basic facts about nuclear dangers and the reasons for various survival preparations, standing, no one can do his best at following any type of survival instructions.

By following the instructions in this book, you and your family can increase the odds favoring survival. If such instructions were made widely available from official sources, and if our Government would fund them during a worsening crisis lasting at least several days, additional millions would be saved. The danger of an attack, even the threat of an attack, could be decreased if an enemy nation knew that we had so greatly improved our defenses in this way.

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#### Chapter 1

##### The Dangers from Nuclear Weapons: Myths and Facts

An all-out nuclear war between Russia and the United States would be the worst catastrophe in human history, so huge it is difficult to comprehend. Even so, it would be far from the end of human life. The dangers from nuclear weapons have been distorted and exaggerated, for varied reasons. These exaggerations and the allying myths, believed by millions of Americans.

While working with hundreds of Americans building expedient shelters and life-support equipment, we have found that many people at first see no sense in talking about details of survival skills. Those who are interested about the dangers from nuclear weapons must first be convinced that nuclear war would not be a quick and easy death and everything worthwhile. Only after they have begun to question the truth of these distorted views, interested, under normal peacetime conditions, in acquiring nuclear war survival skills. Then, in the instructions for making and using survival equipment, we will examine the most harmful of nuclear war dangers, along with some of the grim facts.

° Myth: Fallout radiation from a nuclear war would poison the air and all parts of the environment.

everyone. (This is the demoralizing message of On the Beach and many similar pseudoscientifi

° Facts: When a nuclear weapon explodes near enough to the ground for its fireball to touch the ground, a crater is formed. (See Fig. 1.1.)

Fig. 1.1. A surface burst. In a surface or near-surface burst, the fireball touches the ground and forms a crater. ORNL-DWG 786264 (nw011.tif)

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Many thousands of tons of earth from the crater of a large explosion are pulverized into tiny particles. These particles are contaminated by radioactive atoms produced by the nuclear explosion. Thousands of tons of these particles are carried up into a mushroom-shaped cloud, miles above the earth. These radioactive particles are carried out of the mushroom cloud, or out of the dispersing cloud of particles blown by the winds thus becoming a part of the fallout.

Each contaminated particle continuously gives off invisible radiation, much like a tiny X-ray. The particles continue to give off radiation while descending, and after having fallen to earth. The descending radiation is carried by the winds like the sand and dust particles of a miles-thick sandstorm cloud except that the particles move at lower speeds and in many areas the particles are so far apart that no cloud is seen. The largest particles reach the ground first, in locations close to the explosion. Many smaller particles continue to travel tens to thousands of miles before falling to earth. At any one place where fallout from a nuclear explosion is deposited on the ground in concentrations high enough to require the use of shelters, deposition continues for a few hours.

The smallest fallout particles those tiny enough to be inhaled into a person's lungs are the most dangerous. These tiny particles would fall so slowly from the four-mile or greater heights to which they are carried by recently deployed Soviet warheads that most would remain airborne for weeks to years before falling to earth. At that time their extremely wide dispersal and radioactive decay would make them much less dangerous. However, such tiny particles are promptly brought to earth by rain-outs or snow-outs in scattered showers. These rain-outs or snow-outs, blown about by the winds, would these invisible particles constitute a long-term and relatively great danger.

The air in properly designed fallout shelters, even those without air filters, is free of radiation. You can breathe except in a few rare environments as will be explained later.

Fortunately for all living things, the danger from fallout radiation lessens with time. The lessening is called decay, is rapid at first, then gets slower and slower. The dose rate (the amount of radiation received per hour) decreases accordingly. Figure 1.2 illustrates the rapidity of the decay of radiation from fallout two days after the nuclear explosion that produced it. R stands for roentgen, a measurement of radiation exposure to gamma rays and X rays. Fallout meters called dosimeters measure the dose received. The number of R. Fallout meters called survey meters, or dose-rate meters, measure the dose rate, the number of R being received per hour at the time of measurement. Notice that it takes about 48 hours for the dose rate to decay from 1000 roentgens per hour (1000 R/hr) to 10 R/hr (48 hours) as it takes 7 hours to decay from 100 R/hr (7 hours). (Only in high-fallout areas would the dose rate 1 hour after the explosion be 1000 R/hr.)

number of R being received per hour at the time of measurement. Notice that it takes about 48 hours for the dose rate to decay from 1000 roentgens per hour (1000 R/hr) to 10 R/hr (48 hours) as it takes 7 hours to decay from 100 R/hr (7 hours). (Only in high-fallout areas would the dose rate 1 hour after the explosion be 1000 R/hr.)

Fig. 1.2. Decay of the dose rate of radiation from fallout, from the time of the explosion.

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If the dose rate 1 hour after an explosion is 1000 R/hr, it would take about 2 weeks for the dose rate to be reduced to 1 R/hr solely as a result of radioactive decay. Weathering effects will reduce the dose rate. For example, rain can wash fallout particles from plants and houses to lower positions on or close to the ground. Surrounding objects would reduce the radiation dose from these low-lying particles.

Figure 1.2 also illustrates the fact that at a typical location where a given amount of fallout is deposited later than 1 hour after the explosion, the highest dose rate and the total dose will be less than at a location where the same amount of fallout is deposited 1 hour after the explosion. If particles have been airborne before reaching the ground, the less dangerous is their radiative effect.

Within two weeks after an attack the occupants of most shelters could safely stop using them and go outside the shelters for an increasing number of hours each day. Exceptions would be in areas where fallout might occur downwind from important targets attacked with many weapons, especially large cities. To know when to come out safely, occupants either would need a reliable fall-out meter to measure changing radiation dangers, or must receive information based on measurements made nearby by a portable instrument.

The radiation dose that will kill a person varies considerably with different people. A dose of 400 R exposure of the whole body to fallout radiation is often said to be the dose that will kill a person, although most studies indicate that it would take somewhat less. 1 (Note: A number of writers refer the reader to a source listed in the Selected References that follow Appendix D.) Although people confined to expedient shelters after a nuclear attack would be under stress and without clear medical aid to fight infections. Many also would lack adequate water and food. Under these unprecedented conditions, perhaps half the persons who received a whole-body dose of 350 R within a few days would die. 2

Fortunately, the human body can repair most radiation damage if the daily radiation doses are kept low. As will be explained in Appendix B, a person who is healthy and has not been exposed in the past to a large radiation dose of more than 100 R can receive a dose of 6 R each day for at least two months without becoming incapacitated.

Only a very small fraction of Hiroshima and Nagasaki citizens who survived radiation doses that were nearly fatal have suffered serious delayed effects. The reader should realize that to do anything about a nuclear attack, many survivors must be willing to receive much larger radiation doses than they have received. Otherwise, too many workers would stay inside shelter too much of the time, and work that must be done for recovery could not be done. For example, if the great majority of truckers were so fearful of receiving incapacitating radiation doses that they would refuse to transport food, additional millions of people would starve alone.

° Myth: Fallout radiation penetrates everything; there is no escaping its deadly effects.

° Facts: Some gamma radiation from fallout will penetrate the shielding materials of even the best shelters and reach its occupants. However, the radiation dose that the occupants of an excellent shelter can receive in this shelter can be reduced to a dose smaller than the average American receives during his lifetime from other radiation exposures normal in America today. The design features of such a shelter include a sufficient thickness of earth or other heavy shielding material. Gamma rays are like X rays, but they are stopped more readily by matter. Figure 1.3 shows how rapidly gamma rays are reduced in number (but not in their ability to penetrate) by matter. Each of the layers shown is one halving-thickness of packed earth- about 3.6 inches thick. The halving-thickness is the thickness of a material which reduces by half the dose of radiation passing through it.



shelters.)

Fig. 1.5. Undamaged earth-covered family shelter in Nagasaki, (nw015.jpg)

Many earth-covered family shelters were essentially undamaged in areas where blast and firings. Figure 1.5 shows a typical earth covered, backyard family shelter with a crude wooden door. The shelter was essentially undamaged, although less than 100 yards from ground zero at Nagasaki. 4 The blast pressure (pressure above the normal air pressure) was about 65 pounds per square inch (65 psi). A small shelter without a blast door would have been killed by blast pressure at this distance. However, in a recent blast test, 5 an earth-covered, expedient Small-Pole Shelter equipped with

at 53 psi. The pressure rise inside was slight not even enough to have damaged occupants' lives. Available field tests have indicated that many families can build such shelters in a few

The great life-saving potential of blast-protective shelters has been proven in war and calculations. For example, the area in which the air bursting of a 1-megaton weapon would kill people with blast doors is about 2.7 square miles. Within this roughly circular area, practically all people in shelters would be killed by blast, carbon monoxide from fires, or radiation. The same blast would kill people who were using basements affording 5 psi protection, over an area of about 58 square miles.

° Myth: Because some modern H-bombs are over 1000 times as powerful as the A-bomb that destroyed Hiroshima, these H-bombs are 1000 times as deadly and destructive.

Facts: A nuclear weapon 1000 times as powerful as the one that blasted Hiroshima, if exploded under the same conditions, produces equally serious blast damage to wood-frame houses over an area up to 1000 times as large, not 1000 times as large.

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For example, air bursting a 20-kiloton weapon at the optimum height to destroy most buildings would damage houses out to about 1.42 miles from ground zero. 6 The circular area of at least several cities would be about 6.33 square miles. (The explosion of a 20 kiloton weapon releases the same amount of energy as 20,000 tons of TNT.) One thousand 20-kiloton weapons thus air burst, well separated to avoid overlap, would destroy or severely damage houses over areas totaling approximately 6,330 square miles. The air bursting of one 20-megaton weapon (equivalent in explosive power to 20 million tons of TNT) would severely damage the great majority of houses out to a distance of 16 miles from ground zero. The area would be about 800 square miles - not 6,330 square miles.

Today few if any of Russia's huge intercontinental ballistic missiles (ICBMs) are armed with nuclear warheads. Now a huge Russian ICBM, the SS-18, typically carries 10 warheads, each having a yield of 100 kilotons, programmed to hit a separate target. See Jane's Weapon Systems, 1987-88.

° Myth: A Russian nuclear attack on the United States would completely destroy all American cities.

Facts: As long as Soviet leaders are rational they will continue to give first priority to their own cities and other military assets that can damage Russia and kill Russians. To explode enough nuclear warheads to completely destroy American cities would be an irrational waste of warheads. The Soviets could not afford most of the warheads that would be required to completely destroy American cities; the warheads probably already are targeted to knock out our retaliatory missiles by being surface burst over their hardened silos, located far from most cities and densely populated areas.

Unfortunately, many militarily significant targets - including naval vessels in port and fighters on the ground, air base and airport facilities that can be used by bombers, Army factories - are in or close to American cities. In the event of an all-out Soviet attack, would be destroyed by air bursts. Air bursting (see Fig. 1.4) a given weapon subjects about blast effects severe enough to destroy "soft" targets as does surface bursting (see Fig. 1.5). Unfortunately for Americans living outside blast and fire areas, air bursts produce only very tiny amounts of radioactive fallout. Extremely small radioactive particles remain airborne for so long that their radioactive decay before reaching the ground make them much less life-endangering than the promptly deposited large surface and near-surface bursts. However, if you are a survival minded American you should expect heavy fallout wherever you are. Unpredictable winds may bring fallout from unexpected directions. You may be in a "hot spot" of life-endangering fallout caused by a rain-out or snow-out of both surface and air bursts. Or the enemy may use surface or near-surface bursts in your part of the country to delay or otherwise disrupt U.S. retaliatory actions by producing heavy local fallout.

Today few if any of Russia's largest intercontinental ballistic missiles (ICBMs) are armed with a nuclear warhead. A huge Russian ICBM, the SS-18, typically carries 10 warheads each having a yield of 100,000 tons of TNT, programmed to hit a separate target. See "Jane's Weapon Systems. 1987-1988." However, in March 1986, Director William Webster told the U.S. Senate Armed Services Committee that ".... The USSR's

modernization program continues unabated," and that the SS-18 Mod 5 can carry 14 to 20 nuclear warheads are generally assumed to be smaller than those of the older SS-18s.

° Myth: So much food and water will be poisoned by fallout that people will starve and die where there is enough food and water.

Facts: If the fallout particles do not become mixed with the parts of food that are eaten, and water in dust-tight containers are not contaminated by fallout radiation. Peeling fruit removes essentially all fallout, as does removing the uppermost several inches of stored grain onto the ground. Water from many sources - such as deep wells and covered reservoirs, tanks, and cisterns - is not contaminated. Even water containing dissolved radioactive elements and compounds can be made safe by simply filtering it through earth, as described later in this book.

° Myth: Most of the unborn children and grandchildren of people who have been exposed to nuclear explosions will be genetically damaged will be malformed, delayed victims of nuclear war.

Facts: The authoritative study by the National Academy of Sciences, A Thirty Year Study of Hiroshima and Nagasaki, was published in 1977. It concludes that the incidence of abnormal children later conceived by parents who were exposed to radiation during the attacks on Hiroshima and Nagasaki is the incidence of abnormalities among Japanese children born to un-exposed parents.

This is not to say that there would be no genetic damage, nor that some fetuses subjected to radiation would not be damaged. But the overwhelming evidence does show that the exaggerated fears of genetic damage to future generations are not supported by scientific findings.

° Myth: Overkill would result if all the U.S. and U.S.S.R, nuclear weapons were used meaning that the two superpowers have more than enough weapons to kill all of each other's people, but also enough weapons to exterminate the human race.



Facts: Statements that the U.S. and the Soviet Union have the power to kill the world's population over are based on misleading calculations. One such calculation is to multiply the deaths ploded over Hiroshima or Nagasaki by an estimate of the number of kilotons in either side's arsenal (one kiloton is one that produces the same amount of energy as does 1000 tons of TNT.) The unstated assumption is somehow the world's population could be gathered into circular crowds, each a few miles in diameter with a population density equal to downtown Hiroshima or Nagasaki, and then a small (Hiroshima-sized) nuclear explosion ploded over the center of each crowd. Other misleading calculations are based on exaggerated long-lasting radiation and other harmful effects of a nuclear war.

° Myth: Blindness and a disastrous increase of cancers would be the fate of survivors of a nuclear war. The nuclear explosions would destroy so much of the protective ozone in the stratosphere that ultraviolet light would reach the earth's surface. Even birds and insects would be blinded. People could not go outside for time for years without dark glasses, and would have to wear protective clothing to prevent skin cancer. Plants would be badly injured and food production greatly reduced.

Facts: Large nuclear explosions do inject huge amounts of nitrogen oxides (gasses that destroy ozone) into the stratosphere. However, the percent of the stratospheric ozone destroyed by a given amount of nuclear explosions has been greatly overestimated in almost all theoretical calculations and models. For example, calculations of stratospheric nuclear test explosions of large weapons in 1952-1962 were calculated by Foley and Ruderman to show a reduction of more than 10 percent in total ozone. (See M. H. Foley and M. A. Ruderman, "Stratospheric Ozone After Past Nuclear Explosions", Journal of Geophysics, Res. 78, 4441-4450.) Yet observations show no significant reductions in ozone. Nor did ultraviolet increase. Other theoreticians calculated sizable increases in ultraviolet and interpreted the observational data to indicate either no reduction, or much smaller reductions than those calculated.

A realistic simplified estimate of the increased ultraviolet light dangers to American survivors of a nuclear war equates these hazards to moving from San Francisco to sea level at the equator, where the incidence of skin cancers (seldom fatal) is highest- about 10 times higher than the incidence at San Francisco. In a nuclear war, thousands of American survivors might get skin cancer, but little or no increase in skin cancer deaths.

After a nuclear war, post-attack world deliberate sun tanning and going around hatless went out of fashion. Furthermore, today's warheads are smaller than those exploded in the large- weapons tests mentioned above. They inject much smaller amounts of ozone-destroying gasses, or no gasses, into the stratosphere, where they persist for years. And nuclear weapons smaller than 500 kilotons result in increases (due to destruction of stratospheric ozone). In a nuclear war, these increases would partially compensate for the ultraviolet decreases-as explained by Julius S. Chang and Donald J. Wuebbles of Lawrence Livermore National Laboratory.

° Myth: Unsurvivable "nuclear winter" surely will follow a nuclear war. The world will be plunged into darkness. Tons (less than one percent of all nuclear weapons) are used to ignite cities. World-enveloping dust from surface bursts will prevent almost all sunlight and solar heat from reaching the earth's surface. Global darkness for weeks ! Sub-zero temperatures, even in summertime ! Frozen crops, even in the tropics ! Worldwide famine ! Whole species of animals and plants exterminated ! The survival of only a few million people.

Facts: Unsurvivable "nuclear winter" is a discredited theory that, since its conception in the 1970s, has frightened additional millions into believing that trying to survive a nuclear war is a waste of time. The theory that only by ridding the world of almost all nuclear weapons do we have a chance of survival is also a myth.

Non-propagandizing scientists recently have calculated that the climatic and other environmental effects of an all-out nuclear war would be much less severe than the catastrophic effects repeatedly

tronomer Carl Sagan and his fellow activist scientists, and by all the involved Soviet scientists. From these recent, realistic calculations are summarized in an article, "Nuclear Winter Revisited," in the 1986 summer issue of *Foreign Affairs*, the prestigious quarterly of the Council on Foreign Relations. Starley L. Thompson and Stephen H. Schneider, are atmospheric scientists with the National Center for Global Change Research. They showed "that on scientific grounds the global apocalyptic conclusions of the nuclear winter hypothesis can now be relegated to a vanishing low level of probability."

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Their models indicate that in July (when the greatest temperature reductions would result) the United States would be reduced for a few days from about 70 degrees Fahrenheit to approximately 40 degrees Fahrenheit.

(In contrast, under the same conditions Carl Sagan, his associates, and the Russian scientists predicted a global average temperature of about 10 degrees below zero Fahrenheit, lasting for many weeks!)

Persons who want to learn more about possible post-attack climatic effects also should read the summer issue of *Foreign Affairs*. This issue contains a long letter from Thompson and Schneider which further discusses the possibility of catastrophic "nuclear winter." Continuing studies indicate there will be even smaller temperature reductions than those calculated by Thompson and Schneider.

Soviet propagandists promptly exploited belief in unsurvivable "nuclear winter" to increase tensions and war, and to demoralize their enemies. Because raging city firestorms are needed to pump smoke into the stratosphere and thus, according to one discredited theory, prevent almost all sunlight from reaching the ground, the Soviets changed their descriptions of how a modern city will burn if blasted by a nuclear weapon.

Figure 1.6 pictures how Russian scientists and civil defense officials realistically described "nuclear winter" - the burning of a city hit by a nuclear weapon. Buildings in the blast zone will be reduced to scattered rubble - mostly of concrete, steel, and other noncombustible materials. They will not burn in blazing fires. Thus in the Oak Ridge National Laboratory translation (ORNL) of a Russian civil defense manual, *Civil Defense: Second Edition* (500,000 copies), Moscow, 1970, by Egorov, Shlyakhov, and Alabin, we learn that "in zones of complete destruction . . . that are characterized by an overpressure exceeding 10 psi, because rubble is scattered and covers the burning structures. As a result the rubble only smolders and does not occur."

Fig. 1.6. Drawing with Caption in a Russian Civil Defense Training Film Strip. The blazing surface burst are shown in standing buildings outside the miles-wide "zone of complete destruction." Outside the zone, blast-hurled "rubble only smolders."

Translation: [Radioactive] contamination occurs in the area of the explosion and also along the edge of the smoke cloud which forms a radioactive track. (nw018.jpg)

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Firestorms destroyed the centers of Hamburg, Dresden, and Tokyo. The old-fashioned buildings contained large amounts of flammable materials, were ignited by many thousands of small fires, and burned as quickly as standing structures well supplied with air. No firestorm has ever injected smoke into the stratosphere, and thus caused appreciable cooling below its smoke cloud.

The theory that smoke from burning cities and forests and dust from nuclear explosions would cause a "nuclear winter" is a myth.

freezing temperatures was conceived in 1982 by the German atmospheric chemist and environmentalist and continues to be promoted by a worldwide propaganda campaign. This well funded campaign included televised scientific -political meetings in Cambridge and Washington featuring American and Soviet scientists. A large number of newspaper and magazine articles followed, including a scaremongering article by Carl Sagan in the 30, 1983 issue of Parade, the Sunday tabloid read by millions. The most influential article was in the November 23, 1983 issue of Science (the weekly magazine of the American Association for the Advancement of Science), "Nuclear winter, global consequences of multiple nuclear explosions," by five scientists,

P. Ackerman, J. B. Pollack, and C. Sagan. Significantly, these activists listed their names as "the taps," the bugle call proclaiming "lights out" or the end of a military funeral.

Until 1985, non-propagandizing scientists did not begin to effectively refute the numerous assumptions, and computer modeling weakness' of the TTAPS and related "nuclear winter" hypothesis. It is that government organizations, private corporations, and most scientists generally avoided political controversies, or making statements likely to enable antinuclear activists to accuse them of war dangers, thus undermining hopes for peace. Stephen Schneider has been called a fascist by some supporters for having written "Nuclear Winter Reappraised," according to the Rocky Mountain News. Three days later, this paper, that until recently featured accounts of unsurvivable "nuclear winter," again defended Thompson and Schneider in its lead editorial, "In Study of Nuclear Winter, Scientists Find Hope." In a free country, truth will out - although sometimes too late to effectively counter the propaganda.

Effective refutation of "nuclear winter" also was delayed by the prestige of politicians and scientists and scientific organizations endorsing the TTAPS forecast of worldwide doom. Full exploration of the TTAPS hypothesis could not be effectively explored until adequate Government funding was available to cover costs of lengthy, expensive studies, including improved computer modeling of interrelated meteorological phenomena.

Serious climatic effects from a Soviet-U.S. nuclear war cannot be completely ruled out. However, the possibility of uncertain climatic effects are a small danger compared to the incalculable millions of deaths from starvation caused by disastrous shortages of essentials of modern agriculture sure to result from American nuclear war, and by the cessation of most international food shipments.

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## Chapter 2

### Psychological Preparations

#### LEARNING WHAT TO EXPECT

The more one knows about the strange and fearful dangers from nuclear weapons and about the weakness' of human beings when confronted with the dangers of war, the better chance one has of surviving. A self-destructive emotion, is almost always the result of unexpected danger. Some people in the world was upon them if they happened to be in an area downwind from surface bursts of nuclear weapons that sucked millions of tons of pulverized earth into the air. They might give up all hope if they saw. People are more likely to endure and survive if they learn in advance that such a disaster, if combined with smoke from great fires, may turn day into night as have some volcanic forest fires.

People also should expect thunder to crash in strange clouds, and the earth to shake. The flickering purples and greens of "artificial auroras" caused by nuclear explosions, especially the earth.

## FEAR

Fear often is a life-saving emotion. When we believe death is close at hand, fear can inspire us to do harder and longer. Driven by fear, we can accomplish feats that would be impossible otherwise. Weak legs, and cold sweat do not mean that a person has become ineffective. Doing hard, nevertheless, are the best ways to keep one's fears under control.

Brave men and women who are self-confident admit their fears, even when the threat of death is present. They plan and work to lessen the causes of their fears. (When the author helped Charles A. Lindbergh build a reinforced-concrete blast shelter for his family and neighbors, Lindbergh frankly admitted his fears of attack and being trapped. He was able to lessen both of these fears by building an excellent escape openings.)

## TERROR

If the danger is unexpected enough or great enough, normal persons sometimes experience terror. Terror prevents the mind from evaluating dangers and thinking logically. It develops in two stages as described by Dr. Walo von Gregerz, a physician with much war experience, in his book *Psychology of War*. The first stage is apathy: people become indifferent to their own safety and are unable even to care for their families. The second stage is a compulsion to flee.

Anxiety, fear, and terror can result in symptoms very similar to those caused by radiation sickness: extreme trembling, diarrhea. Dr. von Gregerz has described terror as being "explosively contagious." Persons who learn to understand the nature of our inherent human traits and behavior and symptoms of terror become terrorized and ineffective in the event of a nuclear attack.

## EMOTIONAL PARALYSIS

The most common reaction to great danger is not terror, but a kind of numbing of the emotions. This may be helpful. Dr. von Gregerz calls this "emotional paralysis." This reaction allows many persons, in the face of great danger, to avoid being overwhelmed by compassionate emotions and horrible sights. They think clearly and act effectively.

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## HELP FROM FELLOW AMERICANS

The atomic explosions that destroyed most of Hiroshima and Nagasaki were air bursts and there was no deadly local fallout. So we cannot be sure how people would behave in areas subjected to both air and surface bursts. However, the reactions of the Japanese survivors are encouraging, especially among them the relative number of horribly burned people was greater than is likely to be in the United States that expects a nuclear attack and takes any sort of shelter. Dr. von Gregerz summarizes: "The Japanese were, of course, apathetic and often incapable of rational action, but open panic or extreme terror occurred only in exceptional cases among the hundreds of thousands of survivors of the two atomic attacks. Also encouraging: "... serious permanent psychological derangements were rare after the at-

as they were after the large-scale conventional bombings."

#### ATOM BOMB SURVIVORS

Some maintain that after an atomic attack America would degenerate into anarchy and every-struggle for existence. They forget the history of great human catastrophes and the self-human beings are capable of displaying. After a massive nuclear attack starvation would afflict America's grain-producing regions still would have an abundance of uncontaminated food. Here Americans in the food-rich areas would help the starving. Like the heroic Russians who drove Leningrad through bursting Nazi bombs and shells. 7 many Americans would risk radiation and truckloads of grain and other necessities to their starving countrymen. Surely, an essential ration for surviving a modern war is a well-founded assurance that many citizens of a state help each other and will work together with little regard for danger and loss.

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### Chapter 3

#### Warnings and Communications

##### IMPORTANCE OF ADEQUATE WARNING

When Hiroshima and Nagasaki were blasted by the first nuclear weapons ever to be used in war, tens of thousands of Japanese killed or injured were inside their numerous air raid shelters caught them by surprise. People are not saved by having shelters nearby unless they receive their shelters and unless they heed that warning.

##### TYPES OF WARNINGS

Warnings are of two types, strategic and tactical.

Strategic warning is based on observed enemy actions that are believed to be preparations for attack. For example, we would have strategic warning if powerful Russian armies were advancing into western Europe. Soviet leaders were threatening massive nuclear destruction if the resisting nations should refuse to clear weapons. With strategic warning being given by news broadcasts and newspapers over a wide area, Americans in areas that are probably targeted would have time to evacuate. Given a day or more, millions of us could build or improve shelters and in other ways improve our chances of survival.

By doing so, we also would help decrease the risk of attack.

Tactical warning of a nuclear attack on the United States would be received by our highest authorities after missiles or other nuclear weapons had been launched against our country. Radar, sophisticated means of detection would begin to feed information into our military warning system. Raw information would have to be evaluated, and top-level decisions would have to be made. The information would have to be transmitted down to communities all over America.

Tactical warning (attack warning) of an out-of-the-blue, Pearl-Harbor-type attack would be received by the average American later than would an attack warning given after recognized strategic warning. The short time (only 15 to 40 minutes) that would elapse between missile launchings and the

on targets in the United States would make it difficult for even an excellent warning system to warn Americans in time for them to reach the best available nearby shelter.

Strengths and weaknesses of the present official warning system are summarized in the following table. Then the life-saving warnings that the first nuclear explosions would give, especially to civilians, are described.

#### OFFICIAL WARNING SYSTEM

The U.S. official warning system is designed to give civilians timely warning by means of radio and television announcements. The National Warning System (NAWAS) is a wire-line network which transmits attack information to official warning points nationwide. NAWAS is not protected against electromagnetic pulse (EMP) effects from nuclear explosions. When the information is received at warning points, officials responsible, they will sound local sirens and initiate radio and TV emergency broadcasts. Officials at NAWAS warning points include many local civil defense directors. NAWAS receives information from constantly improving military warning and communications systems.

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#### SIREN WARNINGS

The Attack Warning Signal is a wavering, wailing sound on the sirens lasting three to five seconds, followed by short blasts on whistles or horns. After a brief pause, it is repeated. This signal means that an enemy attack is imminent. Action- go promptly to the best available shelter. Do not try to telephone for information.

The Attack Warning Signal will be broadcast after you reach shelter. It is Federal policy that the Attack Warning Signal will be broadcast if an enemy attack on the United States has been detected. However, since local authorities may vary, the reader is advised to check the plans in his community before a crisis arises.

The following limitations of attack warnings given by sirens and broadcasting stations should be noted:

- ° Only a relatively small fraction of urban Americans could hear the sirens in the present system. In the past, most urban citizens had evacuated during a crisis.
- ° Except in a crisis threatening the outbreak of nuclear war at any moment, most people who hear the warning signal either would not recognize it or would not believe it was a warning of actual attack.
- ° A coordinated enemy attack may include the detonation of a few submarine-launched ballistic missiles at high altitudes over the United States within a few minutes of the launching of hundreds of intercontinental ballistic missiles (ICBMs). Such high-altitude bursts would produce electromagnetic pulse effects primarily intended to knock out or disrupt U.S. military communications. These EMP effects also could disrupt public power necessary to sound sirens and could put most unprotected broadcasting stations out of service.

Radio warnings and emergency communications to the general public will be broadcast by the Emergency Broadcast System (EBS). This system uses AM broadcasting stations as the primary means to reach the general public. FM and TV stations are included for backup. All stations during a crisis plan to use their assigned frequencies.

EBS stations that are not put off the air by EMP or other effects of early explosions will continue to broadcast siren warnings of a nuclear attack. They will try to give information to listeners in the

and whistles cannot be heard. However, EMP effects on telephones are likely to limit the stations. The functioning EBS stations should be able to warn listeners to seek the best shelter for most of these listeners to reach such shelter before ICBMs begin to explode. Limitations of the Emergency Broadcasting System in February 1986 included the fact that EMP protection had been complete for approximately 2,771 radio stations in the Emergency Broadcast System. One hundred and ten Emergency Operating Centers also had been protected against EMP effects. Many of the protected stations were knocked out by blast; most do not afford their operating personnel fallout protection that would allow broadcasts for long in areas subjected to heavy fallout.

#### WARNINGS GIVEN BY THE ATTACK ITSELF

The great majority of Americans would not be injured by the first explosions of a nuclear attack, the early explosions would give sufficient warning for most people to reach nearby shelters within minutes or more before big intercontinental ballistic missiles (ICBMs) blasted our cities, major industrial areas, most citizens would see the sky lit up to an astounding brightness, would hear the sound of distant explosions, or would note the sudden outage of electric power and most communications. Warnings would result from the explosion of submarine-launched ballistic missiles (SLBMs). Many ICBMs. The SLBM warheads would explode on Strategic Air Command bases and on many civil airports and runways that are long enough to be used by our big bombers. Some naval bases and high-priority communication centers would also be targeted.

The vast majority of Americans do not know how to use these warnings from explosions to help save their lives. Neither are they informed about the probable strategies of an enemy nuclear attack.

One of the first objectives of a coordinated enemy attack would be to destroy our long-range bombers. Each surviving U.S. bomber would be one of our most deadly retaliatory weapons. Once bombers are well away from their runways, they are difficult to destroy.

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To destroy our bombers before they could get away, the first SLBMs would be launched at the first ICBMs would be fired from their silos in Europe and Asia. U.S. surveillance systems would transmit warnings within a very few minutes. Since some enemy submarines would be only a few miles from their targets, some SLBMs would explode on American targets about 15 or 20 minutes before they were hit.

Some SLBMs would strike civilian airport runways that are at least 7000 Ft long. This is the length required by B-52s; there were 210 such runways in the U.S. in 1977. During a crisis, big bombers would destroy many of these long runways, and enemy SLBMs would be likely to target and hit these runways to destroy the maximum number of bombers.

Today most Soviet SLBMs have warheads between 100 kilotons and one megaton. See Jane's Weapons, 1987-88. Within 10 to 15 minutes of the beginning of an attack, runways 7000 feet or longer would be destroyed by airbursts, to destroy U.S. aircraft and airport facilities. Later cratering explosions would destroy long runways, or at least to produce local fallout so heavy that they could not be used for landings and refueling our bombers. Therefore, homes within about 4 miles of a runway at least 7000 feet long would be destroyed before residents receive warning or have time to reach blast shelters away from the runway. Miles away could be lightly damaged by such a warhead, with the blast wave from a 1-megaton warhead about 22 seconds after the warning light. Some windows would be broken 40 miles away. But few citizens would be injured by these early SLBM attacks. These explosions would be life-

ings to most Americans, if they have been properly informed.

Sudden power and communications failures caused by the electromagnetic pulse (EM P) effects also could serve as attack warnings in extensive areas. An EMP is an intense burst of energy generated by a nuclear explosion. The strong, quick-rising surges of electric current induced in power lines and long antennas could burn out most unprotected electrical and electronic equipment. Computers, some motors of modern autos, trucks, and tractors may be put out of commission. Metal bodies are damaged whereas plastic bodies give little.

The usual means of protecting electrical equipment against surges of current produced by lightning is ineffective against EMP. The protective measures are known, but to date all too few civilian buildings are protected against EMP effects. Three or four nuclear weapons skillfully spaced and detonated in the United States would produce EMP effects that might knock out most public power, most radio and television stations lacking special protection against these effects, and most radios connected to power lines. Explosions on or near the ground may produce damaging EMP effects over areas somewhat larger than the blast effects. Such equipment and buildings would be damaged by the blast effects.

#### HOW TO RESPOND TO UNEXPECTED ATTACK WARNINGS

Although a Pearl-Harbor-type of attack is unlikely, citizens should be prepared to respond to unexpected warnings.

These warnings include:

- ° Extremely bright lights -more light than has been seen before. The dazzling, bright light from explosions on targets in many parts of the United States would be seen by most Americans. On the ground, determine the source of light and heat, because there is danger of the viewer's eyes being damaged from a large explosion at distances as far as a hundred miles away, in clear weather. Look for the probable source, and quickly get behind anything that will shield you from most of the intense light. A thermal pulse delivers its heat and light for several seconds- for more than 10 seconds from a 10-megaton surface burst and for approximately 44 seconds if from a 20- megaton surface burst.

If you are at home when you see the amazingly bright light, run out of rooms with windows. Go into a hallway or down into the basement. If you have a shelter close to your house, but separate from the house, the best cover in your home to run outdoors to reach the shelter; wait until about two minutes after the light.

If outdoors when you see the bright light, get behind the best available cover.

It would be impossible to estimate the distance to an explosion from its light or appearance. Run for cover for about two minutes. A blast wave initially travels much faster than the normal speed of sound (about 1 mile in 5 seconds). But by the time its overpressure has decreased to 1 pound per square inch, its thunderous sound have slowed down and are moving only about 3% faster than the normal speed of sound.

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If no blast or sound reaches you in two minutes, you would know that the explosion was over and you would not be hurt by blast effects, unless cut by shattered window glass. After two minutes, get to the best cover in your home and get a radio. Turn the dial to the stations to which you normally listen.



information. Meanwhile, quickly make preparations to go to the best shelter you and your family have. Within a few minutes the probable time interval before the first ICBMs start to explode.

At no time after an attack begins should you look out of a window or stay near a window. Under atmospheric conditions, windows can be shattered by a multi-megaton explosion a hundred miles away.

° The sound of explosions. The thunderous booms of the initial SLBM explosions would be heard in all parts of the United States. Persons one hundred miles away from a nuclear explosion may reach their shelter by hearing it about 7-1/2 minutes later. Most would have time to reach nearby shelter before the explosion.

Loss of electric power and communications. If the lights go out and you find that many radios have suddenly gone off the air, continue to dial if you have a battery-powered radio, and try to find a radio station broadcasting.

#### HOW TO RESPOND TO ATTACK WARNINGS DURING A WORSENING CRISIS

If an attack takes place during a worsening crisis, the effectiveness of warnings would be greatly reduced if the government did not order an evacuation of high-risk areas, millions of Americans would already be in those areas if they had learned that the enemy's urban civilians were evacuating or that tactical nuclear weapons were being employed overseas. Many prudent citizens would sleep inside the best available shelter during most of their waking hours. Many people would have made or improved family car small-group shelters. Many would have supplied them with most essentials. The official warning systems would have been fully tested.

During such a tense crisis period, neighbors or people sheltered near each other should have their own radios and radio stations at all times of the day and night. If the situation worsened or an attack warning was given, they could alert the others.

One disadvantage of waiting to build expedient shelters until there is a crisis is that many people would be outdoors improving their shelters when the first SLBMs are launched. The SLBM warheads would be in the air that the civilian warning systems cannot respond in time. To reduce the risk of being burned or killed by falling debris when expecting an attack should wear shirts, hats, and gloves. They should jump into a ditch or behind a nearby shielding object at the first warning, which may be the sudden cut-off of some radio station.

#### REMAINING INSIDE SHELTER

Curiosity and ignorance probably will cause many people to come out of shelters a few hours after the first warning, if no blast or obvious fallout has endangered their area. This is dangerous, because by then most all missiles have been launched the first enemy bombers may strike. Cities and other areas not spared because missiles malfunctioned or missed are likely to be destroyed by nuclear bomb fallout several days after the first attack.

Most people should stay inside their shelters for at least two or three days, even if they are not the probable target and even if fallout meter readings prove there is no dangerous fallout. Except for the people who would need to improve shelters or move to better shelters. Such persons should take a small risk during the interval between the ICBM explosions and the arrival of enemy bomber strikes. After fallout deposition a few hours, later.

Fallout would cover most of the United States within 12 hours after a massive attack. People should stay in their shelters on information received from distant radio stations regarding changing fallout dangers and not go outside until they could go outside their shelters. Weather conditions such as wind speed would cause fallout to vary with distance. If not forced by thirst or hunger to leave shelter, they should depend on their own resources.

or on radiation measurements made by neighbors or local civil defense workers.

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#### HOW TO KEEP RADIOS OPERATING

Having a radio to receive emergency broadcasts would be a great advantage. The stations that are on the air after an attack would probably be too distant from most survivors to give them reliable information on local, constantly changing fallout dangers. However, both morale and the prospects of long-term survival improved in shelters with a radio bringing word of the large-area fallout situation, food and medical supplies, and what the government and other organizations were doing to help. Radio communication probably can be maintained after an attack if you remember to:

- ° Bring all of your family's battery-powered, portable radios with you to shelter along with you.
- ° Protect AM radios by using only their built-in short loop antennas. The built-in antennas are too short for EMP to induce damaging surges of current in them.
- ° Keep antennas of FM, CB, and amateur radios as short as practical, preferably less than 1/4 wavelength. Induced by EMP, a danger that may continue for weeks after the initial attack because of repeated surges, do not add a wire antenna or connect a short radio antenna to a pipe. Remember that the resulting surge from EMP especially can damage diodes and transistors, thus ending a radio's useful range of reception.
- ° Keep all unshielded radios at least six feet away from any long piece of metal, such as pipes, found in many basements and other shelters. Long metal conductors can pick up and carry large induced current to surge in nearby radios and damage them.
- ° Shield each radio against EMP when not in use by completely surrounding it with conductive material within six feet of a long conductor through which powerful currents produced by EMP might flow. A radio shielded against EMP by placing it inside a metal cake box or metal storage can, or by covering it with aluminum foil or metallic window screen.
- ° Disconnect the antenna cable of your car radio at the receiver or at least ground the antenna by connecting it with a wire to the car frame. Use tape or clothespins to assure good metal-to-metal contact. If an outside mirror is a convenient grounding-point. Park your car as near to your shelter as possible. After fallout has decayed sufficiently you may be able to use the car radio to get distant station reports.
- ° Prevent possible damage to a radio from extreme dampness (which may result from long occupancy in low ground shelters) by keeping it sealed in a clear plastic bag large enough so the radio is not crowded. An additional precaution is to keep a plastic-covered radio in an air-tight container with a layer of wallboard gypsum, as described in Appendix C.
- ° Conserve batteries, because after an attack you may not be able to get replacements for some time. Listen periodically, to the stations you find give the most useful information. The batteries of transistor radios last twice as long if the radios are played at reduced volume.

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#### CHANGED EVACUATION REQUIREMENTS

The most threatening Soviet nuclear warheads in the mid- 1970s were multi-megaton, such as approximately 20 megatons carried by each of over 250 SS- 1 8s. About half of these huge F have hit within a quarter of a mile or less of their intended targets - close enough to de silo. Today's improved Russian warheads have a 50-50 probability of hitting within a few h points. With such accuracy, multi- megaton warheads are not needed to destroy very hard ta in their blast-protective silos.

Soviet strategy continues to stress the destruction of military targets, in order to minim retaliatory strikes. This logical, long-established Soviet strategy is emphasized in numer books, including the three editions of Soviet Military Strategy by Marshall of the Soviet

One result of this logical strategy has been the replacement of huge Soviet warheads by nu much more accurate warheads. In 1990 almost all large missiles have several Multiple Indep try Vehicles (MIRVed) warheads. Soviet warheads - especially the 10 warheads of 500 kiloto SS- 1 8s - could destroy almost all important U.S. fixed military installations, and also control facilities, airport runways longer than 7,000 feet, major seaports, and the factor basis of our military power. (Although an all-out Soviet attack could destroy almost all n them, a first-strike attack is deterred in part by the possibility that most U.S. missiles warning and would be in space, on their trajectories toward Russian targets, before Soviet silos.)

How should your plans either to evacuate during a worsening crisis, or to remain in your h by the dramatic changes in the Soviet nuclear arsenal? Some of these changes are indicated rates information on the dimensions of the stabilized clouds of one megaton and 200 kiloto 6, The Effects of Nuclear Weapons, 1977, and similar information on a 20-megaton cloud der page 20 of The Effects on the Atmosphere qf a Major Nuclear Exchange, by the Committee on Effects of Nuclear Explosions, National Research Council, National Academy Press, Washingt NRC graph is based on Ballistic Missile Organization 83-5 Part 1, dated 29 September 1983, erally available.)

The air bursting of one of the probably few 20-megaton warheads carried by Soviet ICBMs w American homes up to about 16 miles from ground zero. In contrast, the air bursting of an warhead - one of the large warheads in today's Soviet arsenal - would destroy most homes w area having a radius of "only" about 5 miles. So, if you take into consideration the advan their largest ICBMs with several very accurate smaller warheads, each capable of destroyin target, you may logically conclude that unless your home is closer than 10 miles from the need not evacuate to avoid blast and fire dangers.

Y our planning to avoid incapacitating or fatal exposure to fallout radiation will involve your plans to avoid blast and fire dangers. The high altitude winds that carry fallout far blow from west to east. Therefore, in most areas your chances of avoiding extremely danger improved if

Fig. 4. 1 . Stabilized radioactive fallout clouds shown a few minutes after air-burst explosion from Ground Zeros at which the wood frames of typical homes are almost completely collapsed. Surface or near-surface bursts are almost as large, but the distances of blast damage are much smaller. (nw028.tif)

you evacuate westward to an area away from likely nearby targets. However, since no one can predict in what directions future winds will blow, your plans to remain where you live, or your plans to include building, improving, or utilizing high-protection-factor shelter, as explained in

If you live near a target the destruction of which has high priority in Soviet war-winning strategy, so ago it quite likely was targeted by a 20-MT warhead. Fig. 4. 1 shows the awesome size of the cloud from a 20-MT air burst. This cloud would expand in minutes to this huge size in the sky. It would contain only extremely small particles almost all of which would remain airborne for hours. The result in no fallout deposition that would promptly incapacitate exposed people.

A 20-MT surface burst or near-surface burst would produce a stabilized radioactive cloud expanding in all directions from GZ as would a 20-MT air burst. Its tremendous fireball would "suck up" pulverized rock and would contaminate those particles with its radioactive material. Fallout would fall from the stabilized cloud to the ground in minutes. Very heavy fallout could be expected upwind from such a 20-MT explosion, with heavy fallout, capable of causing fatalities with downwind for several hundred miles.

A 1 -MT surface burst, Fig. 4.1. would produce a stabilized fallout cloud unlikely to result in the upwind or crosswind directions from GZ beyond the range of the explosion's home-destructive radius. Clearly, the risk of your being endangered by very heavy fallout if you remain 6 miles from the burst, and happen to be upwind or crosswind from GZ, is less than the risk you would have had if you had stayed 18 miles upwind or crosswind from the same target, which had been destroyed by a near-surface burst.

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#### HIGHEST-RISK AND HIGH-RISK AREAS

Highest-risk areas are those in which buildings are likely to be destroyed by blast and/or fire. In the open for the first two weeks after fallout deposition would receive a total radiation dose of 1000 rads or more. The largest highest-risk areas would be those within our five Minuteman missile fields and around them, and for up to about 150 miles downwind. These huge highest-risk areas are indicated by the largest black fallout patterns on Fig. 4.2.

Fig. 4.2 is an outdated, computer-drawn fallout map based on a multi-megaton attack considered likely some time ago. (An updated, unclassified fallout map of the United States, showing radiation doses to be available.) This outdated attack included 113 surface bursts of 20 megatons each on urban targets. This is an unlikely assumption similar to those used in making some official civil defense risk-area maps. Employing all surface bursts makes little sense to the military. The same weapons would destroy most military installations, as well as factories and other urban targets over approximately twice as large an area.

As will be explained later, to survive in such areas people would have to stay inside very good shelters for weeks, or, after two weeks or more, leave very good shelters and drive in a few hours to areas free of fallout dangers. A "very good" fallout shelter is one that reduces the radiation dose received

than 1 /200th of the dose they would have received outdoors during the same period. If the were 20,000 R, such a shelter with a protection factor of 200 (PF 200) would prevent each dose greater than 100 R – not enough to incapacitate. Even a completely below ground home ly improved as described in Chapter 5, would give entirely inadequate protection.

High-risk fallout areas are those where the two- week dose outdoors is between 5,000 and 1 areas, good fallout shelters would be essential, supplied at least with adequate water and Furthermore, survivors would have to remain inside shelters for most of each day for sever

The radiation dangers in the shaded areas of the map are shown decreasing as the distances increase. This generally is the case, although sometimes rain or snow' carries radioactive ducing "rainouts" of exceptionally heavy fallout farther downwind. Furthermore, this compu Oak Ridge National Laboratory does not indicate the very dangerous fallout near the isolat

the most dangerous fallout would be carried by high- altitude winds that usually blow from fied fallout patterns as those shown should be used only as rough guides to help improve c probable blast area or very heavy fallout area and going to a less dangerous area. Wind di an enemy's targeting can be unexpected; weapons can miss. A prudent citizen, no matter whe build a shelter that gives excellent protection against fallout radiation.

A major disadvantage of all types of risk-area maps is the fact that poorly informed peopl them and conclude that if they are outside a mapped risk area, they are relatively safe fr ly fallout dangers.

Another reason for not placing much reliance on risk-area maps like Fig. 4.2 is that such able in 1986 are based on the largest attacks considered possible a decade ago. In 1986 th are much smaller, their numbers are much larger, and their total megatonage and cap abilit about the same as 10 years ago.

The outdated attack scenario used in producing Fig. 4.2 also involved the surface bursting heads totaling 3, 190 megatons on military targets, including over 2,000 megatons logical Minuteman missile fields. Such an attack on our missile fields would produce about the sam shown in Fig. 4.2. Today, however, heavy fallout from our missile fields would extend some downwind, because of the lower heights of the stabilized radioactive clouds from one-megat and near-surface bursts, as compared to those of multi-megaton warheads that would have be ago, at a time when a 20-megaton warhead was typical of the Soviet nuclear ICBM arsenal.

In 1986 hundreds of targets besides those indicated in Fig. 4.2 might be hit, but the tota subjected to lethal fallout probably would be less than is shown in Fig. 4.2. To maximize and fire, most targets in urban and/or industrial areas would be attacked with air bursts, no promptly lethal or incapacitating fallout – except perhaps in scattered "hot spots" whe could bring huge numbers of tiny, very radioactive particles to earth within hours after t kiloton-range Soviet warheads. And since most Americans live far away from "hard" targets missile silos, downwind from which extremely heavy fallout is likely – most of us living i ably would be endangered primarily by blast and fire, not fallout, in the event of a Soviet

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Fig. 4.2 Simplified, outdated fallout patterns showing total radiation doses that would be

on the surface and in the open for the entire 14 days following the surface bursting of 50 targets indicated, if the winds at all elevations blew continuously from the west at 25 mph.

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#### WHETHER TO EVACUATE

Let's assume that Russian cities were being evacuated, or that tactical nuclear weapons were used in what had been an overseas conventional war involving the United States, and that such a war would be fought in such a way that Americans could improve their survival chances by getting out of the highest-risk and high

U.S. capabilities for war-crisis evacuation are poor and tending to worsen. Several years ago, approximately 3,100 evacuation plans required nationwide, about 1,500 had been made, and these covered only one-third of Americans living in risk areas. By 1986 some cities and states had abandoned their plans; most still have plans that would save millions if ordered in time during a crisis to be completed before the attack. Who would order an evacuation under threat of attack, and under what conditions? These remain unanswered questions. Furthermore, compulsory evacuation during a war crisis was not an official American evacuation plan. So, if you believe that a nuclear attack on the United States would improve your chances of surviving, then well before a desperate crisis arises you had better get out to improve your and your family's survival chances at or near where you live, or plan and

Spontaneous evacuations, in which Americans would make their own decisions without the aid of official recommendations, probably would occur during a worsening war crisis. Traffic jams and

are less likely to occur if citizens start leaving high-risk areas on their own, over a period of several days, rather than if almost everyone, on receiving recommendations from officials, at once evacuates. (Spontaneous evacuation by Gulf Coast residents, begun under threat of an approaching hurricane, lessened subsequent traffic problems in the evacuations ordered or recommended by officials.)

Except in areas where the local civil defense war-crisis evacuation plans are well developed, evacuation farther than 10 miles from the nearest probable separate target probably can best improve survival in the event of a nuclear attack by preparing to remain at or near their homes and there to make or improve their defenses. Exceptions include those living in the vicinity of targets of great military importance to the United States, such as missile fields, on which many warheads would be surface or near-surface bursted, producing explosions that could extend up to 150 miles downwind. Americans living in these greatly endangered areas would do well to stick with the local official civil defense evacuation plans, at least regarding directions. Areas far from such targets are likely to be endangered by heavy fallout.

Nuclear submarine ports, Strategic Air Command bases, and Air Force installations with long runways would be destroyed by even a limited Soviet counterforce or disarming attack. These prime strategic targets could be blasted by Submarine Launched Ballistic Missiles (SLBMs) in the first 15 or 20 minutes of a war. SLBMs are not as accurate as ICBM warheads, and air bursts can destroy bombers and submarines in large areas as if these same weapons are exploded at or near the surface. Therefore, Strategic Air Command bases would be air bursted on these prime "soft" targets, with little or no local fallout. (In a war, long runways are likely to be cratered by accurate ICBM warheads and by bombs, to make sure that aircraft carriers could not use them.)

On the following page are listed considerations, favorable and unfavorable, to evacuation. These may help you and your family make a more logical decision regarding evacuation:

#### Evacuation Factors Table:

##### Favorable to Evacuation:

You live in a highest-risk or high-risk area.

You have transportation (this means a car and enough gasoline), and roads are open to a considerably lower-risk area.

You are in fairly good health or can evacuate with someone capable of taking care of you.

Your work is not of the kind that your community depends on (such as a policeman, fireman, or telephone operator).

You have some tools with which to build or improve a fallout shelter. You also have water containers, food, clothing, etc., adequate for life in the area to which you would go.

##### Unfavorable to Evacuation:

You live outside a highest-risk or high-risk area and could build an expedient fallout shelter and make other survival preparations where you live.

You have no means of transportation or you believe that roads are likely to be blocked by the time you make your decision.

You are sick, decrepit, or lack the will to try to survive if things get tough.

You cannot suddenly leave your home area for several days without hurting others.

You lack the tools, etc., that would be helpful - but not necessarily essential - to successful evacuation.

Instructions for building expedient fallout and blast shelters and for making expedient li given in following chapters. The reader is advised to study all of this book carefully bef ing basic survival action.

#### THE NEED FOR AN EVACUATION CHECKLIST

A good flyer, no matter how many years he has flown, runs through a checklist covering his off. Similarly, a citizen preparing under crisis pressures to do something he has never do use a checklist to be sure that he takes with him the most useful of his available possess

A family planning to use an expedient shelter or basement at or near home also should use list on the following page to make sure needed survival items are not overlooked.

The family of six pictured in Fig. 4.3 used the Evacuation Checklist given below to select could be carried in and on their small car. They assembled categories of items in separate items to take with them from each pile. They were able to leave their home 76 minutes after Checklist. (Following chapters of this book include descriptions of this family's success Covered Trench Shelter, and living in it continuously for 77 hours.)

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## EVACUATION CHECKLIST

(Includes items for building or improving shelters)

Loading Procedure: Make separate piles for each category (except categories 1 and 5). Then some items from each category, taking as much as can be safely carried and being careful to passengers.

### A. THE MOST NEEDED ITEMS

Category 1 . Survival Information: Shelter building and other nuclear survival instruction small battery-powered radios and extra batteries, a fallout meter such as a homemade KFM (C), and writing materials.

Category 2. Tools: Shovel, pick, saw (a bow- saw is best), ax or hatchet, file, knife, pliers specified in the building instructions for the shelter planned. Also take work gloves.

Category 3. Shelter-Building Materials: Rain- proofing materials (plastic, shower curtains specified in the instructions for the type of shelter planned Also, unless the weather is shelter- ventilating pump such as a KAP, or the materials to build one (see Appendix B ).

Category 4. Water : Small, filled containers plus all available large polyethylene trash pillow cases, water-purifying material such as Clorox, and a teaspoon for measuring.

Category 5. Peacetime valuables: Money, credit cards, negotiable securities, valuable jewelry the most important documents kept at home. (Evacuation may be followed not by nuclear war, continuing unstable nuclear peace.)

Category 6. Light: Flashlights, candles, materials to improvise cooking-oil lamps (2 clear size, cooking oil, cotton string for wicks (see Chapter 11, Light), kitchen matches, and storing matches.

Category 7. Clothing: Cold-weather boots, overshoes, and warm outdoor clothing (even in su



attack these would be unobtainable), raincoats and ponchos. Wear work clothes and work shoes.

Category 8. Sleeping Gear: A compact sleeping bag or two blankets per person.

Category 9. Food: Food for babies (including milk powder, cooking oil, and sugar) has the

Compact foods that require no cooking are preferred. Include at least one pound of salt, a can and bottle opener, a knife, and 2 cooking pots with lids (4-qt size preferred). For each person, a bowl, and large spoon. Also, a bucket stove, or minimum materials for making a bucket stove. 10 all- wire coat hangers, a nail, and a cold chisel or screwdriver (see Chapter 9, Food).

Category 10. Sanitation Items: Plastic film or plastic bags in which to collect and contain waste; plastic container for urine; toilet paper, tampons, diapers, and soap.

Category 11. Medical Items: Aspirin, a first-aid kit, all available antibiotics and disinfectants (if essential to a member of the family), potassium iodide (for protection against radiation, see Chapter 13), spare eyeglasses, and contact lenses.

Category 12. Miscellaneous: Two square yards of mosquito netting or insect screen with which to cover shelter openings if insects are a problem, insect repellents, a favorite book or two.

B. SOME USEFUL ITEMS (To take if car space is available):

1. Additional tools.

2. A tent, a small camp stove, and some additional kitchen utensils.

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#### EVACUATING BY CAR

The small car shown in Fig. 4.3 was skillfully loaded for a safe evacuation trip. To make room, the back seat was left at home. The load on top of the car included blankets, a small rug, and springy materials which kept the load from becoming compacted and working loose under the weight. The load was tightened around it. The two loopended ropes went over the load and around the top of the car, and the closed doors.

#### USING MUSCLE POWER

Flazards of evacuation would include highways blocked by wrecks and stalled vehicles. If help were provided, the muscle power of people usually could quickly clear a highway. During the evacuation before advancing Japanese armies in World War II. I observed Chinese, using only their own muscle power, clear a mountain road of wrecks and other obstructions. Americans can do the same, if some one shows them how they can do it, as proved by a wintertime episode on Monarch Pass over the Continental Divide. For 100 vehicles were held up after a large wrecking truck overturned on the icy highway. The road was closed until I told them how the Chinese handled such a situation. The patrolmen then called among the delayed motorists to lift the overturned truck back onto its wheels, and, less than an hour later, they had combined their muscle power and opened Monarch Pass to traffic.

Citizens should take direct action to keep traffic moving during a crisis evacuation.

## MAKING AN EXPEDIENT OR PERMANENT SHELTER INSTEAD OF EVACUATING

Millions of Americans have homes within very large urban-industrial areas, probably not all subjected to blast and fire dangers. Many, whose homes are in the suburbs or adjacent town areas, could logically decide not to evacuate, but to build earth-covered shelters at or near their homes and supply them with life-support essentials.

Fig. 4.3. Six members of a Utah family arriving at a rural shelter-building site 64 miles from their home (nw034.jpg)

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Likewise, people living even as close as 5 miles from an isolated probable target may decide to stay near their supplies, rather than to evacuate. This is a good idea, provided that (1) their homes are far enough from probable aiming points to make such shelters practical, and (2) enough time, space, and resources are available.

The photo (Fig. 4.4) shows a family with no adult male that built an expedient shelter that provided better protection from fallout, blast, and fire than almost any home. They succeeded, despite the necessary winter days with snow flurries. The top two inches of earth were frozen and the next two feet had to be loosened with their dull pick. No member of this family had done any serious digging. The shelter that would have given about 100 times as much protection against fallout radiation as a frame house and at least 25 times as much as a typical home basement.

(Fallout shelters are designed for protection against radiation from fallout particles. Although they have blast doors and other means for keeping out blast, the better types would prevent their occupants from being killed by blast effects in extensive areas where people in houses would have little chance of survival. The term "expedient shelter" generally means an expedient fallout shelter.)

Even as simple an earth-covered fallout shelter as this Door-Covered Trench Shelter, if built in areas with flammable buildings, usually would save its occupants' lives in extensive areas devastated by fire. The area of probable survival in a good earth covered fallout shelter would extend from where the shelter is located but fires likely to be numerous, inward toward GZ to where most homes would be collapsed by fire. This ring-shaped area of probable survival from blast and/or fire effects of a 1-MT air burst would be from about 8 miles from GZ inward to approximately 5.5 miles. Its area would be about 105 square miles, or about the 95 square miles in the circular area with a radius of 5.5 miles centered on GZ and within which most of the other types of earth-covered expedient shelters described in this book have been tested. (Door-Covered Trench Shelter with 2 feet of earth on its roof, 34 hours after receiving the building instructions. Test explosions conducted by the Defense Nuclear Agency.)

Fig. 4.4. This family completed their Protection Factor 200 (PF 200) fallout shelter, a Door-Covered Trench Shelter with 2 feet of earth on its roof, 34 hours after receiving the building instructions.

In many areas, this and even better types of expedient fallout shelters affording considerable protection could be built by untrained families, following the written, field-tested instructions in this book (see Appendix D, Expedient Blast Shelters) within a few days a small but significant fraction of the population could have expedient blast shelters complete with expedient blast doors and providing at least 15-psf protection.

## Chapter 5

### Shelter, the Greatest Need

#### ADEQUATE SHELTER

To improve your chances of surviving a nuclear attack, your primary need would be an adequate shelter for many days of occupancy. A shelter that affords good protection against fallout radiation is adequate in more than 95% of the area of the United States. However, even in almost all areas, protection against blast and fire during a massive nuclear attack, the fallout protection provided by most existing shelters is inadequate if the winds blew from the wrong direction during the time of fallout deposition.

To remain in or near cities or other probable target areas, one would need better protection against fallout than is provided by most shelters in buildings. Blast tests have proved that the existing shelters described in this book can survive blast effects severe enough to demolish most houses.

This chapter is concerned primarily with expedient shelters that give excellent protection against fallout. These earth-covered fallout shelters could be built in 48 hours or less by tens of millions of people using the tested, written instructions. 8 Expedient blast shelters are discussed in Appendix D. The design features needed for effective blast protection require more work, materials, and skilled labor than expedient fallout shelters.

If average Americans are to do their best when building expedient shelters and life-support equipment for themselves, they need detailed information about what to do and about why it is to their advantage. Most people accustomed to blindly following orders. Unfortunately, during a crisis threatening the nation, it is long to read instructions explaining why each important feature was designed as specified. Detailed instructions are included in the step-by-step, illustrated instructions given in Appendix A for building expedient shelters during a crisis.

In this chapter, reasons will be given for designing a Pole-Covered Trench Shelter as specified in the National Laboratory instructions given in Appendix A.2. The two pages of drawings and plans in Appendix A.2 show the parts of this shelter, -except for the essential shelter- ventilating trench. The following account of how an urban family, after evacuating, used these instructions in less than 36 hours also includes explanations of various radiation dangers and of simple methods of protection against these dangers.

This family, like scores of other families recruited to build shelters or life-support equipment for about equivalent to laborers' wages if its members completed the experiment within a specified time, began the moment the family received the written, illustrated instructions preparatory to the experiment mentioned in the preceding chapter. Like the other test families, this family was paid for all expenses. They all worked hard and completed the project in half the specified time, it was paid a cash bonus. The workers were guided only by the written instructions, which were improved after each successful experiment.

The successful outcome of almost all the shelter building experiments indicates that tens of millions of people in a nuclear war crisis would work hard and successfully to build earth-covered expedient shelters.

them better protection against fallout, blast, and fire than would all but a very small fraction. However, this belief is dependent on two conditions: (1) that in a desperate, worsening crisis there would be strong, motivating leadership; and (2) that Americans would have received well-thought-out building and other practical, tested survival instructions.

#### SHELTER AGAINST RADIATION

The family previously pictured evacuating by car (Fig. 4.3) drove 64 miles to build a shelter (Fig. 5.1). Although the August sun was very hot in this irrigated Utah valley, the family moved to the

shade of nearby trees. To avoid digging through roots, they carried the poles about 150 feet to the edge of the cornfield.

The father and the oldest son did most of the work of making the shelter. The mother and sons had no experience with such problems; the two youngest children were not accustomed to work.

The family followed an earlier version of the plans and instructions given in Appendix A for a Pole-Covered Trench Shelter. Because the earth was firm and stable, the trenches were dug with straight walls. If the earth had been less stable, it would have been necessary to slope the walls increasing the depth of the trench from 3-1/2 to 5 feet.

Before placing the roof poles, the workers assured themselves a more comfortable shelter by building up the walls. They had brought a large number of the plastic garbage bags required in their home for trash; they opened the bags to make wall coverings. Bed sheets or other cloth could have been used.

The room of this 6-person shelter was 3-1/2 feet wide, 4-1/2 feet high, and 16-1/2 feet long. A small hole was dug at one end, so each tall occupant could stand up and stretch several times a day.

The trenches for entry and emergency exit were dug only 22 inches wide, to minimize radiation entering through these openings. One wall of these two narrow trenches was an extension of the main room, as shown in Fig. 5.1.

Fig. 5.1. Placing 9-foot poles for the roof of a Pole-Covered Trench Shelter, (nw037.jpg)

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The family sat and slept along the left wall, to be better shielded from radiation coming from the right.

This shelter was designed so that its main trench could be enlarged to make a much more livable room without disturbing its completed roof. For this reason, the 9-foot roofing poles were placed off-center, resting on the ground to the right of the main room.

Whenever practical, expedient shelters should be built so that they can be readily enlarged into permanent living quarters. After it becomes safe to emerge for limited periods, occupants can spend part of their waking time in such a rainproof dugout that affords excellent protection against weather, living in a dugout like this is more comfortable than living in a tent or shack. When the radiation rate outdoors has decayed to less than about 2 R per hour, the small vertical entry could be replaced by a steeply inclined stairway.

The importance of giving inexperienced shelter builders detailed instructions is illustrated by the family's experience.

done by the young women shown in Fig. 5.2. They had agreed to try to build a Pole-Covered ing unassisted and using only hand -tools. Because the summer sun in Utah was hot, they se a large tree. The brief instructions they received included no advice on the selection of ging out the numerous roots was very difficult for them and required several of the 22 hou working.

Another disadvantage of making a shelter under trees is that more of the gamma rays from f leaves and branches would reach and penetrate the shelter than if these same particles wer gamma rays from fallout particles on the ground would be scattered or absorbed by striking trunks, or houses before reaching a below ground shelter.

Fig. 5.2. Two non-athletic college girls who completed a 4-person Pole-Covered Trench Shel 35-1/2 hours, despite tree roots, (nw038.jpg)

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#### TYPES OF SHIELDING

Shelters provide protection against radiation by utilizing two types of shielding: barrier shielding.

° Barrier shielding is shown by Fig. 5.3, a simplified illustration. (In a real fallout ar would have fallout particles all over and around him.) The 3 -foot thickness of earth show concrete) will provide an effective barrier, attenuating (absorbing) about 99.9%, of all c the illustration, only a single fallout particle 3 feet from the edge of the trench is cor of 1000 could penetrate the 3 feet of earth shown and strike the person in the trench. Ray away than 3 feet would be negligible; rays from particles closer than 3 feet would be atte thickness of earth between the fallout particle and the man in the trench.

However, the man in the trench would not be protected from "skyshine," which is caused by ing after striking the nitrogen, oxygen, and other atoms of the air. The man's exposed hea level, would be hit by about one-tenth as many gamma rays as if it were 3 feet above groun lout could be kept out of the trench and off the man and every part of the ground within 3 trench, skyshine from heavy fallout on the surrounding ground could deliver a fatal radiat open trench.

Skyshine reaches the ground from all directions. If the man were sitting in a deeper tren of this scattered radiation, but not all of it. For good protection he must be protected c rier shielding.

The barrier shielding of the Pole-Covered Trench Shelter shown in Fig. 5.4 was increased b earth onto its "buried roof." After father and son had mounded earth about 1 8 inches deep roof poles, a large piece of 4-mil-thick polyethylene was placed over the mound. This wate "buried roof" after it was covered with more earth. Any rainwater trickling through the ea have run off the sloping sides of the "buried roof" and away from the shelter.

Geometry shielding reduces the radiation dose received by shelter occupants by increasing tween them and fallout particles, and by

Fig. 5.3. Simplified illustration of barrier shielding and skyshine (scattered gamma radiation provides poor protection. ORNL.DWG 78.7205 (nw039.tif)

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Fig. 5.4. Increasing the barrier shielding over a Pole-Covered Trench Shelter, (nw040.jpg)

providing turns in the openings leading into the shelter. Figure 5 .5 is a sectional drawing of the Utah family.

The farther you can keep away from a source either of light or of harmful radiation, the less radiation will reach you. If fallout particles are on the roof of a tall building and you are much smaller radiation dose from those particles than if they were on the floor just above you. If visible light or gamma rays are coming through an opening at the far end of a passageway, less radiation enters at the other end if the passageway is long than if it is short.

Turns in passageways are very effective in reducing the amount of radiation entering a shelter. A right-angle turn, either from a vertical or horizontal entry, causes a reduction of about 90%.

Note: Fallout shelters need not provide additional shielding to protect occupants against radiation that is emitted from the fireballs of nuclear explosions. (See Figs. 1 . 1 and 1 .4.) Large fallout shelters were employed in an attack on the United States. The initial nuclear radiation from the sizes of the explosions that would be great enough to destroy the United States would be greatly reduced in passing through the miles of air between the fireballs and the shelters far enough away to survive the blast effects. The smaller an explosion, the larger the radiation dose it delivers at a given blast overpressure distance from ground zero. (For a discussion of the shielding requirements of blast shelters that would enable occupants to survive blast effects, see Chapter 6 and Appendix B and therefore would be subjected to much larger exposures of initial nuclear radiation, see Chapter 6 and Appendix B.)

Figure 5.6 shows the completed shelter after it was occupied by the family of six just 32% of the way through the shelter-building instructions and beginning preparations to evacuate. (This family worked on the shelter for 36 hours and also a larger bonus given if all members then stayed inside continuously for 36 hours.)

get a better idea of how six people can live in such a small shelter, look at the drawings of the shelter.

In warm or hot weather, shelters, especially crowded ones, must be well ventilated and cooled. The volume of outdoor air pumped through them. This family had built an efficient homemade air pump as described in Chapter 6 and Appendix B .

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Fig. 5.5. Skyshine coming into a shelter through a vertical entry would be mostly absorbed by the earth and traveling down the entryway trench. ORNL.DWG 78-7204 (nw041a.tif)

Fig. 5.6. Earth mounded over a 3 foot-wide Pole-Covered Trench Shelter. The canvas canopy would protect the vertical entry against both fallout and rain. (A smaller canopy over the air entrance at the other end is obscured by the mounded earth.) (nw041b.jpg)

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All of the earth excavated in digging the trenches was mounded over the roof poles, making deep. This shelter had a protection factor (PF) of over 300; that is, persons inside would receive the gamma-ray dose of fallout radiation that they would receive if they were standing outside.

To have made the roof covering more than 36 inches thick would not have increased the protection very much, unless the entry trench and the air duct-emergency exit trench had been covered. Field tests have shown that some families, given only 48 hours, cannot dig the longer trenches and shovel on the additional earth necessary for a shelter that would offer significantly better protection than shown here. The Pole-Covered Trench Shelter and the other shelters described in Appendix B can be dug by untrained families within 48 hours, the minimum time assumed to be available to Americans in case of attack if the Russians should begin to evacuate their cities.

#### EARTH ARCHING USED TO STRENGTHEN SHELTERS

Several types of expedient shelters can be made to withstand greater pressures if their roofs are covered with enough earth to attain "earth arching." This arching results when the earth causes part of the load carried by the roof to be shifted to the overlying earth particles in such a way that an arch is formed. This arch carries the load to surrounding supports that supports often include adjacent earth that has not been disturbed.

To attain earth arching, the earth covering the yielding roof must be at least as deep as the distance between its supports. Then the resultant earth arch above the roof carries most of the load.

(A familiar example of effective earth arching is its use with sheet metal culverts under roads. A few feet of earth over a thin-walled culvert prevents it from being crushed by the weight of the road above.)

Figure 5.7 shows how a flexible roof yields under the weight of 30 inches of earth mounded over it. Earth arching develops. After the arch is formed, the only weight that the yielding roof supports is the thickness of earth between the roof and the bottom of the arch.

Protective earth arching also results if a shelter is covered with a material that compresses. The whole roof or the whole shelter can be pushed down a little without being broken.

Fig. 5.7. Earth arching over a yielding roof enables a shelter to withstand much greater pressure. (ORNL.DWG 78.7441 (nw042.tif))

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#### SHELTER AGAINST BETA AND ALPHA PARTICLES

In addition to the invisible, light-like gamma rays, fallout particles radiate two types of particles: beta and alpha particles. These radiations would be minor dangers to informed people, but they are a danger to those who had entered almost any kind of shelter before the fallout began to be deposited.

° Beta particles are high-speed electrons given off by some of the radioactive atoms in fallout. High-energy beta particles can penetrate more than about 10 feet of air or about 1/8 inch of tissue. Any building that keeps out fallout particles will prevent injury from beta radiation.

The only frequently serious dangers are from (1) internal beta-radiation doses from fallout or drink, and (2) beta burns from fresh fallout particles. Fresh fallout particles are not therefore very radioactive. If fresh particles remain for at least several tens of minutes beta burns are likely to result. If only thin clothing separates fresh fallout particles from the skin, a short time will elapse before their radiation causes beta burns.

In dry, windy weather, fresh fallout particles might get inside one's nose and ears, along with the skin. This could cause beta burns if not promptly washed off or otherwise removed.

Prompt washing will prevent beta burns. If water is not available, brushing and rubbing the skin will help.

If a person is exposed outdoors where there is heavy, fresh fallout for a long enough time of gamma radiation, the highest-energy beta radiation given off by fresh fallout particles presents a relatively minor danger to his eyes and skin. Even ordinary glasses give good protection to the eyes from gamma radiation, and ordinary clothing gives good protection to the skin.

Ordinary clothing will shield and protect the body quite well from all but the highest-energy beta radiation by fresh fallout deposited on the clothing. Fallout-contaminated clothing should be removed and at least brushed and beaten before entering a shelter room, to rid it of as many fallout particles as possible. Particles that are many days old will not cause beta burns unless large quantities are on the skin.

Most of the knowledge about beta burns on human skin was gathered as a result of an accident at the U.S. H-bomb test in the tropical Pacific. The wind blew the fallout in a direction not anticipated. Five hours after the multimegaton surface burst, some natives of the Marshall Islands noticed a white dust being deposited on everything exposed, including their bare, moist skin. Unknown to them, the dust was fresh fallout. (Most fallout is sand-like, but fallout from bursts that have cratered coral reefs and limestone, is powdery or flaky, and white.) Since the natives knew nothing about fallout, the white dust was at first thought of as ashes from a distant volcanic eruption. For two days, until they were removed and cared for by doctors, they paid practically no attention to the white dust. Living in unconstructed homes, they received from the fallout all around them a calculated gamma-ray dose of about 100 rads in two days they were exposed.

The children played in the fallout-contaminated sand. The fallout on these islanders' scalps and tops of their bare feet caused itching and burning sensations after a time. Days later, began to appear extreme discoloration of the skin. Beta burns are not deep burns; however, if not treated with proper medical attention, developed into ulcers. (No serious permanent skin injury resulted.)

For survivors confined inside crowded, unsanitary shelters by heavy fallout, and without means of escape, the problem could be a worse problem than were similar burns to the Marshall Islanders.

All of the Marshall Islanders unknowingly ate fallout-contaminated food and drank fallout-contaminated water for two days. Mainly as a result of this, radioactive iodine was concentrated in their thyroids, and thyroid malignancies developed years later. (There is a simple, very low-cost means of attaining almost complete immunity to this delayed hazard: taking minute prophylactic doses of a salt, potassium iodide. This will prevent the thyroid from concentrating radioactive iodine.)

In dry, dusty, windy areas the human nasal passages usually filter out much dust. A large amount of dust may be hazardous if the dust is contaminated with fallout. Under such dry, windy conditions, a nosebleed may be caused by large amounts of dust lodged inside the nasal passages. Breathing through a cloth mask will help.



would give good protection against this localized hazard. In conclusion: persons under considerable effort to protect themselves from beta radiation.

° Alpha particles, identical to the nuclei of helium atoms, are given off by some of the radioactive elements. These particles have very little penetrating power: 1 to 3 inches of air will stop them. They can get through unbroken skin; they cannot penetrate even a thin fabric. Alpha particles from sources that emit them (such as the radioactive element plutonium) enter the body and are retained in other parts of the body. Any shelter that excludes fallout particles affords excellent protection. Unless survivors eat or drink fallout-contaminated food or water in considerably large quantities, completely uninformed natives of the Marshall Islands, danger from alpha particles would be minimal.

#### PROTECTION AGAINST OTHER NUCLEAR WEAPONS EFFECTS

° Flash burns are caused by the intense rays of heat emitted from the fireball within the explosion. This thermal radiation travels at the speed of light and starts to heat or burn before the arrival of the blast wave. Thermal radiation is reduced but not eliminated if it is blocked by clouds, or thick smoke. On a clear day, serious flash burns on a person's exposed skin can occur from an explosion that is 25 miles away.

A covering of clothing preferably of white cloth that reflects light can reduce or prevent burns. If people are in a large part of an area in which thermal radiation is a hazard. However, in areas close to the explosion for severe blast damage, the clothing of exposed people could be set on fire and their bodies burned.

° Fires ignited by thermal radiation and those resulting from blast and other causes especially in people pinned down by fallout while in or near flammable buildings. Protective measures against fires, from fire, carbon monoxide, and toxic smokes are discussed in Chapter 7.

Flash blindness can be caused by the intense light from an explosion tens of miles away in the open. Although very disturbing, the blindness is not permanent; most victims recover within seconds. At Hiroshima and Nagasaki survivors (people who had been in the open more than persons expected to be), there were a number of instances of temporary blindness that lasted as long as 48 hours. In one case of permanent retinal injury was reported.

Flash blindness may be produced by scattered light; the victim of this temporary affliction is not looking directly at the fireball. Flash blindness would be more severe at night, when the pupils are dilated. Permanent injury, can result at great distances if the eye is focused on the fireball.

People inside any shelter with no openings through which light can shine directly would be protected from burns and eye damage. Persons in the open with adequate warning of a nuclear explosion can protect themselves from both flash blindness and retinal burns by closing or shielding their eyes. They should get into a shadow quickly.

#### SKIN BURNS FROM HEATED DUST (THE POPCORNING EFFECT)

When exposed grains of sand and particles of earth are heated very rapidly by intense thermal radiation, they explode like popcorn and pop up into the air. While this dust is airborne, the continuing high temperatures that may be as high as several thousand degrees Fahrenheit on a clear day in areas near the explosion.

the shock wave and blast winds arrive and can carry the burning-hot air and dust into an open shelter. Open shelters have been singed and seriously burned in some of the nuclear air-burst tests.

Thus Japanese working inside an open tunnel-shelter at Nagasaki within about 100 yards of the blast were burned on the portion of their skin that was exposed to the entering blast wind, even though they made one or two turns in the tunnel. 4. 4 (None of these Japanese workers who survived the blast suffered serious radiation injuries, which they certainly would have suffered had they been subjected to the thermal pulse and the intense initial nuclear radiation from the fireball.)

Experiments conducted during several nuclear test explosions have established the amount of radiation that must be delivered to exposed earth to produce the popping effect. 6 Large air bursts have been burned by hot dust and heated air produced at overpressure ranges as low as 3 or 4 psi.

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However, calculations indicate that the large surface bursts most likely to endanger American occupants of small, open shelters being burned by these effects except at somewhat higher overpressures.

Protection is simple against the heated dust and very hot air that may be blown into an open shelter. When expecting an attack, occupants of an open shelter should keep towels or other cloths handy. Against the bright light from an explosion, they should cover their heads and exposed skin. If time is available, much better protection is given by making expedient blast doors, as described in Appendix B. Against the very bright light from a large explosion miles away, they can close and secure such doors before the blast wave several seconds later.

#### ESSENTIAL LIFE-SUPPORT EQUIPMENT

Shelters can be built to give excellent protection against all nuclear weapon effects, except in areas very close to cratered areas. But most shelters would be of little use in areas of heavy fallout without enough life-support equipment to enable occupants to stay in the shelters until conditions improve. In heavy fallout areas most high-protection-factor shelters would be crowded; except in cold climates, a ventilating pump to remove warmed air and bring in enough cooler outdoor air to maintain comfortable humidity conditions. Means for storing adequate water is another essential life-support requirement. Other essential or highly desirable life-support needs are covered in following chapters.

#### BASEMENT SHELTERS

The blast and fire effects of a massive, all-out attack of the magnitude possible in 1987 would endanger most American homes and other buildings and endanger the occupants of shelters inside them. In areas not hit and/or fire areas, the use of shelters inside buildings would not be nearly as hazardous. The target areas into which large numbers of urban Americans had evacuated before the attack are not believed to be included in Soviet strategy.

Earth-covered expedient shelters in a blast area give better protection against injury from fallout than do almost all basements. But during the more likely kinds of crises threatening nuclear war, including those who would evacuate into areas outside probable blast areas, probably would not have the space, determination, physical strength, or time required to build good expedient shelters. Shelters dug into the ground and covered with earth. As a result, most unprepared urban citizens would have to use shelters in existing structures, for want of better protection.

Shelters in buildings, including basement shelters, have essentially the same requirements adequate shielding against fallout radiation, strength, adequate ventilation-cooling, water hygiene, etc. Sketches and short descriptions of ways to improve the fallout protection are to be found in widely distributed civil defense pamphlets, including two entitled "In "Protection in the Nuclear Age. " In 1987, millions of copies of these pamphlets are stock during a crisis. Unfortunately, most of such official instructions were written years ago, tonnage and the number of Soviet warheads were small fractions of what they are today. Off instructions now available to average Americans do not inform the reader as to what degree of fallout radiation (what protection factor) is given by the different types of do-it-yourself mention of dependable ways to provide adequate cooling-ventilation, an essential requirement is to be occupied by several families in warm or hot weather. Outdated or inadequate water, food, the improvement of shelter in one's home, and other survival essentials.

No field-tested instructions at present are available to guide householders who may want to over a home basement so that it can safely support 2 feet of shielding earth piled on it. strengthening often would be needed to safely support adequate overhead shielding, especially

jarred by a light shock from a distant explosion. In the following paragraphs, a way to get protection afforded by a typical home basement is outlined. If improved in this manner, a cellent fallout protection for several families.

First, earth should be placed on the floor above to a depth of about one foot. Earth can be using sacks or pillowcases, using the techniques described in Chapter 8 for carrying water

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If earth is not available because the ground is frozen or because of the lack of digging tools (containers of water, heavy furniture, books, etc.) should be placed on the floor above. To enough to produce a loading of about 90 pounds per square foot about the same weight as the initial loading of the floor joists causes them to carry some of the weight that otherwise posts that then are to be installed.

Next, a horizontal beam is installed so as to support all of the floor joists under their

Figure 5.8 shows a beam and one of its supporting posts. Such a supporting beam preferably 2x6s securely together. (Three 2x4s would serve quite well.)

Fig. 5.8. Supporting beam and one of its posts installed to increase the load of shielding carried safely by the floor above a home basement, (nw046.tif)

Cut posts to fit exactly under the beam. If trees at least 4 inches in diameter are not available, use ing boards together. Position the two outermost posts within 2 feet of the ends of the beam intervals, with each post under a floor joist. A post under every third joist is ideal; the between posts of about 4-1/2 feet. If the basement is 20 feet long, 5 posts are enough. Nail secure the bases of each with brace boards laid on the basement floor, as illustrated.

Finally, place a second 1-foot-thick layer of earth on the floor above. If the basement walls are boards and if all but a part of one window and all the aboveground parts of the basement walls are 2 feet thick, the basement shelter will have a protection factor of several hundred against

Adequate ventilation and cooling should be assured by using a homemade air pump (a KAP), as described in Appendix B. Forced ventilation is especially necessary if more than one family is sheltered in warm or hot weather.

More work and materials are - required to improve a home basement in this manner than are required for a covered-trench shelter for one family. An earth-covered shelter separate from buildings will provide protection against radiation, better protection against blast, and much better protection against fire.

If a family cannot build a separate, earth-covered shelter outdoors, often it would be advisable to build a small shelter in the most protected corner of the basement. Such an indoor shelter should be 6 to 8 feet high (6 feet for tall people) and no wider than 3 feet. Its walls can readily be built of chairs, boxes, or drawers. Interior doors make an adequately strong roof. Expedient shielding materials, to be used on the two exposed sides, can be ordinary water containers and bureau drawers, boxes, and pillows or other heavy materials. Or, if heavy-duty plastic trash bags or 4-mil polyethylene film are available, line the water containers and use them for shielding. To do so, first line bureau drawers, boxes, pillows, and other heavy materials with plastic. Place the lined containers in position to shield your shelter, then fill the containers with drinkable water (see Chapter 8).

As demonstrated by hot-weather occupancy tests of such very small indoor shelters, a small KAP must be operated to maintain a forced flow of air through such a crowded shelter, to maintain comfortable temperature-humidity conditions. (See Chapter 6 for ventilation-cooling requirements, including the need for an adequately large opening in each end of a shelter.) In some basements a second small KAP will be needed to pump outdoor air through the basement. This KAP could be operated by pulling a rope from a small shelter, using an improvised pulley" as described in Appendix B .

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## PUBLIC SHELTERS

In the event of an unexpected attack, many unprepared Americans should and would take refuge in marked public shelters. Throughout the populated areas that would not be subjected to blast, the use of public shelters could save millions of lives. All persons concerned with survival should know the location of the majority of officially surveyed and marked shelters give better protection against radiation than home basements.

Persons preparing to go to public shelters should be aware that many lack forced ventilation and fans of most forced ventilation systems would be stopped by loss of electric power due to power outages or by other effects of nuclear explosions on electrical systems. A blast wave at a pressure as low as 1 psi (144 pounds per square foot) would wreck most shelter- ventilating fans. In 1987, a survey of public shelters found that 10% of the shelters were not stocked. A person who brought to a public shelter 10 large plastic trash bags and 10 pillowcases, each containing 6 gallons of water, could store 60 gallons of water (as described in Chapter 8), would provide a source of water for himself and for other shelter occupants. If he hoped to share the basement in a strange family's home, his shelter would be improved if he brought a small homemade shelter- ventilating pump and other supplies. A small pump would be impractical in a large public shelter. An Oak Ridge National Laboratory survey in 1978 found that if all citizens were to go to National Shelter Survey (NSS) shelters within 10 minutes of their homes, 69% of those who found space would be in shelters rated for 1000 or more occupants. The average space per person in this largest class of public shelters was 31.79 square feet. The prospect of living in a shelter with this many unprepared people each of whom would have only 10 square feet of floor space is not very attractive.

tion to work hard to build and equip a small, earth-covered shelter.

#### DECIDING WHAT KIND OF SHELTER TO BUILD OR USE

Before deciding what kind of shelter you and your family should build or use, it is best to read the chapters on life-support equipment. Your final decision should include consideration of ways to provide life-support equipment. At this stage, however, the reader will find it helpful to review important reasons why shelters offer the best hope of survival to different people, in different areas, and under different conditions.

This book is written primarily to improve the survival chances of people who cannot or do not have access to shelters. The information which follows will help you select the best expedient or available shelter for your situation.

#### SHELTER NEAR OR IN YOUR HOME

If your home is 10 or more miles from an average target such as a major airport with long runways, or more miles from a great city with several strategic targets, you are fortunate: you can probably build a shelter close to home. No one can foretell accurately which way the winds will blow or where weapons will fall. If practical, you should build a shelter that gives better protection against fallout, blast, and fire. Most people living outside targeted areas could build such a shelter in two days or less, using the earth-covered expedient shelters detailed in Appendix A.

Even if you plan to evacuate, you should decide where you would take shelter nearby in case you cannot evacuate. There is always a chance that an attack may be launched without warning, giving insufficient time to evacuate. Or the missile aimed at the area in which you live may miss its target. If your targeted home area is subject to extremely heavy fallout might be the only danger; even an improved basement shelter would be better than no shelter at all.

#### EARTH-COVERED EXPEDIENT FAMILY SHELTERS

Advantages of earth-covered, expedient family shelters:

- \* Better protection against heavy fallout, blast, and fire than afforded by the great majority of existing buildings.
- \* The possibility of building in favorable locations, including places far removed from targets, where it is impractical to build or to improve large group- shelters giving good protection.

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- \* The opportunity for men, women, and children to work together to provide good protection.
- \* A better chance to benefit from thoughtful preparations made in advance than would be the case with shelters where water, food, etc. must be shared.
- \* Less risk of personality clashes, hysteria under stress, exposure to infectious diseases that arise when strangers are crowded together for days or weeks.

Disadvantages of earth-covered, expedient shelters:

- \* It may be difficult to meet the requirement for time, space, people able to work hard, and

to get all these together at the building site.

\* Building is difficult if heavy rain or snow is falling or if the ground is deeply frozen. Americans have built good fallout shelters with shielding provided by 5 or more feet of packed earth, including a winter version of the Crib-Walled Pole Shelter described in Appendix A. The present use of several Russian designs of snow-covered expedient shelters also has been demonstrated by winter construction tests in Colorado. 1 2 )

\* The fewer occupants of family shelters could not provide as many helpful skills as would public shelters, with tens-to-thousands of occupants.

\* The lack of instruments for measuring changing radiation dangers. However, the occupants can make a homemade fallout meter by following the instructions in Appendix C, or buy a commercial instrument if a rapidly worsening crisis arises.

#### PUBLIC AND OTHER EXISTING SHELTERS

Advantages of the great majority of public and other existing shelters, most of which are

\* Their immediate availability in many localities, without work or the need to supply materials.

\* The provision of fair-to-excellent fallout protection generally much better than citizen homes.

\* The availability in some shelters of fallout meters and occupants who know how to use them and can provide other needed skills.

\* The chance for persons who are not able to carry food or water to a public shelter to share it with the more provident occupants.

Disadvantages of the great majority of public and other existing shelters available to large numbers of people are

\* The location of most of them in targeted areas.

\* Poor protection against blast, fire and carbon monoxide.

\* Lack of water and means for storing it, and lack of stocked food.

\* No reliable air pumps, which are essential in warm or hot weather for supplying adequate fresh air to maintain endurable conditions in fully occupied shelters especially belowground.

\* Uncertainties regarding the availability of fallout meters and occupants who know how to use them.

\* No dependable lights, sanitary facilities, or other life-support equipment, with few exceptions.

\* The crowding together of large numbers of people who are strangers to each other. Under conditions that might continue for weeks, the greater the number of people, the greater would be the spread of infectious diseases and of hysteria, personality clashes, and the development of other psychological problems.

## BELOWGROUND EXPEDIENT EARTH-COVERED FALLOUT SHELTERS

(Appendix A details two designs of belowground shelters, three designs of aboveground shelters that affords excellent protection built either below or aboveground) .

Advantages of belowground, earth-covered expedient fallout shelters:

- \* They afford better protection than do aboveground, earth-covered types.
- \* Less time, work, and materials are required to build them than to build equally protective designs.

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\* If built sufficiently separated from houses and flammable woods, they provide much better fire hazards than do shelters in buildings.

\* If dug in stable earth, even types with unshored earth walls give quite good blast protection ranges of at least 5 psi where most homes and buildings would be destroyed by blast or fire.

Disadvantages of belowground expedient fallout shelters:

- \* They are not practical in areas where the water table or rock is very near the surface.
- \* It is impractical to build them in deep-frozen ground.
- \* They are usually more crowded and uncomfortable than improved basement shelters.

## EXPEDIENT BLAST SHELTERS

Advantages of expedient blast shelters:

\* Occupants of expedient blast shelters described in Appendix D could survive uninjured in areas where fallout shelters would not prevent death or injury.

\* Blast doors would protect occupants from shock waves, dangerous overpressures, blast wind, exposed skin caused by the popcoming effect and heated air.

\* The expedient blast shelters described in Appendix D of this book were built and blast tested in Nuclear Agency blast tests. Their air-supply systems were not damaged by blast effects that blew over or broken off the aboveground, vertical air-supply pipes typical of even expensive Finnish permanent family blast shelters. (Notwithstanding this weakness, such permanent blast shelters save many lives.) The horizontal blast doors of these tested expedient blast shelters were not damaged because they were protected on all sides by spiked-together blast-protector logs surrounding them. (In contrast, the horizontal blast door of the most expensive blast shelter described in a Federal Emergency Management Agency pamphlet (number H-12-3) is unprotected on its sides. Its untested blast door probably would be torn off and blown away if struck by a strong blast wind, and pieces of houses and trees that would be hurled hundreds of feet.)

\* The blast-tested expedient blast valve described in Appendix D will prevent entry of blast wind.

shelter's ventilation pipes and resultant destruction of the ventilation pump and possible

Disadvantages of expedient blast shelters:

- \* They require more time, materials, tools, skill, and work than are needed for building e
- \* Especially expedient blast shelters should be well separated from buildings and woods th to produce dangerous quantities of carbon monoxide and toxic smoke.
- \* Their ventilation openings permit the entry of many more fallout particles than do the v goosenecks and filters of typical permanent blast shelters. (However, deadly local fallout a major danger in the blast areas where the great majority of Americans live, because a ra employ air bursts to destroy the mostly 'soft" targets found in those areas. Air bursts ca militarily significant soft" targets over about twice as many square miles as can the surf bursting of the same weapons. Fortunately, air bursts produce only tiny particles, and onl these, while they still are very radioactive, are likely to be promptly brought to earth i rain-outs and snow-outs. Thus relatively few prompt fatalities or delayed cancer cases fro are likely to result- even from the air bursting of today's smaller Soviet warheads that w their particles into the troposphere at altitudes from which wet deposition can take place

WARNING: Permanent home fallout and blast shelters described in widely available FEMA pamph tion factors in line with the PF 40 minimum standard for public shelters in buildings. In fraction of the occupants of PF 40 shelters will receive radiation doses large enough to i Permanent shelters built specifically to protect against nuclear weapon effects should hav 40.

None of the permanent home or family shelters described in official OCD, DCPA, or FEMA fre struction pamphlets have been built for evaluation and/or testing - a finding confirmed to tired shelter specialist who for some 20 years served in Washington with FEMA and its prec

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## Chapter 6

### Ventilation and Cooling of Shelters

#### CRITICAL IMPORTANCE

If high-protection-factor shelters or most other shelters that lack adequate forced ventil for several days in warm or hot weather, they would become so hot and humid that the occup from the heat if they were to remain inside. It is important to understand that the heat a the bodies of people in a crowded, long-occupied shelter could be deadly if fallout preven

When people enter an underground shelter or basement in the summertime, at first the air f most shelters are fully occupied for a few days without adequate ventilation, the floors, cool, will have absorbed about all the body heat of which they are capable. Some shelters hot in a few hours. Unless most of the occupants body heat and water vapor from sweat are lated through a typical shelter, the heat-humidity conditions will become increasingly dar



er. One of the most important nuclear war survival skills people should learn is how to keep a shelter adequately ventilated in all seasons and cool enough for many days of occupancy in warm or hot weather. For ventilating with homemade devices and for keeping ventilating air from carrying fallout described in Appendices A and B. Instructions for Directional Fanning, the simplest means of causing volumes of air to flow through shelters, are given at the end of this chapter.

#### MAKING AND USING AN EXPEDIENT AIR PUMP

The best expedient way to maintain livable conditions in a shelter, especially in hot weather, is a large-volume shelter-ventilating pump. Field tests have proved that average Americans can build the pump described in Appendix B in a few hours, with inexpensive materials found in most houses.

This simple pump was invented in 1962 by the author. I called it a Punkah-Pump, because its operation is somewhat like that of an ancient fan called a 'punkah', still used by some primitive peoples. (Unlike the punkah, however, this air pump can force air to move in a desired direction and speed.) It was named the Kearny Air Pump (KAP) by the Office of Civil Defense following tests of various designs at the Research Institute, the Protective Structures Development Center, and General American Transportation. These tests confirmed findings first made at Oak Ridge National Laboratory regarding the effectiveness of the pump both as a manually operated pump for forcing large volumes of outdoor air through shelters and for circulating air within shelters and fanning the occupants. See Fig. 6.1.

The air pump instructions given in Appendix B are the result of having scores of families of all ages and abilities, including children, build and use this air pump. They were guided by successively more detailed, written instructions, that include many illustrations (see Appendix B). Some people, especially those building things will find these instructions unnecessarily long and detailed. However, they have shown that the physically stronger individuals, usually the more experienced builders, should do the manual work when shelters are built, and that those less experienced at building should do the planning and making shelter ventilating pumps. These detailed, step-by-step instructions have enabled people who had attempted to build a novel device of any kind to make serviceable air pumps.

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(Photograph) Fig. 6.1. A 6 foot KAP tested for durability at Oak Ridge. After 1000 hours of operation during which it pumped air through a room at a rate of 4000 cubic feet per minute, there were only minor tears in the plastic flaps. (nw05 1a.jpg)

(Photograph) Fig. 6.2. Behind the girl is the homemade air pump that made it possible for six to live in a crowded trench shelter for more than three days. Outside the temperature was 100°F. (nw051b.jpg)

(The air pump instructions given in Appendix B repeat some information in this chapter. They are both to help the reader when he starts to build an air pump and to increase the chances of the instructions being given to local newspapers during some future crisis. The instructions are also photographed, reproduced, and mass-distributed by newspapers.)

Figure 6.2 shows (behind the girl) a 20-inch-wide by 36-inch-high KAP installed in the trench shelter. The father of the Utah family described earlier had made this simple pump at home with tools found in many homes as described in Appendix B. He carried the pump on top of his car to the site. The pendulum-like, flap-valve pump was swung from two cabinet hinges (not shown) secured to a board nailed to roof poles of the narrow entry trench extending behind the girl in the shelter.

was attached to the pump frame below its hinged top and extended along one trench wall for shelter. Any one of the six occupants could pull this cord and easily pump as much as 300 outdoor air through the shelter and through the insect screens over both its entrances. (Numerous mosquitoes in this irrigated area would have made the family's shelter stay very un-

During the 77 hours that the family continuously occupied their narrow, covered trench, the temperature rose as high as 930 F. Without the air pump, the six occupants would have been driven from the shelter by unbearable temperature-humidity conditions during the day. 8

The photo in Fig. 6.2 also shows how the air pump hung when not being operated, partially open, and causing a "chimney effect" flow of air at night. There was a 10-inch space between the pump and the floor, and the resulting flow of air maintained adequate ventilation in the shelter. At night the temperatures dropped as low as 450 F. Cool outdoor air flowed down into the entry and under the pump, replacing the body-warmed air inside the shelter. The entering cool air continuously

forced the warm air out of the shelter room at ceiling height through the emergency crawlway at the other end. When the weather is cool, a piece of plastic or tightly woven cloth could be hung over the pump, designed, narrow shelter, to cause a flow of fresh air in the same manner.

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Numerous shelter occupancy tests have proved that modern Americans can live for weeks in a cooled shelter with only 10 square feet of floor space per person. 1 3 Other tests, such as those conducted near Washington, D.C. during an abnormally cool two weeks in August, 1962, have shown that it is not so difficult even when summertime outdoor air is being pumped through a long-occupied shelter at 3 cubic feet per minute, per person. 1 4 ' 1 5 This is four times the minimum ventilation rate recommended by the Federal Emergency Management Agency (FEMA) for American shelters: 3 cubic feet per minute. 1 cfm is about three times the supply of outdoor air needed to keep healthy people from having to breathe exhaled carbon dioxide. In hot, humid weather, much more outdoor air than 12 cfm per person is needed in a crowded, long-occupied shelter, as will be described in the following section and in Appendix B.

#### MAINTAINING ENDURABLE SHELTER CONDITIONS IN HOT WEATHER

The Navy test mentioned above showed how much modern Americans who are accustomed to air conditioning could learn from jungle natives about keeping cool and healthy by skillfully using hot, humid weather. In jungles from the Amazon to Burma, I observed the methods used by the natives to avoid heat, and like those experienced in the Navy shelter, which was ventilated in a conventional American manner. These methods include the first five of the six cooling methods listed in this section. During 2 years of field research, my colleagues and I have improved upon the cooling methods of jungle people, primarily through thorough field-testing of the homemade KAP described in Appendix B, and of the Directional Air Pump and its instructions at the end of this chapter.

Even during a heat wave in a hot part of the United States, endurable conditions can be maintained in a small, occupied, belowground shelter with this simple pump, if the test-proven requirements listed in Appendix B are followed.

Most basement shelters and many aboveground shelters also can be kept at livable temperatures by using the cooling methods listed below are ALL followed:

° Supply enough air to carry away all the shelter occupants' body heat without raising the temperature ' of the air at the exhaust end of the shelter by more than 2 degrees F. The "

temperature" of the air to which a person is exposed is equivalent to the temperature of a relative humidity that causes the same sensation of warmth or cold. "Effective temperature effects of the temperature of the air, its relative humidity, and its movement. An ordinary thermometer does not measure effective temperature. In occupancy tests of crowded shelters when the outdoor air was hot and dry, shelter occupants have been surprised to find that they felt the air-exhaust end of their shelter, where the temperature reading was lower than at the air-

Their sweaty bodies had acted as evaporative air coolers, but their body heat had raised the temperature, a reliable indicator of heat stress. If 40 cubic feet per minute (40 cfm) per person is supplied and properly distributed, then (even if the outdoor air is at a temperature during the hottest hours during a heat wave in a hot, humid area of the United States) the effect of the shelter air will be increased no more than 2 degrees F by the shelter occupants' body heat and water vapor. Except for a relatively few sick people dependent on air conditioning, anyone in a shelter with air that has an effective temperature only 2 degrees F higher than that of the air outdoor

(There are exceptions to this ventilation requirement when the ceiling or walls of basements or aboveground shelters in buildings are heated by the sun to levels higher than skin temperature. In such shelters, more than 40 cfm of outdoor air per occupant must be supplied. However, if a shelter is buried at least two feet of earth, it will be so well insulated that its ceiling and walls will not heat the occupants.)

- ° Move the air gently, so as not to raise its temperature. In the aforementioned Navy test shelter, the electric ventilating pump and the frictional resistance of pipes and filters raised the temperature of the air supplied to the shelter by 3 degrees F. Under extreme heat wave conditions, an air supply much hotter than outdoor air could be disastrous especially if considerably less than 40 cfm per person is supplied, and body heat raises the air temperature several additional degrees.

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- ° Distribute the air quite evenly throughout the shelter. In a trench shelter, where air is supplied at one end and flows out the other, good distribution is assured. In larger shelters, such as basements, air will move from the air-supply opening straight to the air exhaust opening. Persons out of the air stream will not be adequately cooled. By using one or more additional, smaller KAPs (also described in Appendix B ), fresh air can be distributed easily throughout large shelter rooms, and the air can be gently fanned.

- ° Provide occupants with adequate drinking water and salt. In extremely hot weather, this means 1 quart of water per day per person and 1 tablespoon (10 grams) of salt, including the salt in food.

- ° Wear as few clothes as practical. When the skin is bare, moving air can evaporate sweat for effective cooling. Air movement can keep bare skin drier, and therefore less susceptible to chafing and skin infections. In the inadequately ventilated Navy test shelter, 34 of the 99 initial subjects had heat rash and 23 had more serious skin complaints at the end of their sweaty two-week confinement, although their overall physical condition had not deteriorated. However, during the war, very few shelter occupants would have medicines to treat skin diseases and infections. Simple measures for preventing skin diseases and infections—means proved very effective by jungle natives and trained jungle infantrymen in World War II—are described in the Prevention of Skin Diseases, Chapter 12.

- ° Keep pumping about 40 cfm of air per person through the shelter both day and night during

so that the occupants and the shelter itself will be cooled off at night. In the Navy test of 7 to 12 cfm was not high enough to give occupants the partial relief from heat and sweat normally get at night. 1 5 hi a National Academy of Sciences meeting on protective shelter stated: "Laboratory experiments and field investigations have shown that healthy persons can tolerate daily exposures to ETs [effective temperatures] up to 900 F, provided they can get

sleep in a cooler environment. " 1 4 An effective temperature 900 F is higher than the highest effective temperature during a heatwave in the South or in American deserts.

#### ADEQUATE VENTILATION IN COLD WEATHER

In freezing weather, a belowground shelter covered with damp earth may continue to absorb occupants' body heat for many days and stay unpleasantly cold. In one winter test of such a temperature of the humid air in the shelter remained around 50 degrees F. 1 6 Under such conditions should continue to ventilate their shelter adequately, to avoid the following conditions:

- ° A dangerous buildup of carbon dioxide from exhaled breath, the first symptoms of which are dizziness and deeper breathing.
- ° Headaches from the carbon monoxide produced by smoking. When the ventilation rate is low smoking should not be permitted, even near the exhaust opening.
- ° Headaches, collapse, or death due to carbon monoxide from open fires or gasoline lantern flames and gases into the shelter air.

#### NATURAL VENTILATION

Enough air usually will be blown through an aboveground shelter if sufficiently large openings on opposite sides and if there is any breeze. But if the weather is warm and still and the shelter humidity conditions soon can become unbearable.

Adequate natural ventilation for belowground shelters is more difficult. Even if there is a vertical entry will make a right-angle turn and go down a vertical entry, make another right-angle turn, a trench or other shelter partially obscured by people and supplies.

In cool weather, occupants body heat will warm the shelter air and make it lighter than the outside air. If a chimney-like opening or vent-duct is provided in the ceiling, the warmed, lighter air will rise and exit the shelter, provided an adequate air-intake vent is open near the floor. An Eskimo igloo is a good example of a shelter with very small ventilation openings, skillfully located in the ceiling and at floor level, making chimney-type natural ventilation to supply the 1 cfm per person of outdoor air needed to prevent the shelter air from becoming dangerously concentrated.

In warm weather, chimney-type natural ventilation usually is inadequate for most high-protection shelters that are fully occupied for days. And in hot weather, when as much as 40 cfm per occupant is needed, shelter air is no lighter than the outdoor air. Chimney-type ventilation fails completely

## SHELTER VENTILATION WITHOUT FILTERS

Numerous tests have shown that the hazards from fallout particles carried into shelters by air are minor compared to the dangers from inadequate ventilation. A 1962 summary of the official findings of the National Academy of Sciences on the radioiodine inhalation problem stated this conclusion: "Air filters are not essential for small (family) shelters. Findings have led to the same conclusion for large fallout shelters. A 1973 report by the Subcommittee is that inhalation is far less of a threat than ingestion [eating or drinking] measures such as filters in the ventilating systems of shelters." 18

Recommendations such as those above realistically face the fact that, if we suffer a nuclear attack, the majority of Americans will have only the fallout protection given by buildings and some experience. How best to use available resources must be the primary consideration when planning for protection against dangers of a nuclear attack; relatively minor hazards may have to be accepted. For unprepared fallout particles would be a minor danger compared to being forced out of a shelter because of inadequate ventilation.

The most dangerous fallout particles are those deposited on the ground within the first few miles of the explosion that produces them. Typically, these "hot" particles would be so large and fast-falling

that they would not be carried into expedient shelters equipped with low-velocity air intake openings, such as tents. Nor would these most dangerous "hot" fallout particles be "sucked" into gooseneck air-intake openings of a permanent shelter.

For most shelters built or improved hurriedly during a crisis it will be impractical to provide pumped air. The Over-Trench Shelter pictured in Fig. 6.3 points up the overriding need for pumped air for shelters during warm or hot weather. This simple shelter provides fallout protection about four times as good as given by a typical home basement. After the car was driven over the trench, earth was shoveled into the trunk and on top of its hood. At one end was a combined crawlway entrance/air intake opening. At the other end was a 1-foot-square air exhaust opening. Each opening was covered by a small awning. To keep the car running under the car and into the trench, the upper edges of 5-foot-wide strips of polyethylene were attached with duct tape to the sides and ends of the car, about 2 feet above the ground. The strips were then pulled over the parts of the film strips that were lying on the ground, to secure them. Finally, earth was shoveled over the attached film strips.

Fig. 6.3. Pulling a Small, Stick-Frame KAP to Keep Temperatures Endurable for Occupants of an Over-Trench Shelter in Warm Weather. Enough air also can be supplied with a small Direct-Flow Fan, but more laboriously, (nw054.jpg)

(Placing earth rolls - see page 150 - around the sides of an earth-loaded car provides better protection, but requires more materials and work.)

## INHALATION DANGERS

Only extremely small fallout particles can reach the lungs. The human nose and other air passages filter out almost all particles 10 micrometers [10 microns] [or larger] in diameter, and about 95 percent of particles 5 micrometers in diameter. (See reference 6, page 599.) Five micrometers equal 5 millionths of a meter, or 0.000005 meter.

Using a dust mask or breathing through cloth would be helpful to keep from inhaling larger particles which may cause beta burns in noses, sinuses, and bronchial tubes. Many such retained particles when cleared from one's air passageways by the body's natural protective processes.

As shown below in Fig. 6.4, a relatively "large" particle - 40 microns (40  $\mu$ m) in diameter and sand-like density of most fallout particles - falls about 1300 feet in 8 hours. (A dark-corned diameter is about as small a speck as most people can see with the naked eye.) Most 40- $\mu$ m particles would take a

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(Illustration) Fig. 6.4. Stabilized Radioactive Fallout Clouds Shown a Few Minutes After the explosion with distances that spherical fallout particles having diameters of 40, 50, and 100 microns would take to fall (nw055.tif)

few days to fall from the cloud of a one-megaton explosion down far enough into the troposphere to be scavenged and promptly brought to earth by rain or snow while still very radioactive. In 1945, thousands of deployed Soviet ICBM warheads are 550 kilotons or smaller. (See Jane's Weapons Yearbook 1984.) The stabilized clouds of such explosions would be mostly in the troposphere, and some of the smaller ones would be small enough to be breathed into one's lungs - would be promptly scavenged and deposited in "hot spots." Fortunately, most of the very small and tiniest fallout particles would not be deposited where they would be scavenged which time radioactive decay would have made them much less dangerous. Breathing tiny radioactive particles into one's lungs would constitute a minor health hazard compared to other dangers that would be encountered if subjected to a large scale nuclear attack.

#### SCAVENGING OF RADIOACTIVE PARTICLES

Scavenging is most effective below about 30,000 feet, the maximum height of most rain and snow. Fig. 6.4. Because the Soviets have deployed thousands of ICBMs with warheads of "only" 100 to 500 kilotons, the face increased dangers from very radioactive particles scavenged by rain-outs or snow-outs. Clouds of fallout heavy enough to kill unsheltered people in a few weeks could be scattered even by a strong wind from areas of multiple explosions, especially missile fields. Prudent Americans, even those living many miles from important targets, whenever practical should equip their shelters with adequate air filters.

This potential danger from extremely small fallout particles will be worsened if the United States has ICBMs such as Midget-fan, probably on large military reservations in the West. (The Soviet Union has mobile ICBMs in its nuclear forces.) In the event of a Soviet attack, our hard-to-target mobile targets subjected to a barrage of relatively small warheads air-bursting so as to blanket their deployment areas with large clouds of extremely small radioactive particles in the troposphere usually would be the most life-endangering "hot spots" from rain-outs and/or snow-outs could be scattered clear. Fortunately, even in many expedient shelters completed in a few days, filtered air can be made available to pump air through furnace or air-conditioner filters, as described in the last chapter. To learn how you can supply a shelter at low cost with air so well filtered that essentially all radioactive particles and infective aerosols are removed, see Appendix E, How To Make a Homemade Plywood Shelter with a Fan and Filter.

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These worsening potential dangers from extremely small "hot" fallout particles brought preventing are not likely to endanger nearly as many Americans' lives as would 24-hour fallout from surface and near-surface explosions. Providing enough outdoor air to shelters, rather than to deserve first priority.

#### STOPPING OR RESTRICTING SHELTER VENTILATION

When instrument readings or observations show that heavy fallout has begun to be deposited, the shelter should decide whether to restrict or stop ventilation. If it is windy outside, even some fallout can be blown into a shelter with large ventilation openings. However, ventilation should not be restricted if weaker occupants are on the verge of collapse from overheating, or to result in headache or carbon dioxide.

If a house is burning dangerously close to a separate, earth covered shelter, closing the doors and windows for an hour or two usually will prevent the entry of dangerous concentrations of carbon monoxide or smoke. (Most houses will burn to the ground in less than two hours.)

When an attack is expected, a shelter, occupied or soon to be occupied, should be kept as open as possible, pumping large volumes of outdoor air through it when the outdoor air is cooler than the shelter air, to ensure that the air is fresh and low in exhaled carbon dioxide. Then, if a need arises to stop ventilation, the shelter can be closed for longer than could be done safely otherwise.

#### VENTILATION/COOLING OF PERMANENT SHELTERS

A permanent family fallout shelter, built at moderate cost before a crisis, should have a means to supply adequate volumes of either filtered or unfiltered air, pumped in through an air-inlet and out through an air-exhaust pipe. Provision also should be made for the grim possibility that fallout could be so heavy that the shelter might have to be occupied for weeks, or even part-time for months. A small or medium-sized shelter should be designed so that most of the time after an attack it can have adequate natural ventilation and emergency exit. During hot spells, forced ventilation through these same large air openings should be provided by using a homemade KAP. This manual air pump, described in Appendix B, can force air through low-resistance openings with minimum effort.

Ways to ventilate and cool permanent shelters are described in Chapter 17, "Permanent Family Shelters for Dual Use," and in Appendix E, "How to Make and Use a Homemade Plywood Double-Action Air Filter."

#### WARNING: MANY OFFICIAL INSTRUCTIONS FOR BUILDING AND VENTILATING PERMANENT SHELTERS ARE LIFE-ENDANGERING

The reader is advised not to read this section if pressed for time during a crisis, unless the shelter is an expedient or permanent shelter described in an official civil defense publication.

Because of the worldwide extreme fear of radiation, civil defense specialists who prepare instructions for building shelters have made radiation protection their overriding objective. In New York and Washington who decide what shelter-building and shelter-ventilating instructions - especially instructions for building and improving expedient shelters-do not understand the importance of maintaining endurable temperature/humidity conditions in crowded shelters. It must be recognized that a shelter may have to be occupied continuously for days in warm or hot weather.

Russian small expedient shelters are even more dangerously under-ventilated than are most counterparts, and can serve to illustrate similar ventilation deficiencies of American shelter drawing (with its caption translated) of a "Wood - Earth Shelter" in a Soviet self-help civil defense booklet, "Radiation Shelters in Rural Areas." This booklet, published in a 200,000-copy edition, is for building 20 different types of expedient shelters. All 20 of these shelters have dangerous ventilation, and none of them have air pumps. Note that this high-protection-factor, covered shelter is flowing down through its "Dust Filter with Straw Packing (hay)" and out through its small Damper."

As part of Oak Ridge National Laboratory's participation in Defense Nuclear Agency's "Dice

I built two Russian Pole-Covered Trench Shelters. These were like the shelter shown in Figure 6.5. It lacked a trapdoor and filter. As anticipated, so little air flowed through these essential shelters that temperatures soon became unbearable.

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(Illustration) Fig. 6.5. [Figure 20.] Wood-Earth Shelter without Lining of the Walls for Cold Weather. Occupants: 1 - Trap Door; 2 - Dust Filter with a Straw Packing (hay); 3 - Earth Cover 60-80 cm; 4 - Roofing made of Poles; 5-Exhaust Duct with Damper; 6 - Curtain made of Tightly Woven Cloth; 7 - Container for Wastes; 8- Water Collecting Sump.

NOTE: Bill of materials is: Rough Lumber, 2.7 cubic meters; Nails, 0.12 kilogram; Wire, 0.12 kilogram; Work Requirement, 90-110 man-hours; Shielding Coefficient, 250-300. (nw057.tif)

Russian earth-covered expedient fallout shelters are based on military dugouts designed for surviving a conventional attack. Subsequently, they were improved for fallout protection but were not tested by Soviet civil defense specialists. Apparently these specialists were ignorant of ventilation; certainly they did not field-test small expedient fallout shelters for habitability. Tens of thousands were taught to build such shelters.

Once any bureaucracy issues dangerously faulty equipment or instructions, it rarely corrects the error under pressure. I have experienced this reluctance even during wartime, when trying to improve shelter design that was causing American soldiers to lose their lives. Continuing proofs of such bureaucratic dangerous errors are hundreds of thousands of potentially life-endangering civil defense instructions, especially the several editions of In Time of Emergency - kept nationwide in hundreds of copies for mass distribution.

Some American official instructions for building expedient shelters have been slowly improved. The best are given in the June 1985 edition of Protection in the Nuclear Age, one of the Federal Emergency Management Agency's widely available free booklets. Yet even in this improved edition no mention is made of the

need for forced ventilation during warm weather, nor for expedient, simple means for providing ventilation. In the June 1985 edition of Protection in the Nuclear Age, the second crawlway entry/exit of the Door-Covered Shelter (see Appendix A.4) is replaced by a "4-6" DIA. "PIPE FOR VENTILATION, this very small shelter essentially dead-ended and thereby eliminates adequate ventilation



a 6-inch-diameter air-exhaust opening, not nearly enough air can flow naturally in warm weather. The shelter's room (only about 39 inches wide by 34 inches high). As proved by habitability tests, a KAP or Directional Fan must be used, even with two crawlway entry/exits.

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The essential second crawlway entry/exit of the Aboveground Door-Covered Shelter was eliminated by a recommendation by a contractor for FEMA charged with field testing and evaluating and improving abbreviated shelter-building instructions. No habitability tests were required. So in his 1978 report to FEMA that the second entry/exit should be eliminated because "The building and with this small a shelter a second entry is really not justified."

In peacetime, bureaucracies of all nations tend to divide up responsibilities between special means by which non- prestigious wartime problems can apparently be solved with the least expense.

#### DIRECTIONAL FANNING TO VENTILATE SHELTERS

The Directional Fanning instructions on the following two pages may save more lives than are given in this book for a homemade survival item. I regret that no one rediscovered this yet effective way of manually pumping air until after the original Nuclear War Survival Skills.

In 1980, Dr. William Olsen, a NASA research engineer long concerned with improving self- help, rediscovered one kind of Directional Fanning. Since then, with the assistance of able Americans, he designed and tested several types of Directional Fans. I have field-tested and repeatedly enabled average people to quickly learn how to make and use such fans effectively.

The great advantage of Directional Fanning is that almost anyone who is given the field-tested instructions can quickly make and use one of these simple fans. Only very widely available materials are needed. The disadvantage is that Directional Fanning is a more laborious way to ventilate a shelter than using a fan in Appendix B .

Americans are not likely to receive Directional Fanning instructions from the Federal Emergency Management Agency. FEMA's predecessors, the Office of Civil Defense and the Defense Civil Preparedness Agency, were unable to get the millions of dollars necessary to buy factory-made KAPs and other manual air-pumping devices for specially designated fallout shelters, and FEMA has avoided shelter ventilating controversies. The latest American publication includes instructions for making and using any expedient airpumping device.

Thanks to Congressman Ike Skelton, Democrat of Missouri and strong civil defense advocate, who insisted to demonstrate Directional Fanning to Louis Giuffrida, at that time the Director of FEMA. Since then, the FEMA specialists concerned with shelter ventilation, all of whom have since left FEMA. The Directional Fanning instructions have been reproduced in three private civil defense publications. The metric version of the instructions were distributed to British civil defense professionals. In the United States, Civil Defense and Emergency Planning Officers, FEMA has not even evaluated Directional Fanning.

In contrast, in 1981 I gave copies of instructions for both KAPs and Directional Fans to the Director of the People's Republic of China's National Research and Design Institute of Civil Defense. He started evaluating these simple devices. (At that time I was traveling extensively in China, changing civil defense information.) Dr. Yin, who heads all Chinese civil defense research, sent me with his top ventilation and shelter design specialists to a furniture factory in Beijing. They were to build both a large and a small KAP, and also Directional Fans. Then Dr. Yin and his specialists used air- velocity meters to measure the volumes of air that these simple devices could pump. They also participated in more ventilation tests using KAPs and Directional Fans in tunnel blast shelters.

city of Dalien.

While watching these top Chinese civil defense professionals make and test KAPs and Direct thinking: 'This is the way Thomas Edison and Flenry Ford would have evaluated simple device portance to millions. "

The reader is urged to keep the following two pages of Directional Fanning instructions ready in a crisis. The sections on the small 2-handled Directional Fan and the large 1-Man Fan will be prepared people. Ventilation by pairs of men using Bedsheet Fans is an effective method for moving volumes of outdoor air through tunnels, corridors and mines with ceilings at least 9 feet high and large openings. However, this method requires organization and discipline.

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#### DIRECTIONAL FANNING TO VENTILATE SHELTERS

Directional Fanning is the simplest way to force enough outdoor air through typical basement or expedient shelters to maintain endurable conditions, even in extremely hot, humid weather.

During a worsening nuclear crisis most unprepared citizens probably will not have the time to make a KAP or other efficient shelter- ventilating pump - even if they have the instructions. Surveys with average citizens have indicated that if they have instructions for making and using Directional Fans are a few hours of warning time before the attack, then the majority will be able to ventilate shelters, except some of the largest.

The principal disadvantage of Directional Fans is that they are more laborious to operate than manually powered, pendulum- like air pumps that conserve energy.

#### A. DIRECTIONAL FANNING TO VENTILATE AND COOL SMALLER SHELTERS

A 2-Handled Directional Fan of the size illustrated (14 in. x 16 in. with a 6 in. handle) requires less manual dexterity than does a 1 -handled fan with the same size blade. With this fan one can quite easily can force about 300 cubic feet per minute (300 cfm) of outdoor air through a small shelter. This is enough air for up to 9 adults crowded into a small shelter in extremely hot weather, or enough for about 100 people in cold weather. By fanning vigorously, 500 to 600 cubic feet of air can be forced through a small covered-trench shelter.

Fig. Pg. 59a (Illustration) (nw059a.tif)

To make a durable 2-handled fan, first make its frame out of 2 sticks each 14 inches long and 1 1/2 inches long. See sketch. To strengthen the comers, overlap the sticks about one-half inch, and

When using sticks cut from a tree, select ones with diameters of about 3/4 inch, and make sure the sticks are straight before tying together the 4 comers of the blade. If you do not have strong string, use bedsheets, or other strong cloth, slightly twisted.

If using sawed sticks, be sure to use none smaller than 3/4 inch in cross section. If you use wire brads, use only one to connect each corner: then tie each corner securely. To prevent possible chafing, wrap cloth around the fan handles, or wear gloves.

To cover the fan's blade, any strong, light fabric, such as bedsheet cloth, serves well. If the cloth, first cuts 26 x 30-inch piece. Wrap the 30-inch width smoothly crosswise around notches in the cloth's corners, so that the tied-together parts of the sticks will not make a smooth blade: finally sew securely. (If waterproof construction adhesive is available can be used and the blade can be covered in a very few minutes.)

If time and/or materials are very limited, make a fan with its blade merely a piece of cloth 22-inch-long sticks. This very simple fan is reasonably effective, although tiring to use.

Cardboard covering a blade is likely to become damp and fragile in the humid air of a crowd. Sheetmetal makes a good fan blade and requires only 2 sticks. A blade of 4-inch plywood is

If no sticks are available, a double thickness of heavy, stiff cardboard 22 inches long by almost as much air if used as a handleless fan. The pieces should be securely tied or taped. If tape is available, cover the parts that you will grip with sweaty hands, thus preventing the cardboard from becoming damp.

For maximum ventilation, the air-intake opening of a shelter should be at least as large as the air-exhaust opening. (If the air-exhaust opening of your small shelter is much larger than that shown in the sketch, reduce it to approximately this 24-inch-high by 20-inch-wide size, for more effective use.) The fan should be fanned out of the shelter in the direction in which the air is naturally flowing. For maximum ventilation, about 40 strokes per minute.

With one or more Directional Fans, air inside a shelter can be distributed effectively and evenly. Also, if during the time of maximum fallout dose rate the occupants get close together in the shelter, they often will get unbearably hot unless fanned.

To fan air out through an air-exhaust opening, sit facing the opening with your elbows above the bottom of the opening. Then count 1,2,3 while you:

Fig. Pg. 59b (Illustration) (nw059b.tif)

- 1) Quickly raise the fan to a vertical position close in front of your face and immediately thrust it into the opening - ending the power stroke with your arms fully extended and with the fan blade pointing into the opening and out of the way of air that was "sucked" behind the fan and is still flowing out through the opening.
- 2) , 3) After a slight pause, leisurely withdraw the almost horizontal fan until the bottom of the fan is at your stomach - preparatory to the next power stroke.

To increase the flow of air through a shelter, while fanning the occupants:

Have two or more occupants sitting inside the shelter each use a fan of the size described above. This method as to increase its velocity in the direction in which air already is flowing through the shelter. This is especially effective in increasing the air flow through small, narrow shelters.

To avoid higher radiation exposures near openings, build an essentially airtight partition with a 24-inch-high x 20-inch-wide hole in it through which to fan. By fanning through a board partition built across a doorway inside a U-shaped permanent trench shelter 76 feet

creased by an average of 327 cubic feet per minute.

## B. DIRECTIONAL FANNING TO LARGER SHELTERS VENTILATE AND COOL LARGER SHELTERS

### 1. With a Large 1-Man Fan

Fig. Pg. 59c (Illustration) (nw059c.tif)

To ventilate larger basements, big covered trenches, and other large shelters lacking adequate more large 1-man fans. See sketch. Note that the 20 x 30-inch fan blade is made like a 2-per end of the longer diagonal stick serves as 10-inch handle. The model illustrated is made of boards, one 46 inches long and the other 35 inches long. These boards are connected at a point at their lower ends, first with a single clinched nail, and then by being tied securely. The smooth

The blade frame is covered on both sides with strong bedsheet cloth, that is wrapped around strong cords or wires tied to notches cut in the boards or sticks) near the 4 corners of the available. 4 2-inch-wide strips of strong cloth, slightly twisted, serve well.)

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A durable but laboriously heavy fan can be made in a few minutes using a 20 x 30-inch piece nailed to a single 46-inch-long. 1x2-inch board. Or use a single round stick about 1-1/4 on one side.

A fan with its blade made of two sheets of very heavy cardboard tied on both sides of a 1-edly effective when dry. However, typical cardboard will become soft and worthless in most humid shelters.

To fan directionally, it is best to stand just outside and to one side of a doorway, so that you can direct the air flow. Preferably stand opposite and facing the open door, which should be parallel to its doorway. Hold the fan like a golf club and swing it with your arms extended. This is how you:

1. Make the power stroke with the fan blade broadside until the end of the stroke, when you turn it 90 degrees.

2. Make the pendulum-like return stroke with the fan blade kept edgewise ("feathered") to the end, when you quickly turn it 90 degrees, preparatory to making the next power stroke.

To pump more air, block off the upper part of the doorway with cloth, cardboard, plywood, etc., so that air is flowing back in the wrong direction through the upper part of the doorway. See sketch on page 59c.

Whenever practical, directionally fan the air in the same direction that the air is naturally moving. More air usually can be pumped through a shelter if the fan is used to force air out of the shelter. This reduces the air pressure inside the shelter and causes fresh outdoor air to be drawn in through the air-intake doorway, or through other large air-intake openings. Thus with one minute can be pumped through a fully occupied shelter. This is enough outdoor air - if it is used properly - to maintain tolerable conditions for weeks for 25 occupants during extremely cold weather about 300 occupants during cold weather.

To ventilate and cool a room having only one doorway and no other opening, do not block off the doorway. If air is fanned into such a room through the lower part of its completely open doorway, the air will flow out of the room through the upper part of the doorway. However, this pumps much less air than if a large air-exhaust opening is provided.

To increase the flow of outdoor air through a tunnel-shelter, several fanners equally spaced in the direction of the natural air flow. This procedure was first proved practical in a test that Cresson H. Kearny participated in with Chinese civil defense officials in the past. Five fanners, each with a fan of approximately the size illustrated, forced air from the outdoors into a tunnel between two opened entrances of a typical Chinese tunnel-shelter. The air flow was increased from 290 cubic feet per minute to 3,680 cubic feet per minute. The 5 excellent Chinese fans each consisted of a piece of 3 mm (approx. 1/8th in.) plywood nailed to a single board.

## 2. With a Bedsheet Fan

Use a 2-man Bedsheet Fan to force thousands of cubic feet per minute of outdoor air through a corridor having at least a 9-foot ceiling and a large opening at each end. The most practical construction is a strong double bedsheet cut down to 6-foot width, with the wide hem at its head-end left and a similar-sized hem sewn in its opposite end, to give a finished length of 6 feet. A 6-foot diameter stick (or an approximately 1-1/2 inch diameter stick) was secured inside each end hem of various construction adhesive, or with tacks, or by tying. Before a board was inserted, its edges were smoothed.

Two persons preparing to use a Bedsheet Fan (see sketch) should stand facing each other, at a distance of 6 feet, with the cloth extended horizontally between them. Each fanner holds one end of the cloth with one hand near its "downwind" end and with his other hand near its center.

A pair of Directional Fanners get ready to make a power stroke by leaning in the upwind direction. Then the pair of fanners should count 1,2,3 while they:

1 Make the power stroke by rapidly sweeping their sticks and the attached cloth in an arc, from the upwind direction to the downwind direction and the sticks and cloth are again horizontal. See sketch.

2,3 Hold the sticks and cloth horizontal (to permit air that was "sucked" behind the cloth to flow in the desired direction) while leisurely moving the Bedsheet Fan back to the starting position. After 3 strokes, the fanners change hands, as illustrated. Note that what was the upper side of the fan at the start of the power stroke now has become the lower side.)

Two men thus fanning vigorously produced a net air flow of 5,500 cubic feet per minute through a corridor that is 8 feet wide, has a 9-foot ceiling, and is 194 feet long. The doors at both ends of the corridor adequately ventilate and cool people crowded into a long tunnel in hot weather, a pair of Bedsheet Fans positioned about every 100 feet along its length.

Fig. Pg. 60 (Illustration) (nw060.tif)

The practicality of using Bedsheet Fans to ventilate some very large mines or caves having long narrow corridors was proved by tests with members of the Citizens Preparedness Group of Greater Kansas City. The tests were conducted in 1982 near Kansas City in a huge limestone mine that has a ceiling averaging 15 feet high, corridors about 35 feet wide, columns of unexcavated rock about 15 feet square, and over 1

level, dry floor space. The air inside is "dead", remarkably still, because the only opening is on one side of the mine. Five pairs of Bedsheet Fanners, spaced about 75 feet apart down a corridor, produced a measured air flow of approximately 100,000 cubic feet per minute.

With many more pairs of Bedsheet Fanners working, enough air for at least 10,000 occupants can be fanned into this mine through one of its 17 x 20-foot portals, fanned down a corridor to the far end, fanned through across corridor, and finally fanned back out through the corridor that has its opening at its outer end.

A pair of pre-mechanization coal miners produced a directed airflow' by holding a piece of canvas between them while they quickly walked a short ways in the direction of the desired airflow; the canvas held horizontally between them. Then they repeated.

#### C. ADDITIONAL ADVANTAGES OF DIRECTIONAL FANS

1 . No installation is needed, thus saving working time and materials for making habitable or upgraded during a crisis.

2. Directional Fans enable shelter occupants to quickly reverse the direction of air flow when outdoor wind changes cause the direction of natural air flow' to be reversed.

3. Four or more Directional Fans when used to circulate air within a shelter room can serve simultaneously fanning occupants.

4. Directional Fans are very unlikely to be damaged by blast effects severe enough to wreck other fixed ventilation devices placed at or near air-intake or air-exhaust openings, but injure shelter occupants.

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#### Chapter 7

##### Protection Against Fires and Carbon Monoxide

#### RELATIVE DANGERS

Fire and its consequences probably would be the third-ranking danger to unprepared Americans in the event of a massive nuclear attack. Direct blast effects would be first, covering a large fraction of the population and killing far more people. Considerably fewer fatalities seem likely to result from the secondary effects of radiation.

#### THE FACTS ABOUT FIRE HAZARDS

Firestorms would endanger relatively few' Americans; only the older parts of a few American cities are close enough together, over a large enough area, to fuel this type of conflagration. In cities where less than about 30% of a large area was covered with buildings. 19

In the blast area of Hiroshima, a terrifying fire storm that burned almost all buildings within a few square miles resulted from many fires being ignited almost simultaneously. Many were caused by the fireball. Even more fires were due to secondary effects of the blast, such as the overpressure winds that blew in from all directions replaced the huge volumes of hot air that rose skyward.

Lack of oxygen is not a hazard to occupants of shelters in or near burning buildings or to those who closed tightly to prevent the entry of smoke or fallout. Carbon monoxide, toxic smoke from fires, and concentrations of carbon dioxide from shelter occupants' exhaled breaths would kill occupants before they could escape from lack of oxygen.

#### FIRES IGNITED BY HEAT RADIATION

Figure 7.1 shows a wood-frame house after it was heated for one second by heat radiation from a nuclear weapon exploded in a Nevada test. This test house had no furnishings, but the heat was intense enough to charred exposed upholstery, curtains, bedding, papers, etc. in a typical home.

Heat radiation will set fire to easily ignitable materials (dry newspapers, thin dark fabrics, dry grass) in about the same extensive areas over which blast causes moderate damage to frame and high-speed blast winds will blow out many flames. However, tests have shown that fire can spread within some materials such as upholstery and dry rotted wood, and after a while it often will spread. The burning automobile pictured in Fig. 7.2 is an example of such ignition beyond moderate damage.

The number of fires started by heat radiation in areas where blast is not severe can be reduced by closing the insides of window panes and by removing flammable materials from places in and around the shelter. Protection could reach them. Also, occupants of shelters in some homes that would be only slightly damaged could move quickly to extinguish small fires and throw out smoldering upholstered articles before they catch.

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Fig. 7.1. Heat radiation charred the paint on this house, which had been painted white to reflect heat rays. The charring instantaneously produced the smoke. However, precautions taken to prevent this typical U.S. house from being destroyed by fire, because the test was made to study the effects of blast, rather than fire. The house was demolished by the 5-psi overpressure a few seconds later, but it did not burn, (nw062a.jpg)

Fig. 7.2. Thermal radiation from a nuclear explosion entered the car above through its closed windows and ignited the upholstery. The windows were blown out by the blast a few seconds later. However, the explosion was at such a distance that the blast wave was not severe enough to destroy the car. (nw062b.jpg)

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Earth-covered shelters can be protected against heat radiation from nuclear explosions and against any exposed wood and other combustible materials at shelter openings with a thick coat of whitewash. The World War II firebombing of Kassel was less effective than were

German cities because the roof timbers of buildings had been so treated. 20

Figure 7.3 illustrates the effectiveness of a thick coating of slaked lime in protecting a board from ignition by heat radiation. No flames from the burning logs touched the board. (Before this test the uppermost burning logs of a vertical-sided pile were removed so that the board could be seen.)

Chinese civil defense instructions recommend coating exposed wood with both slaked lime and mud if it is available, a coating of it protects wood quite well. If kept damp, a mud coating is even better. (Keeping all exposed flammable materials damp is helpful.)

In blast areas, cloth or plastic canopies over the openings of expedient shelters usually provide protection from heat and certainly would be blown away by even moderate blast winds. If extra canopies and stakes are kept inside the shelter, these replacements could be quickly erected after blast winds begin. It takes at least 15 minutes after the explosion. If no spare canopies were available, it would be possible to use available canopies and their stakes inside the shelter, if it were not raining.

#### FOREST AND BRUSH FIRES

Unless forests or brushy areas are dry, it is difficult to start even scattered fires. Damp brush is unlikely, except in blast areas where the heat radiation would be very intense. However, if a shelter is to be dug, it would do well to select a shelter site at least as far away from trees as the height of the trees. In dry weather, the shelter because of fire and smoke hazards in dry weather, and because digging a shelter in brushy areas is difficult.

Fig. 7.3. Heat radiation had ignited the flaming half of the board on the ground, while the other half, which had been shovel-painted white with a thick coating of slaked lime-had not even begun to smoke, although it was in the same blast area.

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#### CAUSES OF FIRE

Figure 7.4 pictures the same house shown in Fig. 7. 1 after it had been struck by the blast wave from a nuclear test explosion at the 5-psi overpressure range. (If the house had been hit by the blast wave from a nuclear weapon, with longer-lasting blast winds, it would have been wrecked about as completely at the 5-psi range. At the 3-psi overpressure range, the blast winds from an explosion 1000 times as powerful as the one that wrecked this house would blow 10 times as long. This longer-duration, 100-megaton explosion would increase the damage done by the blast wave. The 3-psi overpressure range from a 20-megaton explosion extends about 10 miles from the center of the crater, and from a one-megaton surface burst, about 4 miles.)

If the blast-wrecked house shown in the illustration had had a furnace in operation when the blast hit, the chances of its being set on fire would have been high. In Hiroshima many of the first fire effects of blast, especially the overturning of stoves, and not from heat radiation. Although the blast caused fallout, firefighters from undamaged, nearby communities were unable to reach most of the damaged areas because of blast debris blocking the roads. Later they were kept from burning areas by the intense heat and smoke, which made water unavailable for fire fighting in certain areas.

In the event of an attack on the United States employing many surface bursts, fallout would be a problem for days to weeks in a large part of the most populated regions.



The basements of many substantial buildings will withstand 5-psi blast effects and can prevent suffering serious injuries from blast. Most home basements can be reinforced with stout beams to give good protection against blast effects up to considerably higher than 5 psi. But considering probable blast areas, it is safer to build an earth-covered shelter well removed from buildings than in shelters inside buildings.

#### CARBON MONOXIDE AND TOXIC SMOKE

If an undamaged building is burning, people inside may be killed by carbon monoxide, toxic Tests have shown that even fast-burning, rubble-free fires produce very high concentrations. In large-scale fires are burning near a shelter, the dangers from both carbon dioxide and carbon monoxide last for as long as 12 hours after ignition. 22 Therefore, the ventilation pipes or openings close to a building that might be expected to burn.

In the smoldering rubble of a large test fire, after 24 hours the carbon monoxide concentration was 1% and the air temperature was 1900 degrees F. A carbon monoxide concentration of only 0.01% (10,000 parts of air) will cause headache, dizziness, and nausea in 45 minutes, and total death in 2 hours.

Fig. 7.4. Unburned wreckage of the same two-story, wood-frame house pictured in Fig. 7.1

after being wrecked by the 5-psi blast effects of a small nuclear test explosion, (nw064.j

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Realization of carbon monoxide dangers to persons in simple fallout shelters and basements led the writers of Soviet civil defense publications to define the "zone of total destruction" as the area where pressure exceeds 7 psi and "residential and industrial buildings are completely destroyed and the ground covers the burning structures," and "As a result the rubble only smolders, and fires as smoldering fires produce more carbon monoxide than do fiercely burning fires. Whether or not the shelters survive the direct blast effects is of little practical importance in those blast areas where fires burn or smolders. So in the "zone of complete destruction," Russian rescue brigades plan to rescue persons trapped inside excellent blast shelters by the rubble.

About 135,000 Germans lost their lives in the tragic city of Dresden during three days of bombardment. Casualties were caused by the inhalation of hot gases and by carbon monoxide and smoke poisoning. It was learned that when these dangers were threatening an air raid shelter, the occupants' best chance was to run outside, even if the bombs were still falling. But in a nuclear war the fallout dose rate outside would be high. Occupants of a shelter threatened by smoke and carbon monoxide might suffer a more certain death if they ran outside. Instead, if they know from instrument readings and their calculations that they can survive the dose before they could reach another shelter, the occupants should close all openings as tight as possible. Carbon monoxide in deadly concentrations would not reach them, nor would they be overcome by respiratory carbon dioxide before the fire dangers ended.

Dr. A. Broido, a leading experimenter with fires and their associated dangers, reached this conclusion: "In building a fallout shelter I would spend a few extra dollars to build it in my backyard rather than in a basement, locating the intake vent as far as possible from any combustible material. In such a shelter one can survive anything except the close-in blast effects." 22

This advice also applies to expedient shelters built during a crisis.

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## WATER AND SALT REQUIREMENTS

Painful thirst has been experienced by very few Americans. We take for granted that we will have enough water to drink. Most of us think of "food and water" in that order, when we think of what should be stored. But if unprepared citizens were confined in a shelter by heavy fallout, they should have given first priority to storing adequate water.

For the kidneys to eliminate waste products effectively, the average person needs to drink enough water so he urinates at least one pint each day. (When water is not limited, most people drink enough water so that additional water is lost in perspiration, exhaled breath, and excrement.) Under cool conditions, a person needs about 3 pints of water a day if he eats but little food and if that food is low in protein. This would be the exception in crowded belowground shelters occupied for many days. Under such conditions, five quarts of drinking water per day are essential in very hot weather, with none allowed for cooking. For a week shelter stay, 15 gallons per person should be stored in or close to a shelter. This allows for some water remaining after two weeks, to prevent thirst in case fallout dangers were to occur.

In a 1962 Navy shelter occupancy test lasting two weeks, 99 sailors each consumed an average of 1.5 liters (about 1/4 oz, or 1/2 cup) of water per day. The test was conducted in August near Washington, D.C.; the weather was cool. The shelter was not air-conditioned except during the last two days of the test.

When one is sweating heavily and not eating salty food, salt deficiency symptoms especially develop within a few days. To prevent this, 6 or 8 grams of salt (about 1/4 oz, or 1/2 teaspoon) should be consumed daily in food and drink. If little or no food is eaten, this small daily salt ration is essential. Under hot conditions, a little salt makes water taste better.

## CARRYING WATER

Most families have only a few large containers that could be used for carrying water to a shelter. Adequate amounts for several weeks. Polyethylene trash bags make practical expedient water containers as waterproof liners inside smaller fabric bags or pillowcases. (Plastic bags labeled as for use with oil or odor-controlling chemicals should not be used.) Figure 8.1 shows a teenage boy carrying more than 80 pounds of water, well balanced front and back for efficient packing. Each of his two 20-gallon polyethylene trash bags, one inside the other. (To avoid possible pinhole leaks, the waterproof bag inside another.)

To close a plastic bag of water so that hardly any will leak out, first spread the top of the bag so the sides of the opening are together. Then fold in the center so that the folded opening is 4 inches wide (Fig. 8.2). Continue smoothly folding in the middle until the whole folded-up opening is only 1 inch wide. Then fold the top of the bag over on itself so the folded-up opening points down. With a string, bind and tie the folded-over part with a bow knot, as illustrated.

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Fig. 8.1 . Carrying 80 pounds of water in two burlap bags, each lined with two larger plastic bags inside the other. (Photograph) (nw067a.jpg)

Fig. 8.2. Folding and tying the mouth of a water-filled plastic bag. (Illustration) (nw067

For long hikes, it is best to tie the water-holding plastic bags so that the openings 'are inside.

To transport this type of expedient water bag in a vehicle, tie a rope around the fabric c that the rope also encircles and holds the plastic liner-bags just below their tied-shut c rope should then be tied to some support, to keep the openings higher than the water level

To use two fabric bags or pillowcases to carry a heavy load of water contained in larger p the two fabric bags as shown in Fig. 8.1.

A small pebble, a lump of earth, or a similar object should be tied inside the opening of tied together, to hold them securely. The bag that is to be carried in front should have t further down from the edge of its opening than the pebble tied in the bag to be carried in from being pressed against the carrier's shoulder by a heavy load.

A pair of trousers with both legs tied shut at the bottoms can be used to carry a balanced fabric bags are not at hand. Such a balanced load can be slung over the shoulder with the than if the same weight were carried in a single bag-like pack on the back. However, trous and do not provide room to carry more than a few gallons.

To prevent water from slowly leaking through the tied-shut openings of plastic bags, the wa kept below the openings.

#### STORING WATER

When storing expedient water bags in a shelter, the water levels inside should be kept bel

Not many expedient shelters would be large enough to store an adequate volume of water for two or more weeks. Plastic-lined storage pits, dug in the earth close to the shelter, are volumes of water using cheap, compact materials.

Fig. 8.3. Vertical section of cylindrical water storage pit lined with two 30-gallon water This pit held about 20 gallons. (Illustration) ORNL- DWG 77- 0423R (nw067c.tif)

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Figure 8.3 shows a cylindrical water-storage pit dug so as to have a diameter about two in flated diameter of the two 30-gallon polyethylene trash bags lining it (one bag inside the placed in such a pit, the ends of roots should be cut off flush to the wall with a sharp k carefully removed.

The best way to keep the upper edges of the pit lining bags from slipping into the pit is cular wire hoop the size of the opening of the bag, and tape it inside the top. In firm gr double bags have been satisfactorily held in place simply by sticking six large nails thro the bags and into the firm earth.

Figure 8.3 shows how to roof and cover a water storage pit so as to protect the water. The

proof material prevents any contamination of the stored water by downward-percolating rain. It also contains bacteria or small amounts of radioactive substances from fallout. The thick earth cover gives excellent blast protection, due to the earth arching that develops under blast pressure. In a clear Agency blast test, a filled water-storage pit of the size illustrated was undamaged in the pressure range which could demolish the strongest aboveground buildings (53 psi).

A simpler way to store water is illustrated in Fig. 8.4. If the soil is so unstable that a vertical sides cannot be dug, the opening of the bag (or of one bag placed inside another) is covered to minimize leakage (see Fig. 8.4). Fill the bag with water, tie it, and place it in the pit to the level of the water inside. A disadvantage of this method is leakage through the tied-shut top of the bag. To lessen leakage, leave an air space between the filled bag and the surrounding loose earth on the bag. To lessen leakage, leave an air space between the filled bag and the surrounding loose earth so that the weight of earth piled on top of the roofing will not squeeze the bag. This is a disadvantage: after the earth covering and the roof are removed, it is difficult to bail out the water. If the water is bailed out, the loose surrounding earth moves inward and squeezes the bag above the

Fig. 8.4. These two 30-gallon polyethylene trash bags, one inside the other, held 16 gallons of water. They were undamaged by 50-psi blast effects while buried in dry, very light soil. The earth placed over the water bag were removed before this picture was taken, (nw068a.jpg)

Fig. 8.5. Post-blast view of plastic-lined water storage pit undamaged at a 6.7-psi overpressure range. This pit held about 200 gallons, (nw068b.jpg)

Large volumes of water can be stored in plastic lined rectangular pits. In order to roof the pits with materials such as ordinary 1/2-inch plywood or small poles, the pits should be dug no wider than the materials used for such a pit: 8 feet long, 27 inches wide, and 30 inches deep. It was lined with a 10-foot-wide plastic sheet. The edges of this plastic sheet were held in place by placing them in shallow trenches around the pit and covering them with earth.

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Earth was mounded over the plywood roof to a depth of about 30 inches, with a "buried roof" and earth cover and its "buried roof" were similar to the pit covering illustrated by Fig. 8.3. The pit held about 200 gallons of water. No water leaked out after the pit had been subjected to blast effects that have flattened most substantial buildings. However, rectangular pits at higher overpressure ranges could cause caving that caused leaks.

In a subsequent blast test by Boeing Aerospace Company, a plastic-lined water pit was undamaged in the overpressure range. First a rectangular pit 4 ft. wide, 12 ft. long, and 2 ft. deep was dug. Then a 2-ft. water-storage pit was dug, and lined with plastic film. After being filled full of water, it was covered with plywood, on which was shoveled 2 ft. of earth.

Plastic garbage cans are usually watertight; most used metal garbage cans are not. If thoroughly treated with a strong chlorine bleach solution, watertight garbage cans can serve for emergency wastebaskets. If new plastic film is available, it can be used as a lining to waterpits. To reduce the chances of the plastic being punctured, rough containers first should be lined with fabric.

If shelter is to be taken in or near a building, water trapped in hot water heaters and toilets might be available after an attack.

SIPHONING

Pouring water out of a heavy water-storage bag is inconvenient and often results in spills and contamination. If a tube or piece of flexible garden hose is available, siphoning (see field-tested method is described below. To prevent the suction end of the tube from being the plastic liner of the bag, tape or tie a wire "protector" to the end, as pictured later

To start siphoning, 'suck on the tube until water reaches your mouth. Next fold over the tube full. Lower its closed end until it is near its position shown in Fig. 8.6. Then release siphoning.

To cut off the water, fold over the tube and secure it shut with a rubber band or string.

(Illustration) Fig. 8.6. Using a tube to siphon water from a fabric bag lined with a larger plastic liner. ORNL-DWG 78-14676 (nw069.tif)

Water can be siphoned from a covered water storage pit into a belowground shelter so that water can be used for weeks, if necessary. The Utah family mentioned earlier siphoned all the water stored in a nearby lined pit. A field-tested method of siphoning follows:

1 . Dig the water storage pit far enough away from the shelter so that the covering mounds can have drainage ditches.

2. Use a flexible tube or hose which is no more than 25 feet long. For a single family, a 1/2-inch inside diameter of inch (such as surgical tubing) would be best. A flexible 2 -inch hose with mobile homes and boats serves well. As indicated by Fig. 8.7, the tube should be long enough from the bottom of the water pit to within about a foot of the shelter floor.

3. Make sure that the end in the water pit will not press against plastic and block the flow. This can be avoided by (1) making and attaching a wire "protector" to the end of the tube, as shown in Fig. 8.8, or (2) taping or tying the end to a rock or other object, to keep the end in the desired position.

4. Protect the tube by placing it in a trench about 4 inches deep. This small trench is begun at either the storage pit or the shelter. Be sure a roof pole or board does not crush the tube. Tie the tube to the earth and tie it so that the end in the storage pit cannot be accidentally pulled out of position.

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Fig. 8.7. Water siphoned into a belowground shelter. ORNL-DWG 78-1 1471

(Illustration) (nw070a.tif)

Fig. 8.8. Two wire "protectors," each made of two pieces of coathanger wire taped to a 1/2-inch diameter and a rubber tube. Shown on the right is a tube closed with a rubber band to stop a siphon.

(Illustration) (nw070b.jpg)

5. To start the flow of water into the shelter, hold the free end of the tube at about the same level as the water in the storage pit, while pulling gently on the tube so that the part in the shelter is

Exhale as much breath as you can, then place the end of the tube in your mouth, and suck hard.

(The longer the tube or hose and the larger its diameter, the more times you will have to suck to get the same amount of water.)

6. Without taking the tube out of your mouth, shut it off airtight by bending it double near the end.

7. Exhale, straighten the tube, and suck again, repeating until you feel a good flow of water. Shut off the flow by bending the tube double before taking it out of your mouth.

8. Quickly lower the end of the tube (which is now full of water) and place the closed end in the storage pit outside and close to where the tube comes into the shelter floor. Finally, open the end to start the siphoned flow.

9. When you have siphoned enough water, stop the flow by bending the tube double. Keep it in this doubled-over, air-tight position with a strong rubber band or string, as shown in Fig. 8.8. To prevent water by accidental siphoning, suspend the end of the tube a couple of inches higher than the water in the storage pit outside and close to where the tube comes into the shelter. (Despite the fact that air may accumulate in the highest part of the tube, blocking a siphoned flow and making it necessary to repeat the siphoning by repeating the sucking.)

#### DISINFECTING WATER

Water-borne diseases probably would kill more survivors of a nuclear attack than would fall-out radiation. Before an attack, if water from a municipal source is stored in expedient containers, the water should be disinfected. For long storage, it is best to disinfect all water, since even a faintly disinfected water may give stored water a bad taste or odor. Properly disinfected water remains safe for drinking. Plastic or glass containers sealed airtight. For multi-year storage do not use thin plastic containers which in time often develop leaks.

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Any household bleach solution, such as Clorox, that contains sodium hypochlorite as its only active ingredient, may be used as a source of chlorine for disinfecting. The amount of sodium hypochlorite, usually given on the label. (In recent years, perhaps as a precaution against drinking undiluted chlorine bleach, many

bleach containers show a warning such as "Not For Personal Use." This warning can be safely ignored. The label states that the bleach contains only sodium hypochlorite as its active ingredient, and that the instructions are used to disinfect water.) Add 1 scant teaspoonful of bleach to each quart of clear water, and stir. Add 2 scant teaspoonfuls if the water is muddy or colored. Wait at least 15 minutes to allow enough time for the chlorine to kill all the microorganisms. 24 Properly disinfected water should be free of chlorine odor.

To disinfect small quantities of water, put 2 drops of household bleach containing 5.25% sodium hypochlorite in each quart of clear water. Use 4 drops if the water is muddy or colored. 24 If a dropper is not available, use a square-ended strip of paper or thin cloth about 1/4 inch wide by 2 inches long. Put one end in the water and the other end hanging down about 2 inch beyond the end of the spoon. Then when bleach is placed in the water, the strip is carefully tipped, drops the size of those from a medicine dropper will drip off the end of the strip.

As a second choice, 2% tincture of iodine can be used. Add 5 drops to each quart of clear water. Wait 15 minutes. 24 If the water is cloudy, add 10 drops to each quart. Commercial water purification tablets may also be used.

directed.

If neither safe water nor chemicals for disinfecting it are available during a crisis, store at hand even muddy river water. Most mud settles to the bottom in a few days; even in often could be found to boil water. Bringing water to a boil for one minute kills all type Boiling for 10 to 20 minutes is required to kill some rarer infective organisms.

#### SOURCES OF WATER IN FALLOUT AREAS

Survivors of a nuclear attack should realize that neither fallout particles nor dissolved compounds can be removed from water by chemical disinfection or boiling. Therefore, water the least radioactive sources available. Before a supply of stored drinking water has been should be located. The main water sources are given below, with the safest source listed f listed in decreasing order of safety.

1 . Water from deep wells and from water tanks and covered reservoirs into which no fallout fallout-contaminated water has been introduced. (Caution: Although most spring water would spring water is surface water that has flowed into and through underground channels without filtered.)

2. Water from covered seepage pits or shallow, hand-dug wells. This water is usually safe fallout-contaminated surface water has been prevented from entering by the use of waterpro by waterproofing the surrounding ground to keep water from running down outside the well c is taken from a Chinese civil defense manual. 2 1 It shows a well dug to obtain safe water contaminated source. If the earth is not sandy, gravelly, or too porous, filtration through effective.

3. Contaminated water from deep lakes. Water from a deep lake would be much less contaminated radioactive material and fallout particles than water from a shallow pond would be, if both amount of fallout per square foot of surface area deposited in them. Furthermore, fallout bottom more rapidly in deep lakes than in shallow ponds, which are agitated more by wind.

4. Contaminated water from shallow ponds and other shallow, still water.

5. Contaminated water from streams, which would be especially dangerous if the stream is n heavy rains after fallout is deposited.

(Illustration)Fig. 8.9. A water- filtering well. This Chinese drawing specifies that this 10 meters (roughly 5 to 10 yards) from a pond or stream, (nw071.tif)

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The first runoff will contain most of the radioactive material that can be dissolved from deposited on the drainage area. 2 3 Runoff after the first few heavy rains following the c likely to contain much dissolved radioactive material, or fallout.

6. Water collected from fallout-contaminated roofs. This would contain more fallout partic runoff from the ground.

7. Water obtained by melting snow that has fallen through air containing fallout particles the ground onto which fallout has fallen. Avoid using such water for drinking or cooking,

#### WATER FROM WELLS

The wells of farms and rural homes would be the best sources of water for millions of survivors after a massive nuclear attack, the electric pumps and the pipes in wells usually would be useless. Areas would be eliminated by the effects of electromagnetic pulse (EMP) from high-altitude bursts of blast and fire on power stations, transformers, and transmission lines. However, enough time would be available to remove these pipes and pumps from wells so that bail-cans could be used to reach water for drinking and basic hygiene.

Fig. 8.10. Lower part of an expedient bail-can. The unattached, "caged" valve can be made of wood or plastic that does not have the springiness of soft rubber. ORNL-DWO 78-669 1R (Illustration) (nw073.tif)

How to make a simple bail-can is illustrated in Fig. 8. 10. An ordinary large fruit-juice can is at least 1 inch smaller than the diameter of the well-casing pipe. A hole about 1 inch in diameter is cut in the center of the can's bottom. The hole should be cut from the inside of the can: this keeps the hole smooth, so it will serve as a smooth seat for a practically watertight valve. To cut the hole in the wood surface and press down repeatedly with the point of a sheath knife, a butcher knife, or a screwdriver.

The best material for the circular, unattached valve shown in Fig. 8. 10 is soft rubber, such as inner-tube rubber. Alternately, the lid of a can about 3/4 inch smaller in diameter than the hole in the can, with several thicknesses of plastic film taped to its smooth lower side. Plastic film about 4 mil thick. The handle of a bail-can should be made of wire, with a loop at the top to which a rope or string can be attached.

Filling-time can be reduced by taping half-a-pound of rocks or metal to the bottom of the bail-can.

#### REMOVING FALLOUT PARTICLES AND DISSOLVED RADIOACTIVE MATERIAL FROM WATER

The dangers from drinking fallout contaminated water could be greatly lessened by using expedient filtration methods to remove fallout particles and most of the dissolved radioactive materials. Heavy fallout, less than 2% of the radioactivity of the fallout particles contained in the water. If nearly all the radioactive fallout particles could be removed by filtering, the health hazards would be likely to result from drinking and cooking with most fallout-contaminated water.

##### 0 Filtering

Filtering through earth removes essentially all of the fallout particles and more of the dissolved radioactive materials than does boiling-water distillation, a generally impractical purification method that does not remove dissolved active iodines. Earth filters are also more effective in removing radioactive iodines than water softeners or charcoal filters. In areas of heavy fallout, about 99% of the radioactivity is removed by filtering it through ordinary earth. 7 3

Fig. 8.11. Expedient filter to remove radioactivity from water. ORNL DWG 77- 18431 (Illustration) (nw073.tif)



To make the simple, effective filter shown in Fig. 8.11, the only materials needed are those found at home. This expedient filter can be built easily by proceeding as follows:

1. Perforate the bottom of a 5-gallon can, a large bucket, a watertight wastebasket, or a similar container with about a dozen nail holes. Punch the holes from the bottom upward, staying within about 2 inches of the center.
2. Place a layer about 1 inch thick of washed pebbles or small stones on the bottom of the container. If pebbles are not available, twisted coat-hanger wires or small sticks can be used.
3. Cover the pebbles with one thickness of terrycloth towel, burlap sackcloth, or other quality cloth in a roughly circular shape about 3 inches larger than the diameter of the can.
4. Take soil containing some clay almost any soil will do from at least 4 inches below the surface of the ground. (Nearly all fallout particles remain near the surface except after deposition on snow or ice.)
5. Pulverize the soil, then gently press it in layers over the cloth that covers the pebbles, forcing it snugly against the sides of the can. Do not use pure clay (not porous enough) or sand (too fine). The soil layer in the can should be 6 to 7 inches thick.
6. Completely cover the surface of the soil layer with one thickness of fabric as porous as possible to keep the soil from being eroded as water is poured into the filtering can. The cloth should be held in place by a dozen small stones placed on the cloth near its edges with care. The cloth should be adequately covered.
7. Support the filter can on rods or sticks placed across the top of a container that is 1 foot higher than the filter can. (A dishpan will do.)

The contaminated water should be poured into the filter can, preferably after allowing it to settle. The filtered water should be disinfected by one of the previously described methods.

If the 6 or 7 inches of filtering soil is a sandy clay loam, the filter initially will deliver about 1 quart per hour. (If the filtration rate is faster than about 1 quart in 10 minutes, remove the soil.) After several hours, the rate will be reduced to about 2 quarts per hour.

When the filtering rate becomes too slow, it can be increased by removing and rinsing the soil, and then replacing the fabric. The life of a filter is extended if muddy water is first allowed to settle for several hours in a separate container, as described above. After 1 quart has been filtered, rebuild the filter by replacing the used soil with fresh soil.

#### ° Settling

Settling is one of the easiest methods to remove most fallout particles from water. Further settling of muddy or murky water before filtering will extend the life of the filter. To settle water:

1. Fill a bucket or other deep container three quarters full of the contaminated water.
2. Dig pulverized clay or clayey soil from a depth of four or more inches below ground surface. Use about a 1-inch depth of dry clay or dry clayey soil for every 4-inch depth of water.

practically all the clay particles are suspended in the water.

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3. Let the clay settle for at least 6 hours. The settling clay particles will carry most of the particles to the bottom and cover them.

4. Carefully dip out or siphon the clear water, and disinfect it.

#### ° Settling and Filtering

Although dissolved radioactive material usually is only a minor danger in fallout-contaminated water, you should use an earth filter even the clear water produced by settling, if an earth filter is available. Finally, disinfect the water.

#### POST-FALLOUT REPLENISHMENT OF STORED WATER

When fallout decays enough to permit shelter occupants to go out of their shelters for short periods, they should try to replenish their stored water. An enemy may make scattered nuclear strikes for weeks after an attack. Some survivors may be forced back into their shelters by the resultant fallout. The containers should be used to store the least contaminated water within reach. Even without fallout, water and stored shortly after the occurrence of fallout will become increasingly safer with time as the decay of radioactive iodines. These would be the most dangerous contaminants of water during the period after an attack.

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#### Chapter 9

##### Food

#### MINIMUM NEEDS

The average American is accustomed to eating regularly and abundantly. He may not realize that food would not be essential for survival during the first two or three weeks following a nuclear attack. He would be infants, small children, and the aged and sick, some of whom might die within a week of the attack. Other things are more important for short-term survival: adequate shelter against fallout, an adequate supply of air, and enough water.

The average American also may not realize that small daily amounts of a few unprocessed staples can keep him to survive for many months, or even for years. A healthy person if he is determined to eat and to prepare and use whole-grain wheat or corn can maintain his health for several months. If fruits and vegetables are substituted for some of the grain, the ration would be improved and could maintain health. The nutritional information given in this chapter is taken from a July, 1979 publication, "Survival During a Prolonged Food Crisis." 26 This book brings together from worldwide sources the information needed to help unprepared people use unaccustomed foods advantageously during the prolonged period following a heavy nuclear attack. The practical know-how which will be given in this chapter requires

cessing and cooking of basic grains and beans is based on old ways which are mostly unknown. These methods have been improved and field-tested by civil defense researchers at Oak Ridge.

#### LOSS OF HIGH-PROTEIN ANIMAL FOODS

A massive nuclear attack would eliminate the luxurious, complicated American system of food processing, and distribution. Extensive, heavy fallout and the inability of farmers to feed the cattle, hogs, and chickens that are the basis of our high-protein diet. The livestock and their owners' inability to care for them would be cattle on pasture. However, these grazing animals would suffer severe radiation damage. 27 Also, they would suffer radiation burns from fallout in the downwind area where the total dose from gamma radiation emitted within a few days from fallout might be only 1 50 R, most grazing animals probably would be killed by the combined effect of gamma radiation, beta burns, and internal radiation. 2 7

#### PRECAUTIONS WHEN EATING MEAT

In areas where the fallout would not be enough to sicken animals, their meat would be safe to eat; however, animals that have eaten or drunk fallout-contaminated food or water will have incorporated radioactive atoms and molecules in their internal organs. The thyroid gland, kidneys, and liver especially are likely to be affected.

If an animal appears to be sick, it should not be eaten. The animal might be suffering from radiation sickness and might have developed a bacterial infection as a result of this dose. Meat from animals infected with bacteria produced by some kinds of bacteria could cause severe illness or death if eaten, even if the meat is cooked.

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Under crisis conditions, all meat should be cooked until it is extremely well done - cooked until it loses the last of its pink color. To be sure that the center of each piece of meat is well cooked, meat should be cut into pieces that are less than 1/2-inch thick before cooking. This procedure saves time and saves fuel.

#### SURVIVAL OF BREEDING STOCK

Extensive areas of the United States would not receive fallout heavy enough to kill grazing animals. The food of surviving animals would provide some food and the fertile breeding stock needed for natural reproduction. The fertility caused by severe radiation doses is rarely permanent. Extensive experiments with the offspring of severely irradiated animals are healthy and fertile. 27

#### LIVING ON BASIC PLANT FOODS

Even if almost all food-producing animals were lost, most surviving Americans should be able to live and multiply on the food that enable most of the world's population to live and multiply: grains, beans, and vegetables. Despite the remarkable productivity of American agriculture, there usually would be enough grain and beans to feed the surviving Americans with sufficient food for at least a year following a heavy nuclear attack. To get the unprocessed foods, which are stored in food-producing regions, to the majority of the population outside these regions.

Surprisingly little transportation would be needed to carry adequate quantities of these vegetables and grains.

vors in famine areas. A single large trailer truck can haul 40,000 pounds of wheat enough feeling hunger pains for a day. More than enough such trucks and the fuel needed to carry areas would survive a massive nuclear attack. 28 It is likely that reasonably strong America would prevail so that, after the first few weeks, millions of the survivors in starving areas would have access to processed foods.

Eating food produced in the years after a large attack would cause an increase in the cancer rate due to its content of radioactive strontium and cesium from fallout-contaminated soil. Over the long term, this increase would be a small fraction of the number of additional cancer deaths that would be caused by radiation. 29 Cancer deaths would be one of the tragic, delayed costs of a nuclear war, but not numerous enough to endanger the long-term survival of the population.

#### LIVE OFF THE LAND?

Very few survivors of a heavy attack would be in areas where they could live off the land and gatherers. In extensive areas where fallout would not be heavy enough to kill human beings, but would die from the combined effects of external gamma radiation, swallowed fallout particles, and radioactive contamination of the soil. Survival plans should not include dependence on hunting, fishing, or gathering wild

#### FOOD FOR SHELTER OCCUPANTS

Most people would need very little food to live several weeks; however, the time when survival would leave their shelters would mark the beginning of a much longer period of privation. Therefore, to maintain physical strength and morale, persons in shelters ideally should have food that provides well-balanced, adequate meals for many weeks.

In most American homes there are only enough ready-to-eat, concentrated foods to last a few days. It would be an important survival advantage to keep on hand a two-week supply of easily transported, non-perishable foods. In case, occupants of shelters would be uncertain about when they could get more food and would have to make decisions about how much to eat each day. (Those persons who have a fallout meter, such as the one described in Chapter 10, could estimate when and for how long they could emerge from the shelter. As a result, these persons could ration their limited foods more effectively.)

During the first few weeks of a food crisis, lack of vitamins and other essentials of a well-balanced diet would be of primary importance to previously well-nourished people. Healthful foods with enough calories and adequate energy would meet short-term needs. If water is in short supply, high-protein foods should be eaten only in moderation, since a person eating high-protein foods requires more water than is required for an equal number of calories from foods high in carbohydrates.

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#### EXPEDIENT PROCESSING OF GRAINS AND SOYBEANS

Whole-kernel grains or soybeans cannot be eaten in sufficient quantities to maintain vigor when boiled or parched. A little boiled whole-kernel wheat is a pleasantly chewy breakfast cereal. The pioneers got sore tongues and very loose bowels when they tried to eat enough boiled whole-kernel wheat to meet even half of their daily energy needs. Some pioneers, however, ate large quantities of whole-kernel wheat and had harmful results after boiling and simmering it for many hours. Even the most primitive people on grains grind or pound them into a meal or paste before cooking. (Rice is the only important grain that Americans know how to process whole-kernel grains and soybeans (our largest food reserves)

rance could be fatal to survivors of a nuclear attack.

Making an expedient metate, the hollowed-out grinding stone of Mexican Indians, proved impractical post-attack conditions. Pounding grain into meal with a rock or a capped, solid-ended pestle was slow work. The best expedient means developed and field-tested for pounding grain or beans was an improvised 3-pipe grain mill. Instructions for making and using this effective grain-pounder are given below.

#### IMPROVISED GRAIN MILL

The grain mill described can efficiently pound whole-grain wheat, corn, etc., into meal and improve digestibility and avoid the diarrhea and sore mouths that would result from eating unground grain.

#### TO BUILD:

- (1) Cut 3 lengths of pipe, each 30 inches long; 3/4-inch-diameter steel pipe (such as ordinary galvanized pipe).
- (2) Cut the working ends of the pipe off squarely. Remove all roughness, leaving the full-diameter working end should have the full diameter of the pipe.
- (3) In preparation for binding the three pieces of pipe together into a firm bundle, encircle the pipes with cushioning, slip-preventing tape, string or cloth in the locations illustrated.
- (4) Tape or otherwise bind the 3 pipes into a secure bundle so that their working ends are in the same plane resting evenly on a flat surface.
- (5) Cut the top smoothly out of a large can. A 4-inch-diameter, 7-inch-high fruit-juice can will do. If you don't have a can, improvise something to keep grain together while pounding it.

Fig. ORNL-DWO 73-11449 (Illustration) (nw077.tif)

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#### TO MAKE MEAL AND FLOUR:

- (1) Put clean, dry grain ONE INCH DEEP in the can.
- (2) To prevent blistering your hands, wear gloves, or wrap cloth around the upper part of the can.
- (3) Place the can (or open-ended cylinder) on a hard, smooth, solid surface, such as concrete.
- (4) To pound the grain, sit with the can held between your feet. Move the bundle of pipes back and forth about 3 inches, with a rapid stroke.
- (5) If the can is 4 inches in diameter, in 4 minutes you should be able to pound 1-1/2 lb. of wheat into 1/5 lb. of fine meal and flour, and 3/10 lb. of coarse meal and fine-cracked wheat.
- (6) To separate the pounded grain into fine meal, flour, coarse meal, and fine-cracked wheat, use a window screen.

(7) To separate flour for feeding small children, place some pounded grain in an 18 X 18 net, gather the edges of the net together so as to hold the grain, and shake this bag-like

(8) To make flour fine enough for babies, pound fine meal and coarse flour still finer, and of cheesecloth or similar material.

As soon as fallout decay permits travel, the grain-grinding machines on tens of thousands should be used for milling grain for survivors. It is vitally important to national recovery back as soon as possible to labor-saving, mechanized ways of doing essential work.

In an ORNL experiment, a farmer used a John Deere Grinder-Mixer powered by a 100-hp tractor samples of wheat and barley. When it is used to grind rather coarse meal for hogs, this machine takes 1 hour. Set to grind a finer meal-flour mixture for human consumption, it ground both hard wheat at a rate of about 9 tons per hour. This is 2400 times as fast as using muscle power to operate a grain mill. With its finest screen installed, this large machine can produce about 3 tons of flour per hour. Unlike wheat and corn, the kernels of barley, grain sorghums, and oats have rough, fibrous hulls from the digestible parts to produce an acceptable food. Moistening the grain will toughen the hulls, making them easier to remove. If the grain is promptly pounded or ground into meal, the toughened hull pieces are smaller than will the hulls of undampened grain. A small amount of water, weighing about 2% of the grain, should be used to dampen the grain. For 3 pounds of grain (about 6 cups), sprinkle 60 grams, or about 2 tablespoons) of water, while stirring constantly to moisten all the kernels. After stirring, the grain will appear dry. The small amount of water will have dampened and toughened the hull parts- inside will have remained dry. Larger pieces of hull are easier to remove after

One way to remove ground-up hulls from meal is by flotation. Put some of the meal-hulls mixture deep in a pan or pot, cover the mixture with water, and stir. Skim off the floating hulls, and discard more hulls. Sunken pieces of hulls that settle on top of the heavier meal can be removed with a spoon. When the water is poured off. To produce a barley meal good for very small children, the meal and hulls can be separated by flotation.

Figure 9.1 illustrates sieving fine, dry barley meal and the smaller pieces of hulls from the larger pieces. The sieve was a piece of window screen that measured 18 inches before its sides were bent to form an open-topped box.

(Illustration ) Fig. 9. 1 . Sieving ground barley through a window screen sieve, (nw078.jpg)

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To lessen their laxative effects, all grains should be ground as finely as possible, and meal should be removed. Grains also will be digested more easily if they are finely ground. The occupants should be especially careful to avoid foods that cause diarrhea.

#### COOKING WITH MINIMUM FUEL

In areas of heavy fallout, people would have to remain continuously in crowded shelters for weeks. They would have to stay in the shelters most of each 24 hours for weeks. Most shelter occupants should have a supply of their ready-to-eat foods; therefore, they should have portable, efficient cook stoves. Another reason: to help maintain morale. Even in warm weather, people need some hot food and a comforting effect and to promote a sense of well-being. This is particularly true when people are in shelters. The Stove pictured on the following pages (Figs. 9.2 and 9.3) was the most satisfactory of the stoves developed at Oak Ridge and later field-tested.

## ° Bucket Stove

If operated properly, this stove burns only about 1/2 pound of dry wood or newspaper to heat from 600 F to boiling.

Materials required for the stove:

- \* A metal bucket or can, 12- to 16-quart sizes preferred. The illustrations show a 14-quart pot.

- \* Nine all-metal coat hangers for the parts made of wire. (To secure the separate parts of coat-hanger wire grate, 2 feet of finer wire is helpful.)

- \* A 6 X 10-inch piece of a large fruit-juice can, for a damper.

Construction:

With a chisel (or a sharpened screw driver) and a hammer, cut a 4- 1/2 X 4-1/2-inch hole in about 1-1/2 inches above its bottom. To avoid denting the side of the bucket when chiseling bucket over the end of a log or similar solid object.

To make the damper, cut a 6-inch- wide by 10-inch-high piece out of a large fruit-juice can metal. Then make the two coathanger-wire springs illustrated, and attach them to the piece hammering the outer 1 inch of the two 6-inch-long sides over and around the two spring wires slid up and down, to open and close the hole in the bucket. The springs hold it in any desired position. If these springs are not available, the air supply can be regulated fairly well by placing a piece of metal so that it will block off part of the hole in the side of the bucket.)

To make a support for the pot, punch 4 holes in the sides of the bucket, equally spaced 4 inches below the bucket's top. Then run a coat-hanger wire through each of the two pairs of holes in the sides of the bucket. Bend these two wires over the top of the bucket, as illustrated, so that they hold the pot centered in the bucket. Pressure on the pot from these four wires does not hinder putting it into the stove or taking it out.

Bend and twist 4 or 5 coat hangers to make the movable grate, best made with the approximate dimensions in Fig. 9.2.

For adjusting the burning pieces of fuel on the grate, make a pair of 12-inch-long tongs constructed as illustrated by Fig. 9.3.

To lessen heat losses through the sides and bottom of the bucket, cover the bottom with ash or earth. Then line part of the inside and bottom with two thicknesses of heavy-duty aluminum foil.

To make it easier to place the pot in the stove or take it out without spilling its contents, attach a handle with a longer piece of strong wire.

Operation:

The Bucket Stove owes its efficiency to: (1 ) the adjustable air supply that flows up through the movable grate that lets the operator keep the maximum amount of flame in contact with

pot, and (3) the space between the sides of the pot and the inside of the bucket that keep contact with the sides of the pot.

In a shelter, a Bucket Stove should be placed as near as practical to an air exhaust opening in it.

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Fig. 9.2. Bucket-stove with adjustable damper and movable wire grate. Photo 1397-78A (nw08

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Fig. 9.3. Bucket-stove with its sliding damper partly closed. Foot-long tongs of coat hanger are useful when burning twisted half-pages of newspaper. (nw08 1 .jpg)

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If wood is to be burned, cut and split dry wood into small pieces approximately 1/2 inch square. Start the fire with paper and small slivers of wood, placing some under the wire grate. To keep the fire burning in a humid shelter, keep it in a large plastic bag.

If newspaper is to be burned, use half-pages folded and twisted into 5-inch-long "sticks," and with wire tongs, feed a paper "stick" into the fire about every half-minute.

Add fuel and adjust the damper to keep the flame high enough to reach the bottom of the pot. The fire should go up the sides of the pot.

To use the Bucket Stove for heating in very cold weather, remove the pot and any insulating material from the bucket; burn somewhat more fuel per minute.

If used with the Fireless Cooker described on the following pages, a Bucket Stove can be used to cook beans, grain, or tough meat in water. Three quarts of such food can be cooked with less fuel than to boil an egg over a small campfire.

#### ° Fireless Cooker

Fig. 9.4. Boiling-hot pot of food being placed in an expedient Fireless Cooker, (nw082.jpg)

A Fireless Cooker cooks by keeping a lidded pot of boiling-hot food so well insulated all around that it cools very slowly. Figure 9.4 shows one of these simple fuel-saving devices made from a bushel basket lined with newspapers, with a towel-lined cavity in the center. The Cavity is the size of the 6-quart pot. The pot goes all around the pot and will be placed over it to restrict air circulation. If the boiler is lined with newspapers about 4 inches thick, the temperature will remain for hours so near boiling that slow-cooking food will be ready to eat.

The essential materials for making an effective Fireless Cooker are enough of any good insulating material (blankets, coats, paper, hay that is dry and pliable) to cover the boiling-hot pot all over.



insulation. A container to keep the insulating materials in place around the pot is useful

Wheat, other grains, and small pieces of tough meat can be thoroughly cooked by boiling them about 5 minutes, then insulating the pot in a Fireless Cooker for 4 or 5 hours, or overnight. Beans boiled for 10 to 15 minutes before they are placed in a Fireless Cooker.

#### COOKING GRAIN AND BEANS WHEN SHORT OF FUEL OR POTS

##### 0 Cooking Grain Alone

When whole grains are pounded or ground by expedient means, the result usually is a mixture of meal, and a little flour. Under shelter conditions, the best way to cook such meal is first to boil 4 parts of water for 1 part of meal). Add 1 teaspoon (5 grams) of salt per pound of dry meal. When the water is on fire (or stop adding fuel to a Bucket Stove) and quickly stir the meal into the hot water. In briskly boiling water, lumping becomes a worse problem.) Then, while stirring constantly, bring to a boil. Since the meal is just beginning to swell, more unabsorbed water remains, so the result is thinner than if the meal were added to cold water and then brought to a boil.

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If any type of Fireless Cooker is available, the hot cereal only has to be boiled and stirred until the watery part remains. This usually takes about 5 minutes. Continue to cook, either in a Fireless Cooker for at least 4 or 5 hours, or by boiling for an additional 15 or 20 minutes.

When it is necessary to boil grain meal for many minutes, minimize sticking and scorching by using at least 4 parts of water. However, cooking a thinner hot cereal has a disadvantage: a larger increased volume of food must be eaten to satisfy one's energy needs.

If grain were the only food available, few Americans doing physical work could eat enough to gain weight at first, until their digestive tracts enlarged from eating the very bulky foods. This was the case with the early pioneers.

months. Small children could not adjust adequately to an all-grain diet: for them, concentrates are needed to provide enough calories to maintain growth and health.

##### ° Cooking Grain and Beans Together

When soybeans are being used to supplement the lower quality proteins of grain and when fuel is scarce, first grind or pound the beans into a fine meal. To further reduce cooking time, soak the meal a couple of hours, keeping it covered with water as it swells. Next put the soaked bean meal in a pot with 3 times as much water as the combined volume of a mixture of 1 part of dry bean meal and 3 parts of grain meal. Gently boil the bean meal for about 15 minutes, stirring frequently, before adding the grain meal.

Stop boiling and add the grain meal while stirring constantly. Again bring the pot to a boil, being careful to avoid burning and scorching, and boil until the meal has swelled enough to have absorbed all the water. Then cook the grain-bean mush for another 15 minutes or more before eating, or put it in a fireless cooker.

Soybeans boiled alone have a taste that most people find objectionable. Also, whole soybeans need to be soaked a couple of hours to soften them sufficiently. But if soybeans are pounded or ground into meal, the objectionable taste of the soybean meal is boiled with 4 parts of meal made from corn or another grain, the scorching taste and the sweetish taste to the resulting mush. The unpleasant soybean taste is eliminated. If cooked in a fireless cooker, the mush can be eaten

beans and other beans or dried peas can be made digestible and palatable with minimum cook

#### 100% GRAIN AND 100% BEAN DIETS

A diet consisting solely of wheat, com, or rice, and salt has most of the essential nutrients would be vitamins A, C, and D. Such a grain -based diet can serve adults and older children for months. Table 9. 1 shows how less than 1-3/4pounds of whole wheat or dry yellow corn satisfy essential nutritional requirements of a long-term emergency ration. [The nutritional values in bold type, to make an easier comparison with the Emergency Recommendations, also printed in bold type, are given in kilocalories (kcal), commonly called calories (Cal).] Expedient ways of saving space from these rations are described in a following section of this chapter.

Other common whole grains would serve about as well as wheat and yellow corn. At least 1/6 cup (about 5 grams) is essential for any ration that is to be eaten for more than a few days, 1/2 tablespoon) should be available to allow for increased salt needs and to make grain and bean diets. Additional salt would be consumed as needed.

To repeat: few Americans at first would be able to eat the 3 or 4 quarts of thick mush that would be required with a ration consisting solely of whole-kernel wheat or corn. Only healthy Americans determined to fare well for months on such unaccustomed and monotonous food as an all-grain diet. Different kinds of grain and cooking in different ways would make an all-grain diet both more palatable and more nourishing.

Not many people would be able to eat 27 oz (dry weight before cooking) of beans in a day, but they could eat a daily ration of almost 23 oz of soybeans. Beans as single-food diets are not recommended because their protein content requires the drinking of more fluids. Roasted peanuts would provide a better alternative.

#### GRAIN SUPPLEMENTED WITH BEANS

People who live on essentially vegetarian diets eat a little of their higher-quality protein with the grain that is their main source of nutrition. Thus Mexicans eat some beans along with their corn, Chinese eat a little fermented soybean food or a bit of meat or fish with a bowl of rice.

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Table 9. 1 . Daily rations of 100% grain, beans, or peanuts a)

Wheat

(dry)

Yellow

Field

Com b)

(Dry)

Emergency

Recommen-dations

Soybeans

(dry)

Red Beans

(dry)

Peanuts

(roasted)

Weight g

790

750

645

760

447

(27.8 oz.)

(26.4 oz.)

(22.7 oz.)

(26.8 oz.)

(15.8 oz.)

Energy, kcal

2,600

2,600

2,600

2,600

2,600

2,600

Protein, g

103

67

55 c)

220

171

117

Fat, g

15

29

30

114

11

218

Calcium, mg

324

165

400

1,458

836

322

Magnesium, mg

1,260

1,100

200-300

1,710

1,240

782

Iron, mg

26

15.7

10

54.2

52.4

9.8

Potassium, mg

2,920

2,130

1,500

2,000

10,500

7,420

vitamin A, RE

0

368

555

52

15

0

Thiamin, mg

4.3

2.8

1

7.1

3.9

1.3

Riboflavin, mg

1

0.9

1.4

2

1.5

0.6

Niacin, mg

34

16.5 d)

17

14.2

17.5

76.4

Vitamin C, mg

0

0

15-30

0

0

0

Vitamin D, ug

0

0

0 e)

0

0

0

a) Salt (1/3 oz., or 10 g., or 3/4 tablespoon) should be available. This would be consumed

b) White com supplies no Vitamin A, whereas yellow corn supplies 49 RE (retinol equivalent value) per 100 g dry weight. Most com in the United States is yellow com.

c) If a diet contains some animal protein such as meat, eggs, or milk, the recommended protein is 41 g per day. If most of the protein is from milk or eggs, only 41 g per day is recommended.

d) The niacin in corn is not fully available unless the corn is treated with an alkali, such as lye (and many Americans) add to the water in which com kernels are soaked or boiled.

e) Infants, children, and pregnant and lactating women should receive 10 g (10 micrograms, or 400 IU) of vitamin D per day. For others, the current recommended daily allowance (RDA) for vitamin D is 200 IU (5 g).

Nutritionists have found that grains are low in some of the essential amino acids that the body needs to build its proteins. For long-term good health, the essential amino acids must be supplied each meal by eating some foods with more complete proteins than grains have. Therefore, in a diet of grains one should strive to eat at every meal at least a little of any higher-quality protein food. These include ordinary beans, soybeans, milk powder, meat, and eggs.

Table 9.2 shows that by adding 7.0 oz (200 g) of red beans (or other common dried beans) to either whole wheat or yellow corn, with salt added, you can produce rations that contain all the important nutrients except vitamin C, vitamin A, vitamin D, and fat. If 5.3 oz (150 g) of the red beans, the fat requirement is satisfied. The 600 g of yellow com contains enough calories to produce more than half the emergency recommendation of vitamin A. The small deficiencies of vitamin A and D will not cause sickness.

Other abundant grains, such as grain sorghums or barley, may be used instead of the wheat in Table 9.2 to produce fairly well-balanced rations. Other legumes would serve to supplement the grains and beans. (Peanuts are the exception: although higher in energy (fat) than any other unprocessed food, their protein is not as high as that of other legumes.)

EXPEDIENT WAYS TO SUPPLY DEFICIENT ESSENTIAL NUTRIENTS

## 0 Vitamin C

A deficiency of vitamin C (ascorbic acid) causes scurvy. This deadly scourge would be the to afflict people having only grain and/or beans and lacking the know-how needed to sprout vitamin C. Within only 4 to 6 weeks of eating a ration containing no vitamin C, the first pear: swollen and bleeding gums. This would be followed by weakness, then large bruises, h that would not heal. Finally, death from hemorrhages and heart failure would result.

The simplest and least expensive way to make sure that you, your family and neighbors do n attack from scurvy is to buy one kilogram (1 ,000,000 milligrams) of pure vitamin C, which acid" form. Unlike vitamin C tablets, pure vitamin C crystals do not deteriorate. An inexp Bronson Pharmaceutical, 4526 Rinetti Lane, La Canada, California 9 101 1 ; in 1988 1 bough postage paid. An ample daily dose is 25 milligrams, about 0.0009 ounce. Ten grams (about c for a whole year for one person who is eating only unsprouted grain and/or other foods pro gram (1,000 mg) of crystalline ascorbic acid is 1/4 teaspoonful. If you do not have a 1/4 spoonful of the crystals on a piece of paper, and divide the little pile into 4 equal part 1,000 mg. One of these 1,000 mg piles can easily be divided into 4 tiny piles, each 250 mg ample daily doses of 25 mg each. If your family has a 1 ,000,000 mg supply, taking a 50 mg line ascorbic acid may be preferred, either sprinkled on food or dissolved in water.

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Table 9.2. Daily rations of whole wheat or yellow com supplemented with soybeans or red be daily salt ration, including salt in food: 3/4 tablespoon (1/3 Oz, or ID g).

600 g (21.1 oz.)  
Whole wheat plus  
200 g (7.0 oz.)  
Red beans (dry wt)

600 g (21.1 oz.)  
Whole wheat plus  
150 g (5.3 oz.)  
Soybeans (dry wt)

Emergency

Recommendations

600 g (21.1 oz.)  
Yellow Com \*' plus  
150g(5.3oz.)  
Soybeans (dry wt)

600g(21.1 oz.)  
Yellow Com a plus  
150g(5.3oz.)  
Soybeans (dry wt)

Energy, kcal



2,666

2,585

2,600

2,693

2,774

Protein, g

123

129

55 a) b)

105

98

Fat, g

15

39

30

50

26

Calcium, mg

466

585

400

471

352

Magnesium, mg

1,286

1,358

200-300

1,280

1,208

Iron, mg

33.6

32.4

10

25.2

26.4

Potassium, mg

4,188

4,736

1,500-2,000

4,220

3,672

vitamin A, RE

4

12

555

306

298

Thiamin, mg

4.3

5

1

3.9

3.2

Riboflavin, mg

1.1

1.2

1.4

1.2

1.1

Niacin, mg

30.4

29.1

17

16.5 c)

17.8

Vitamin C, mg

0

0

15-30

0

0

Vitamin D, ug

0

0

0 d)

0

0

a) White corn supplies no vitamin A, whereas yellow corn supplies 49RE (retinol equivalent, vitamin A value) per 100g dry weight. Most corn in the United States is yellow corn.

b) If a diet contains animal protein such as meat, eggs or milk, the recommended protein is 55g per day. If all the protein is from milk or eggs, only 41g per day is required.

c) The niacin in corn is not fully available unless corn is treated with an alkali, such as added by Mexicans and Americans in the South and Southwest to the water in which they soak corn kernels.

d) Infants, children, and pregnant and lactating women should receive 10 g (10 micrograms) vitamin D. For others, the Current recommended daily allowance (RDA) for vitamin D is 200

One good expedient way to prevent or cure scurvy is to eat sprouted seeds not just the sprouts. This prevented scurvy during a famine in India. Captain James Cook was able to keep his sailors healthy during a three-year voyage by having them drink an unfermented beer made from dried, sprouted seeds. The Chinese have prevented scurvy during the long winters of northern China by consuming sprouted seeds.

Only 10 mg of vitamin C taken each day (1/5 of the smallest vitamin C tablet) is enough to prevent scurvy. A little over an ounce (about 30 grams) of dry beans or dry wheat is sprouted until the sprouts appear. The sprouted seeds will supply 10 to 15 mg of vitamin C. Such sprouting, if done at room temperature, requires about 48 hours. To prevent sickness and to make sprouted beans more digestible, they should be boiled in water for not longer than 2 minutes. Longer cooking will destroy too much vitamin C.

Usual sprouting methods produce longer sprouts than are necessary when production of enough sprouts is the objective. These methods involve rinsing the sprouting seeds several times a day in safe water. In areas not confined to shelters are likely to be short of water, the method illustrated in Fig. 9.5 is preferred. Seeds to be sprouted are picked clean of trash and broken seeds. Then the seeds are covered with water and soaked for about 12 hours. Next, the water is drained off and the soaked, swollen seeds are placed on a cloth or paper, or ajar, in a layer no more than an inch deep. If a plastic bag is used, you should make tiny holes in it. Sprinkle them a little, dampen them, and place them inside the bag, along its sides. As shown in Fig. 9.5, folded paper rolls keep the plastic from resting on the seeds and form an air passage down the center. The paper should be placed in the mouth of the bag or jar so as to leave an air opening of only one inch. If the paper is kept moist, the seeds will remain sufficiently damp while receiving enough circulation. They will sprout sufficiently after about 48 hours at normal room temperature.

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Fig. 9.5. Sprouting with minimum water, (nw086.jpg)

Sprouting seeds also increases their content of riboflavin, niacin, and folic acid. Sprouts are more digestible than raw, unsprouted beans, but not as easily digested or nourishing as are sprouted beans sauteed for a couple of minutes. Sprouting is not a substitute for cooking. Contrary to the claims of some publications, sprouting does not increase the protein content of seeds, nor does it improve digestibility. Sprouting reduces the caloric value of seeds. The warmth generated by germinating seeds reduces the caloric value somewhat, as compared to unsprouted seeds.

° Vitamin A

Well-nourished adults have enough vitamin A stored in their livers to prevent vitamin A deficiencies for several months, even if their diet during that time contains none of this essential vitamin. Infants and children develop deficiencies sooner than adults. The first symptom is an inability to see well in dim light, which then causes changes in body tissues. In infants and children, lack of vitamin A can result in serious problems even blindness. Therefore, a survival diet should be balanced with respect to vitamins, with children having priority.

Milk, butter, and margarine are common vitamin A sources that would not be available to most survivors if they were no longer available, yellow corn, carrots, and green, leafy vegetables (including dandelion leaves) are the best sources. If these foods were not obtainable, the next best source would be sprouted wheat grains if seeds could be sprouted for three days in the light, so that the sprouts are green. While sprouting is a source, sprouting is not a very satisfactory way to meet vitamin A requirements. The development of a survival diet makes 3-day sprouted wheat kernels difficult to eat. And one must eat a large amount of sprouts to satisfy the recommended daily emergency requirements up to 5 1/2 cups of sprouts.

and roots to satisfy the recommended daily emergency requirements up to 5 1/2 cups of sprouts. Survivors of a nuclear attack would wish they had kept an emergency store of multivitamin tablets.

#### ° Vitamin D

Without vitamin D, calcium is not adequately absorbed. As a result, infants and children with rickets (a disease of defective bone mineralization). A massive nuclear attack would cut off the vast majority of survivors from their main source of vitamin D, fortified milk.

Vitamin D can be formed in the body if the skin is exposed to the ultraviolet rays of the sun. However, one must be posed to sunlight very cautiously, initially for only a few minutes especially after a massive nuclear attack possibly could cause atmospheric changes that would permit more ultraviolet light to reach the earth, causing sunburn in the U.S. as severe as on the equator today. In cold weather, maximum exposure should be done in a shallow pit shielded from the wind. Exposure in a shallow pit would give protection from gamma radiation from fallout particles on the surrounding ground.

#### 0 Niacin and Calcium

Niacin deficiency causes pellagra, a disease that results in weakness, a rash on skin exposed to sunlight, diarrhea, and mental deterioration. If a typical modern American had a diet primarily of corn, which normally supply niacin, symptoms of pellagra would first appear in about 6 months. Since corn is the principal crop the U.S. production in 1985 was about 425 billion pounds the skillful treatment of corn is essential for post-attack survival and recovery.

During the first part of this century, pellagra killed thousands of Americans in the South who had corn for their principal staple and ate few animal protein foods or beans. 87

Yet Mexicans, who eat even more corn than did those Southerners – and have even fewer foods than we – do not suffer from pellagra.

The Mexicans' freedom from pellagra is mainly due to their traditional method of soaking corn in a lime-water solution. They use either dry, unslaked lime (calcium oxide, a dangerous material made by roasting limestone) or dry, slaked lime (calcium hydroxide, made by adding water to quicklime). The lime weighing about 1% as much as the dry corn is added to the soak water, producing an alkaline solution. The lime also can be used instead of lime to make an alkali solution. The alkali treatment of corn makes it more palatable and easier to digest.

the human body. Tables 9. 1 and 9.2 show com as having adequate niacin. However, the niacin is not readily available to the body unless the com has received an alkali treatment.

Treating corn with lime has another nutritional advantage: the low calcium content of com is increased.

#### 0 Fat

The emergency recommendation for fat is slightly over 1 ounce per day (30 g) of fat or cooking oil. Fat provides only 10% of the calories in the emergency diet, which does not specify a goal. Fat would be in very short supply after a nuclear attack. This amount is very low when compared with the intake in this country, in which fat provides about 40% of the calories. It would be difficult to consume sufficient calories to maintain normal weight and morale without a higher fat intake; fat should be available as soon as possible. Increased fat intake is especially important for young children for normal growth and development. Oak Ridge National Laboratory field tests have shown that people, especially, prefer considerably more oil added to grain mush than the emergency recommendation.

#### 0 Vitamin B-12 and Animal Protein

Vitamin B-12 is the only essential nutrient that is available in nature solely from animal sources. A person has a 2 to 4-year supply of vitamin B-12 stored in his liver, a deficiency should not develop before enough food of animal origin would again be available.

Many adults who are strict vegetarians keep in good health for years without any animal sources. Grains and beans together. It is more difficult to maintain normal growth and development on vegetarian diets. When sufficient animal sources of food are available, enough should be provided to supply animal protein daily. This could be provided by about 1.4 ounces (38 g) of lean meat, 0.7 cup of milk, or one medium-sized egg. When supplies are limited, young children should be given priority. High-grade supplementary protein foods should be eaten with every meal.

#### 0 Iron

Most people live out their lives without benefit of an iron supplement. However, many pregnant women and some children need supplemental iron to prevent anemia. One tested expedient way to increase iron intake is to use iron pots and pans, especially for cooking acid foods such as tomatoes. And iron nails (not galvanized nails) in vinegar until small amounts of iron begin to float to the surface in a few weeks. Then a teaspoon of iron-vinegar solution will contain about 30 to 60 mg of iron, enough for one day. The emergency recommendation is 10 mg per day. A teaspoon of the iron-vinegar solution is added to water. The iron content of fruit, such as an apple, can be increased by placing iron nails in the fruit.

#### FOOD RESERVES

Russia, China, and other countries that make serious preparations to survive disasters store large quantities of food primarily grain—both in farming areas and near population centers. In contrast, the United States stores grain and soybeans as an unplanned survival resource resulting from the production of more than enough for domestic use. The grain and soybeans can be eaten or sold abroad. The high productivity of U.S. agriculture is another unplanned survival resource. The calories and other essential nutrients for 100 million surviving Americans would necessitate about 12% of our 1985 crop of corn, wheat, grain sorghum, and soybeans — if nothing else was available.

In 1985, the U.S. production of corn, wheat, soybeans, and grain sorghum totaled about 625 7 pounds per day for one year for every American. A total of 2 pounds per person per day of the proportions shown in Table 9.2, would be sufficient to provide the essentials of an adult ration of about 27 ounces. (Grain sorghum is not listed in Table 9.2; it has approximately the same value as the remaining 5 ounces of the 2 pounds would feed enough chickens to meet a survivor's minimum requirement for animal protein.

If corn, wheat, grain sorghum, and soybeans were the only crops raised, the annual production would be only 730 pounds per person. Our 1985 annual production would have supplied every adult, child, and infant in a population of 100 million with 6250 pounds of these four staples. This is more than 8 times the requirement by Chinese standards.

Recovery from a massive nuclear attack would depend largely on sufficient food reserves being available to survivors to concentrate on restoring the essentials of mechanized farming. Enough housing or could be built to provide adequate shelter for the first few crucial years; enough clothing and medical supplies available. But if survivors were forced by hunger to expend their energies attempting primarily survival, many deaths from starvation would occur and the prospects for national recovery would be greatly reduced.

Americans' greatest survival asset at the end of 1985 was about 17 billion bushels (about 100 billion pounds) of wheat, corn, grain sorghums, and soybeans in storage, mostly on farms. If 200 million Americans survived a limited nuclear attack and if only half of this stored food reserve could be delivered to survivors, it would have adequate food for over 3 years, by Chinese nutritional standards.

In view of the crucial importance of large food reserves to the prospects for individual and national survival, it can be hoped that U.S. food surpluses and large annual carry-overs will continue.

A BASIC SURVIVAL RATION TO STORE

A ration composed of the basic foods listed below in Table 9.3 provides about 2600 calories per day, is nutritionally balanced. It keeps better than a ration of typical American foods, requires much less

transport, and is much less expensive. The author and some friends have stored enough of this ration to feed their families several months during a crisis, and have eaten large quantities of these foods over the past 20 years. (A different emergency ration should be stored for infants and very small children, discussed in the following section.) Field tests have indicated that the majority of Americans would find this ration acceptable under crisis conditions. In normal times, however, no one should store this or any other ration until after he has prepared, eaten, and found its components satisfactory.

Unprocessed grains and beans provide adequate nourishment for many millions of the world's population. There is little else to eat. Dry grains and beans are very compact: a 5 -gallon can holds about 38 pounds. When cooked, dry whole grains become bulky and give a well-fed feeling a distinct advantage over short rations during a prolonged crisis.

Table 9.3. A basic survival ration for multi-year storage

Ounces per day

Grams per day

Pounds for 30 days full  
ration

Kilograms for 30 days  
full ration

Whole-kernel hard wheat

16

454

30

13.6

Beans

5

142

9.4

4.3

Nonfat milk powder

2

57

3.8

1.7

Vegetable oil

1

28

1.9

0.9

Sugar

2



57

3.8

1.7

Salt (iodized)

1/3

IQ

0.63

03

Total Weights

10

748

49.5

22.5

Multi-vitamin pills:

1 pill each day

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This basic ration has two disadvantages: (1) it requires cooking, and (2) Americans are unaccustomed to a grain-based diet. Cooking difficulties can be minimized by having a grain-grinding device, a bucket stovetop, or newspapers for fuel, and the know-how to make a "fireless cooker" by using available materials such as extra clothing. The disadvantage of starting to eat unaccustomed foods at a stressful time is that eating more whole grains and beans in normal times thereby, incidentally, saving money and improving the American diet by reducing fat and increasing bulk and fiber.

When storing enough of this ration to last for several months or a year, it is best to select a variety for variety and improved nutrition. If soybeans are included, take into account the different types of common beans, as noted earlier in this chapter.

In many areas it is difficult to buy wheat and beans at prices nearly as low as the farmer's prices for these commodities. However, in an increasing number of communities, at least one store sells whole-grain flour in large sacks at reasonable prices. Mormons, who store food for a range of possible personal needs, are often the best sources of information about where to get basic foods in quantity, at reasonable prices. In such a case, bulk foods should be removed from sacks (but not necessarily from sealed-plastic li

metal containers or in thick-walled plastic containers for storage. Especially in the more humid States, grain and beans should be frequently checked for moisture. If necessary, these foods should be freed of insects as described later in this chapter.

Vegetable oil stores as well in plastic bottles as in glass ones. The toughness and lightness of plastic makes them better than glass for carrying when evacuating or for using in a shelter. Since a pound of oil contains four times as much energy as does a pound of sugar, dry grain, or milk powder, storing additional oil is an efficient way to improve a grain diet and make it more like the 40%-fat diet of typical American diets.

All multivitamin pills providing 5000 International Units (1500 mg retinol equivalent) vitamin A, 400 to 800 units of vitamin D, and 50 to 100 mg of vitamin C, must meet U.S. Government standards, so the least quality is quite adequate. Storage in a refrigerator greatly lengthens the time before vitamin pills expire. Because vitamin C is so essential, yet very inexpensive and long-lasting, it is prudent to have a supply of it.

It would be wise to have on hand ready-to-eat, compact foods for use during a week or two of emergency. In addition to those normally kept in the kitchen. It is not necessary to buy expensive "survival" foods or dehydrated foods carried by many backpackers. All large food stores sell the following concentrated foods: condensed milk, condensed vegetable soup, condensed tomato soup, condensed vegetable powder, canned peanuts, compact ready-to-eat dry cereals such as Grape Nuts, canned meat and vegetables, and vegetable oil in plastic bottles, iodized salt, and daily multivitamin pills. If shelter is not available (see Figs. 9.2 and 9.3, Bucket Stove), it is advisable to include rice, noodles, and an "instant" oatmeal or wheat along with coffee and tea for those who habitually drink these beverages.

Parched grain is a ready-to-eat food that has been used for thousands of years. Whole-grain rice can be parched by the following method: Place the kernels about 1/4-inch deep in a pan and cook over a low flame while shaking it over a flame, hot coals, or a red hot electric burner. The kernels will pop and become parched. These parched grains are not difficult to chew and can be pounded to a meal more easily than whole kernels. Parched grain stores well if kept dry and free of insects.

#### EMERGENCY FOOD FOR BABIES

Infants and very small children would be the first victims of starvation after a heavy nuclear attack. If emergency preparations are made on their behalf. Our huge stocks of unprocessed foods, which could protect the lives of unprepared survivors from dying of hunger, would not be suitable for the very young. They are too hard, concentrated and less rough. Most American mothers do not nurse their infants, and if a few months' supply of baby foods were exhausted the parents might experience the agony of seeing their baby slowly starve.

Few Americans have watched babies starving. In China, I saw anguish on starving mothers' faces as they pressed and squeezed their flat breasts, trying to get a little more milk into their weak babies' mouths. This was a tragedy in the midst of tens of thousands of Chinese evacuating on foot before a ruthless Japanese invasion in World War II.

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Years later, my wife and I stored several hundred pounds of milk powder while our five children were growing. I believe that parents who fear the use of nuclear weapons will be glad to bear the small expense of storing emergency baby foods listed in Table 9.4, below. (More detailed descriptions of these and instructions for their use, are given in an Oak Ridge National Laboratory report, Maintaining Family Health During A Prolonged Food Crisis, ORNL-5352, 1979. This report may be purchased for \$6.50 from the Technical Information Service, U.S. Department of Commerce, 5385 Port Royal Road, Springfield, Virginia 22151.)

To make a formula adequate for a 24-hour period, the quantities of instant non-fat dry milk and sugar listed in the "Per Day" column of Table 9.4 should be added to 4 cups of safe water prepared daily in cool weather or when a refrigerator is available. In warm or hot weather conditions, it is safer to make a formula 3 times a day. To do so, add 1/3 cup plus 2 teaspoons of instant nonfat milk powder to 1-1/3 cups (2/3 pint) of boiled water, and stir thoroughly (about 1/3 ounce, or 9 grams) of vegetable oil and 2 teaspoons of sugar, and stir. (If required, 1/4 cup is enough when making one-third of the daily formula, 3 times a day.) If baby's milk can be spoon-fed to an infant.

Especially during a war crisis, the best and most dependable food for an infant is mother's milk. If mother is assured an adequate diet. The possibility of disaster is one more reason why a mother should store food for a full year. Storing additional high-protein foods and fats for a nursing mother usually is better than against her infant getting sick or starving than keeping adequate stocks of baby foods and formula for sanitary feeding after evacuation or an attack.

To give a daily vitamin supplement to a baby, a multivitamin pill should be crushed to a fine powder, mixed with spoons and dissolved in a small amount of fluid, so that the baby can easily swallow it. If the baby does not get adequate amounts of vitamins A, D, and C, he will develop deficiency symptoms in 1 to 3 months. Vitamin C deficiency, the first to appear, can be prevented by giving the baby 1/3 of a 50-mg vitamin C tablet, pulverized) or customary foods such as orange juice. Lacking these sources, the juice squeezed from sprouted grains or legumes

or pills or foods rich in vitamin D are available, exposure of the baby's skin to sunlight will provide vitamin D. It would be wise to wait about 30 days after an attack before exposing the baby to sunlight. Exposures would be safe except in areas of extremely heavy fallout. As a further precaution, the baby should be in an open, shallow pit that will provide shielding from radiation given off by fallout particles. Exposure should be very short, no more than 10 minutes.

Table 9.4. Emergency food supply for one baby

Ingredients

Per Day

Per Month

Per 6 Months

Volumes and Ounces

Grams

Pounds Kilograms

Pounds Kilograms

Instant non-fat dry milk

1 cup + 2 tablespoons  
(2-3/4 oz)

8

6

2.72

32

15

Vegetable cooking oil

3 tablespoons ( 1 oz)

30

2

0.9

12

55

Sugar

2 tablespoons (0.7 oz)

20

1.3

0.6

8

3.6

Standard daily multi-vitamin pills

1/3 pill

10 pills

60 pills

If sufficient milk is not obtainable, even infants younger than six months should be given bies must be pureed to a fine texture. Using a modern baby food grinder makes pureeing qui crisis conditions, a grinder should be cleaned and disinfected like other baby-feeding ute section.

Several expedient methods are available: the food can be pressed through a sieve, mashed w squeezed through a porous cloth. Good sanitation must be maintained; all foods should be b ing to insure that the food is safe from bacteria.

A pureed solid baby food can be made by first boiling together 3 parts of a cereal grain a they are soft. Then the mixture should be pressed through a sieve. The sieve catches the t kernels and the skins from the beans. The grain-beans combination will provide needed calc supplemented protein. The beans also supply the additional iron that a baby needs by the t Flours made from whole grains or beans, as previously described, also can be used; however rough material.

Some grains are preferable to others. It is easier to sieve cooked corn kernels than cooke wheat is the grain most likely to cause allergies, it should not be fed to an infant until grains, such as rice or com, are available.

Small children also need more protein than can be supplied by grains alone. As a substitut should be provided at every meal. If the available diet is deficient in a concentrated ene ar, a child's feedings should be increased to 4 or 5 times a day, to enable him to assimil small child should have a daily diet that contains at least one ounce of fat (3 tablespoor This would provide more than 10% of a young child's calories in the form of fat, which wou

If under emergency conditions it is not practical to boil infant feeding utensils, they ca solution. Add one teaspoon of ordinary household bleach to a quart of water. (Ordinary hou 5.25% sodium hypochlorite as its only active ingredient and supplies approximately 5% avai strength of the bleach is unknown, add 3 teaspoons per quart.) Directions for safe feeding

#### The Utensils

(Include at least one 1-quart and one 1-pint mason jar, for keeping prepared formula steri

1 . Immediately after feeding, wash the inside and outside of all utensils used to prepare the infant.

2 .

Fill a covered container with clean, cold water and add the appropriate amount of chlorine

3. Totally immerse all utensils until the next feeding (3 or 4 hours). Be sure that the bc bleach solution. Keep container covered.

## At Feeding Time

- 1 . Wash hands before preparing food.
2. Remove utensils from the disinfectant chlorine solution and drain, but do not rinse or
3. Prepare formula; feed the baby.
4. Immediately after feeding, wash utensils in clean water and immerse again in the disinfectant
5. Prepare fresh chlorine solution each day.

## STORAGE OF FOODS

Whole grains and white sugar can be stored successfully for decades; dried beans, non-fat vegetable oil can be stored for several years. Some rules for good storage follow:

Keep food dry. The most dependable way to assure continuing dryness is to store dry grain in airtight containers, such as ordinary 5-gallon metal storage cans or 55-gallon metal drums with gas tight lids. 5-gallon cans are light enough to be easily carried in an automobile when evacuating.

Particularly in humid areas, grain which seems to be dry often is not dry enough to store for years. To be sure that grain is dry enough to store for years, use a drying agent. The best drying agent is silica gel with color indicator. The gel is blue when it is capable of absorbing water and turns pink when it has been heated to become an effective drying agent again.

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Silica gel is inexpensive if bought from chemical supply firms located in most cities. By heating it in a can over a fire until it turns blue again, silica gel can be used repeatedly for years.

The best containers for the silica gel used to dry grain (or to determine its dryness) are envelopes large enough for a heaping cupful of the gel. A clear plastic window should be secured on the envelope which color changes can be observed. Put an envelope of silica gel on top of the grain in the container to within a couple of inches of its top. Then close the can tightly. Even a rather loose-fitting lid can be made tight by taping it tightly with tape. If after a few days the silica gel is still blue, the grain is dry enough. If it has turned pink, repeat the process with fresh envelopes until it can be seen that the grain is dry.

0 Keep grains and beans free of weevils, other insects, and rodents. Dry ice (carbon dioxide)

is the safest means still widely available to the public for ridding grain and beans of insects. To use dry ice, place a block of dry ice on top of the grain in a 5-gallon metal container. Put the lid on somewhat loosely so that the grain can be driven out of the can. (This will happen as the dry ice vaporizes and the heat from the grain gas sinks into the grain and displaces the air around the kernels.) After an hour or two, remove the lid and tape the lid with tape. After one month, all insects in this carbon-dioxide atmosphere will have died.

° Store foods in the coolest available place, out of the light. Remember that the storage life of foods is cut in half by an increase of 18 F (10 C) in storage temperature. Thus 48 months of storage at 60 F is equivalent to 24 months at 70 degrees F, and to 12 months at 88 degrees F.

Illustrative of the importance of cool storage are my experiences in storing nonfat milk powder covered, cool shelter. In steel drums I stored unopened 100-pound bags of compact, non-fat I bought from bakeries. The cost per pound was much less than I would have paid for the same sold in supermarkets. After 7 years storage at temperatures of about 50 degrees F the year powder was still good as good as it would have been if stored in a normally air-conditioned for about 3 years.

° Do not place stored metal containers directly on the floor. To avoid possible condensation and the rusting that results, place containers on spaced boards. For long term storage in shelters or damp basements, use solid-plastic containers with thick walls.

Rotate stored foods. Eat the oldest food of each type and replace it with fresh food. Although and non-fat milk powder remain edible after several years of storage at room temperature, other dry foods are more nourishing and taste better if stored for no more than 2 years. Most taste better if kept no more than one year. Exceptions are whole grains and white sugar, which last decades if stored properly.

Store plenty of salt, for our modern world salt is so abundant and cheap that most Americans think that in many areas soon after a major nuclear attack salt would become a hard-to-get essential. Persons working hard without salt would suffer cramps and feel exhausted within a few days. Relief shipments of grain probably would not include salt. So store enough salt both to save for months and to trade for other necessities.

## SEEDS

For thousands of years storing seeds has been an essential part of the survival preparation of prudent people fearing attack. Seeds are hopes for future food and the defeat of famine, through wars.

Among the most impressive sounds I ever heard were faint, distant rattles of small stones, freezing night in 1944. An airraid was expected before dawn. I was standing on one of the mountains, China, trying to pinpoint the sources of lights that Japanese agents had used just to guide attacking bombers to blacked-out Kunming. Puzzled by sounds of cautious digging started, I asked my interpreter if he knew what was going on. He told me that farmers walked most of the night that no one was following them, and were burying sealed jars of seeds in secret places, fearing that probably no one would hear them digging.

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My interpreter did not need to tell me that if the advancing Japanese troops succeeded in ruthlessly strip the surrounding countryside of all food they could find. Then those prudent people would find and hope in a starving land.

If you doubt that enough of our current "oversupply" of stored whole grains, soybeans, milk, etc. will reach you after a nuclear attack, you should store seeds known to grow well in your area.

When getting your supply of survival seeds, remember:

° Grains and beans are the best plant sources of energy and protein.

° Even if you have enough vitamins for several months, you may not be able to buy more until after a nuclear war.

° The deadly curses of scurvy, vitamin A deficiencies, and pellagra can be prevented by eating seeds, and sprouted seeds described earlier in this chapter.

° Plants grown from hybrid seeds give larger yields, but do not produce as productive seeds from good non-hybrid seeds.

Seeds of proven productivity in your locality may be more valuable than money after a major disaster.

° You should get mostly non-hybrid and store seeds, after learning from experienced local gardeners what is best.

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## Chapter 10

### Fallout Radiation Meters

#### THE CRITICAL NEED

A survivor in a shelter that does not have a dependable meter to measure fallout radiation lacks someone who knows how to use it will face a prolonged nightmare of uncertainties. He cannot smell, taste, hear, or see fallout radiation. A heavy attack would put most radio stations out of commission. If there is an electromagnetic pulse, blast, fire, or fallout from explosions. Because fallout intensifies with distance, those stations still broadcasting would rarely be able to give reliable information about changing radiation dangers around a survivor's shelter.

Which parts of the shelter give the best protection? How large is the radiation dose being received? When is it safe to leave the shelter for a few minutes? When can one leave for an hour? How much water is needed? As the fallout continues to decay, how long can one safely work each day outside? How long can the shelter be left for good? Only an accurate, dependable fallout meter will enable answers to these or-death questions.

Gamma radiation is by far the most dangerous radiation given off by fallout particles. Gamma rays are only more penetrating and harmful. The roentgen (R) is the unit most commonly used to measure gamma rays, or to X rays, and most American civil defense instruments give readings in roentgens per hour (R/hr). Therefore, for simplicity's sake, in this book almost all radiation doses are given in roentgens per hour (R/hr). This simplification is justified because for whole-body gamma radiation from fallout, the numerical value of an exposure or dose given in roentgens is approximately the same as the numerical value given in rems or rads. (For information on the seriousness and probability of injuries likely to be suffered as a result of receiving different doses of radiation, see "Lifetime Risks from Radiation", a section of Chapter 13.)



The dose (the quantity) of radiation that a person receives, along with the length of time received, determine what injuries, if any, will be suffered as a result of the dose. Of people who receive a dose of 350 roentgens under nuclear war conditions, about half will die. Doses are measured by instruments called dosimeters, either by directly reading the dose between the time at which the meter is read and the time of a subsequent reading, or by calculating by subtraction the dose received in the interval. To avoid receiving a lethal or sickening dose, the most useful instrument is a dose rate meter. The Academy of Sciences' Advisory Committee on Civil Defense in 1953 concluded: "The final effect of a nuclear war depends upon the occupants of any shelter having simple, rugged, and reliable dose rate meters. The dose rate outside the shelter."

With a reliable dose rate meter you can quite quickly determine how great the radiation danger is at various places, and then promptly act to reduce your exposure to these unseen, unfelt dangers. For example, if you find an excellent fallout shelter and learn by reading your dose rate meter that you are being exposed at a rate of 30 R/hr, that if you stay there for one hour you will receive a dose of 30 R. But if you go back in after 2 minutes, then while outside you will have received a dose of only 1 R.

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(2 minutes  $\frac{2}{60}$  of an hour =  $\frac{1}{30}$  hr; and receiving a dose at the rate of 30 R/hr for  $\frac{1}{30}$  hr =  $30 \text{ R/hr} \times \frac{1}{30} \text{ hr} = 1 \text{ R}$ .) Under nuclear war conditions, receiving an occasional dose of 1 R will be of little concern, as explained in Chapter 13 and 18.

#### WARNINGS FOR BUYERS OF FALLOUT METERS

You are "on your own" when buying a dose rate meter or dosimeter because:

- ° No U.S. Government agency or other Government facility advises the public regarding sources of

available radiation-measuring instruments for use in time of war, or warns concerned individuals that many of the instruments are either incapable of measuring adequately high dose rates or doses for wartime use or are dangerously inaccurate. For example, a dose rate meter that in 1982 sold nationwide was tested at Oak Ridge National Laboratory to determine its accuracy for measuring gamma radiation. It was reasonably accurate at low dose rates, but at the high dose rates of life or death importance in nuclear war its readings were dangerously low: When it should have read 150 R/hr, it read 13.9 R/hr. Another dose rate meter of this same model, tested in California by Dr. Bruce Clayton, read only 16 R/hr when it should have read 400 R/hr. Obviously, if this model were used and trusted by a person doing rescue work outdoors in heavy fallout, while believing that he was receiving a non-incapacitating dose rate, he could be getting a fatal dose!

- ° Instruments that measure only milliroentgen-range dose rates are sold for war use by some companies. Most Americans have no idea what size of radiation doses would incapacitate or kill them, but they know that a milliroentgen is  $\frac{1}{1000}$  of a roentgen, so some people buy instruments that are capable of measuring maximum dose rates of only one roentgen or less per hour. For example, an American company that in 1986 sold for \$370.00 a dose rate meter that has a maximum range of "0 1000 mR/hr" (1000 milliroentgens per hour) in that company's listing of "Radiation Detection Products for the General Public" as...., applicable for use in case of nuclear war." The highest dose rate that it can measure is 1000 mR/hr, which is far too low to be of much use in a nuclear war.

- ° Used and surplus dose rate meters and dosimeters are likely to be inaccurate or otherwise

few buyers have access to a radiation source powerful enough to check instruments for accurate full ranges of measurements. My education regarding bargain fallout meters began in 1961, dosimeters of a model then being produced by a leading manufacturing company and purchased by the Office of Civil Defense. Within a week after receiving these instruments, one of them was charged. The other was found to be inaccurate. Later I learned that the manufacturing company had publicized its instruments that did not pass Government quality tests.

Most Federal and State organizations do not criticize faulty civil defense products, are not charged with this responsibility and want to avoid angering manufacturers and sellers who pressure Congressmen or Legislators to seek redress for lost sales.

In this book I am not giving the names of any of the companies that sell or have sold potential survival items. To do so would reduce the chances of this book being distributed or advocated by government agencies.

#### WAR RESERVES OF FALLOUT METERS

One of Americans' most important assets for surviving a nuclear war is the Federal Emergency Management Agency's (FEMA's) supply of fallout meters. These instruments include approximately 600,000 about 3,300,000 dosimeters, all suitable for wartime use. In 1986 almost all of these old instruments reportedly still are in good working condition. Because of continuing inadequate funding for many years most of FEMA's instruments have been serviced, calibrated, and, if necessary, repaired many years. In a few localities these instruments are no longer being serviced.

Most of these critically important instruments are kept in cities, in buildings likely to be damaged in the event of a massive Soviet attack. If there were a sufficiently long, officially recognized attack, it might be possible during such a worsening crisis to move a large fraction of the population to the areas of probable blast or fire damage, and to place them in officially designated fallout shelters. An unlikely development would not provide private family shelters with instruments.

Most families need their own fallout meters. This need is greatest for families living in areas likely to be damaged by blast or fire, and for those planning to evacuate to such less hazardous localities in a crisis.

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#### COMMERCIALLY AVAILABLE FALLOUT METERS

In 1987 an American does not have many choices if he wants to buy an off-the-shelf dose rate meter for measuring the high levels of fallout radiation that would result from a nuclear attack. Although many meters and dosimeters have been under development by the military services and civil defense agencies for the past 15 years, they have not been produced commercially for sale to the public. Field tests have not been completed at this writing.

#### Dose Rate Meters

The best radiation-measuring instrument for wartime use available in the United States is the Survey Meter RD-10, manufactured in Finland by Alnor Oy. It is sold in the United States by the Federal Bureau of Investigation, 2585 Washington Road, Suite 120, Pittsburgh, Pennsylvania 15241. In 1988 the FOB price was \$1,200. The RD-10 accurately measures gamma and X rays from very close to natural background radiation levels.

in two ranges (0.03 - 300 m R/hr, and 0.03 - 300 R/hr). It meets Finnish Army standards for operation in sub-zero cold (down to -250C, or -130F); it has an illuminated scale for night signal, and is built to withstand electromagnetic pulse (EMP) effects. (A few of my friends own Finnish instruments of an earlier model, the RD-8; they still are in excellent working order.)

A less expensive dose rate meter designed for rugged wartime use is the Portable Radiological PDRM 82, manufactured in England by Plessey Controls Limited, Sopers Lane. Poole, Dorset Bournemouth. This instrument is the current standard issue of the British armed forces and civil defense units. Its life of at least 20 years, is microcomputer controlled, EMP-proof, and displays "FAIL" if it fails. Sometimes, occasionally a PDRM 82 does fail. One bought by a friend in 1987 and tested by a friend read 86 centigrays per hour when it should have read 300, and failed to display "FAIL." Maysey Controls finally replaced it with another new PDRM 82.) The only consequential disadvantages compared to more expensive dose rate meters, are that it reads in centigrays per hour (cGy/hr, or Rads/hr, or R/hr) and does not measure dose rates lower than 0.1 cGy/hr (100 mR/hr). In 1987 a digit-liquid-display dose rate meter is sold by Plessey Controls for 250 British pounds plus shipping, pre-paid. To learn the latest delivery date and the latest price delivered direct by air, write to Maysey Controls.

However, in a nuclear war 100 mR/hr will be a low dose rate in most life-threatening fallout situations. I ordered a PDRM 82 direct from England in 1984, my objective being to have it tested at Oak Ridge National Laboratory. I learned that such testing was unnecessary, since U.S. Army specialists already had tested it and found it excellent.)

Technical Note. Conversion of readings of most foreign and scientific radiation-monitoring instruments to radiation units usually given by American civil defense instruments, or used in the U.S. in assessing radiation hazards:

#### ABSORBED RADIATION DOSE

1 gray (1 Gy) = 100 Rads 1 centigray (1 cGy) = 1 Rad

(As explained in the first section of this chapter, for practical civil defense work 1 Rad is approximately equivalent to 100 mR.)

1 sievert (1 Sv) 100 Hems

1 millisievert ( 1 mSv) = 0.1 Rem

#### ACTIVITY

1 becquerel (1 Bq) 27 picocuries (Radiation contamination of milk and water are given in picocuries per liter. One picocurie is one millionth of one millionth of a curie; 1 curie is 3.7 x 10<sup>10</sup> Bq.)

No wonder that most newspaper and television accounts of radiation accidents and hazards are so misleading.

I have not been able to find an American-made, modern dose rate meter that is designed for being sold in 1987. Among those designed for peacetime use that may be satisfactory in war is the RO-2A, manufactured by Eberline, P.O. Box 2108, Santa Fe, New Mexico 87504-2108. The RO-2A is a portable chamber instrument used to measure beta, gamma, and X-ray radiation from 50 mR/hr to 50 R/hr. It costs \$950.00. In Eberline's summary specifications and in the specifications that I have read of dose rate meters, no mention is made of the instruments' being EMP-proof.

## Dosimeters

Several reliable dosimeters and dosimeterchargers are sold in the United States. Among the sources is Dosimeter Corporation, P.O. Box 42377, Cincinnati, Ohio 45242. Its DCA Model No. 909, cost \$90.00; one charger can be used to charge several dosimeters.

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A more expensive direct reading 600 R dosimeter is model 019-006 of Atomic Products Corporation, 1157, Center Moriches, New York 11934. It sells for \$120.00; dosimeter charger 020-001, "Direct- Reading Dosimeters", costs \$98.00.

To keep them dependable, all commercially available dosimeters and dose rate meters should (1) be checked with fresh batteries for charging or operating, (2) checked with a strong enough radiation source (3-year intervals) to see if they still are measuring radiation accurately, and (3) repaired if a dose rate meter still is functioning, use a radioactive check source such as Dosimeter Corporation Source (Model 3001), that contains 5 microcuries cesium-137 and sells for \$35.00. This type of instrument only that your instrument measures dose rates slightly above normal background radiation; if the instrument could accurately measure the much higher dose rates that will be of vital concern in a fallout situation, instrument companies will properly calibrate a radiation measuring instrument that is sent to them. Dosimeter Corporation charges \$50.00 to calibrate a dose rate meter or dosimeter, and makes a nominal cost.)

The reader is advised to buy at least a good commercial dose rate meter, with which to quickly estimate the levels of fallout radiation -if he can afford one. A family that has a reliable dose rate meter can estimate almost all of the time during which fallout dose rates outdoors are dangerously high, can estimate the accuracy of the accumulated doses received by its members. To do this, a continuous record must be kept of the times at which those measurements are made. (Having a reliable dosimeter eliminates the need for such detailed records and making these calculations, but if only one instrument can be afforded, a dose rate meter.) A good commercial instrument, if properly maintained and periodically calibrated, to check its accuracy, probably will be serviceable for years.

A prudent owner of even an excellent dose rate meter would do well to make and learn to use a dependable homemade fallout meter briefly described later in this chapter, with complete instructions for using it given in Appendix C. Then during a period of heavy war fallout you can check the accuracy of your instrument by comparing them with those of your KFM, and, if the complex instrument is giving reasonable readings, your KFM will meet your basic need.

### A HOMEMAKEABLE DOSE RATE METER, THE KFM

What is a KFM?

The only do-it-yourself fallout meter that is accurate and dependable was invented in 1977 (Kearny Fallout Meter); one is pictured in Fig. 10.1.

Fig. 10.1. A homemade KFM, an accurate dose rate meter for measuring dose rates from 30 mR/hr up to 43 R/hr. (nw097.jpg)

This simple instrument has undergone rigorous scientific testing in several laboratories, National Laboratory; its accuracy and dependability were confirmed. Many hundreds of KFMs untrained people, ranging from members of junior high school science classes to grandmothers, children and grandchildren. These successful makers have been guided only by thoroughly fitted patterns not quite as good as the improved ones given in Appendix C of this updated book.

Only common materials found in millions of homes are needed to build a KFM. (If all of the those for a dry-bucket, have to be purchased, their total cost in 1986 is less than seven dollars as an accurate dose rate meter when used in conjunction with a watch and the KFM's attached in readings in listed time intervals to dose rates.

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No radiation source is needed either to initially calibrate a KFM or subsequently to check for accuracy were completed at Oak Ridge National Laboratory and are the basis of the KFM. KFM is more accurate than most civil defense instruments, and its accuracy is permanently physics applicable to the specified dimensions and other characteristics of its parts, and each other- provided that it is made and maintained according to the instructions. Unlike measuring civil defense instruments that are reliable and available today, a KFM is charge battery is needed.

#### Additional Advantages of KFMs

\* A KFM combines the provenly practical radiation measuring functions of an electroscope and a chamber having a specified volume. Electroscopes were the basic radiation measuring instruments of scientists, including Nobel Laureate Lord Rutherford, who pioneered studies of atomic nuclei. The author is indebted to another Nobel Laureate physicist, Dr. Luis W. Alvarez, for the idea of a homemade electroscope with two thread-suspended, aluminum- foil leaves, to measure fallout radiation. Excellent and unavoidably expensive dose rate meters, including civil defense instruments, chamber devices.

\* A KFM, used in conjunction with a watch, does not have to be charged to any specified initial reading, discharged by exposure to radiation to any specified final reading, to accurately measure a time interval specified on its attached table. Fig. 10.2 illustrates this operational advantage.

Fig. 10.2 Normalized Calibration Points for Two KFMs, Showing the Straight-Line Relationship Between Milliroentgen Radiation DOSES and Resultant Readings. The complete instructions for using a KFM (see Appendix C) explain how an operator with a watch can use this instrument to measure DOSE RATES.

(Text in Graph) ADJUSTED CALIBRATION CURVES OF KFM 20 G AND KFM 20 I  
(BOTH HAVE 8-PLY LEAVES OF STANDARD ALUMINUM FOIL)

THE CALIBRATION READINGS HAVE ALL BEEN ADJUSTED GRAPHICALLY TO SIMULATE ALL INITIAL LEAF SEPARATIONS BEING EQUAL. ORNL-DWG 76-6548 (nw098.tif)

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#### Additional Information on Accuracy and Dependability

\* Readers who want additional technical information on the KFM are advised to buy a copy of Ridge National Laboratory report on this instrument, The KFM, A Flomemade Yet Accurate and Fallout Meter (ORNL-5040, CORRECTED), by Cresson H. Kearny, Paul R. Barnes, Conrad V. Ches and Margaret W. Cortner. Date published: January 1978. Copies are sold by the National Technical Information Service, U.S. Department of Commerce, 5285 Port Royal Road, Springfield, Virginia

Since the price continues to increase, it is best to write first, to learn the postage -paid

\* Civil defense professionals of foreign countries also have concluded that KFMs have life June 1978 Special Issue of The Journal of the Institute of Civil Defence ("The Premier Social Studies", with headquarters in London) was entirely devoted to the KFM, and gave international the original complete instructions and cut-out paper patterns. The interest of Chinese civil the KFM and my other low cost survival inventions led to my making, with White House approval trips in China as an official guest. In eight Chinese cities I acquired survival know-how defense information with top civil defense officials.

#### A Major Disadvantage: A KFM Looks Like a Toy

\* This instrument appears too simple to be trusted to measure deadly radiation, a frighter people. Typical modems are accustomed to pushing buttons and twisting dials to get information from instruments they do not understand. Most feel that a dependable radiation-monitoring be complex. However, especially during a worsening nuclear crisis many typical Americans will KFMs if they become convinced of the accuracy and dependability of this homemakeable instrument they can understand, use intelligently', and repair if necessary.

#### Caution:

Earlier versions of KFM-making instructions, written when common sewing threads were good sewing threads for suspending a KFM's leaves. Now most sewing threads are anti-static treated and are unsatisfactory for use in KFMs. Makers of KFMs should use the instructions in this recommend widely available, excellent insulators for suspending a KFM's leaves, and that include design improvements.

#### Instructions for Making and Using KFMs

Appendix C gives the latest field-tested instructions (with patterns) to enable you to make how to use it.

The great need for civil defense instruments is likely to be fully recognized only during Therefore, in this edition the KFM instructions and patterns are printed on only one side at the end of the text, and with two pages at the very end to expedite the rapid reproduction Timed printing tests by two newspapers have proved that, with the help of these two pages newspaper can paste up and photograph all pages of the KFM instructions, print a 12-page tabloid to start distributing the tabloid-all in less than one hour. Thus, if you have a copy of this nuclear crisis, you may be able to give these instructions to a newspaper and help thousands obtain the information that they need to make fallout meters for themselves.

#### Advice on Building a KFM

The reader is urged to set aside several hours in the near future for making a KFM and for

During field tests, average American families have needed about 6 hours to study the instrument, to make this simple instrument, and to learn how to use it. These several hours may not be a crisis. Higher priority work would be the building of a high-protection factor shelter,

ventilating pump, and the storing of adequate water, in a crisis it might not be possible for a KFM.

It is very difficult to concentrate on unfamiliar details during a nerve-racking crisis, and hands that may become unsteady. The best time to build and learn to use a KFM is in peacetime. Then this long-lasting instrument should be stored for possible future need.

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## Chapter 1 1

### Light

#### THE NEED FOR MINIMUM LIGHT

Numerous disasters have proved that many people can remain calm for several days in total darkness. Occupants of a shelter full of fearful people probably would go to pieces if they could see out. It is easy to imagine the impact of a few hysterical people on the other occupants of a shelter under wartime conditions, even a faint light that shows only the shapes of nearby people and things. The difference between an endurable situation and a black ordeal.

Figure 11.1 shows what members of the Utah family saw in their shelter on the third night. The family's flashlights and other electric lights had been used until the batteries were exhausted. Candles at home and failed to bring the cooking oil, glass jar, and cotton string included in the kit. These materials would have enabled them to make an expedient lamp and to keep a small light burning for weeks, if necessary.

At 2 AM on the third night, the inky blackness caused the mother, a stable woman who had never before experienced darkness, to experience her first claustrophobia. In a controlled but tense voice she suddenly said, "I have to get out of here. I can't orient myself." Fortunately for the shelter-occupancy experiment, the mother overcame her fears and lay down to sleep on the floor near the entrance.

Conclusion: In a crisis, it is especially bad not to be able to see at all.

Fig. 11.1. Night scene in a trench shelter without light, (nw 100.tif)

#### ELECTRIC LIGHTS

Even in communities outside areas of blast, fire, or fallout, electric lights dependent on public power probably would fail. Electromagnetic pulse effects produced by the nuclear explosions, power stations and transmission lines, would knock out most public power.

No emergency lights are included in the supplies stocked in official shelters. The flashlight people would bring to shelters probably would be insufficient to provide minimum light for

A low-amperage light bulb used with a large dry cell battery or a car battery is an excellent continuous light. One of the small 12-volt bulbs in the instrument panels of cars with 12-volt light for 10 to 15 nights,

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without discharging a car battery so much that it cannot be used to start a car.

Making an efficient battery-powered lighting system for your shelter is work best done before a crisis you should give higher priority to many other needs.

Things to remember about using small bulbs with big batteries:

- o Always use a bulb of the same voltage as the battery,
- o Use a small, high-resistance wire, such as bell wire, with a car battery,
- o Connect the battery after the rest of the improvised light circuit has been completed.
- o Use reflective material such as aluminum foil, mirrors, or white boards to concentrate light as needed.
- o If preparations are made before a crisis, small 12-volt bulbs (0.1 to 0.25 amps) with sockets can be bought at a radio parts store. Electric test clips for connecting thin wire to a car battery can be bought at an auto parts store.

#### CANDLES AND COMMERCIAL LAMPS

Persons going to a shelter should take all their candles with them, along with plenty of matches in a container such as a Mason jar. Fully occupied shelters can become so humid that matches in open containers cannot be lighted after a single day.

Lighted candles and other fires should be placed near the shelter opening through which air enters to avoid buildup of slight amounts of carbon monoxide and other headache-causing gases. If the shelter is closed for a time for any reason, such as to keep out smoke from a burning house nearby, all fires in the shelter should be extinguished.

Gasoline and kerosene lamps should not be taken inside a shelter. They produce gases that can cause illness or even death. If gasoline or kerosene lamps are knocked over, as by blast winds that would affect extensive areas, the results would be disastrous.

#### SAFE EXPEDIENT LAMPS FOR SHELTERS

The simple expedient lamps described below are the results of Oak Ridge National Laboratory tests started with oil lamps of the kinds used by Eskimos and the ancient Greeks. Our objective was to develop dependable, long-lasting shelter lights that can be made quickly, using only common household materials. Field tests have proved that average Americans can build good lamps by following the instructions (11.2).

These expedient lamps have the following advantages:



- o They are safe. Even if a burning lamp is knocked over onto a dry paper, the flame is so extinguished if the lamp fuel being burned is a cooking oil or fat commonly used in the kitchen. The wick is not much larger than 1/16 inch in diameter.

- o Since the flame is inside a jar, it is not likely to set fire to a careless person's clothing or a breeze.

- o With the smallest practical wick and flame, a lamp burns only about 1 ounce of edible oil.

- o Even with a flame smaller than that of a birthday candle, there is enough light for reading. To make such a small flame, attach aluminum foil to three sides and the bottom of the jar, and suspend the jar over you and your book, just high enough not to block your vision. (During the long, anxious days of waiting for fallout to decay, shelter occupants will appreciate having someone read aloud to them.)

- o A lamp with aluminum foil attached is an excellent trap for mosquitoes and other insects. It is a common problem in an unscreened shelter. They are attracted to the glittering light and fall into the jar.

- o Two of these lamps can be made in less than an hour, once the materials have been assembled. There is no reason to wait until a crisis arises to make them. Oil exposed to the air deteriorates, so make the lamps filled with oil or to keep oil-soaked wicks for months.

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Fig. 1 1.2. Safe Expedient Lamps (nwl02.tif)

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## Chapter 1 2

### Shelter Sanitation and Preventive Medicine

#### AN OUNCE OF PREVENTION

Should fallout force Americans to stay crowded into basements and expedient shelters for days, we should protect ourselves against the spread of infectious diseases by taking both accustomed and preventive measures. Thousands of our jungle infantrymen in World War II learned to practice the preserving techniques described in this chapter. If modern medical facilities were temporarily unavailable, the prevention of diseases would become much more important to all of us.

The following infection-preventing measures are simple, practical, and require some self-discipline. I have observed their practice and have used them while exploring and soldiering in a number of jungle regions. I also have used these measures while field-testing nuclear war survival skills in the desert.

Basic first aid also would be of increased importance during a major confrontation or war. First aid instructions are available in practically all communities, so most first aid information is readily available.

#### DISPOSAL OF HUMAN WASTES

To preserve health and morale in a shelter without a toilet or special chemicals for treating urine, human wastes should be removed before they produce much gas. A garbage can with a lid of plastic will not hold the pressurized gas produced by rotting excrement. The following alternatives are listed in increasing order of effectiveness.

- ° Use a 5-gallon paint can, a bucket, or a large waterproof wastebasket to collect both urine and feces and keep it near the air-exhaust end of the shelter. Keep it tightly covered when not in use. The top keeps out insects and reduces odors. When such waste containers are full or beginning to overflow, put them outside the shelter still covered to keep out flies.

For some people, especially the aged, bringing a toilet seat from home would be justified. The bucket also helps those who have to sit down. An improvised seat of plywood or board s

If only one container is available and is almost filled, periodically dump the wastes outside before they are deposited. Before an anticipated attack, people who plan to stay in a shelter should do not have sufficient waste containers for weeks of shelter occupancy. The pit should be located outside the shelter in the down-wind direction. This usually will be the air exhaust end of an earthen shelter. The pit should be surrounded by a ring of mounded, packed earth about 6 inches high, to keep surface water from running into it.

Quickly putting or dumping wastes outside is not hazardous once fallout is no longer being deposited. Assume the shelter is in an area of heavy fallout and the dose rate outside is 400 R/hr external dose in about an hour to a person exposed in the open. If a person needs to be exposed to dump a bucket, in this 1/360th of an hour he will receive a dose of only about 1 R. Under 1-R dose is of little concern.

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If the shelter design does not permit an occupant to dispose of wastes without running outside, slip on a plastic shoe cover over his shoes before going out, and remove these coverings in the entry before going back in. This precaution will eliminate the chance of tracking "hot" fallout particles into the shelter. Someone getting a tiny beta burn in this way.

- ° Have all occupants only urinate in the bucket, and defecate into a piece of plastic. Urine and feces are non-hazardous and can be safely dumped outside.

Two thicknesses of the thin plastic used to cover freshly dry-cleaned clothes will serve for several persons. Gather the plastic around the excrement to form a bag-like container. Tie the upper edges with a string or narrow strip of cloth. Do not tie it so tightly as to be gas-tight.

The bag should be gently tossed outside. As the excrement rots, the gas will leak out of the tied bag. Flies will be attracted in swarms, but they will not be able to get into the plastic to collect eggs. And because rotting excrement is so attractive to flies, shelter occupants will be bothered by flies and other pests.

If you have prudently kept a can of modern fly bait in your survival supplies, a little spraying can kill literally thousands of flies. The most effective fly baits, such as Die Fly, are sold in farm supply stores.

Fig. 12. 1. A 5-gallon paint can used for a hose-vented toilet can, with a plastic trash

liner, (nwl04.jpg)

° Use a hose-vented, 5-gallon can or bucket lined with a heavy plastic bag: cover tightly use. Figure 12. 1 shows this type of expedient toilet.

The vent-hose runs through a hole near the top of the paint can shown and is taped to seal can be quite easily cut with a chisel or a sharpened screwdriver. The hose is long enough ter. Its outer end should be secured about 6 inches above ground level, to prevent water f heavy rain. When a toilet-can is tightly covered, foul gases can escape through the hose t

With its opening tied shut, a large plastic trash bag containing as much as 30 pounds of w toilet-can and disposed of outside the shelter.

The 6-member Utah family described in preceding chapters used a home-like expedient toilet shelter stay. Figure 12.2 pictures the toilet seat they took with them, placed on a hose-v ground. The toilet was at one end of the shelter. A person sitting on this toilet could pu up hole" and be more comfortable.

The blanket shown hanging on the left in Fig. 12.2 could be drawn in front of the toilet f head was the emergency crawlway-ventilation trench. When the toilet was being used, the sh pumped air under the blanket-curtain and out the ventilation trench, resulting in very lit ter.

Vomiting is certain to cause both morale and health problems, especially for unprepared sh first dramatic symptom of radiation sickness. Nervousness, combined with the effects of ur ter, will cause even some healthy persons to vomit. In a crowded shelter, the sight and sm throw up. Plastic bags, well distributed throughout a shelter, are the best means to catch off the floor. Buckets, pots, or a newspaper folded into a cone also will serve.

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Fig. 12.2. The hose-vented expedient toilet used by the Utah family for over 3 days. (The telephone was brought along as a joke.) (nwl05.jpg)

#### DISPOSAL OF DEAD BODIES

In large shelters which are occupied for many days, someone may die even when no occupants by blast, fire, or radiation. The sight or the sickly-sweet stink of a decaying human body civil defense workers have theorized that the best way to take care of a corpse in a shelt outdoors is low enough to allow burial is to seal it in a large plastic bag. A simple test impractical: gas pressure caused the bag to burst. One solution is to put the corpse outsi dent. First, if possible, place it in a bag made of large plastic trash bags taped togethe pinholes.

#### CLEAN WATER AND FOOD

Disinfecting water by boiling (preferably for at least 10 minutes) or by treating it with described in Chapter 8, Water.

When water is first stored, it should be disinfected by the addition of 1 scant teaspoon c

bleach for each 10 gallons.

To avoid contaminating water when removing small quantities from a container such as a wat simplest way is first to pour some into a pot or other medium-sized container, from which poured into individual cups. Dipping water with a cup runs more risk of contamination. The quantities of water out of a container is to siphon it with a flexible tube, as described

Sanitary storage of food in expedient shelters is often difficult. Although almost any pap keep fallout particles from food, shelter dampness can cause paper containers to break. An can cut through paper or plastic coverings to reach food inside. Placing paper containers suspending the bags from the ceiling of the shelter entryway gives good protection against protection against moisture for a few weeks. (Do not obstruct the air flow through an entr A small amount of insect repellent or grease smeared on the suspending string or wire will and strong plastic containers with tight lids protect food best.

The hygienic preparation and serving of food in a shelter, especially in hot weather, requ eaten promptly. It is best to eat within half-an-hour after cooking. Canned foods should b opening. The cleaning and disinfecting of utensils, bowls, etc. , should be done promptly, plying and to lessen the chances of ants and other insects being attracted into the shelte cereals in the cooking pot, to avoid spilling.

In Oak Ridge National Laboratory shelter tests, only a few infants and toddlers have been cupants. Feeding infants and small children over a piece of plastic would be one good way lage from complicating shelter life.

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To avoid using dishes, most foods can be served on squares of plastic. Spoons and such pla clean after eating, then disinfected by boiling or by dipping them into chlorine bleach so of Clorox-type bleach to a quart of water.

A shelter occupant without a spoon can eat very thick grain mush in a sanitary manner by p plastic held in his hand, forming it into a ball, and taking bites. .Although Chinese peas in their bare hands, experiments have indicated -not unexpectedly that Americans do not li

Cooking without oil or fat makes disinfecting utensils much easier when water and fuel are reals and sugar are easy to wash off with a little water, without soap.

#### CONTROL OF INSECTS

Insect sprays used in high-protection-factor shelters are likely to cause more problems th sonous insecticides should be used with caution. Insect repellents on the skin and clothin likely to be in sufficient supply to last for weeks or months. Some insect problems and si them are described below.

Mosquitoes would multiply rapidly after an attack, because normal control measures would r insect screen or mosquito netting to cover the ventilation openings of a shelter is the be flies, and all larger insects. The lack of insect screening-when it would be too late to c harassment, discomfort and possible disease than most people accustomed to modem living ar However, if the shelter has no air pump, it is impractical to use screens that obstruct th except in cold weather.

The fly population would explode after a nuclear attack. Radiation doses several times larger would kill people do not sterilize insects, in extensive rural areas where almost all people to be safe from fallout, most domestic animals and wild creatures would be killed. Trillions of dead bodies.

If you have prudently kept a can of modern fly bait in your survival supplies, a little spray covering can kill literally thousands of flies.

Shelter occupants should make every effort to prevent flies from reaching disease-spreading

Ants, especially in the warmer parts of the country, could drive people out of expedient shelter. A convention is to try to find a shelter-building site that is

not near an ant nest, if shelter occupants are careful in storing food and eating, ants are not a problem.

Ticks and chiggers are usually found on grass and low bushes. To avoid carrying these pests, do not bring grass or dead leaves into your shelter for bedding except in freezing weather. Check the ground: few pests live in tall vegetation.

#### PERSONAL POSSESSIONS

Toothbrushes are not boiled or otherwise disinfected after being used, because we all develop resistance to our own infective organisms. For the same reason, each individual should have a cup, bowl, and spoon. They should be cleaned as well as possible and kept covered when not

#### PREVENTION OF SKIN DISEASES

In crowded shelters, especially during hot weather, skin diseases are likely to be a more generally recognized problem. The importance of learning how to prevent skin diseases was made apparent in a few shelter-occupancy tests to be conducted in the summer without air conditioning. This summer 99 men lived for 12 days in an underground shelter cooled only with outdoor summer air. 11 complaints were high, even though medical treatment was available on a daily basis. The total sick call was 560; 34 of these 99 healthy young men contracted heat rash and 23 had other fungus infections. However, these sailors lived in an inadequately ventilated shelter and should use the other methods listed below for preventing skin troubles.

Even in shelters that are skillfully ventilated with adequate outdoor air, skin diseases were especially in hot weather unless special hygiene measures are followed. Humid heat and heat contribute to skin diseases. Most of the following measures for preventing skin diseases have been used for thousands of years.

° Wash off sweat and dead skin. (When it is hot and humid, dead skin is continuously rubbing off and starting to decay.) Many jungle natives rinse their bodies several times a day. Bathing with soap is harmful in humid heat: the rapid loss of normal skin oils is one of the causes of

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Your skin can be kept fairly clean by rinsing off each day with just a cup of water, while using a very small cloth. A 6-inch square of bedsheet cloth serves well. So that you can dispose of

afterwards, wash yourself while standing on a piece of plastic with its edges held up slightly by narrow boards under the edges.) Use about two-thirds of the precious water for the first wash, the face down and gently rubbing neck, armpits, stomach, groin, buttocks, and feet with a cloth. Use the remaining water to rinse off again, using bare fingers. If boiling water is available, wash every day by boiling them for a few minutes.

- ° Sleep as cool and bare as practical, to dry the maximum skin area.
- ° If practical, sit and sleep only where other members of your family do and avoid use of other members of one family.
- ° Avoid infection from toilet seats by disinfecting with a strong chlorine solution and then wiping with paper, or by not sitting down.
- ° Wash or disinfect clothing as often as practical, especially underwear and socks. Disinfecting and laundering it, is the most important health objective under difficult shelter conditions. Boiling water disinfects it. Unless plenty of water is available for rinsing, do not disinfect in a chlorine bleach solution.
- ° Wear shoes or sandals when walking about, to prevent fungus infections of the feet.

## RESPIRATORY DISEASES

The spread of respiratory and other diseases transmitted by coughing and sneezing would be a major problem in long-occupied shelters. Adequate ventilation would help in disease prevention. In small shelters, if persons who are sneezing or coughing could stay near the opening being used for air exchange, many occupants, the risk of one or more of them having a disease that is easily spread obviously increases in a small shelter.

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## Chapter 13

### Surviving Without Doctors

#### A TEMPORARY RETURN TO SELF-HELP

Most doctors, hospital facilities, and medical supplies are located in cities. An all-out nuclear war would destroy these modern blessings. Even if medical assistance were nearby, only a few of the survivors in the most sheltered areas of heavy fallout would be able to get needed medicines or the help of a doctor. For the next months, most unprepared survivors would be forced to live under medical conditions almost never experienced by the majority of mankind for all but the past few decades of human history.

#### BENIGN NEGLECT

Life without modern medical help would be less painful and hazardous for those survivors who lack knowledge of what should be done or not done under primitive, unsanitary conditions. Intelligence and hygienic precautions can be obtained from widely available Red Cross and civil defense manuals. This knowledge, with a stock of basic first aid supplies, would reduce suffering and prevent death. However, first aid instructions do not include advice about what to do for serious injuries if no first aid or effective medicines are available.

Where There Is No Doctor, 32 the excellent self- help handbook ;recommended by Volunteers of America, gives much information that goes far beyond the scope of first aid. But even this handbook recommends getting professional medical help whenever possible for serious injuries and illnesses.

Fortunately, the human body has remarkable capabilities for healing itself, especially if the patient and his companions practice intelligent "benign neglect. " Such purposeful non-interferentive processes were called "masterful inactivity" by Colonel C. Blanchard Henry, M.D., a surgeon who worked on mass casualty evacuation and treatment. Colonel Henry was one of the first medical officers in Japan and Nagasaki after their destruction and was an experienced analyzer of civil defense preparations in other countries.

The following is a brief summary of Colonel Henry's medical advice for nuclear war survivors who are in primitive conditions and unable to get the help of a doctor or effective medicines. 33 (Adapted from brackets, is from a medical publication. 34 )

° Wounds: Apply only pressure dressings to stop bleeding unless an artery has been cut, as with a piece of glass. If blood is spurting from a wound, apply both a pressure dressing and a wide tourniquet. Loosen the tourniquet pressure about every 15 minutes, to allow enough blood to flow beyond the tourniquet and keep it alive. There is a fair chance that clotting under the pressure will stop blood loss before it becomes fatal.

° Infected wounds: Do not change dressings frequently. The formation of white pus shows that white blood corpuscles are mobilizing to combat the infection. In World War I, wounded soldiers in hospital died in agonies having their wounds cleaned and dressed frequently: many died as a result of such treatment. In contrast, before antibiotics became available late in World War II, casts and dressings or bandages sometimes were not changed for weeks. (The author saw this treatment in China and India and the stench resulting from such "benign neglect" of American soldiers' wounds— neglect that helped save many lives.)

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° Pieces of glass deeply embedded in flesh: Do not probe with tweezers or a knife in an attempt to remove them. Most glass will come out when the wounds discharge pus.

° Burns: Do not apply grease, oil or any other medicine to the burned area. Cover the area with a clean, dry dressing or folded cloth. Do not change the dressing frequently. [For most burns the dressing should not be removed until the tenth to fourteenth day. Give plenty of slightly salted water: about 1 gm of salt per quart (or liter), preferably chilled, in amounts of 1 to 3 liters daily. 4

° Broken bones: Apply simple splints to keep the bones from moving. Do not worry about deformities; they can be corrected later by a doctor. Do not attempt traction setting of broken bones.

° Shock: Keep the victim warm. Place blankets or other insulation material under him. Do not use so many blankets that he sweats and suffers harmful fluid losses. Give him plenty of slightly salty water [about a teaspoon of salt in a liter (or quart) of water] .

° Heat prostration: Give adequate fluids, including slightly salty water.

° Simple childbirth: Keep hands off. Wait until the mother has given birth. Do not tie and do not use a potent disinfectant is available. Instead, use the primitive practice of wrapping the cord around the infant until they dry. Avoid the risk of infecting the mother by removing the cord. Urge the mother to work to expel it.

° Toothache: Do not attempt to pull an aching tooth. Decaying teeth will abscess and fall out, but seldom fatal process — one which was endured by most of our remote ancestors who reached America.

#### VETERINARIAN ANTIBIOTICS

People who for decades have used antibiotics to combat their infections have not produced antibodies, and have subnormal resistance to many infections. People who have not been dependent on antibiotics have these antibodies. In the aftermath of a massive nuclear attack, most surviving American areas: many would need antibiotics. A large part of their need could be met by the supplies kept on livestock and chicken farms, at feed mills, and in small towns. Many animals are given antibiotics on short lives than most Americans receive in theirs. Hogs, for example, are given antibiotic-controlling medicines in their feed each day. In many farming areas, veterinary antibiotic supplies are a larger supply than are those for people. Realistic preparations to survive an all-out attack must include antibiotic supplies.

#### RADIATION SICKNESS

For the vast majority of Americans who would receive radiation doses from a massive attack, antibiotics, blood transfusions, etc., would not be of life-or-death importance. Very few people (received within 24 hours) of less than 100 R would become sick, even briefly. All of those between 100 R and 200 R should recover from radiation effects. However, under post-attack stresses and privations, some who receive acute radiation doses of 100 R to 200 R may die because of their reduced resistance. If total doses this size or even several times larger are given in a few months in small doses of around 6 R per day, no incapacitating symptoms should result. The body can repair almost all radiation damage if the daily doses are not too large.

The majority of those with acute doses of less than about 350 R will recover without medical aid. All of those receiving acute doses of over 600 R would die within a few weeks, even if they were in a typical hospital during peacetime. If all doctors and the equipment and drugs needed medically were to survive an attack and persons suffering from radiation sickness could reach medical aid, many additional lives could be saved.

The most effective way to reduce losses of health and life from radiation sickness is to protect people from exposure to radiation. Adequate shelter and essential life-support items are the best means of protection. The following information on radiation sickness is given to help the reader understand the symptoms and to help him distinguish between symptoms of common illnesses and first symptoms of radiation sickness.



The first symptoms of radiation sickness are nausea, vomiting, headache, dizziness, and a illness. 6 These symptoms begin several hours after exposure to acute doses of 100 R to 2 or less after receiving a fatal dose. A source of probable confusion is the fact that one experienced by many people when they are first exposed to great danger, as in an air raid al bombardment.

The occupants of a shelter might worry unnecessarily for weeks, mistaking their early emot initial phase of radiation sickness. This would be particularly true if they had no depend radiation, or if none of them knew how to use such an instrument.

The initial symptoms end within a day or two. Then follows the latent phase of radiation s the patient experiences few, if any, symptoms. If the dose received was in the non-fatal r may last as long as 2 weeks.

In the final phase, the victim of serious or fatal radiation sickness will have reduced re likely to suffer diarrhea, loss of hair, and small hemorrhages of the skin, mouth, and; or common causes may be confused with the onset of radiation sickness, but hemorrhages and lo clear indications of having received serious, but not necessarily fatal, radiation exposur for one to two months. Any available antibiotics should be reserved for this critical phas

Doses of 1 000 R to 5000 R result in bloody diarrhea, fever, and blood circulation abnorma symptoms beginning within less than 30 minutes after exposure and the final phase occurrin ter. Death results within 2 to 14 days. The victim of a dose of over 5000 R dies a hard de radiation damage to the central nervous system.

Recovery from most cases of radiation sickness will be more likely for patients who receiv rest, freedom from stress, and clean surroundings. But most patients, even without these a proved by the survival of thousands of Hiroshima and Nagasaki citizens who suffered seriou Nursing radiation victims is not hazardous. Even persons dying from a dose of 5000 Rare no radiation by wartime standards, and radiation sickness is not contagious.

#### LIFETIME RISKS FROM RADIATION

The large radiation doses that many survivors of a nuclear attack would receive would resu risks of death from cancer, but the lifetime risks from even large wartime radiation doses ple believe. Significantly, no official U.S. estimates have been made available to the pub deaths to be expected if America is subjected to a nuclear attack. However, reliable stati numbers of additional fatal cancers suffered by persons who received large whole -body rad and in other disasters, and who lived for months to decades before dying. Dr. John N. Auxi leading health physicist at Oak Ridge National Laboratory, was one of the American scienti Japanese scientists studying the Hiroshima and Nagasaki survivors, and currently is workin with International Technology Corporation-in 1986 summarized for me the risk of excess fat whole-body radiation doses: "If 1 ,000 people each receive a whole-body radiation dose of 100 R], about 10 additional fatal cancers will result." These 10 fatal cancers will be in cers that normally will develop among these 1 ,000 people during their lifetimes. This ris thus, if 1,000 people each receive a dose of 200 rems, about 20 additional lethal cancer c

"Rem" is an abbreviation for "roentgen equivalent (in) man." The rem takes into account th different kinds of radiation. For external gamma-ray radiation from fallout, the numerical

given in roentgens is approximately the same as the numerical value given in rems or in rad. Radiation energy absorption in any material and applies to all kinds of nuclear radiations. For the sake, this book gives both instrument readings (exposures) and doses in roentgens (R).

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The reader desiring good information on the long-term and worldwide effects of radiation is referred to the authoritative reports of the National Academy of Sciences, Washington, D.C. 20006: The Effect of Exposures to Low Levels of Ionizing Radiation (The BEIR Report made by the NAS Committee on Effects of Ionizing Radiation) (November 1972); and Long-Term Worldwide Effects of Multiple Detonations (1975).

From the standpoint of basic survival know-how, these and other complicated scientific studies estimate lifetime risks from radiation, after a nuclear attack people should:

- ° Provide the best protection against radiation for pregnant women and young children, since the very young are the most likely to be hurt by radiation.

- ° Realize that, with the exception of lung cancer, older people are no more susceptible to radiation than are those in the prime of life. Also, a 65-year-old probably will not live long enough to take 20 years or more to develop. Many older people, if they know realistic risk estimates, will continue essential outdoor work and take non-incapacitating radiation doses in order to spare their families the risk of getting cancer decades later.

#### PREVENTION OF THYROID DAMAGE FROM RADIOACTIVE IODINES

There is no medicine that will effectively prevent nuclear radiations from damaging the human body in the event of a nuclear strike. However, a salt of the elements potassium and iodine, taken orally even in very small amounts a day before radioactive iodines are swallowed or inhaled, prevents about 99% of the damage that otherwise would result. The thyroid gland readily absorbs both non-radioactive and radioactive iodine and retains much of this element in either or both forms. When ordinary, non-radioactive iodine is in the blood for absorption by the thyroid gland before any radioactive iodine is made available, it retains so much that it becomes saturated with non-radioactive iodine. When saturated, the thyroid can absorb only about 1% as much additional iodine, including radioactive forms that later may become available. The rest is said to be blocked. (Excess iodine in the blood is rapidly eliminated by the action of the kidneys.)

An excess of ordinary iodine retained in the thyroid gland is harmless, but quite small amounts of radioactive iodine retained in the thyroid eventually will give such a large radiation dose to the thyroid that it will result in thyroid cancer. These would include loss of thyroid function, nodules in the thyroid, or thyroid cancer. In 1954, the Marshallese Islanders on Rongelap Atoll were accidentally exposed to radioactive fallout produced by a nuclear test explosion on Bikini Atoll, about 100 miles away. Twenty-two of them developed thyroid abnormalities within a few years. 6 In the two days before they were taken out of the fallout area, these completely unsuspecting islanders, who were essentially outdoors, had received estimated whole-body gamma-ray doses of about 175 R from the fallout. They absorbed most of the radioactive iodine retained by their thyroid glands as a result of eating a fallout-contaminated food and water during their two days of exposure. (Because of unusual weather conditions at the time of fallout deposition, some of the retained radioactive iodine may have come from the fallout itself.)

An extremely small and inexpensive daily dose of the preferred non-radioactive potassium iodide (KI), if taken 1/2 hour to 1 day before exposure to radioactive iodine, will reduce later absorption of radioactive iodine by the thyroid to only about 1% of what the absorption would be without this preventive measure.

imentation and study have led to the Federal Drug Administration's approval of 130-mg tablets for this preventive (prophylactic) use only. 36 ' 37 A 130-mg dose provides the same daily amount of iodine as the tablet that English authorities for years have placed in the hands of the police near nuclear power plants to the surrounding population in the very unlikely event of a major nuclear accident. The 130-mg sized dose is in the Russian "individual, standard first-aid packet. " According to a comprehensive handbook, 3 x this first-aid packet contains "anti-radiation tablets and anti-vomiting tablets (metoprolol)." ."

° Prophylactic use of potassium iodide in peacetime nuclear accidents.

When the Three Mile Island nuclear reactor accident was worsening and it appeared that the containment structure might rupture and release dangerous amounts of radioactive iodines and other radionuclides into the atmosphere, the Government rushed preparation of small bottles of a saturated solution of potassium iodide. The reactor's containment structure did not rupture.

the atmosphere, the Government rushed preparation of small bottles of a saturated solution of potassium iodide. The reactor's containment structure did not rupture.

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The 237,013 bottles of saturated KI solution that were delivered to Harrisburg, Pennsylvania, would have been effective if the Three Mile Island accident had become an uncontained meltdown -were they in a warehouse, and were never used.

Since this famous 1979 accident, that injured no one, the Governors of the 50 states have taken steps for protecting Americans against radioiodines by providing prophylactic potassium iodide. In Tennessee have Americans, other than some specialists, been given potassium iodide tablets. At Three Mile Island reactor some 7,500 residents have been given the officially approved KI tablets, to assure that they have it if a nuclear accident occurs.

In April of 1982 the Bureau of Radiological Health and Bureau of Drugs, Food and Drug Administration, Department of Health and Human Services released "FINAL RECOMMENDATIONS, Potassium Iodide As A Thyroid Blocking Agent In A Radiation Emergency: Recommendations On Use". These lengthy recommendations are summarized in the FDA's "mandated patient product insert". (See a complete copy in the following pages). It is packed with every bottle of non-prescription KI tablets sold. However, the lengthy FDA recommendations mention many facts not mentioned in this required insert, including the following: "Based on the findings of the NCRP [National Council on Radiation Protection and Measurements] and an estimated 48 x 10<sup>6</sup> [48 million] 300-mg doses of potassium iodide administered each day, the NCRP [National Council on Radiation Protection and Measurements] estimated an additional risk of cancer from 1 in a million to 1 in 10 million doses." (Note that this extremely low adverse reaction is as large as the 130-mg prophylactic dose.)

FDA PATIENT INFORMATION USE OF 130-MG SCORED TABLETS OF POTASSIUM IODIDE

FOR THYROID BLOCKING

(Potassium Iodide Tablets, U.S.P.)

(Pronounced poe-TASS-e-um EYE-oh-dyed)

(Abbreviated KI)

TAKE POTASSIUM IODIDE ONLY WHEN PUBLIC HEALTH OFFICIALS TELL YOU. IN A RADIATION EMERGENCY, RADIOACTIVE IODINE COULD BE RELEASED INTO THE AIR. POTASSIUM IODIDE (A FORM OF IODINE) CAN HELP PROTECT YOU.

IF YOU ARE TOLD TO TAKE THIS MEDICINE, TAKE IT ONE TIME EVERY 24 HOURS. DO NOT TAKE IT MORE OFTEN. MORE WILL NOT HELP YOU AND MAY INCREASE THE RISK OF SIDE EFFECTS. DO NOT TAKE THIS DRUG IF YOU KNOW YOU ARE ALLERGIC TO IODINE (SEE SIDE EFFECTS BELOW).

#### INDICATIONS

THYROID BLOCKING IN A RADIATION EMERGENCY ONLY

#### DIRECTIONS FOR USE

Use only as directed by State or local public health authorities in the event of a radiation emergency.

#### DOSE

ADULTS AND CHILDREN ONE YEAR OF AGE OR

OLDER: One (1) tablet once a day. Crush for small children.

BABIES UNDER ONE YEAR OF AGE: One-half (1/2) tablet once a day. Crush first.

DOSAGE: Take for 10 days unless directed otherwise by State or local public health authorities.

Store at controlled room temperature between 150 and 300C (59 degrees to 86 degrees F). Keep container tightly closed and protect from light.

#### WARNING

POTASSIUM IODIDE SHOULD NOT BE USED BY PEOPLE ALLERGIC TO IODIDE. Keep out of the reach of children. In case of overdose or allergic reaction, contact a physician or public health authorities.

#### DESCRIPTION

Each iOSAT ' 1 Tablet contains 130 mg. of potassium iodide.

#### HOW POTASSIUM IODIDE WORKS

Certain forms of iodine help your thyroid gland work right. Most people get the iodine through their diet from iodized salt or fish. The thyroid can "store" or hold only a certain amount of iodine.

In a radiation emergency, radioactive iodine may be released in the air. This material may

enter the thyroid gland and damage it. The damage would probably not show itself for several weeks. People who are likely to have thyroid damage.

If you take potassium iodide, it will fill up your thyroid gland. This reduces the chance will enter the thyroid gland .

#### WHO SHOULD NOT TAKE POTASSIUM IODIDE

The only people who should not take potassium iodide are people who know they are allergic potassium iodide even if you are taking medicines for a thyroid problem (for example, a thyroid drug). Pregnant and nursing women and babies and children may also take this drug.

#### HOW AND WHEN TO TAKE POTASSIUM IODIDE

Potassium iodide should be taken as soon as possible after public health officials tell you every 24 hours. More will not help you because the thyroid can "hold" only limited amounts will increase the risk of side effects. You will probably be told not to take the drug for

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#### SIDE EFFECTS

Usually, side effects of potassium iodide happen when people take higher doses for a long not to take more than the recommended dose or take it for longer than you are told. Side effects of the low dose and the short time you will be taking the drug.

Possible side effects include skin rashes, swelling of the salivary glands, and "iodism" (itching and throat, sore teeth and gums, symptoms of a head cold, and sometimes stomach upset and

A few people have an allergic reaction with more serious symptoms. These could be fever and swelling of parts of the face and body and at times severe shortness of breath requiring immediate

Taking iodide may rarely cause overactivity of the thyroid gland, underactivity of the thyroid gland (goiter).

#### WHAT TO DO IF SIDE EFFECTS OCCUR

If the side effects are severe or if you have an allergic reaction, stop taking potassium iodide and call your doctor or public health authority for instructions.

#### HOW SUPPLIED

Tablets (Potassium Iodide Tablets, U.S.P.): bottles of [number of tablets in a bottle] tablets ( ). Each white, round, scored tablet contains 130 mg. potassium iodide.

Note that this official FDA required insert given above prudently stresses the name, the chemical formula (KI) of these Government- approved 130-mg potassium iodide tablets. Perhaps this information will keep some alarmed Americans (misinformed in a future crisis by the media that

the Chernobyl nuclear accident that "iodine tablets" were being given to people endangered the burning reactor) from getting and taking iodine tablets, widely sold for water purification.

Strangely, neither in official information available to the general public on the prophylaxis above-mentioned FDA "Final Recommendations" is any mention made of the much greater need for war—even for Americans during an overseas nuclear war in which the United States would not

Also note that this official insert contains no instructions for giving a crushed KI tablet to children. Nor is there any mention of the fact that the KI under the tablet's coating is a more potent form than most people ever have taken. This omitted information is given in the next to last section.

° Protection against radioactive iodine in fallout from a nuclear war fought outside the United States.

Most strategists believe that a nuclear war fought by nations other than the United States is more threatening than a nuclear attack on America. Several of the Soviet and Chinese nuclear test experiments have shown very light fallout deposition and some contamination of milk by radioactive iodine in many areas. In a serious contamination of milk, fruits, and vegetables could result if war fallout from many areas were carried to an America at peace. These potential dangers and effective countermeasures are discussed in 18, Trans-Pacific Fallout.

If a nuclear war were to be fought in northern parts of Asia, or in Europe, or in the Middle East, a large fraction of the fallout would come to earth on parts or all of the United States. 40 This would be an overwhelming catastrophe to Americans, although the long-term health hazards would be serious and the economic losses would be great. 40 The dangers from radioactive iodine in milk and fallout contaminated feeds or drank fallout contaminated water would be minimized if Americans had a supply of potassium iodide tablets for several weeks after the arrival of war fallout. Safe milk and other basic foodstuffs would be in very short supply. The parents of babies and young children would be especially thankful they had made this very inexpensive preparation, which provides protection to the thyroid. All members of families with a supply of potassium iodide could survive before those without it could do so.

The most dangerous type of radioactive iodine decays rapidly. At the end of each 8 day period it emits only half as much radiation as at the start of that period. So at the end of 80 days it emits only one-tenth as much radiation per hour as at the beginning of these 80 days. Because of this rapid decay, a 100-day supply should be sufficient if a nuclear war, either overseas or within the United States, were to occur.

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The probability of most Americans being supplied with prophylactic potassium iodide during a nuclear war appears low. Under present regulations the decision concerning whether to stockpile potassium iodide tablets rests solely with each state's governor. 41

° Need for thyroid protection after a nuclear attack on the United States.

After a nuclear attack, very few of the survivors would be able to obtain potassium iodide tablets. When to start taking it or stop taking it. In areas of heavy fallout, some survivors without potassium iodide receive radiation doses large enough to destroy thyroid function before modern medical treatment is

available. Even those injuries to the thyroid that result in its complete failure to function

times, but under post-attack conditions thyroid damage would be much more hazardous.

° Ways to obtain potassium iodide for prophylactic use.

\* By prescription.

With a prescription from a doctor, a U.S.P. saturated solution of potassium iodide can be obtained today. (In a crisis, the present local supplies would be entirely inadequate.) The smallest amount of a compound that prevents it from deteriorating significantly for a few years is in a small glass bottle with a solid, non-metallic cap that screws on liquid-tight. A separate medicine bottle can be kept at the same place. An authoritative publication 16 of the National Committee on Radiation Protection states: "Supplies of potassium iodide can be stored in a variety of places, including home

In 1990 the price of a 2-ounce bottle of U.S.P. saturated solution of potassium iodide, which is available only, ranges from about \$7.00 to \$11.00 in Colorado. A 2-ounce bottle contains about 500 mg, the daily dose of 130 mg for adults and for children older than one year. For babies less than one year the dose of a saturated solution is two drops (65 mg). Thus approximately 99% effective protection against the uptake of radioactive iodine by the thyroid can be gotten by taking saturated potassium iodide by prescription, today the recommended daily dose costs 6 to 9 cents.

\* Without prescription.

In 1990 the leading company selling 130-mg potassium iodide tablets without prescription in the United States is ANBEX, Inc., P.O. Box 861, Cooper Station, New York, N.Y. 10276. Two bottles of fourteen 130-mg potassium iodide tablets, cost \$10.00. Thus the cost per 24-hour dose is \$0.71. To my knowledge, the company in the U.S. that in July of 1990 is selling 130-mg KI tablets with the lowest price is Preparedness Products, 3855 South 500 West, Bldg. G, Salt Lake City, Utah 84115. Their 130-mg tablets, in a brown, screw-cap glass bottle, for \$3.50, postpaid, including shipping charge, makes the price \$2.50 per bottle.

After the disastrous Russian nuclear power reactor accident at Chernobyl in May of 1986, Poland soon sold all of their 130-mg potassium iodide tablets and Poland limited its inadequate supplies of potassium salts to the protection of children. In California, pharmacists reported abnormally large sales of tincture of iodine- apparently due to the buyers' having been misinformed by the media that they were taking "iodine" for protection.

Individuals can buy chemical reagent grade potassium iodide, that is purer than the pharmaceutical grade, from some chemical supply firms. No prescription or other authorization is necessary. In 1990 the lowest price which I am aware of is NASCO, 901 Jamesville Avenue, Fort Atkinson, Wisconsin 53538. The price of 100,000 mg (100,000 mg) in 1990 is \$10.50, plus \$2.00 to \$4.00 for shipping costs. Thus the cost in 1990 is less than 2 cents. NASCO sells 500 grams (500,000 mg- about one pound) for \$35.50, plus \$2.00 for shipping-making the cost per standard daily dose only one cent.

For years of storage, crystalline or granular potassium iodide is better than a saturated solution. Potassium iodide should be stored in a dark bottle with a gasketed, non-metallic cap that screws on tightly. Small bottles, filled with dry potassium iodide as described below, are good sizes for a family. Larger bottles should be kept with stored bottles.

Thus at low cost you can buy and store enough potassium iodide for your family and large number of friends and neighbors- as I did years ago.

° Practical expedient ways to prepare and take daily prophylactic doses of a saturated solution of potassium iodide.

To prepare a saturated solution of potassium iodide, fill a bottle about 60% full of crystalline potassium iodide. (A 2-fluid-ounce bottle, made of dark glass and having a solid, non-metallic, screw-on cap, is best for this purpose.)

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About 2 ounces of crystalline or granular potassium iodide is needed to fill a 2-fluid-ounce bottle.

Next, pour safe, room-temperature water into the bottle until it is about 90% full. Then close the bottle and shake it vigorously for at least 2 minutes. Some of the solid potassium iodide should remain at the bottom of the bottle; this is proof that the solution is saturated.

Experiments with a variety of ordinary household medicine droppers determined that 1 drop of potassium iodide contains from 28 to 36 mg of potassium iodide. The recommended expedient solution (approximately 130 mg for adults and children older than one year, and 65 mg for children under one year) are as follows:

\* For adults and children older than one year, 4 drops of a saturated solution of potassium iodide.

\* For babies younger than one year, 2 drops of a saturated solution of potassium iodide.

Potassium iodide has a painfully bad taste, so bad that a single crystal or 1 drop of the solution in a child's mouth would make him cry. (A small child would be screaming in pain before he could swallow the crystalline KI to make him sick. Some KI tablets are coated and tasteless.) Since many people dislike tasting medication, especially if no short-term health hazards are likely to result from not taking it, several methods of taking a saturated solution are recommended:

\* Put 4 drops of the solution into a glass of milk or other beverage, stir, and drink quickly. Do not add the beverage with nothing added. If only water is available, use it in the same manner.

\* If bread is available, place 4 drops of the solution on a small piece of it; dampen and roll it into a ball the size of a large pea, about 3/4 inch in diameter. There is almost no taste if this "pill" is dissolved in water. (If the pill is coated with margarine, there is no taste.)

As stated before, 4 drops of the saturated solution provide a dose approximately equal to 130 mg of potassium iodide.

° Preparing potassium iodide tablets to give to infants and small children.

The official FDA instructions for using KI tablets state that one half of a 130-mg tablet, given every 24 hours to "babies under one year of age", and that a whole tablet should be given to children older than one year.

Putting even a small fraction of a crushed or pulverized potassium iodide tablet on one's tongue causes a burning sensation. A slightly burnt sensation continues for hours. Therefore, do not make this experiment where her children cannot see her.

To eliminate the painfully bad taste of a crushed or pulverized KI tablet, first pulverize the tablet in a mortar and pestle.



for a minute into at least 2 ounces of milk, orange juice, or cold drink, to make sure the KI is dissolved. Then the taste is not objectionable. If only water is available, stir the powder into 4 ounces of water.

KI is a corrosive salt, more injurious than aspirin to tissue with which it is in direct contact. Take KI tablets after meals, except when so doing would delay taking the initial dose due to nausea. Recognize that taking a dilute solution of KI is easier on the stomach than taking the same dose of tablets. It may be a consequential consideration when taking KI for weeks during a prolonged nuclear war emergency.

#### ° WARNINGS

\* Elemental (free) iodine is poisonous, except in the very small amounts in water disinfectant or a few drops of tincture of iodine. Furthermore, elemental iodine supplied by iodine tablets or tincture of iodine dropped into water is not effective as a blocking agent to prevent thyroid damage. Do not have any potassium iodide, DO NOT TAKE IODINE TABLETS OR TINCTURE OF IODINE.

\* DO NOT MAKE A FUTILE, HARMFUL ATTEMPT TO EAT ENOUGH IODIZED SALT TO RESULT IN THYROID BLOCKING. Iodized salt contains potassium iodide, but in such a low concentration it is impossible to eat enough iodized salt to be helpful as a blocking agent.

#### OTHER WAYS TO PREVENT THYROID DAMAGE

Besides the prophylactic use of potassium iodide, the following are ways to prevent or reduce thyroid damage during peacetime or wartime conditions:

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\* Do not drink or otherwise use fresh milk produced by cows that have consumed feed or water that is consequentially contaminated with fallout or other radioactive material resulting from a nuclear explosion or from nuclear explosions in a war.

\* As a general rule, do not eat fresh vegetables until advised it is safe to do so. If under official advice is obtainable, avoid eating fresh leafy vegetables that were growing or exposed to fallout deposition; thoroughly wash all vegetables and fruits.

\* If a dangerously radioactive air mass is being blown toward your area and is relatively close to a possible nuclear power facility accidents), and if there is time, an ordered evacuation of the area is unnecessary even to take potassium iodide.

\* For protection against inhaled radioactive iodine, the FDA Final Recommendations (which are in the preceding section) state that the following measures "should be considered": "...sheltering indoors can significantly reduce inhaled doses, evacuation, respiratory protection, and/or potassium iodide."

Research has been carried out in an effort to develop a thyroid protection procedure based on topical solutions which are used as disinfectants. Since iodine solutions such as tincture of iodine are dangerous poisons if taken orally, these experiments have utilized absorption through the skin on bare skin.

All reported experimental topical applications on human skin have given less thyroid protection than oral potassium iodide.

oral administration of potassium iodide. Moreover, undesirable side effects of skin application for these reasons researchers to date have not recommended a procedure for the use of ordinary fallout protection.

Potassium iodide, when obtained in the crystalline reagent form and used as recommended above and 15, is safe, inexpensive, and easy to administer. Prudent individuals should obtain adequate supply of potassium iodide well in advance of a crisis.

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## Chapter 14

### Expedient Shelter Furnishings

#### IMPORTANCE OF ADEQUATE FURNISHINGS

Throughout history, people have endured being crowded together while living and sleeping. At times of war and privation, people have lived in such conditions for much longer periods than in peacetime. In shelter occupancy due to fallout, 42 Realistic basement-shelter-occupancy tests conducted by the U.S. Office of Civil Defense (now the Federal Emergency Management Agency) have shown that Americans can live and sleep for two weeks on a concrete floor. In some of these tests, only 15 square feet of space was provided for each person; only pieces of corrugated cardboard 3/16-inch thick lay between sleeping and sitting on concrete. 1 3

Nevertheless, shelters should be adequately furnished whenever possible, for these reasons:

- ° More people can occupy a properly furnished shelter for weeks, if necessary if adequate ventilation is supplied for the additional occupants.
- ° Cleanliness, health, and morale are better if well designed furnishings are used. More serious discomforts than discomfort are likely to result if occupants have to huddle together on a bare floor that is damp earth.
- ° Persons occupying a shelter made relatively comfortable by its furnishings are more likely to remain in the shelter long enough to avoid dangerous exposure to fallout radiation.

#### CHAIRS, BENCHES, AND BUNKS

The father of the previously described Utah family of six knew that the members of his family would find the shelter uncomfortable and probably would have sore backs if they spent the required 72 hours of confinement huddled on the floor. (Their shelter room was only 3-1/2 feet wide and 16-1/2 feet long.) He brought home four folding chairs and two pieces of plywood (each 2 1/2 inches wide by 6 feet long) and placed them on top of the family car. Four small wooden boxes served as food containers during the drill. In the shelter, the boxes were used to support the ends of two narrow plywood bunks.

The family's system of sleeping and sitting in shifts worked reasonably well. There were complaints that they found the two plywood bunks too narrow, and the plywood was so hard that all the family members used sleeping bags for padding rather than for needed warmth on chilly nights. The father and oldest son, who was during normal waking hours, had trouble sleeping in such a small shelter while the liv-

awake.

Note that in the shelter pictured in Fig. 14. 1 the earth walls are covered with plastic film. Covering earth walls with plastic or bed sheets makes for a cleaner shelter, with less earth falling on the floor. Bedsheets on the walls make a shelter brighter, but are flammable and plastic film prevented the earth walls from drying and crumbling as a result of the hot, dry weather during the shelter during the day.

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Fig. 14.1. Bunks and folding chairs furnished this Pole-Covered Trench Shelter. (Note the suspended transistor radio. Reception is good in all types of expedient shelters tested to date.)

Fig. 14.2. Benches with overhead bunks in a skillfully designed Small-Pole Shelter of Russian design. Three rural families in a wooded area of Tennessee built this expedient blast shelter in the time spent felling trees and making furnishings, (nw 1 1 8b.jpg)

Benches with overhead wooden bunks are shown in Fig. 14.2. These were installed in a Small-Pole Shelter wide with a ceiling almost 7 feet high.

A well-designed expedient shelter should be as small as practical, with all space used very efficiently. Designers should make the heights and widths of benches and bunks as specified in the detailed drawings of those for the Small-Pole Shelter given in Appendix A.3.

Serious difficulties can result from failure to use specific dimensions that may appear unimportant in field tests at Fort Bragg, N.C., 48 airborne infantrymen, working only with hand tools, constructed 24-man Small-Pole Shelters in less than 24 hours. 43 The men did not think it necessary to use the specified dimensions when they made the furnishings. As a result, they built the benches too high and the overhead bunks too close to the ceilings. This error forced the men to sit for hours in hunched positions. Even these able-bodied men have developed very sore backs and would have wanted very much to leave their shelters if they had to sit in a bent-over position for days.

Figure 14.2 shows a good example of the importance of using dimensions which have been the result of experience when building essential parts of a shelter. Note the small air-exhaust opening above the overhead bunk at one end of the shelter.

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This opening led to a small, chimney-like, air-exhaust duct made of boards, with its cross-section as shown in Russian civil defense handbooks for natural ventilation of small expedient shelters. With this opening- only 4 square inches (10 square centimeters) per person a fully occupied shelter can become dangerously overheated in warm or hot weather, even though a good low-pressure expedient fan pump (a KAF or a Directional Fan) were to be used. A much larger air-exhaust opening is needed. See Appendix A.3.

BEDSHEET-HAMMOCK AND CHAIR

On the last night of the Utah family's shelter stay it was clear that the six members would prefer them for their 72-hour occupancy of the shelter starting immediately after they completed the author showed them that night how to make boat-shaped hammocks out of bedsheets. (Any right size can be used.) They were shown how to hang these short, yet stable, hammocks secured to the shelter roof. With three members sleeping in hammocks, two on the plywood bunks, and one could sleep at the same time. Figure 14.3 shows part of this sleeping arrangement.

Fig. 14.3. Girl resting in a boat-shaped hammock. Her brother slept on the upper bunk of the 3-1/2-ft-wide trench shelter, (nwl 19a.jpg)

In a shelter this size without bunks, hanging four short hammocks at slight angles to the wall would permit four occupants to sleep comfortably. An additional two persons could sleep on the floor of the shelter, the floor was made comfortable by covering the damp earth with pieces of polyethylene then placing strips of shag rug over the plastic.

In shelters with ceilings at least 6 feet high, one hammock can be hung above another. In a shelter that is 6 feet wide, a greater number of people can sleep or sit comfortably at the same time if Bedsheet-Chairs are used rather than benches or a combination of benches and overhead bunks. how BedsheetHammocks can be used like double-deck bunks.

Fig. 14.4. Bedsheet-Hammocks hung one above the other across the room of a Russian-type Small- Pole Shelter made of lumber, (nwl19b.jpg)

Detailed instructions for making a BedsheetHammock and a Bedsheet-Chair are given at the end of the Book Page: 120

In an evacuation during a real crisis, carrying comfortable folding chairs and the materials to make them would not be advisable. If the family car were loaded instead with an equivalent weight of

clothing, the members' prospects of surviving would be improved. But in an actual crisis everyone entering to occupy a shelter with a strong roof should take along a bedsheet for each member. The instructions described in the instructions for making a Bedsheet-Hammock and a Bedsheet-Chair also should be followed. Using the instructions at the end of this chapter, a comfortable hammock can be made and quickly converted into a comfortable suspended chair when not needed for sleeping.

Hammocks hung high off the floor and above other sleepers must be strong, securely suspended so that it is impossible to fall out accidentally. This is why the instructions emphasize using strong material and folding the cloth so as to make the hammock boat-like, with high sides.

In a cold shelter, keeping warm in a hammock is somewhat difficult. Easily compressible materials, such as those used in a sleeping bag, are squeezed so thin under a person's body that they lose most of their insulating value. Old newspapers about an inch thick, protected by cloth coverings, will reduce heat losses. A blanket, fastened to the underside of a hammock by attaching it with rows of stitches every few inches, will keep each other.

Figure 14.5 shows a Bedsheet-Hammock that had been quickly converted into a Bedsheet-Chair against the shelter wall. It occupies less than half the floor space used by the two hammocks shown in

If a reclining seat is desired, the two support-points on the ceiling to which the chair is located farther out from the wall.

Enough padding material should be placed in the bowl-shaped seat of a Bed sheet-Chair to make it comfortable. Extra clothing or a folded blanket can be used. The three cords suspending the chair should be adjusted so that the sitter's feet can rest on the floor and the edge of the chair seat does not press against the legs. (Such pressure cuts off circulation. During the London Blitz of World War 1 1 , many of the people who spent the night in shelters on folding chairs with canvas seats developed serious leg conditions. Avoiding such chairs into shelters.)

CAUTION: To prevent skin infections and other diseases from spreading, a person's hammock should not be used by others. This precaution is particularly important if the shelter is hot and its occupants are crowded.

Fig. 14.5. A Bedsheet-Hammock converted into a comfortable suspended Bedsheet-Chair.

(nwl20.jpg)

#### HOW TO MAKE A BEDSHEET-HAMMOCK AND CONVERT IT TO A

#### SUSPENDED BEDSHEET-CHAIR

1. PURPOSE: To enable more people to occupy a shelter more comfortably.

2. ADVANTAGES:

- \* The hammock can be made in a few minutes, once you have the materials and the knowhow.

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- \* The only materials required are a strong double-bed sheet (or an equally large piece of fabric), a few feet of rope (or a piece of strong fabric from which expedient "rope" can be made), a few large nails, and some wire.

- \* It is difficult to fall out of the hammock because its sides are each made about 8 inches wide in the lengthwise mid-section, so as to produce a boat-like shape.

- \* It provides room for head and shoulders close to either end; thus it is practical to hang it between supports that are as close together as 6 feet. See Fig. 14.6.

Before beginning work, someone should read aloud all of the instructions for making the hammock. This will help to avoid mistakes.

Fig. 14.6. The author lying in a Bedsheet Hammock. (Note that he is pulling the operating

homemade shelter- ventilating pump, a KAP.) (nwl21.tif)

#### MAKING A BEDSHEET-HAMMOCK

#### A. HOW TO MAKE THE RED SHEET:

1. Select a strong double-bed sheet (one containing polyester is best) and use a ruler or guessing at measurements.
2. Fold the bedsheet lengthwise down its center line, so that pairs of corners are together.
3. With the sheet folded, mark the center of each of the two folded ends; then hold one end.
4. Starting at one corner of one end of the folded bedsheet, make accordion-like pleats. Make the pleats 2 inches wide; make the left corner of each pleat about 1 inch lower than the left corner of the preceding pleat when the sheet is being held as illustrated. Use your left hand to hold the completed pleats and make new pleats with your right hand.

(Illustration) ORNL-DWG 77-17387 (nwl21.jpg)

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5. When one-half of the upper end of the sheet has been folded into pleats almost to the CENTER MARK, adjust the pleats so that the CENTER MARK is about 4 inches below the STARTING CORNER.
6. Continue making 2-inch pleats on past the CENTER MARK, but make the right corner of each pleat 1 inch higher than the right corner of the preceding pleat. When the pleat-folding is completed, the STARTING CORNER and the other corner should be at the same height (4 inches) above the CENTER MARK.
7. Tie the hammock-supporting rope tightly around the end of the sheet about 3 inches below the CENTER MARK. (If a rope strong enough to support at least the weight of two men is not available, use an expedient "rope" by tearing a 16-inch-wide strip from a sheet or other strong cloth and rolling it crosswise to its length to make a tight roll several feet long. Then tie string or string around it, spaced 4 to 6 inches apart, to keep the rolled-up cloth from unrolling.)
8. Bend the pleat-folded end of sheet downward around the hammock rope, so that the knot or fold of the sheet is uppermost.
9. To keep the sheet from being pulled through the encircling hammock rope, bind the double-folded sheet with cord (or with narrow strips of cloth) about 1 inch below the rope. Tie the binding cord three times around, knotting it each time around.

Fig. Pg. 122 (3 Part Illustration) ORNL-DWG 77-17390 (nwl22.tif)

10. Repeat the procedure (4 through 9) with the other end of the double-folded sheet, thus completing the boat-shaped hammock with its two sides each about 8 inches shorter than its lengthwise center.

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#### A. HOW TO HANG THE HAMMOCK:

1. To suspend a hammock from a strong wooden roof such as the poles of a Pole-Covered Trellis, drive two strong nails (at least 3-1/2 inches long) into the wood at approximately 45° angles, facing each other. Bind the two nails together with wire. To prevent a hammock rope from being rubbed against fixed metal, make a loose loop of strong wire (best if doubled) through the crossed ends of the hammock rope to this free-moving wire loop.

2. To suspend the hammock from a wooden wall, use the same type of crossed-nails supports, driven in one above the other.

3. For comfort and safety, hang the hammock with the head end 18 inches higher than the foot end 24 inches higher than the center.

4. To make sure that the hammock is strong enough, two persons should place their open hands on the hammock and put all of their weight on the hammock.

5. To suspend hammocks and hammock-chairs from a pole roof that is not being built under fire attack, use loops of strong wire around the poles at the planned support points. (The correct wire loops takes considerable time and delays completion of the shelter.) To reduce stress breakage, the loops should be loose, as illustrated.

Fig. Pg. 123ab ORNL-DWG 77-17394 (nwl23.tif)

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#### MAKING A SUSPENDED BEDSHEET- CHAIR

A Bedsheet-Hammock may be quickly converted into a comfortable Suspended Bed sheet-Chair occupant can sit comfortably, yet occupy less floor space during the daytime. Follow these

1 . Select one end of the hammock to be the top of the back of the chair.

2. From this end, measure 52 inches (4 ft 4 in.) along each side-edge of the hammock (see these two spots.

3. About 2- 1/2 inches in from these two marks (toward the centerline of the hammock), make Fig. Pg. 124 ab (2 Illustrations) ORNL-DWG 77-18432, ORNL-DWG 77-17395 (nwl24.tif)

4. To make an attachment point for a chair "arm", hold a pebble (or a lump of earth) under the marks, pull the double-thickness cloth tight around the pebble, and tie it in place. (See the other side-edge of the hammock.

5. Tie the end of one rope (or "rope" made of 10- inch-wide strip of strong cloth) to one end of another rope to the other attachment point.

6. Suspend the top of the back of the chair to a suspension point on the ceiling at least 12 inches from the wall, and adjust the length of this suspending rope so that the chair arms will be about 12 inches from the floor as the arms of an easy chair (see sketch).

7. Suspend the arms of the chair from two suspension points 20 inches apart and 20 inches from the wall than the suspension point of the back of the chair. (Study the illustration.)

8. Fold the unused end of the hammock up and back into the "seat" of the chair; fill the hammock with coats, a blanket, or anything else soft, to make it comfortable.

9. Adjust the lengths of all three suspending ropes so that the 'chair seat is the right height in it. When both feet are flat on the floor, the front edge of the seat should not press against the thighs.

10. To simplify repeated conversion of the hammock to a chair, mark the spot on each of the ropes where each is tied to its suspension point on the ceiling; also mark the spot on each suspension point on a chair arm where each is tied to its suspension point on a chair arm. If enough light rope is available, the easiest and quickest way to connect and disconnect the arm supports is to use a knot with a double strand of rope, looped around an attachment point as illustrated by the sketch of the suspension point.

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## Chapter 15

### Improvised Clothing and Protective Items

#### BASIC PRINCIPLES OF COLD WEATHER CLOTHING

If Americans would learn to use skillfully the ordinary clothing, towels, cloth, newspaper, etc., in their homes, they could keep warm enough to stay healthy even under much colder conditions than they are now without specialized outdoor winter clothing. Efficient cold-weather clothing can be improvised if the principles of conserving body heat are understood and used:

Trap "dead" air. Covering enough of your body with a thick layer of trapped "dead" air is the key to keeping warm. Figure 15.1 shows how efficient body insulation works: Both the air next to the skin and the water vapor from evaporated perspiration flow outward into the insulating layer. The insulating layer breaks up and separates air into spaces no more than 1/8 inch

Fig. 15.1 Efficient body insulation. ORNL-DWG 77-18426 (nwl25.tif)

across has efficient cells of "dead" air. Air that is within 1/16 inch of any surface whether it is a wall or of a piece of paper is slowed down by "sticking" to that surface and becoming hard to move. As air moves outward very slowly, carrying heat away from the body at a slow rate thus minimizing convection.

Use windbreaker materials. An outer windbreaker layer of clothing that is essentially air-tight, like a paper bag worn over a knit wool cap, prevents the escape of warmed air and results in an insulating layer of "dead" air. A single layer of good windbreaker material also prevents cold outside air from penetrating the insulating material and displacing warmed air (Fig. 15.1).

The best windbreaker materials permit very little air to pass through them, while at the same time they allow water vapor to escape. Perspiration that cannot be felt or seen on cool skin continually evaporates close to the skin. This moisture escapes outward through good insulating and windbreaker materials. Underlying body insulation remains dry and efficient. Water vapor can pass readily through good insulating or unglazed brown paper, although not enough wind can flow through a single sheet to be felt.

° Prevent excessive heat losses by conduction. Body heat also is lost by conduction the direct contact with cold surfaces.



into a colder material. For example, if one sleeps in an excellent goose-down sleeping bag body weight will compress the down to a small fraction of an inch. This barrier to heat flow prevents the body to lose heat rapidly to the cold earth. Likewise, the soles of ordinary shoes are standing on frozen ground sometimes results in frozen feet.

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#### MINIMIZING HEAT LOSSES FROM HEAD AND NECK

The head and neck of the girl pictured in Fig. 15.2 are insulated almost as well as if she wore a skin-side-out Eskimo parka. She folded a large, fluffy bath towel and placed it over her head and neck. A brown paper bag was worn over the towel. The edges of the face hole cut to prevent tearing. A strip of cloth was tied around the part of the bag over her neck. Such a method is the most efficient way to insulate the head and neck but also prevents air warmed by the body from rising upward around the neck. The girl also is wearing a man's shirt large enough to cover and button around her body and arms.

Fig. 15.2. A bath towel and a paper bag used to efficiently insulate head, neck, and shoulders.  
(nw 126a. jpg)

It is very important to prevent heat losses from the head and neck, which have many blood vessels on the surface. Heat losses from these vital parts cannot be sensed nearly as well as heat losses from the body.

Furthermore, blood vessels near the surface in the head and neck do not automatically constrict to prevent heat losses, as they do in other parts of the body when heat is being lost faster than it can be replaced. When a person is in the cold particularly when inactive he should keep his hands, feet, and head warm by insulating his head and neck very well. (One difficulty in following this advice is that a person may feel unpleasantly warm even sweaty before one's body temperature rises enough to increase blood flow to hands and feet.)

#### INSULATING THE WHOLE BODY

Occupants of freezing-cold shelters can keep warm enough to sleep without blankets by skillfully using indoor clothing plus paper and pieces of cloth to insulate their whole bodies. The girls pictured in Fig. 15.3 were out a blanket in a frozen Door-Covered Trench Shelter while the night temperature outdoors was below zero. The shelter's ventilation openings were adjusted so that the inside temperature remained a few degrees above freezing to prevent frozen earth from melting into icy mud. These girls had insulated themselves well. They wore cotton shirts and pants with 10 thicknesses of newspaper wrapped around their bodies and tucked into their shoes. Then around each arm and leg they wrapped and tied 8 sheets of newspaper, thus insulating 16 thicknesses.

Fig. 15.3. Girls wearing expedient clothing are prepared for sleeping in the freezing cold shelter. (nwl26b.jpg)

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As an outer covering over their legs, they wrapped wide strips torn from a bedsheet. Their feet were of the type described in a following paragraph. Their heads and necks were insulated

brown paper bags. Old cotton raincoats allowed water vapor to pass through and helped hold newspapers, which extended to cover the girls' bare hands.

The girls slept on newspapers spread about an inch thick over the gravel floor of the trench. In cold or frozen ground, it is best to place newspapers or other insulation on top of a layer that drying air can circulate under the bedding. A sheet of plastic under bedding will keep a wet floor but will not prevent it from being dampened after a few days by condensed water from the body.

Newspaper and other paper through which water vapor can pass are such good windbreakers that they can be used under any loose-fitting outer garment even one through which air can pass quite freely to provide good insulation. Figure 15.4 shows the author coming out of an icy shelter at sunrise. Newspaper covered my body and arms and extended like cuffs from my sleeves. A porous cotton blanket under the newspapers and helped hold them in place. Because so little heat was lost through this clothing, heat continued to flow to my bare hands, ridding my body of excess heat by radiation.

#### IMPROVISED WINTER FOOTWEAR

Cold-weather footwear that is warmer than all but the best-insulated winter boots can be made. The trick is learning how to tie the several insulating layers securely in place, so that you can keep them necessary.

For use in dry snow, first tie a porous insulating layer such as two bath towels or 10 big sheets of paper over each shoe. If you have no low-heeled shoes, make a paper sole by folding 3 large newspapers together to make a sole that has 72 thicknesses of paper. Then proceed in the following manner:

1. Place your foot and the sole on 10 newspaper sheets, as pictured in Fig. 15.5.

Fig. 15.4. The author emerges after a night's sleep in freezing cold temperatures inside a Car-Over-Trench Shelter. Expedient clothing, primarily newspaper insulation, kept him warm. (nwl27.jpg)

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2. Fold all the sheets over the top of your foot while keeping the sole in the proper place. (nwl27.jpg)

3. Use a strip of cloth about 3 inches wide and 5 feet long to tie the papers in front of your foot with an overhand knot (half of a square knot). With the same strip, tie another single overhand knot behind the ankle. Finally, tie a bow knot in front of the ankle.

4. Cover the insulating layer with a tough fabric, such as canvas or burlap sack material; tie the strip of cloth and tie as described above.

If the snow is wet, place a piece of strong plastic film or coated fabric outside the insulation with the first strip of cloth. The outer protective covering should be tied over the waterproofing of cloth securing both it and the waterproofing. (When resting or sleeping in a dry place,

layer in the foot coverings, to let your feet dry.) Figure 15.6 shows a test subject's water-er, held in place as described above, after a 2-mile hike in wet snow. His feet were water to tighten or adjust the cloth strips.

Fig. 15.6. Expedient water-proofed foot- covering, over a newspaper sole and other newspaper insulation. (nwl28a.jpg) (nwl28b.jpg)

Persons who have not worked outdoors in icy weather seldom realize the importance of winter. Russian civil defense manuals direct urban citizens to take winter boots with them when summer.

Fig. 15.5. Insulating a foot with a folded newspaper sole and 10 sheets of newspaper, (nwl Book Page: 129

#### KEEPING WARM WITHOUT FIRE

- ° If occupants of a cold room or shelter lack adequate clothing and bedding, all should lie
- ° Always place some insulating material between your body and a cold floor. (Pieces of sheet excellent.) Plastic film should be placed under the insulating material if the ground is cold
- ° Go to bed or put on all your body insulation before you begin to feel cold. Once the loss blood vessels in your hands and feet to constrict, it often is hard to get these vessels to dilation again.
- ° Do not jump up and down or wave your arms to get or to keep warm. The wind-chill factor air movement over your skin; rapid body movements always cause some such air movement. If down and cover up; then do muscular tension exercises by repeatedly tightening all your muscles you tremble.
- ° Prevent sweating and the dampening of insulation by taking off or opening up clothing as exercise, before you begin to sweat.
- ° If you are getting cold, don't smoke. Nicotine causes blood vessels to constrict and the and feet to be reduced.
- ° Don't drink an alcoholic beverage to warm yourself. Alcohol causes increased blood flow surface, resulting in rapid loss of body heat. It is impossible for alcohol to make up for

#### RAINWEAR

All that is needed to make serviceable, improvised rainwear is waterproof material and water film from large trash bags will do; 4-mil polyethylene is better; tough, lightweight, coat tape is the best widely available tape.

Figure 15.7 shows a pair of improvised rain chaps. Rain chaps are separate leg coverings,

suspend it from one's belt and usually made large enough to be pulled on and off over the chaps are better than waterproof trousers for working or walking while wearing a poncho or movements cause drying air to be literally pumped under the chaps. This air keeps trousers fore warmer.

Fig. 15.7. Improvised rain chaps made of trash- bag polyethylene and freezer tape. (nwl29a

In the same way, a poncho or rain cape will allow plenty of air to reach the garments under working. When exercise is stopped, clothing underneath will stay dry and warm for some time.

#### SANDALS

Shoes are almost always in short supply for years following a disastrous war. Except in sandals can be made to serve quite well. The best sandal designs for hard work and serious work the heel and in front of the ankle, with no thong between the toes (Fig. 15.8).

Fig. 15.8. A Ho Chi Minh Sandal, excellent Vietnamese expedient footwear. Rubber bands cut an inner tube have been inserted into a sole of auto-tire tread, (nwl29b.jpg)

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Such sandals also have the advantage of enabling one to wear socks and other foot insulation.

#### CLOTHING TO PROTECT AGAINST BETA BURNS

If fresh and very radioactive fallout particles remain for long on the skin or extremely close result. Any clothing that keeps fallout off the skin helps greatly. The best expedient protection of easily removable clothing similar to the improvised rainwear previously described, but neck and providing plastic trousers instead of chaps. Removable shoe coverings are highly effective coverings should be removed before entering a shelter, or removed in the entryway before room.

If a person has fallout particles on clothing that he must continue to wear, he should vigorously clean clothing before entering a shelter room. If fallout particles are washed off, rinsed off, skin within a few minutes, no beta burns will result.

A few days after a nuclear explosion, fallout particles are not radioactive enough to cause heavy fallout, the danger from external doses of gamma radiation from fallout on the ground than will the risk of beta burns from some of these same fallout particles.

The gamma rays given off by fallout particles brought into a shelter on clothing or bodies of occupants to gamma doses so small as to be of no significance by wartime standards. Nor would be endangered by radiation from the body of a person who, before reaching shelter, had received enough to kill him many times over. Except in science fiction stories, the body of such a person is not radioactive."

#### FALLOUT MASKS

For the majority of Americans in most fallout areas, means for filtering fallout particles would not be essential survival equipment. 1 8 Most fallout particles tiny enough to enter

so slowly that they would reach ground thousands of miles away from the explosions. By the time they would make them much less dangerous, and their deposition would be spread out over much of the country.

In past years American-endangering Soviet warheads typically' were between 20 and one megaton. This large would inject almost all fallout particles into the stratosphere, high above rain-out levels. Deployed Soviet ICBM warheads are between 550 and 100 kilotons. (See Jane's Weapons Systems Yearbook.) Surface bursts and air bursts of today's smaller warheads would inject most of their radioactive material into the troposphere, from whence rain-outs and snow-outs would bring huge numbers of even tiny particles scattered across America.

Persons living in dry, windy areas often wear dust masks and goggles to protect their noses and eyes from sand particles. If fallout particles are mixed with dust and sand that is being blown in dry, windy areas who occupy small below-ground expedient shelters should cover their noses and eyes with thicknesses of towels or other cloth. Those who have dust masks should wear them, especially in dry, windy weather soon after fallout deposition. Other than whole-body exposure to gamma rays, the greatest danger to well informed people would be from possible beta burns caused by fresh, "hot" fallout particles passing through nasal passageways, and from swallowed fallout particles. (Much of the material continually passes through the mouth and throat is swallowed. ) In some fallout "hot spots" a secondary danger would be breathing fallout particles into one's lungs, after a rain-out of tiny fallout particles from fallout from kiloton-range Soviet warheads. In some areas "hot" particles would be dried and blown about for hours of their deposition in rain showers.

Making a homemade dust and fallout mask is still not a high priority survival project. In the meantime, buy and store good masks and goggles. The following instructions for making a homemade mask are based on a Russian design (Fig. 15.9). This mask is the best of several

Fig. 15.9. A Russian-type homemade fallout mask. For most Americans this will continue to be the best available item as long as Russian warheads are large. ORNF-DWG 78-11921 (nwl30.tif)

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homemade types, and the following instructions for making and using it have been field-tested.

#### AN INDIVIDUALLY -FITTED FALLOUT MASK

##### Materials Needed:

- 1 . Three rectangular pieces of fluffy toweling (terry cloth preferred), each piece approximately 12 inches by 18 inches (or use 10 men's handkerchiefs).
2. Elastic. (The best expedient elastic is from the waistband of a man's undershorts.)
3. Clear plastic (from a photo album, billfold, plastic storm window, etc.).
4. Sewing materials.

##### Measurements:

- 1 . Tie a string vertically around your head and face, passing it 1/2 inch in front of each eye and tie it quite tight.
2. Tie a second string horizontally around your head, crossing your forehead 1-1/4 inches above your eyebrows.

eye sockets. These two strings should cross each other at points X and X 1 , over your tem

3. Measure the distance X-to-X 1 across your forehead and the distance X-to-X 1 going unde  
(around your lower jaw and next to your throat), as indicated by Fig. 15.9.

#### Construction:

1. Cut out 3 pieces of terry cloth, making the width of each piece equal to X-to-X 1 (the  
your forehead), and the height of each piece equal to the distance X-to-Y plus 1/2 inch -  
the distance X-to-X 1 (measured under your chin) plus 1/4 inch. See Fig. 15.9.

2. Cut the lower edge of each piece as illustrated.

3. Stitch the 3 pieces of terry cloth together, one on top of the other, thus making the m  
Stitch around all four edges of the cloth rectangle and down the centerline.

4. Mark and cut out the eye holes, as illustrated. Make the mask's dimensions smaller for

5. Cut one rectangle of clear plastic measuring 5- 1/2 x 2- 1/4 inches, and sew this plast  
eye holes, stitching the plastic around its edges and down the centerline.

6. Fold the three pieces along their vertical centerline and stitch the lower side togethe  
line Y-to-Y 1 . Stitch 1/2 inch from the lower edge. Then sew a second row of stitches.

7. Sew on the elastic head bands, making them short enough to hold the mask tightly around  
elastic is from the waistband of a man's undershorts, use a doubled elastic both over the  
around the back of your head. Make these elastic pieces so short that you can just put all  
comfortably between the elastic and your head when the pieces are fully stretched. If usin  
be sure to adjust the lengths to a tight fit, to prevent air leaks under the edges of the  
thickness of material where the elastics are connected to the upper corners of the mask, i  
to do this stitching by hand.

8. To keep fallout particles off your head and neck, sew a loose-fitting piece of bedsheet  
the edges of the mask that fit around your face. This cloth should extend back over your h  
your collar, over which it can be tied.

#### Use:

Put on the mask by first placing it over your chin, then pulling the back elastic down to  
head.

#### CAUTION:

To avoid spreading infections, each mask should be labeled and worn by only one person.

## Chapter 16

### Minimum Pre-Crisis Preparations

Your chances of surviving a nuclear attack will be improved if you make the following long before a serious crisis arises. Once many Americans become convinced that a nuclear attack is imminent, they will rush to stores and buy all available survival supplies. If you wait to prepare until the crisis is upon you, you will be among the majority who will have to make-do with inadequate supplies of water, food, and shelter. Furthermore, even if you have the necessary materials and instructions to make the most of your supplies, you and your family are not likely to have time to make all of them during a few days of tense preparation.

The following recommendations are intended primarily for the majority who live in areas likely to be hit by nuclear blast, fire, or extremely heavy fallout. These people should plan to evacuate to a safer area as soon as possible. Those outside high-risk areas, especially homeowners with yards, can and should make better preparations.

These would include building high -protection-factor permanent shelters covered with earth or concrete.

#### SHELTER

Keep on hand the tools and materials your family or group will need to build or improve a permanent shelter: One or more shovels, a pick (if in a hard-soil area), a bow-saw with an 8-ft blade, a 4-mil polyethylene film for rainproofing your planned shelter. Also store the necessary materials for the kind of shelter you plan to build.

Keep instructions for shelter-building and other survival essentials in a safe and convenient place.

#### VENTILATION- COOLING

Make a homemade shelter- ventilating pump, a KAP, of the size required for the shelter you plan to build.

#### WATER

Keep on hand water containers (including at least four 30-gallon untreated polyethylene tanks or pillowcases for each person), a pliable garden hose or other tube for siphoning, and a solution of sodium hypochlorite bleach (such as Clorox) for disinfecting water and utensils.

#### FAT EOT IT METER

Make one or two KFMs and learn how to use this simple instrument.

#### FOOD

Store at least a 2-week supply of compact, nonperishable food. The balanced ration of basic foods in Chapter 9, Food, satisfies requirements for adults and larger children at minimum cost. For infants or small children, be sure to store more milk powder, vegetable oil, and sugar.

Continuing to breast-feed babies born during an impending crisis would greatly simplify the situation if the crisis develop and worsen.

For preparing and cooking basic foods:

° Make a 3-Pipe Grain Mill like the one described in Chapter 9, Food, or buy a small hand-

which grinds more efficiently than other expedient devices.

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- ° Make a Bucket-Stove as described in Chapter 9. During evacuation, the stove can be used. Store some kitchen-type wooden matches in a waterproof container.

Keep essential containers and utensils on hand for storing and transporting food and for use in a shelter.

#### SANITATION

A hose-vented 5-gallon can, with heavy plastic bags for liners, for use as a toilet. Include bags and toilet paper with these supplies. Tampons.

Insect screen or mosquito netting, and fly bait. See Chapter 12.

#### MEDICINES

- ° Any special medications needed by family members.
- ° Potassium iodide, a 2-oz bottle, and a medicine- dropper, for prophylactic protection of against radioactive iodines. (Described in the last section of Chapter 13, Survival Without Shelter.)
- ° A first-aid kit and a tube of antibiotic ointment.

#### LIGHT

- ° Long-burning candles (with small wicks) sufficient for at least 14 nights.
- ° An expedient lamp, with extra cotton-string, wicks, and cooking oil as described in Chapter 12.
- ° A flashlight and extra batteries.

#### RADIO

A transistor radio with extra batteries and a metal box in which to protect it.

#### OTHER ESSENTIALS

Review the EVACUATION CHECKLIST (developed primarily for persons who make no preparations for a crisis) and add items that are special requirements of your family.

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## Permanent Family Fallout Shelters for Dual Use

### THE NEED

Having a permanent, ready-to-use, well supplied fallout shelter would greatly improve millions' chances of surviving a nuclear attack. Dual use family shelters - shelters that also serve as the ones that Americans are most likely to build in normal peacetime and to maintain for use in a nuclear war.

The longer nuclear peace lasts, the more difficult it will be, even during a recognized crisis, to thinkable war is about to strike us and that we should build expedient shelters and immediate actions. The lifesaving potential of permanent, ready-to-use family shelters will increase.

Americans who decide to build permanent shelters need better instructions than can be obtained from sources or from most contractors. This chapter brings together fallout shelter requirements, shelter components that have been built and tested in several states and nations. The emphasis is on shelters that many Americans can build for themselves. The author believes that millions could build permanent fallout shelters or have local contractors build them - if they learn the shelter requirements from the following sections of this chapter and the facts about nuclear weapon effects and protection from preceding chapters. Builders can use their skills and available local resources to construct shelters at affordable cost.

Requirements for a permanent, dual-use family fallout shelter follow.

### A HIGH PROTECTION FACTOR, AT AFFORDABLE COST

A permanent fallout shelter should be built - and can easily be built - to have a high enough protection factor to prevent its occupants from receiving fatal or incapacitating radiation doses, and also from receiving enough to seriously worsen their risks of developing cancer in the years following an attack. A protection factor of 40 (PF 40) meet the minimum standard of protection for public shelters throughout the country. Permanent family fallout shelters described in official pamphlets provide at least PF 40 protection. In areas, PF-40 shelters would prevent occupants from receiving fatal or incapacitating radiation doses. However, in areas of heavy fallout the occupants of PF-40 shelters could receive enough to significantly contribute to the risk of contracting cancer years later. Furthermore, the longer you receive while in a shelter, the smaller the dose you can receive after you leave shelter will be. You will be killed by your total dose.

If you build a permanent shelter, you would be foolish to build a shelter with a PF of only 40. Protection is so easy to obtain. By making a shelter with a 6-inch-thick concrete roof covering earth, and with other easily attained design features shown in Figs. 17.1 and 17.2, you can obtain a protection factor of about 1000. (An occupant of a PF 1000 shelter will receive a radiation dose as he would receive if he were standing outside in an open field during the same time interval.)

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Fig. 17.1. Permanent Family Fallout Shelter for Dual Use. (nwl35.tif)

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Fig. 17.2. Permanent Family Fallout Shelter for Dual Use. (nwl36.tif)

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To attain PF 1000 protection near the inner door of the illustrated shelter, its occupants water and/or other good shielding material against the door. They can do this easily and supplied with filled water containers such as described in the Water section of this chapter.

The illustrated shelter room has 106 square feet of floor space - room enough for 5 adults. If additional occupants will need for long occupancy, if shelter furnishings like those described in the additional occupant, increase this shelter's length by 2 feet. To increase room size, increase retains maximum roof strength at minimum cost.

Note in Figs. 17.1 and 17.2 the 12-inch-thick concrete wall between the landing at the front end of the shelter room. Only a very small fraction of the radiation coming through the outside stairs will make the 90 degree turn through the inner door, and most of this radiation will be stopped by the place containers filled with water and other shielding material against the door.

Also note the homemadeable, low cost Double- Action Piston Pump and filter, shown in Fig. 17.3. A heat wave will supply adequate air through the 5-inch-diameter air- intake pipe - all described in the Air section of this chapter.

Few survival-minded Americans, before a recognized international crisis arises, either care or they can afford to build a permanent family fallout shelter costing around \$10,000 in 1987 dollars. A concrete, below-ground shelter of the type specified in official Federal Emergency Management Administration about \$100 per square foot of floor space, if built by a contractor in a typical suburban area, with skills and time can save about half of this cost by doing their own work. Also, at many building drainage of the earth around a shelter's walls can be assured and hydrostatic earth pressure prevented, no steel reinforcement in the poured concrete or concrete block walls is needed except local building code.

Caution: Steel reinforcement in the walls and floor is needed in some clay soils that swell under sufficient inward and upward pressure to crack unreinforced walls and floor slabs. Consult a geologist learned from experience whether wall and floor reinforcement is needed in the type of soil. If needed, a grid of 1/2-inch rebars, spaced at 12 inches, usually is adequate.

To save money on steel reinforcement, check prices in salvage yards for used rebars and materials such as junked cable and small pipes.

How to safely pour a shelter's concrete roof slab without using a contractor's usual forms is indicated by Figs. 17.1 and 17.2. These drawings show 8-ft.-long sheets of 3/4-inch plywood supporting the shelter walls 7 feet-6 inches apart. Preparatory to pouring the concrete, the plywood sheets are placed on the centerline of the shelter by 4"x4"s and other lumber, which can be used later to build the shelter. Plywood left on the ceiling reduces condensation and heating problems in cold weather, but in hot outdoor air that must be pumped through the shelter to maintain tolerable temperatures when the weather is hot. This was clearly demonstrated in the summer of 1963 when the author used SIMOCS (Simulated Man) that produce heat and water vapor like people) to determine the habitability of a six -room shelter, with a reinforced concrete roof that had been built in this manner by six New Jersey men. Ordinary 3/4-inch exterior plywood on the ceiling. Because the hollow concrete wall-blocks under the floor also kept heat from escaping into the surrounding soil, and because only the air was provided, the temperature/humidity became dangerously high within a few hours.

Insulating a shelter's walls and ceiling can be disadvantageous, because insulation makes the shelter "sink" of the shelter and its surrounding earth. In hot weather insulation reduces the time

be stopped or restricted without disastrously overheating the occupants.

Today, for such a shelter it would be better to use pressure-treated, rot-proof plywood and leading building codes. For information, write to the American Plywood Association, P.O. Box 9841 1, inquiring about rot-proof plywood, dimensional lumber, and other material used in Wood Foundation. Most lumber yards will obtain treated plywood on order, and sell it for a nary exterior grade plywood.

Big savings in shelter construction costs are made by using salvaged and/or used materials. cast reinforced concrete beams and floor sections often sell rejects for very little.

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Most salvage yards have steel beams and other material that make excellent roofing for ear (Shortly before the Cuban Missile Crisis, when living near New York, I built for myself at

shelter on a well drained hillside. I made it almost entirely of steel channels bought and yard.) Used cylindrical steel tanks with closed ends often make good low-cost permanent sh low-cost family shelters that I ever went into was on a hillside overlooking San Francisco vaged steel brewing tank that had been installed after a vertical cylindrical entrance wit it. The tank's exterior had been protected with a bituminous coating. Its survival-minded Hungarian refugee who, as a boy, had survived in a deep wine cellar throughout the Russian dapest. Nothing equals war experience to teach the lifesaving value of shelters.

#### MINIMIZATION OF FIRE AND CARBON MONOXIDE DANGERS

Many shelter designers and builders do not realize the probable extent of fires after a ma do the big majority of them provide shelters built under or close to a house with adequate of deadly amounts of carbon monoxide if the house bums. Although the areas of fires result sion generally will be about as extensive as the areas of significant blast damage, 6 on a 8 miles from ground zero can be ignited by the thermal pulse of a 1-megaton airburst. 6 Fi photograph of a car set on fire by a nuclear explosion so far away that the car was not ev photograph indicates how a thermal pulse can go through window glass and ignite curtains, even if flammable material outdoors is too damp to be ignited. Furthermore, fires from any cially in fallout areas following a nuclear attack when firefighters may be unable or unwi outdoors to radiation.

Good protection against carbon monoxide is provided by a permanent earth-covered shelter, outdoors and well separated from a house and other flammable structures, and constructed s gas-tight. Both the air-intake and the air-exhaust pipes should be installed so that they as with screw-on fittings. Such closures should be kept well greased and securely attached be used. If a shelter's entry is through a passageway from a house, a gasketed steel fire side, should be installed near the house end of the passageway. The shelter should be furt gas. tight door, to prevent the entry of carbon monoxide and smoke if heat from the burnin on the firedoor. If special firedoor gaskets are not available, rubber weather-stripping w carbon monoxide being pumped into the shelter if the house burns and air must be pumped in fire still is smoldering, the air-intake and pump should be at the far end of the shelter, emergency exit should be near the passageway from the house.

Be sure to seal electrical conduits leading from a basement to a connected shelter, so tha

bon monoxide can not flow through the conduits into the shelter when fresh outdoor air is shelter. The author, while conducting ventilation and habitability tests of an earth-cover through a tunnel to a house's basement, observed air flowing out through unsealed conduits sible life-endangering flow of carbon monoxide. When the shelter was being maintained at a cranking its blower to pump in outdoor air, a little air flowed through the unsealed condu

For ways to minimize carbon monoxide dangers arising from cooking, heating, lighting, and sections in this chapter.

Remember that air contaminated with only 0.16% carbon monoxide can kill you in 2 to 3 hours. Carbon monoxide causes frontal headaches and nausea in 2 to 3 hours. The Navy sets its safe allowable concentration in air at 0.01%. (Shelter Habitability Studies - The Effect of Oxygen Depletants of Shelters, by J. S. Muraoka, Report NCEL-TR-144, July 1961.)

#### PREVENTION OF CRACKS AND WET SHELTER PROBLEMS

If wet basements are a problem in your locality, below ground shelters are likely to be wet. Unless appropriate preventative measures are taken during their design and construction. When you consult local builders of satisfactory basements. You also should question persons who at have observed excavations, holes, and/or basements in your immediate vicinity, or have not places or springs.

The most difficult type of shelter to keep dry for decades is one that is wholly or partially above ground for part or all of the year. Concrete is not completely watertight. Waterproof coatings are applied during construction, or deteriorate with age.

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Shelter walls sometimes crack due to settling and earth movements. Metal shelters usually rust before they become dangerously weakened.

A 100-occupant, below ground shelter, built in 1984 near Dallas, Texas as a prototype blast shelter for workers, was flooded when the water table rose. The poorly sealed opening of this shelter was at low ground level, and after heavy rains was below the water table. A prudently designed shelter should have an emergency exit slightly above the surface of the surrounding earth, as illustrated in Fig. 1. Conduits leading down into a shelter must be well sealed to prevent entry of water.

To prevent a wholly or partially below water-table shelter from becoming wet inside sooner, install a sump and an automatic sump pump to discharge water to the ground outside. If at any time the sump pump is discharging an appreciable amount of water, you may have a serious wet-shelter problem. It fails after an attack.

A manually operated bilge pump and a sufficiently long discharge hose can be bought for about \$10. Marine supply mail order firms, including West Marine Products, Box 5189, Santa Cruz, California 95061, and Marine Industries, Inc., P.O. Box 820, New Rochelle, New York 10802-0820. (Long established marine supply firms also are good sources of use-proven chemical toilets, first aid kits, lights, rope, etc.)

Shelter roof surfaces should be gently sloped, no matter how good a waterproof coating is. When building a concrete shelter roof as little as 1 inch thicker along its centerline than along its sides, the prospects are improved for having an earth-covered coated roof that will not leak.

Figures 17.1 and 17.2 illustrate the following ways to prevent having a wet shelter:

- \* Put a layer of gravel or crushed rock in the bottom of the excavation, and install perforated gravity drainage if practical.
- \* Cover the gravel or crushed rock in the floor area with a plastic vapor barrier before pouring concrete.
- \* Coat the outer surfaces of roof and walls with bituminous waterproofing or other coating most effective in your locality.

Backfill with gravel or crushed rock against the walls, to keep the soil from possibly becoming compacted. Saturated soil exerts hydrostatic pressure against walls, and may crack them and cause them to leak. It is more economical to cover a shelter's coated walls with a subsurface drainage matting (such as Dimplex, manufactured and sold by BASF Corporation, Fibers Division, Enka, North Carolina 28728). This will prevent water from seeping into the shelter. Backfilling with gravel or crushed rock.

#### NON-FLOATABLE SHELTERS

In most localities the water table usually is below the depth of excavation needed to build a shelter. In some areas, however, after rainy periods the water table may rise until it is at the surface. Then a watertight shelter may float upward through the surrounding saturated soil. The weight of its covering earth is sufficient to withstand its buoyancy. (In many places swimmers are prevented from being cracked by uneven buoyant forces if the water table rises.)

Dramatic examples of floating shelters were steel blast shelters, guaranteed by contractors to be installed under the lawns of some Houston, Texas homes shortly after the Cuban Missile Crisis. After the water table rose after heavy rains, these shelters came up to the surface like giant mushrooms, to the surprise and satisfaction of anti-defense newspaper feature writers.

The most expensive permanent family fallout shelter described in a Federal Emergency Management Agency free handout (pamphlet H-12-1) is floatable. It is designed to be built of reinforced concrete on a raised patio only about 3 inches above the ground. Its 6-inch-thick concrete roof is covered by 18 inches of sand and flagstone. Like the rest of FEMA's permanent shelters, no prototype of this shelter was built, nor is there any record of anyone actually having built this shelter, but the shelter illustrated by Figs. 17.1 and 17.2 would not float even if it did not have assured gravel backfill through the perforated drain pipes in the gravel on which it rests, because this shelter is built on a thick earth berm on its roof.

#### ADEQUATE, DEPENDABLE VENTILATION/COOLING

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Basic facts that you need to provide adequate, dependable ventilation for a small or medium shelter are given in Chapter 6, Ventilation and Cooling of Shelters. A good permanent shelter has two ventilation systems:

- ° The primary ventilation system of a small permanent shelter should utilize a manually operated blower, or a homemade Plywood Double-Action Piston Pump. (See Appendix E.) A satisfactory system is capable of supplying adequate outdoor air through an air-intake pipe, a filter, and an air

and 17.2.) "Adequate outdoor air" for a small shelter means at least 15 cubic feet per minute in cooler parts of the United States, and at least 30 cfm per occupant in most of the country. Many homes have centrifugal blowers that can not deliver an adequate volume of air in hot weather for

For a medium sized permanent shelter, installing two or more manually powered air pumps is possible and more economical than providing an emergency generator and its engine to supply power. The Swiss, who have made the world's best and most expensive per capita preparations for nuclear war, use two or more hand-cranked centrifugal blowers to ventilate most of their shelters.

° The multi-week and/or emergency ventilation system of a permanent shelter that has an emergency generator should depend on a homemade KAP, made before a crisis and kept ready to use. (See Appendix D and Use a Homemade Shelter- Ventilating Pump, the KAP. ) By opening both the entrance and the shelter is provided with two large, low- resistance openings through which a KAP can pump air with minimum work.

Warning: Keep screen doors and/or screen panels ready to protect all openings against the mosquitoes likely to become dangerous pests in fallout areas. Use your KAP to pump adequate air. Window screens greatly reduce natural ventilation, as the author first noted in Calcutta when the stifling hot ward of a hastily constructed Army hospital. Because there were no fans or blower, air or circulate the air inside, window screens were opened in the daytime when malaria mosquitoes were common. The doctors correctly concluded that the temperature reduction when the screens were opened was more than they were endangered by the entering filthy flies. Adequate ventilation is more important than protection from flies, but with a KAP and insect screens you can have both. Flies, mosquitoes, and other insects are kept out very effectively by occasionally spraying or painting screens and other alighting surfaces with insecticides containing permithin and sold in many farm stores.

#### ADVICE ON VENTILATION OPENINGS AND FITTINGS

° Install ventilation pipes large enough to reduce resistance to airflow, thus increasing the volume of air the shelter's pump can deliver. A shelter with a 200-cfm pump (such as the homemade Plywood Box Pump described in Appendix E) should have 5-inch galvanized steel pipe. The pumps that family shelters deliver only about 100 cfm or less. Four-inch pipe is adequate for use with pumps that deliver less than 100 cfm. The pipes have no more than two right-angle turns below each gooseneck. (A 90-degree turn has a resistance to airflow as 12 feet of straight pipe.)

° Make and install a gooseneck with its mouth about twice the diameter of the pipe, as indicated in Fig. 6.4. The purpose of such a gooseneck is to prevent more of the larger descending fallout particles from entering the shelter. For example, if 200 cfm of air is being pumped into a shelter through a 5-inch pipe with its mouth also 5 inches in diameter, the velocity of the air up into the mouth of the pipe is about 10 feet per minute, and sand-like spherical particles smaller than approximately 500 microns in diameter are "sucked" up. But if the 5-inch gooseneck's mouth is 10 inches across, then the upward air velocity is about 5 feet per minute, and only particles smaller than approximately 180 microns across are "sucked" up.

Particles in the 180 to 500 micron-diameter range are relatively large and fall to earth in about 40 to 70 seconds after leaving the base of the mushroom cloud of a 1-megaton explosion. (See Fig. 6.4.) These particles are still very radioactive, and their radioactivity has had little time to decay, and should be kept out of a shelter's ventilation system. Large particles retained in a shelter's filter restrict airflow sooner than do small ones. If a 5-inch steel pipe cannot be found in a salvage yard or elsewhere, a good welder can use a substitute gooseneck with two 90-degree turns.)

° Do not use air intake hoods on a permanent shelter's pipes, because hoods are not as effective in preventing fallout particles from entering ventilation pipes.

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Also, pumping a given volume of air through a hood is more work than pumping the same volume through a neck with equal cross-sectional areas.

° Never install any screen inside a gooseneck or air intake hood, because spider webs and other debris will greatly reduce airflow. The author saw a gooseneck with a blast valve built inside it. Attached trash on this blast valve had consequentially reduced the volume of air that this shelter could handle.

Of course a screen is much more easily obstructed than a blast valve. Yet FEMA's widely distributed manual for permanent Home Fallout Shelter (H-12-1) and on a Home Blast Shelter (H-12-3) continue to give detailed instructions for making an air intake hood with a screen soldered inside it. Such shelters have been built and have much more serious weaknesses, including the likelihood that the above-ground ventilation pipes of the blast shelter would be bent over or broken off by blast-wind-hurled debris, even in suburbia. (350 mph is the maximum velocity of the blast wind where the blast overpressure is 1-megaton air burst. 6 The ventilation openings of the blast-tested expedient blast shelter are much more likely to remain serviceable after being subjected to severe blast effects, because they are placed around their openings, that are only a few inches above ground level.)

#### PREVENTION AND CONTROL OF CONDENSATION

A shelter can be watertight, yet at certain seasons of the year its walls, ceiling, and floor can have water condensed from entering outdoor air. (This is a serious problem, except in arid parts of the country. In winter months a shelter and its surrounding earth get cold; then especially on some spring days the temperature of the outdoor air is higher than the temperature of the shelter's interior surfaces. As a result, moisture in those surfaces that are cooler than the dew point of the pumped-through air.

The most dramatic example that I have observed of the seriousness of this condensation problem was the ceiling and wet walls of a reinforced-concrete, family-sized shelter that I inspected on a college campus in Yorkshire, England.

Such condensation also can occur in above-ground structures. Before World War II I had a bedroom in Queen's College, Oxford University. My bedroom's outer wall was made of concrete blocks about 15 inches thick, simply plastered and painted on the inside. On some spring days the inside of the cold outer wall, ran down, and collected in little puddles on the floor. The humidity in my adjacent study was barely sufficient to gradually evaporate the water and reduce my bed to prevent mold from forming. This often-repeated occurrence proves the inadequacy of mere ventilation or light turned on for heat to prevent condensation inside an uninsulated shelter built in a cold climate.

Condensation and resultant 100 percent humid air can rust and eat away most steel pipes. Ventilation pipes should be made of galvanized steel or other materials that are undamaged by seasonal condensation and humidity. In Connecticut I saw shelter ventilation pipes made of steel protected with two coats of paint. The pipes badly rusted only two years after installation.

Operating a dehumidifier with automatic controls is the most practical way for most people to prevent mold and other dampness problems in a shelter during peacetime. Almost all of the Chinese cities and six cities are dual-use shelters and typically are equipped with large dehumidifiers. A small family shelter can be bought from a dependable mail order company, such as Sears, for about \$100.

To save floor space and facilitate removal of water, a dehumidifier should be installed near the shelter. Then water from the dehumidifier can be disposed of most easily through a pipe or tube protruding from the shelter to the outdoors, second best to the sewer. (After a nuclear attack the sewer system may back up and flow into a belowground room having pipes that normally discharge into the sewer. If there are no check valves.) If the above mentioned ways of removing water are not possible, a sump pump discharging water to the ground outdoors can solve the peacetime water disposal problem.

After an attack, electric power can be expected to fail and shelter humidity will have to be controlled as much as possible by ventilation with outdoor air. A simple way to learn when to ventilate a shelter is to use a Russian article on shelter management: Keep several small cans of water in the shelter.

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They will be at about the same temperature as the shelter walls when the shelter is closed. If a filled can is exposed to outdoor air for about 10 minutes and moisture condenses on its outside (as a iced drink "sweats" on a humid summer day), do not ventilate the shelter. If no moisture condenses, ventilate.

#### A WALK-IN ENTRANCE

Only a small fraction of permanent family shelters without a walk-in entrance have been kept for many years. Permanent family shelters with vertical or crawl-in entrances are found everywhere. The majority of owners do not use them even for rotated food storage. In normal peacetime, most people concerned with protecting their families conclude that only a shelter skillfully designed can justify the cost of building and maintaining it. Significantly, Chinese civil defense has come to the conclusion that permanent public shelters : those that have been built, and that still are being built, are not suitable for peacetime and in wartime.

If a family can use its shelter without having to go outside and be exposed to rain, snow, or sun, its dual use shelter will be a more valuable peacetime asset than a shelter not directly connected to the outdoors. Furthermore, a directly connected shelter can be entered more quickly in a crisis, and probably will reduce exposures to fallout radiation received by persons carrying things into the shelter, or by persons entering the shelter to attack to protect the home. The main disadvantages of a directly connected shelter are the lack of protection against heat, smoke and carbon monoxide if the house burns and that it is more easily detected than an earth-covered shelter with an outdoor walk-in entrance, such as the one illustrated in Figure 1.

#### AN EMERGENCY EXIT THAT ALSO PROVIDES A SECOND LARGE VENTILATION OPENING

Having an emergency exit in a fallout shelter is not as important as having one in a blast shelter if the shelter is under or connected to a building. (Since buildings are likely to burn, it is important to have an escape.) However, an emergency exit makes any shelter more practical to live in than a shelter with only one large opening - especially in a heavy fallout area where it may be necessary to stay inside most of the time. Occupants of a shelter with only one large opening will not have adequate natural ventilation and will keep laboriously pumping air - at least intermittently - through the air-supply pipes to maintain acceptable temperature and/or humidity conditions.

By opening both a shelter's entrance and its emergency exit, and taking measures to prevent wind-blown snow, and all but the smallest fallout particles (see Appendix F), natural ventilation will be adequate except in hot, calm weather. With two large openings, a homemade KAP (see Appendix B ) can provide enough outdoor air through the shelter, with much less work than is required to pump less



pressure pump through typical ventilation pipes.

#### A DEPENDABLE, HIGH-PROTECTION FACTOR EMERGENCY EXIT

Occupants of a shelter with a dependable emergency exit will have less fear of being trapped and blocked. If they open the exit, they also will be able to ventilate the shelter with natural air through the exit, or with forced ventilation by operating a KAP.

To provide excellent radiation protection, a typical high-PF emergency exit is filled with sand. A steel plate is bolted on its bottom inside the shelter, and an easily cut, waterproof, plastic film is placed which is a few inches below ground level. Obvious disadvantages of this typical sand-filled exit are the difficulty of safely removing the bottom plate while sand is pressing down on it, and the need for a plastic film over the top of the exit without having fallout-contaminated earth fall on the film and into the shelter room. Furthermore, if the earth covering the top of an emergency exit is too hard, it is unable to break through it and get out.

An improved design of sand-filled emergency exit was conceived by Dr. Conrad V. Chester of the Radiation Laboratory and in 1986 was further improved, built, and tested in a full-scale model. In Fig. 17.2, the sand in this 26 x 26-inch square vertical exit was supported by a piece of plywood. The plywood should be of the previously mentioned rot-proof type, pressure impregnated with wood preservative.

In the center of this 25- 1/2 x 25-1/2-inch plywood sand-support was an 8-inch-diameter hole. The hole had thicknesses of strong nylon cloth.

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The double thickness of cloth was firmly attached to the plywood by being folded over four rods, that were stapled to the plywood with galvanized poultry-netting staples. (See Figs 17.2 and 17.3) The cloth covering the hole can be easily and quickly cut with a knife, permitting sand to fall into the shelter room.

Fig. 17.3. Plan of Vertical Emergency Exit, Before Filling with Sand, (nwl43.tif)

As shown in Figs. 17.2 and 17.3, the square 26 x 26-inch plywood sand-support rests on a ledge around the 23 x 23-inch square hole in the shelter's reinforced concrete roof slab. When a man has cut out the nylon cloth covering the hole and all but about 30 lbs of sand has fallen or blown into the shelter room, he can tilt the plywood sand-support up into a vertical position, and the 26-inch width of the sand-support permits it to be easily lifted down through the 32.5-inch square hole in the shelter's roof slab.

Note the North-pointing arrow on the plywood sand-support shown in Fig. 17.3. Because the sand-support was not made perfectly square, the sand-support could be tilted most easily from its horizontal position if it was oriented so that its side marked "N" was to the north. A similar 'N' arrow on the sand-support enabled a man below it, after the exit's lid was tied down and before the exit was opened, to position the sand-support for easy tilting and removal when the last of the sand was being removed. The sand support should be cut to fit the exit after the exit is completed, because it is difficult to cut to specified dimensions.

The top of the emergency exit should be a few inches above the earth around it, to prevent ground from possibly running in. Securely cover the exit's top with a lid of 1/8-inch galv plug-hole (cut from a steel barrel) welded over a hole cut in the lid's center. Such a lid of a concrete exit by oiling it and pressing it against the concrete before the concrete b

Before filling the exit with sand, while inside the exit tie the lid down on each of its f 3/32-inch nylon cord repeatedly tightened between four galvanized steel loops welded to th four loops set in the shelter's walls and roof slab, as indicated in Fig. 17.2. Use pliers nylon cord while making the loops. Then after the plywood sand- support is positioned in t exit can be filled by pouring dry sand through the plug-hold in the steel lid . Finally, t that it cannot be opened, and then made waterproof. To make doubly sure that the lid will ment paint, and then with whatever color outdoor paint you want.

A shelter occupant can cut out the nylon cloth covering the hole in the sand-support, remo wood sand-support from this exit, cut the four tightened nylon-cord multiple loops holding and then climb up and step outside - all in less than 5 minutes, even if damp sand has bee make removal of the sand more difficult, in his tests the author used damp sand that does necessary to loosen it repeatedly, with one's hand and arm thrust up through the 8-inch ho sand-support.)

In a well designed blast shelter this sandfilled emergency exit will provide excellent pro blast. Blast tests have proved that a 1/8-inch steel lid (the equivalent of a blast door) stand blast effects of at least 50 psi. Furthermore, sand arching will transfer blast load the slightly downward deflecting plywood sand-support to its edges, and thence to the supp rim or ledge around the 23 x 23-inch hole in the shelter's roof slab. (See Fig. 17.2.)

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A small shelter with an emergency exit near its far end has additional advantages: It can quate natural ventilation most of the time, with easy forced ventilation by a KAP when for and with daylight illumination. Means for attaining these advantages are described in Appe

#### ADEQUATE STORAGE SPACE FOR ESSENTIALS

As will be shown in the following sections of this chapter, about 20 square feet of shelte member is needed for shelter furnishings and to store adequate water for a month, a year's foods, cooking and sanitary equipment, blankets, tools, and other post-attack essentials. member also provides enough space per person to store winter clothing and footwear, campin normally kept on hand and rotated as consumed in the course of ordinary family living. Add you plan to use your shelter as a workshop, or as a fallout shelter to save a few of your endangering your own family.

#### SEATS/BUNKS/SHELVES

Seats and overhead bunks built like those specified in Fig. 1 7.4 and pictured in Fig. 17. pants to sit and sleep more comfortably in less space than when using any other shelter fu thor. Note that the seat has an adjustable backrest. This backrest is similar to the finem tested in a prototype of an expensive blast shelter designed and furnished at Oak Ridge Na

Shelter habitability tests have proved the need for backrests. Even some young German sold

backs after sitting on ordinary benches during a 2-week shelter occupancy test.

Backrests should be installed in peacetime and kept ready for crisis use, rolled up against and covered. Then easily removable storage shelves, if needed, can be installed between the bunks.

To sit or sleep comfortably, pull the bedsheet forward and then sit on about the outer foot.

Although this 5 -occupant shelter is illustrated with a bunk for each family member, it is

Fig. 17.4. Seat with Overhead Bunk Proportioned for a Shelter Room with a 7 -Ft. or Higher

with a backrest should be 24 inches wide, (nwl44a.tif)

Fig. 17.5. Backrests of Bedsheet Cloth and 2-Inch-Thick Pads Make Sitting and Sleeping Comfortable

(nwl44b.jpg)

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prudent to count on stored supplies making some bunks unusable, and to realize that in a crisis it is hard to refuse shelter to an unprepared good neighbor. Note that with four persons seated on the floor and one person on the overhead bunk, five persons can be quite comfortably accommodated on 25 square feet of space, including plenty of room for the sitters' legs.

To store the most supplies in a shelter, you should install shelves after you know the height of the space to be stored. Often the space between an overhead bunk and the ceiling can best be used by making additional shelves.

## WATER

° Water needs: Even most well-maintained shelters do not have enough water for prolonged occupancy. A permanent shelter should provide each occupant with at least 30 gallons of safe water - enough for drinking and bathing except in very hot weather.

° Containers: The most practical water container for shelter storage is a 5 -gallon rigid plastic container with a handle, a large diameter opening for quick filling and emptying, and a small spout for pouring. A 5-gallon water can of this type sells for about \$5 in discount stores.

The plastic bottles that household chlorine bleach is sold in also are good for multi-year storage, as are glass jugs. Plastic milk jugs are not satisfactory, because after a few years they become brittle and crack.

Some shelter owners do not realize that, although a shelter can be kept dry in peacetime, its air is likely to become extremely humid after a few days of crowded occupancy. Very high humidity weakens cardboard containers of food and flexible water bags.

° Disinfecting for multi-year storage: To store safe water and keep it safe for years, first disinfect the container by rinsing it with a strong solution of chlorine bleach. Then rinse it with safe water before filling with water to be stored. Next, disinfect by adding household bleach that contains 5.25% sodium hypochlorite to the water.

tive ingredient, in the quantities specified in this book's Water chapter for disinfecting gallons, add 1 scant teaspoon of bleach. Finally, to prevent possible entry of air contain faulty closures, seal the container's closures with duct tape.

° Making efficient use of storage space: A 5- gallon rigid plastic water container typical inches, including the height of its spout. Nine such 5-gallon containers can be placed on Twenty- seven 5-gallon containers, holding 135 gallons of water, can be stored on and over you make a water storage stand 24 x 42 x 48 inches high, built quite like the seat with ov Seats/Bunks/ Shelves section of this chapter. This easily moveable storage stand should ha one 24 inches and the other 48 inches from the floor.

° Using filled containers for shielding Filled 5-gallon water containers can be moved quic additional shielding where needed to increase the protection factor of all or part of a sh inner door of the shelter illustrated by Figs. 17.1 and 17.2 the protection factor is less water containers are placed so as to cover the door with almost the equivalent of an 18- i water, the PF of the part of the shelter room near the door can be raised to about PF 1000

If a shelter has twenty-seven 5-gallon cans of water stored on and under the above-describ age stand, then in less than 3 minutes 2 men can shield the shelter door quite adequately. the 18 cans off the shelves, put 9 of them on the floor against the door and doorway, move 9 door- shielding cans, and place the remaining 18 cans on the stand's 2 shelves.

Equally good doorway shielding can be provided by placing at least a 24-inch thickness of food, such as whole- grain wheat or sugar, against the doorway. Two 55-gallon drums of whe 400 pounds, can be quickly "walked" on a concrete floor and positioned so as to shield the Heavy containers on the floor can provide a stable base on which to stack other shielding

° A water well: The best solution to the water problem quite often is a well inside the sh water table is less than 50 feet below the surface, and a 50-foot well, cased with 6-inch completed for about \$2000. Well drilling should be done after the shelter excavation has b crete shelter floor has been poured. An in-shelter well would be of vital importance not c family shelter, but later on probably to nearby survivors.

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Even if only a gallon or so an hour could be bailed from a well too weak to be useful in p mendous family asset post-attack. If infective organisms are found in water from a well dr ing and after an attack, safe water for months can be assured by merely storing a few gall bleach.

If enough water for worthwhile peacetime use can be pumped from a well, install a submersi with plastic pipe in the well. Then in an emergency the pipe and the attached pump can be well, with only a saw being needed to cut off lengths of the plastic pipe as it is pulled. cleared of pump, pipe and wires, a homemade 2-foot-long bail-bucket on a nylon rope can be water. (See Chapter 8, Water, for instructions.)

Local water sources: Most Americans' normal piped water would not run for months after a l tack. A month's supply of water stored in your shelter should be adequate, because, even i lout, in less than a month radioactive decay will make it safe to haul water from nearby s

An important part of your shelter preparations is to locate nearby wells, ponds, streams, when dry frequently have water a foot or two below the surface. The author has found that apparently dry streambeds often supplies enough filtered water to satisfy several families' sides of a water pit dug in unstable ground from caving in, you should drive a circle of s outside of your planned pit before starting to dig. With more work and materials, you and tered water could dig a well like the Chinese water source shown on page 7 1 . If you are drinking water from a water pit you should filter it through clayey soil to remove fallout active isotopes, as described in the Water chapter. Of course it is prudent to chlorinate surface water after it has been filtered.

## FOOD

° Advantages of a one-year supply: A family that expends the money and work to build and p shelter should store a year's supply of long-lasting foods. If post-attack conditions enab making a living near your home, having a year's food supply will be a tremendous advantage flicted with such dangerous, continuing fallout radiation and/or other post-attack conditi residents are soon forced to evacuate to better areas, then your and your family's chances does not force you to move during the chaotic first few months after a nuclear attack.

° Costs of a one year food supply for a family shelter: Table 17. 1 shows the wide range c basic survival foods for multi-year storage that are listed and explained on page 88.

The delivered costs listed in the right hand column include UPS shipping charges in the ne sive of UPS's 8 zones. For UPS shipping costs to the most distant points in the 48 states, delivered.

All the foods in this survival ration, if stored in moisture-proof and insect-proof contain der should be in nitrogen-packed cans), will provide healthful nutrition for at least 10 y vitamin tablets, which should be replaced every 2 or 3 years, depending on storage tempera spends about \$300 per member on such a one year survival ration can consider that each of covered by famine insurance for \$30 a year.

Scurvy will be the first incapacitating, then lethal vitamin deficiency to afflict unprepa A multi-vitamin tablet contains enough vitamin C to fully satisfy the daily requirement. F owner also should store vitamin C tablets, that keep for years. One hundred 500-mg generic mg of vitamin C - in 1987 typically cost about \$1 .20 in a discount store; a 10 mg daily c nuclear war in some areas vitamin C will be worth many times its weight in gold.

° Sources of the basic foods listed in Table 17.1:

\* Wheat: If you live in a wheat producing area, the least expensive sources of ready-to- s are local seed-cleaning firms. A hundred pounds or more of hard wheat, dried, bagged, and proof containers, costs about 10 cents a pound. (Today the wheat farmer receives about 4.5 loads, usually straight out of the field, not dry enough to store except in well ventilate

that makes weevil infestations more likely.) In some communities a few stores sell big bag wheat for 20 to 35 cents a pound.

Table 17. 1 . A basic one-year survival ration for one adult male.

Component

Ounces  
per day

Pounds  
per day

Range of 1987 costs  
prices in dollars per  
pound. FOB or picked  
up

Range of 1987 costs  
prices in dollars for a  
one year supply FOB  
or picked up

Range of 1987 costs prices in  
dollars for a one year supply  
delivered and in containers for  
multi-year storage

Whole -kernel hard  
wheat

16

365

0.10 to 0.40

36.50 to 135.00

56 to 175

Beans, pinto

5

114

0.25 to 0.79

28.50 to 90. 11

38 to 110

milk powder

2

46

1.68 to 3.86

77.28 to 177.56

83 to 184

Vegetable oil

1

23

0.36 to 0.57

8.28 to 13.11

9 to 14

Sugar

2

46

0.25 to 0.30

11.50 to 13.80

16 to 20

Salt, iodized

1/3

8

0.17 to 0.20

1.36 to 1.60

2 to 3

Multi-vitamin tablets  
(low generic, and an  
expensive brand)

1 per day

365 per  
year

1.3 to 11.8 cents per  
tablet

4.75 to 43.07

6 to 44

Total \$ costs

\$210 to \$550

Shelter owners who are unable or unwilling to obtain wheat from such sources can buy high prices from health food stores and a few mail order companies. The lowest mail order author for hard western wheat in 1990 is \$3 . 17 for 10 pounds, in a vacuum-packed metal containers used for some U.S. Army combat rations. This long-lasting wheat, as well as other sold by Preparedness Products, 3855 South 500 West, Salt Lake City, Utah 84115. Another reseller of wheat and other dry foods packaged for multi-year storage is The Survival Center, 5555 na, Ohio 44266.

\* Beans, like wheat, in many communities can be purchased from local farmers' co-ops or local lower delivered cost than from mail order firms. In one small Colorado town the co-op sold in a polyethylene bag for \$ 11 .25-45 cents a pound. Local supermarkets sell bulk pinto beans a pound.

\* Non-fat milk powder in 1990 is sold nationwide in the larger cardboard packages for around a pound. A better buy for multi-year storage is the instant non-fat milk powder sold by Preparedness Products. A Mormon-owned firm's 1990 FOB price is \$57.95 for a case of 6 nitrogen-packed cans, containing 60 pounds. The author bought a case three years ago and found this non-fat instant milk powder a pound, plus UPS shipping charges, the cost is considerably less than for comparable milk powder for multi-year storage and sold by other companies.

\* Vegetable oil, sugar, salt, and multi-vitamin tablets are best bought at discount supermarkets. Such stores often sell large plastic containers of vegetable oil as "loss leaders" to attract customers at wholesale. Vegetable oil prices in small communities typically are much higher. For an emergency, buy the lowest priced vegetable oil. Remember that one of the worst post-attack nutritional deficiencies is chronic shortages of fats (including oils), and that babies and little children cannot survive on grains and beans, with no oil or fat. See the Food chapter.

° CAUTIONS: Typical health food stores and most firms that specialize in survival foods sell high prices, especially grains, beans, and milk powder. Investigate several other sources before buying.

To make sure that an advertised "one year supply" of survival foods actually will keep an



a whole year, require the seller to inform you by mail what his "one year supply" provides

calories, k cal; (2) protein, g; and (3) fat, g. Then you can use the values in the "Emergency" column in Table 9.1 on page 84 to determine whether the advertised "one year supply" is adequate.

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**Transitional foods:** The emotional shock of suddenly being forced by war to occupy your shelter is worse if you have to adapt suddenly to an unaccustomed diet. It would be a good idea to occupy only your survival rations for a day or two, and to store in your shelter a two week supply of foods similar to those your family normally eats. Then it will be easier if war forces you to make the transitional foods that you store should be rotated and replaced as needed, depending on the situation.

° **A hand-cranked grain mill:** Whole kernel grains and soybeans must be processed into meal for factory use as principal components of a diet. If unprepared America suffers a nuclear attack, food reserves and/or famine relief shipments will consist mostly of unprocessed wheat, corn, and soybeans. A manual grain mill will have a survival advantage and will be a neighborhood asset.

Many health food stores at least have sources of hand-cranked grain mills. Mills with steel gears are more efficient and less expensive than "stone" mills. A mail order firm that still sells hand-cranked mills makes that the author has bought and found to be well made and efficient is Moses Country Hardware.

4th N.W., Albuquerque, New Mexico 87107. It sells a serviceable Polish mill (Model "OB" H-100) for \$35.73 FOB, plus UPS shipping charges. This mill cranks easily and grinds wheat into coarse meal more efficiently than any of the American manual grain mills that the author has bought and used. Unfortunately, manual grain mills no longer are manufactured in the U.S. Before buying a grain mill that grinds corn, our largest food reserve. In 1987 the author bought an advertised West German mill. When it by mail order did he learn by reading the accompanying directions that it is not made for corn.

## COOKING AND HEATING

**Safety precautions:** The first rule for safe cooking and/or heating in a shelter is to do it safely. The exhaust opening. If the fire is under an exhaust pipe, install a hood over the fire. Cover the pump when cooking, unless a natural airflow out through the exhaust opening can be observed. Keep flammable materials, especially clothing, well away from any open fire.

**Hazardous fuels:** Charcoal is the most hazardous fuel to burn in confined spaces, because it produces carbon monoxide. In a crowded shelter there are obvious dangers in using the efficient little charcoal stoves and in storing their easily vaporized and ignited fuels.

"Canned heat" : These convenient fuels are expensive. Sterno, widely used to heat small quantities of food or drink, typically retails in 7-ounce cans for what amounts to about \$9.00 per pound.

° **Wood:** The safest fuel to burn in a shelter is wood, the most widely available and cheapest. Wood smoke is irritating enough to usually alert shelter occupants to sometimes accompanying dangers. Scrap lumber cut into short lengths, made into bundles and stored in plastic bags, can stay dry. Keep a saw and a hatchet in your shelter.

**Bucket Stove:** The most efficient, practical and safe stove with which to cook or heat for a family shelter is a Bucket Stove, that burns either small pieces of wood or small "sticks" of coal.

the Food chapter.) Especially if you believe that you may have to live in your shelter for fuel will not be available after a nuclear attack, you should make and store at least two

An improved Fireless Cooker: To save a great deal of fuel and time, particularly with slow beans, make a very well insulated Fireless Cooker similar to the expedient one described in chapter. Make a plywood box, first measuring carefully to insure that, when completely lined with foam, the styrofoam will fit closely around a large, lidded pot wrapped in a bath towel. A war survival asset that also is useful for peacetime cooking.

(To boil about twice as much wheat flour-meal in a given pot as can be boiled when making use the minimum amount of water and fuel, salt a batch of the flour-meal, add enough water to make a stiff dough, then make dough balls about 1-1/2 inches in diameter, and roll them in flour

into enough boiling water to cover them, and boil at a rolling boil for 10 minutes. Then place in the insulated Fireless Cooker for several hours. Corn balls can be made and boiled in this manner without the constant stirring required when boiling a mush made of home-ground flour-meal.)

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° A sturdy work bench: In the corner under the emergency exit build a work bench, secured to the wall to cook and to which you can attach your grain mill. A bench 36 inches high, 42 inches wide will serve. (The other corner at the air-exhaust end of the shelter should be the curtained-off

° Very warm clothing, footwear, and bedding: Heating a well ventilated shelter usually is not a problem in freezing weather if the occupants have these essentials for living in the cold, or if they make at least as good expedient means for retaining body heat as are described in Chapter 10. The warm hands of little Chinese children wearing padded clothing while living in their below ground shelters was scarcely enough straw, grass, and twigs to cook with. If you store plenty of strong thermal blankets in your shelter, you can make warm clothing out of blankets - as some frontier settlers did in the mountains of Montana.

## LIGHT

° White paint: To make a little light go a long way, paint the walls, ceiling, floor, and

Candles: The most dependable and economical lights for a family shelter are long-burning candles. The candle tested by the author is the 15-hour candle manufactured by the Reed Candle Company of Salt Lake City, sold by the millions in New Mexico each Christmas season for use in outdoor decorative "lantern" type, short candle is not perfumed and has a wick supported by a small piece of metal attached to the base so the wick remains upright and continues to burn if the wax melts and no longer supports it. If used in the lamps described below, this candle gives enough light to read by for 15 hours.

The author has been able to find only one mail order source of 15-hour candles like those described above. New Mexico: Preparedness Products, 3855 South 500 West, Salt Lake City, Utah 84115. A case of 15 candles for \$49.00, FOB in 1990. When UPS shipping charges are added, the delivered cost is between \$50 and \$60 each 15-hour candle, depending on the distance shipped.

Persons who stock a permanent shelter should realize that after a nuclear war fats, oils, and candles, sources of light will be in extremely short supply for at least many months. Light at a few cents per hour will be a bargain blessing.

\* The second-best shelter candle tested by the author is a standard Plumber's Candle, that cents in many stores nationwide, and that burns for about 10 hours with a brighter flame t

Two types of expedient candle-lamps were proved by multi-day tests to be the most practica

\* The best expedient candle-lamp is made of a 1 -pint Mason j ar, identical to the cooking page 102 of the Light chapter except for the substitution of a short candle for the oil ar dle base on a glass jar's rounded bottom, cover it with hot candle wax. To be able to ligh inside the glass jar, make tongs out of an 18-inch length of coathanger wire bent in the m ends bent inward 90 degrees and cut off to make 1/4-inch- long candle grippers.

\* The second-best candlelamp is made by cutting a standard aluminum pop or beer can, as il sharp small knife. To make a stable base for a short candle, put enough sand across the ca barely cover its center. The first candle burned will saturate and then harden the sand, m base or holder.

(Illustration of a candle holder made of a aluminum pop can) (nw 149.tif)

Caution: Although candles are the safest non-electric lights for shelter use, they produce ide to cause headaches in a poorly ventilated, long-occupied shelter. In a 2-week habitabi Princeton, the father and mother did not smoke, yet they had persistent headaches. Special

their headaches were caused by the small amounts of carbon monoxide produced by the candle and to heat food and drink.

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Expedient lamps: The very economical lamp that burns cooking oil in a 1 -pint glass j ar i better of the two expedient lamps pictured on page 102. For use in a shelter, however, lon serviceable after decades of storage, are more practical. Among the disadvantages of exped untreated, soft cotton string needed to make excellent wicks is no longer available in som wire-stiffened wicks is a time consuming chore.

° Other light sources: See the Light chapter.

#### NYLON HAMMOCK-CHAIRS

To enable the maximum number of additional people to occupy a shelter for days to months, Chairs should be made before a war crisis arises, and should be kept stored for emergency model for use in shelters that are at least 7 feet wide is similar to the expedient Flammc 120 through 124, is made of strong nylon cloth, and is 40 inches wide by 9 feet long along its long sides being 8 feet-4 inches long. At each end is a curved hem 3-1/2 inches wide, nylon hammock rope is run, to draw the slung hammock into a boat-like, secure shape with a der room. The breaking strength of a hammock rope should be at least 600 pounds.

To attach the nylon ropes that support the suspended chair's two "arms," a loop of nylon w stitched nylon cloth, should be sewn to each side edge of the hammock 52 inches from the e will serve as the top of the back of the chair. See the illustrations on page 1 24. Especi

ters, support points both for the hammocks, and for the hammocks when they are converted it should be made when the shelter is being built.

#### OFTEN OVERLOOKED SANITATION AND OTHER SHELTER NEEDS

Store enough soap to last your family for at least a year. After a major nuclear attack the in past generations to make soap, will almost all be eaten. Production of detergents is by venerable chemical industries not likely to be restored for years.

A chemical toilet would help bridge the gap between modern living and surviving in a crowded months-long occupancy, however, a more practical toilet is likely to be a 5-gallon can with for its removable liner, a piece of plastic film for its tie-on cover, and a hose to vent 104. Store at least 200 large plastic trash bags.

The author knows from his experiences in primitive regions that using anything other than paper or cloth is hard to get accustomed to. A hundred pounds of newspaper, stored in plastic from getting damp, takes up only about 3 cubic feet of storage space and would be useful for

Keep several thousand matches, in Mason jars, so that they will be sure to stay dry even in very humid post-attack.

Store most of your radiation monitoring instruments in your shelter, along with paper and keep records of radiation exposures, etc. A steel or reinforced concrete shelter should have extra batteries, and a vertical pipe through which an antenna can be run up to improve reception

No one can remember all the needed survival facts and instructions given even in this one War Survival Skills and other survival guidance in your shelter, and also other books that your family's morale and survival chances.

#### PRACTICE SHELTER LIVING

A family that spends a weekend living in its completed small shelter will learn more about needs - and more about each other - than members are likely to learn if they all read civil whole days. Furthermore, after this educational experience family members old enough to help

know for sure that anti-defense activists are talking nonsense when they maintain that if they would become less heedful of nuclear war dangers and more likely to support aggressive

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#### Chapter 18

##### Trans-Pacific Fallout

#### POTENTIAL DANGERS TO AMERICANS

Many strategists believe that if a nuclear war is fought in the next few decades it probably explosions on any of our 50 states. Perhaps the first nuclear war casualties in the United lout from an overseas nuclear war in which our country is not a belligerent. As the number

weapons increases - especially in the Middle East - this generally unrecognized danger to Trans-Pacific war fallout, carried to an America at peace by the prevailing west-to-east wind world, could be several hundred times more dangerous to Americans than fallout from the worst nuclear power reactor accident, and many times more dangerous than fallout from a very important power reactor accident as lethal as the disastrous Chernobyl accident was to Russians.

Fig. 1 is a map showing fallout from a single above ground Chinese nuclear test explosion (300 kiloton") on December 28, 1966.

Fig. 1. The Fifth Chinese Nuclear Test was Detonated on Dec. 28, 1966. It "involved thermonuclear material," and, according to the AEC press release, was a nuclear test in the atmosphere at their test site near Lop Nor." As indicated above, by the end of Dec. 31, 1966, the leading edge of its fallout cloud extended as far east as the dotted line shown running from the Great Lakes. ORNL DWG. 73-4611 (nwl51.tif)

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It produced fallout that by January 1, 1967 resulted in the fallout cloud covering most of China. The Chinese explosion produced about 1.5 million curies of iodine-131 - roughly the same amount of iodine-131 into the atmosphere from the Chernobyl nuclear power plant disaster. (The Lawrence Livermore Laboratory's preliminary estimate is that 10-50 million curies of iodine-131 were released in the Chernobyl disaster; in contrast, its estimate of the iodine-131 released during the Three Mile Island plant accident, the worst commercial nuclear power plant accident in American history, is

Fig. 1 is from an Oak Ridge National Laboratory report, Trans-Pacific Fallout and Protective Measures (ORNL-4900), written by the author of this book in 1970, but not published until 1973. It was used in any version of this report, that summarized findings of the unclassified Trans-Pacific Seminar funded by the U.S. Atomic Energy Commission. This seminar was attended by experts who came from various search organizations and deliberated at Oak Ridge National Laboratory for two days in March 1970.

Later in 1970 a final draft of this report was submitted to Washington for approval before it was promptly classified. Publication without censorship was not permitted until after it was declassified in 1973. None of the recommendations in this pioneering report were acted upon, but many of them are in this chapter.

The findings and conclusions of the above mentioned 1970 Oak Ridge National Laboratory Trans-Pacific Seminar, summarized in the 1973 report, were confirmed by a later, more comprehensive study, Control of the Transoceanic Fallout Threat, by H. Lee and W. E. Strobe (1974; 117 pages), Stanford Research Institute.

Fallout from the approximately 300 kiloton Chinese test explosion shown in Fig. 1 caused milk on pastures near Oak Ridge, Tennessee and elsewhere to be contaminated with radioiodine, and was found to be hazardous to health. However, this milk contamination (up to 900 picocuries of radioiodine per liter) did not cause the measured dose rates from the gamma rays emitted from fallout particles deposited in the United States indicate that trans-Pacific fallout from even an overseas nuclear war in which "only a few megatons would be exploded could result in tens of thousands of unprepared Americans suffering from thyroid injury from radioiodine, and few could get it during the several days that it would take to

reach the United States. Fortunately, removal of even a cancerous thyroid rarely is fatal

em medical facilities.

Only about 7,500 Americans (people living within a few miles of a nuclear power plant in T given prophylactic potassium iodide to keep in their homes. No government organization has living near other nuclear facilities to buy and keep any kind of prophylactic medicine to a peacetime nuclear accident. As expected, official warnings and advice to the public cont preparations that individual Americans could make to protect themselves and their families from trans-Pacific war fallout deposited on an America at peace, or as a result of war fal jected to a nuclear attack.

The worst danger to Americans from trans- Pacific fallout from a large nuclear war would b gamma radiation doses that millions would receive from fallout particles deposited on the buildings. Protective countermeasures would include both sheltering some pregnant women an "hot spot" areas of abnormally high rain-out of fallout, and evacuating others. Unless suc ing and expensive countermeasures were taken, several thousand additional Americans might following 20 or 30 years. The largest total doses would be received by people who would li for the first month or so after fallout deposition, while the dose rates would be highest.

Several thousand additional cancer deaths would be extremely difficult to detect, if cause radiation from fallout deposited nationwide, with these scattered deaths occurring over th Trans-Pacific war fallout disaster. For during these same decades about 1 5 million Americ from cancers indistinguishable from those caused by wholebody radiation from war fallout c peace.

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An authoritative risk estimate of getting cancer from low doses of radiation is given in F 1984) of the National Council on Radiation Protection and Measurement, "Exposures from the Emphasis on Radon and Its Daughters": "The low dose model for total excess cancer mortalit per million people exposed to one rem uniform whole body radiation. This would make the ov average individual in the population about one in ten thousand per rem, i.e., if ten thous dose of one rem (one thousand mrem), one excess fatal cancer would be expected within the

Many radiation specialists have concluded from studies of the effects of extremely low dos similar conservative estimates of excess cancer deaths overestimate the number of fataliti radiation doses, such as would be received by millions of Americans from trans-Pacific war

TO PROTECT YOURSELF AGAINST TRANS PACIFIC FALLOUT, START BY REALIZING  
THAT:

° The dangers from trans-Pacific war fallout have been increased by the continuing trend t of more accurate, smaller, more numerous nuclear weapons, because:

\* A large nuclear explosion (half a megaton, or more) injects most of its fallout particle the stratosphere, above the tops of clouds and above the altitudes at which quite prompt r contaminants from the atmosphere by scavenging takes place. Very small particles in the st not reach the ground before they are blown at least several thousand miles. Most of these remain airborne for weeks to years, are very widely dispersed, and are blown around the wc many times before being deposited. By then the radioactivity of iodine- 131 (that has a h little more than 8 days) is so greatly reduced that it is not nearly as dangerous as is re deposited much sooner with the fallout from smaller weapons of several hundred kilotons, c explosive power.

\* Nuclear explosions smaller than about half a megaton (500 kilotons) inject all or most of their fallout at lower altitudes - within the troposphere, below the stratosphere. Most of such fallout is in the radioactive cloud's first world-circling trip, when even quite rapidly decaying radioisotopes are dangerously radioactive. This greater danger from smaller nuclear weapons has been proved by numerous measurements of fallout from many nuclear test explosions, both foreign and American.

° The dangers from trans-Pacific fallout produced by peacetime nuclear accidents are not nearly what many Americans have been led to believe. For example, the Chernobyl nuclear power reactor injected as much radioactive iodine into the atmosphere as would the explosions of several nuclear weapons totaling perhaps as much as half a megaton in explosive power. But not nearly as much of the radioactivity caused by this reactor accident reached the United States as would reach fallout from nuclear explosions in the same area, capable of injecting an equal amount of radioactivity, because:

\* The cloud from the steam explosion that blew off the roof and otherwise damaged the Chernobyl reactor building, may have risen quite soon to 20,000 feet or more and was partially blown eastward toward Asia and the Pacific Ocean. However, the top of the radioactive smoke cloud over the Chernobyl plant, that burned for days, rose only about 3,000 feet above the ground. As a result, much of the Chernobyl radiation stayed at relatively low altitudes where scavenging (removal) of smoke particles and gasses is most effective and rapid, due to aggregation on cloud droplets, rain, and dry deposition. In contrast, almost all of the fallout particles and radioactive gasses from nuclear weapons are injected much higher, to altitudes where scavenging is less effective; there, the general west-to-east winds promptly start transporting very small particles and radioactive gasses around the world in the mid-latitudes of the northern hemisphere) around the world.

\* Variable winds for days carried much of the Chernobyl radioactive material northward to European countries, then westward and southward to other European countries. The resultant wide dispersal of fallout allowed time for both scavenging and radioactive decay before a small fraction of the radioactive clouds rose and also were blown eastward by the prevailing high-altitude winds. Only a small fraction of the winds carried an extremely small fraction of the radioactive emissions from the burning Chernobyl reactor clear across Asia and the Pacific to America.

° The media habitually exaggerate dangers from nuclear accidents, and exploited the Chernobyl example, when Dr. Robert Gale, the leading bone marrow transplant specialist who helped save Chernobyl victims, first returned from Russia, an Associated Press article quoted him as saying that he can say there are at least 50,000 to 100,000 people who have had some dose of radiation which is of long-term concern.

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There will, unfortunately, be additional casualties. We hope the number will be small. "The New York Times headlined" 100,000 SOVIETS TO SUFFER FROM RADIATION, DOCTOR SAYS". Mary McGrory, the syndicated liberal columnist, also misinterpreted Dr. Gale's risk estimate and misled readers by writing: "He [Dr. Gale] estimated that there could be 100,000 cases of radiation sickness. Such dramatic news items give the impression that 100,000 Russians - not just a small fraction - are likely to suffer sickness or death from the Chernobyl radiation. So additional typical American misinformation of this type and knowing very little about statistical evaluations of risks have had their worst nuclear fears strengthened.

The public's exaggerated fears of extremely small amounts of radiation also are worsened by without explanations of very small units of radiation measurement, including the picocurie (express the radioactive contamination of milk, water, etc., and is only one millionth of a curie [ 1/1,000,000,000,000] of a curie.) One episode in which fears of radiation were thus worsened was the invisible fallout cloud from the Chernobyl disaster first reached the United States. Shortly after when a radio announcer merely stated that milk samples in northwest Oregon showed 118 picocuries

of radioactive iodine. Few Americans know that they will not be advised to stop using fresh milk until the level is 15,000 picocuries or more per liter - as specified in the Food and Drug Administration's "Protective Action Guidance", published in the Federal Register of October 22, 1982.

The maximum measured radioactive contamination of milk in the United States by iodine-131 after the disaster was in milk produced by cows grazing on pasture in Washington: 560 picocuries per liter. The potential danger from trans-Pacific war fallout is brought out by the fact that the approximate test explosion of December 28, 1966 resulted in worse iodine-131 contamination of milk produced on pasture near Oak Ridge, Tennessee: 900 picocuries per liter. Even a small overseas nuclear war of kiloton-range nuclear explosions could cause high enough contamination of milk to result in advising Americans to refrain from using fresh milk. Most Americans would heed this warning and would not otherwise use fresh milk for weeks. In addition, a small overseas nuclear war possibly would cause casualties years to decades later.

## TWO SUMMARY CONCLUSIONS

1. Trans-Pacific war fallout deposited on an America at peace surely would be a disaster, an overwhelming one. The economic and psychological impact probably would be more damaging than the impact of health and life.

2. Prudent individuals should make preparations to enable them to use the low cost protection described in this book, especially those in Chapter 13. Some of the most effective countermeasures are getting enough prophylactic potassium iodide to prevent thyroid damage even if war fallout from nuclear explosions in the United States or overseas were to continue for a couple of months, cannot be avoided after even an overseas nuclear war begins.

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## Appendix A

### Instructions for Six Expedient Fallout Shelters

#### SHELTER-BUILDING INSTRUCTIONS

The following step-by-step instructions for building 6 types of earthcovered expedient shelters were trained families to build even the most difficult of these shelters in less than 2 days. The few who were delayed by very heavy rains. Each of these shelters had been built by different families or groups of families. Only widely available materials and hand tools had been built under simulated crisis conditions in environments typical of large regions of the country. Trench types have been built in forested clay hills of Tennessee, in a bare Colorado valley



irrigated Utah valley in hot August. Most of the aboveground shelters were built by families. The water table is within 18 inches of the surface.

All of these test families used instructions that contained general guidance to help inexperienced families build any type of earth-covered shelter. In this appendix, general instructions which apply to all types of shelters will be given first, to avoid repetition. (However, if the instructions for building one type of shelter are given separately, the pertinent parts of these general instructions should be given before the step-by-step instructions for that shelter.)

#### WARNING

Earth-covered shelters built of green poles can become unsafe within a few months, because of rotting and breeding insects. In the damper parts of the U.S., earth-covered shelters built of dry poles or untreated logs are safe after several months. An exception is very dry areas of the West, where some pioneers have built earth-covered dugouts with pole roofs.

#### GENERAL INSTRUCTIONS FOR BUILDING AN EXPEDIENT SHELTER

1. Read all the instructions and study the drawings before beginning work. (Most families will prefer first to read the instructions aloud and then to discuss problems and work assignments.)

2. Sharpen all tools, including picks and shovels. Dull tools waste time and energy, if not sharpened by rubbing them hard on concrete or a rough stone.

3. Wear gloves from the start. Blisters can lead to serious infections, especially if antiseptics are not used.

4. Whenever practical select a building site that:

- ° Will not be flooded if heavy rains occur or if a large dam farther up a major valley is destroyed by a nuclear explosion.

- ° Is in the open and at least 50 ft away from a building or woods that might be set afire by a nuclear pulse from an explosion tens of miles away. (Keep well away from even a lone tree; it is hit by the pulse through roots.)

- ° Has earth that is firm and stable, if the planned shelter is to be a trench type with unsupported

earth walls. To make sure that the earth is firm and stable enough so that the walls will not cave in, make a "thumb test" by digging an 18-in.-deep hole and trying to push your thumb into the undisturbed earth at the bottom. If you can push your thumb no farther than 1 in., the earth should be safe enough. If the earth does not pass the thumb-test, move to another location and repeat the test. Or build a below ground shelter with shored walls, or an aboveground shelter.

- ° Has a sufficient depth of earth above rock or the water table for a trench to be dug to the required depth.

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(To find out, try to dig a pit to the required depth before excavating the whole trench. If there is no water-table problem, try driving down a sharpened rod or pipe to the required depth.)

places along the planned length of the trench.)

5. If the shelter must be built on sloping ground, locate it with its length crosswise to

6. Before staking out the shelter, clear the ground of brush, weeds and tall grass over an 10 ft beyond the planned edges of the excavation. (If loose earth is shoveled onto tall pl difficult to shovel the second time when covering the completed shelter roof.)

7. Stake out the complete shelter, and then dig by removing layers of earth.

8. When digging earth that is too firm to shovel without first breaking it up, start picki shovel) in a line across the center of the trench. Next, shovel out a narrow trench 6 or 8 across the width of the trench. Then with pick or shovel break off row after row of earth the width of the trench, as illustrated.

(Illustration of pick axe chipping away a layer of dirt) ORNL-DW G78-16212 (nw 1 5 6a.tif)

9. When digging a trench, to avoid having to move the excavated earth twice more to get it pile all earth about 8 ft away from the trench. Later, pile additional earth you are excav away.

10. Never risk a cave-in by digging into lower parts of an earth wall. It is dangerous to overhanging section of wall or to dig a small, cave-like enlargement of a shelter.

1 1 . When making a "sandbag" of a pillowcase or sack to hold earth shielding in place arc openings or along the edges of a shelter roof, fill it so that it will be only about two-t is tied shut securely. Avoid dropping the sandbag.

12. If sufficient sandbags are not available, make earth-filled "rolls." Bed sheets or any or plastic film can be used to make these rolls as described below. (To make a longer roll illustrated below, several persons should make one together, standing side- by-side.)

To make an 8-in.-high earth-filled roll:

(1) Select a piece of cloth at least as strong as a new bed sheet, 2 ft longer than the si to be protected, and 5 ft wide.

(2) Place 2 ft of the width of the cloth on the ground, as illustrated.

(Illustration of man placing earthfilledroll.) ORNL-DWG 78-16213 (nwl56b.tif)

(3) While using both hands to hold up and pull on 3 ft of the width of the cloth and press cloth with your body, have another person shovel earth onto and against the cloth.

(4) While still pulling on the cloth, pull the upper part down over the earth that covers the cloth.

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(5) Cover the upper part of the cloth with earth so as to form an earth-filled 'hook" near as illustrated.

(Illustration of finished earth-filled roll.) ORNL-DWG 78-16214 (nwl57a.tif)

(6) If a greater thickness of rolls is needed, level the earth on top of a roll; then make roll on this level surface.

13. Cut and haul poles and logs more easily by doing the following:

(1) Take time to sharpen your tools before starting to work no matter how rushed you feel.

(2) When sawing green trees that have gummy resin or sap, oil your saw with kerosene or di you don't have these, use motor oil, grease, or even soap.

(3) When felling a small tree, the following method will help make a square cut, keep your being pinched, and help make the tree fall in the desired direction: (a) Saw the tree about through on the side toward which you want it to fall, (b) Then start sawing the opposite s another person pushes on the tree with a 10-ft push- pole, pressing the end of the push-po the tree about 10 ft above the ground. A push-pole with a forked end or with a big nail on best when you get ready to move it, or to use it for building your shelter. Make and use a stick to speed up measuring and cutting poles and logs to the right lengths.

(4) After a tree has been felled, trim off all limbs and knots so that the pole or log is require no additional smoothing

Limb Cut Off To Hook Over The Square-Cut End Of The Role. (ORNL-DWG 78-16210) (nwl57b.tif)

(5) It usually is best first to cut the poles exactly two or three times the final length used in the shelter.

(6) When you are ready to move the poles to the shelter site, drag them rather than trying on your shoulders. Shouldering them is more tiring, and you could injure yourself severely should trip.

To drag a log or several poles by hand: (a) Cut a stick 2 to 2-1/2 in. in diameter and abo short piece of 1/4-in. (or stronger) rope to the center of the stick; (c) Make a lasso-lik rope, so that when it is looped around the log and two people are pulling (see illustratio raised about 6 in. above the ground. The loop should be tightened around the log about 2 f end of the log cannot hit the backs of the legs of the two people pulling it.

(Illustration of 2 men pulling a log.) ORNL-DWG 78-16211 (nwl57c.tif)

CAUTION : If you drag a log down a steep hill, one person should tie a rope to the rear en the dragger, ready to act as the brake if needed.

(7) When you get the poles or logs to the location where you will build the shelter, cut t minimum diameters and specified lengths, and put all those of one specified type together.

Be sure that the diameter of the small end of each pole of one type is at least as large as the minimum diameter specified for its type. Make and use a measuring stick, as previously described.

14. Use snow for shielding material for aboveground shelters if the earth is so deep-frozen that it is impractical. For a Ridge-Pole Shelter (see Appendix A.5), cover the entire shelter with 5 ft of well-packed snow. For a Crib-Walled Shelter (see Appendix A.6), fill the cribs and then cover the entire shelter with snow at least 5 ft thick. With wetted or well-packed snow 5 ft thick, the shielding factor is about 50. Families have completed these winter shelters within 2 days.

Several hundred pounds of snow can be moved at a time by sledding it on a piece of canvas or other strong material 6 to 8 ft wide. Attach a stick across one end of the material and tie a rope to the other end, so as to form a "Y" bridle on which a person can pull.

To keep the occupants of a snow-covered expedient shelter dry and tolerably warm in subfreezing weather, provide sufficient ventilation openings to maintain inside temperatures at a few degrees above freezing. (See Chapter 14, Expedient Clothing.)

15. Make a reliable canopy over the shelter entry. By following the instructions given in Appendix A.7, you can make a dependable canopy that ordinary winds will neither tear nor blow down and that will not catch rainwater even if you have no waterproof material stronger than 4-mil polyethylene.

16. Take to your shelter enough window screen or mosquito netting to cover its openings. In warm weather, flies and mosquitoes would soon become a problem in most localities soon after an attack.

17. Work to complete (1) an expedient ventilating-cooling pump (a KAP) and (2) the storage of 100 gallons of water per person. This work should be accomplished by the time your shelter is completed. Especially in an area of heavy fallout during warm or hot weather, an earth-covered, high-sided shelter when full of people would be useless unless adequately ventilated and cooled and provided with enough water.

18. In cold weather, restrict air flow through the shelter by hanging curtains of plastic or other material by otherwise partially obstructing its two openings. Always be sure to leave at least a few inches open at the floor level of one opening and at the ceiling height of the other, to provide ventilation and prevent a harmful concentration of exhaled carbon dioxide. To prevent exhaled water vapor from wetting clothing and bedding and reducing its insulating value, keep the ventilation openings as small as possible without causing shelter temperatures to be intolerably cold.

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Fig. A. A dependable canopy to keep fallout and rain out of a vertical entry. ORNL-DWG 77-1000 (nwl59.tif)

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## Appendix A.1

### Door-Covered Trench Shelter

(See illustration at the end of Appendix A. 1)

#### PROTECTION PROVIDED

Against fallout radiation: Protection Factor 250 (PF 250) a person in the open outside the shelter receives 250 times as much fallout radiation as he would if inside.

Against blast: Better protection than most homes, if built in very stable earth. Blast tests indicated that this shelter would be undamaged up to at least the 5-psi overpressure range. Without blast doors, the shelter's occupants could be injured at this overpressure range,

Against fire: Excellent, if sufficiently distant from fires producing carbon monoxide and

#### WHERE PRACTICAL

In a location where at least one hollow-core door per occupant is available, and where there is a dry hole or trench 4 1/2 feet deep can be dug without difficulty. (A family evacuating in a station wagon can carry enough doors, with doorknobs removed. Strong boards at least 6 feet long and 1 inch thick, or plywood at least 3/4-inch thick, also can be used to roof this 36-inch-wide head earth shielding.)

Warning: Some doors with single-thickness panels if loaded with earth will break before they result in protective earth arching.

#### FOR WHOM PRACTICAL

For a typical family or other group with two or more members able to work hard for most of the day. (More than-average families with most members able to work hard have completed this type of shelter after receiving step-by-step, well illustrated instructions.)

#### CAPACITY

The shelter illustrated is roofed with 3 doors and is the minimum length for 3 persons. (If more than 3 doors, or boards and sticks at least 3 ft long, make the entry way trenches 3 or 4 ft long and pressed for time.)

For each additional person, add an additional door. (If more than about 7 persons are to be housed, use more separate shelters.)

#### BUILDING INSTRUCTIONS

- 1 . Before beginning work, study the drawings and read ALL of the following instructions.
2. Divide the work; CHECK OFF EACH STEP WHEN COMPLETED.
3. By the time the shelter is finished, plan to have completed (1) a ventilating pump (a Kleanair pump is best, 12 in. high), essential for this shelter except in cool weather, and (2) the storage of at least 1 gallon of water per occupant (see Appendix B and Chapter 8).
4. Start to assemble materials and tools that are listed for the illustrated 3-person shelter.

#### A. Essential Materials and Tools for a 3-Person Shelter

Three hollow-core doors.

A shovel (and a pick, if the earth is very hard).

- ° Two to three square yards per person of waterproof materials for rainproofing the roof. materials such as 4-mil polyethylene film, shower curtains, plastic table-cloths, plastic covers, or canvas.

- ° Two pieces of plastic or tightly woven cloth (each about 6- 1/2 X 6-1/2 ft) to make canopies over two shelter openings. Also sticks and cords or strips of cloth to support the canopies as Fig. A of the introductory section of this appendix.

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- ° Materials and tools for building a simple shelter-ventilating pump, a KAP 16 in. wide air pump. (See Appendix B.) Only in cold or continuously breezy, cool weather can tolerable temperatures and humidities be maintained in a crowded underground shelter without an air pump.

- ° Containers for storing adequate water. (See Chapter 8.)

#### B. Useful Materials and Tools

- ° Large cans, buckets, and; or pots with bail handles in which to carry earth and later to dispose of water or human wastes.

- ° Two pillowcases and one bedsheet per person-to make "sandbags" around shelter openings and to cover trench walls. (If available, large sheets of 4-mil polyethylene are better than bedsheets because they keep earth walls damp and stable. They also help keep shelter occupants dry and prevent earth from falling into their eyes.)

- ° File, knife, pliers, hammer.

- ° Measuring tape, yardstick, or ruler.

- ° Expedient life-support items.

5. To save time and work, SHARPEN ALL TOOLS AND KEEP THEM SHARP.

6. Wear gloves from the start even tough hands can blister and become painful and infected from digging.

7. Check to be sure the earth is stable and firm enough so that a trench shelter with unsupported vertical earth walls will be safe from cave-ins. (Interior doors are not strong enough to support a trench wider than 3 ft.)

As a test of the stability of earth, dig a small hole about 18 in. deep. Remove all loose earth from the bottom of the hole. Then make a "thumb test" by pushing your bare thumb into the undisturbed bottom of the hole. If you can push your thumb into the earth no farther than one inch, the

suitable for this type of shelter. If the earth does not pass the "thumb test," move to another location and repeat the test again. Continue to relocate and repeat until suitable earth is found, or build an aboveground shelter.

8. Prepare to dig a vertical-walled trench 4-1/2 ft deep and 3 ft wide. To determine the width of the trench, measure the widths of all the doors to be used for roofing it, then subtract 8 in. from the total. (To avoid arithmetical errors, it is best to lay all the doors side by side on the ground.)

9. Clear any brush, grass, or weeds that are more than a few inches high from the area where the trench is to be dug. Also clear the ground around all sides of the trench, to a distance of about 8 ft from the trench.

10. Stake out a rectangular trench 36 in. wide, with its length as determined above. Also stake out the entryway trench, as illustrated in Fig. A. 1 at the end of Appendix A.1, and the ventilation trench, as illustrated in Fig. A. 2.

11. Dig the main trench, the entryway trench, and the ventilation trench. Place the excavated earth on the lengthwise sides of the trench, starting at the outside edges of the cleared space. Be sure the earth is not closer than 3 ft to the sides of the trench.

12. To be sure that unstable, unsafe earth is not encountered at depths below 18 in., repeat the test at intervals of 18 in. the trench is deepened an additional foot. If the earth does not pass the test, do not dig deeper; try another location.

13. To keep each trench its full width as it is dug, cut a stick 36 in. long and another 18 in. long. Use the sticks repeatedly from the start to check the widths of the main trench and the entry trenches. Keeping the trenches full width will save much work and time later.

14. Carefully level and smooth the ground to a distance of 2-1/2 ft from the sides of the trench. The ground will lie flat on the ground up to the edges of the trench.

15. If plenty of sheets, bedspreads, plastic, and other materials are available, cover the trench walls. Wall coverings should stop one inch from the floor of the trench to prevent them from being stepped on. Plastic wall coverings keep some types of damp earth walls from drying out and crumbling.

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16. To be able to place an adequate thickness of shielding earth all the way to and around the ventilation hole, stack improvised "sandbags" around these two openings before placing the roofing. Or use cloth or plastic material to make rolls of earth, as illustrated in the introductory section.

17. Shovel earth around the rolls, sandbags, or other means used to raise the level of the shelter openings. Slope this earth outward, and pack it, so that rainwater on the ground outside the shelter.

18. To rainproof the shelter and to prevent the roofing doors from being dampened and weakened, use waterproof materials as follows:

a. If the earth is dry, the easiest and best way to make a rainproof roof is to place the

the ground, with each of the end doors overlapping an end of the main trench by 4 or 5 in. again to level the ground surface as you place each door, so that each lies flat against the way to the edges of the trench.) Next, mound dry earth over the doors. First place a foot of earth on the doors near their ends: then mound it about 12 in. deep above the centerline of the trench. Slope the earth to both sides so as to just cover the ends of the doors. Next, smooth the earth mound, being careful to remove sharp stones that might puncture rainproof materials. Then place waterproof material over the smooth mound, making the "buried roof" shown in Fig. A. Finally, carefully mound an additional 12 to 15 in. of earth on top of the buried roof, again first over the doors near their ends. The earth over the trench should be at least 2 ft thick. The effective earth arching will support most of the weight of the earth covering and will provide considerable protection if struck by blast.

b. If the earth is wet, place the waterproof material directly on top of the doors, to keep them strong. To make water run off this waterproof covering and to keep water from collecting on the horizontal surface and leaking through, slope the doors toward one side of the trench by making one side of the trench about 3 in. higher than the other side. A way to raise one side without increasing the distance the doors must span is to place an earth-filled roll of bed sheet material along one edge of the trench. To keep the waterproof material used to cover the doors from sliding down the slope of the doors when earth is shoveled on, tuck the upper edge of the waterproof material under the higher ends of the doors. Finally, mound earth over the doors, first placing it on the higher ends. The mound should be at least 2 ft deep above the centerline of the roof and about 3 or 4 ft high at both ends of the doors.

If more waterproof material is available than is required to make a buried roof (or to cover the trench) and to make the illustrated canopies over the two shelter openings, use this excess material to cover the wet ground on which the doors are placed.

19. Dig small drainage ditches around the completed shelter, to lead runoff water away.

20. To keep rain and/or sand-like fallout particles from falling into the shelter openings, make a canopy over each opening, as illustrated in Fig. A, shown in the introduction section of A.

21. Install the air pump (a KAP) in the shelter opening into which air is already naturally flowing.

22. If the shelter has a KAP, protection against radiation can be increased by placing containers of heavy foods, or bags of earth, so as to partially block the openings. This would still permit air to be pumped through the shelter, except in very hot weather.

23. For seats, place water and food containers, bedding, etc. along the side of the trench, away from the off-center entry trenches. If the trench floor is damp, covering it with a waterproof material and a brush will help.

24. Fill all available water containers, including pits which have been dug and lined with available materials. If possible, disinfect all water stored in expedient containers, using a chlorine bleach, such as Clorox, for each 10 gallons of water. Even if only muddy water is available. If you do not have a disinfectant, it may be possible to boil water when needed.



25. Put at least your most useful emergency tools inside your shelter.

26. As time and materials permit, continue to improve your chances of surviving by doing the following things as possible:

(1) Make a homemade fallout meter, as described in Appendix C, and expedient lights. (Prud will have made these extremely useful items well ahead of time.)

(2) Install screens or mosquito netting over the two openings, if mosquitoes or flies are a problem. Remember, however, that screen or netting reduces the air flow through a shelter — even when air is pumped through with a KAP.

(3) Dig a stand-up hole near the far end of the shelter. Make the hole about 15 in. in diameter, deep enough to permit the tallest of the shelter occupants to stand erect occasionally.

Fig. A. 1. Door-Covered Trench Shelter. ORNL DWG 73-1 1769R (nwl63.tif)

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## Appendix A.2

### Pole-Covered Trench Shelter

#### PROTECTION PROVIDED

Against fallout radiation: Protection Factor 300 (PF 300) a person in the open outside the shelter receives 300 times more fallout radiation than he would if he were inside.

Against blast: Quite good protection if built in stable earth. Blast tests have indicated that stable earth, would not be seriously damaged by blast effects of large explosions at least 1000 ft range. (At 7 psi, most buildings would be demolished.) Without blast doors, occupants of the shelter would be killed although probably not fatally at this overpressure.

Against fire: Excellent, if sufficiently distant from fires producing carbon monoxide and other toxic gases.

#### WHERE PRACTICAL

In wooded areas with small trees, for builders who have an ax or a bow saw, crosscut or chainsaw tools. Or in any location where the necessary poles may be obtained.

In stable earth, where the water table or rock is more than 4-1/2 ft below the surface.

#### FOR WHOM PRACTICAL

For a typical family or other group with two or more members able to work hard for most of the day. More than-average families with almost all members able to work hard have completed this type of shelter after receiving step-by-step, well illustrated instructions, before traveling to the wooded areas to cut trees and haul poles.)

#### CAPACITY

The shelter illustrated is the minimum length recommended for 4 persons. For each additional 2-3/4 ft to the length of the shelter room. If more than about 10 persons are to be sheltered, use multiple shelters.

#### BUILDING INSTRUCTIONS

1 . Before beginning work, study the drawings and read ALL of the following instructions. Remember that some people will be digging while others are cutting and hauling poles. CHECK ONE EACH TASK WHEN COMPLETED.

2. By the time the shelter is finished, plan to have completed: (1) a ventilating pump, and at least 15 gallons of drinking water per occupant (see Appendix B and Chapter 8).

3. Start to assemble materials and tools. Those listed are for the illustrated 4-person shelter. Adjust for longer shelters.

##### A. Essential Materials and Tools

- ° Saw (bow saw or crosscut preferred) and or ax for cutting poles to the lengths and diameters illustrated.

- ° Shovels (one for each two workers).

- ° Pick (if the ground is hard).

- ° Rainproof roof materials (very important in rainy, cold weather). At least 2 square yards of material per person would be required; 3 square yards per person would be better. Shower caps, plastic tablecloths, plastic mattress covers, canvas, and the like can be used. Also needed are pieces of plastic or tightly woven cloth, each about 6' X 61.2 ft, to make canopies over shelter openings.

- ° Materials and tools for building a simple shelter- ventilating pump, a KAP 22 in. wide air pump. (See Appendix B.) Only in cold or continuously breezy, cool weather can tolerable temperatures and humidities be maintained for days in a crowded underground shelter that lacks an air pump.

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- ° Containers for storing adequate water. (See Chapter 8.)

##### B. Useful Materials and Tools

- ° Large cans, buckets, and / or pots with bail handles in which to carry earth and later to dispose of water and human wastes.

- ° Two bedsheets and two pillowcases per person for covering cracks between roofing logs, making "sandbags," and improvising bedsheet-hammocks and bedsheet-chairs.

- ° A file.

- ° A measuring tape, yardstick, or ruler.
- ° Rope, or strong wire (100 ft) to make earth-retaining pole walls close to the shelter or explained in step 19) and for hammock supports, etc.
- ° Chain saw, pick-mattock, hammer, hatchet, pliers.
- ° Kerosene, turpentine, or oil to keep hand saws from sticking in gummy wood.
- ° Expedient life-support items recommended in this book.
- ° Mosquito netting or window screen to cover the openings, if mosquitoes or flies are like problem.

4. To save time and work, SHARPEN ALL TOOLS AND KEEP THEM SHARP.

5. Wear gloves from the start even tough hands can blister after hours of chopping and dig painful and infected.

6. If possible, select a location for the shelter that is in the open and at least 50 ft from. Remember that on a clear day the thermal pulse (flash of heat rays) from a very large nuclear cause fires as far away as 25 miles.

7. If the site chosen is on a steep slope, locate the shelter with its length crosswise to the slope.

8. Stake out the outlines of the trench, driving stakes as indicated in Fig. A.2.1 at the corners. If more than 4 persons are to be sheltered, increase the length of the shelter room by 2 ft for each additional person.

9. Clear the ground of saplings and tall grass within 10 ft of the staked outlines so that the ground can be easily shoveled back onto the completed shelter roof.

10. Start digging, throwing the first earth about 10 ft beyond the staked outlines of the trench. One of the family should do the easier digging, near the surface. Those members who can use an axe should cut and haul poles. See the introductory section of this appendix for the know-how to make work easier.

11. Pile all excavated earth at least 2 ft beyond the edges of the trench, so roofing poles can be driven into the ground. To make sure that the trenches are dug to the specified full widths at the bottom, use two sticks one 42 in. long and the other 22 in. long to check trench widths repeatedly.

12. At the far end of the shelter dig the ventilation trench-emergency exit, making it 22 in. wide. This will help provide essential ventilation and cooling. In cold weather or when fallout protection or plastic curtains should be hung in the two openings to control the air flow.

13. Make and install threshold boards, to keep the edges of earth steps and earth ledges from crumbling. (In damp earth, it is best to install threshold boards before roofing the shelter.) If board are not available, use poles.

14. Unless the weather is cold, build a shelter- ventilating pump a KAP 20 in. wide X 36 in. high.

is cold, building a KAP can be safely delayed until after the shelter is completed.) A KAP before a crisis, or, if possible, before leaving home.

15. Obtain fresh-cut green poles, or, as a second choice, sound, dry, untreated poles. Use diameter than those specified in the accompanying drawings. For ease in hauling, select poles 50% larger in diameter than those specified.

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16. Lay the poles side by side over the trench. Alternate the large and small ends to keep across the trench. If roof poles 9 ft long are being used to roof a 5-ft-wide trench, be sure poles so that their ends extend 2 ft farther beyond one side of the trench than beyond the other. enable shelter occupants, after the stoop-in shelter is completed, to widen the shelter roof. First, it can be widened to provide a 2-ft-wide sleeping ledge. Later, it can be further widened to provide space for additional expedient hammocks or for double -bunk beds of poles or boards built over the shelter.

17. For ease and safety later when hanging expedient bedsheet-hammocks and bed sheet-chair hammock shelter, place loose loops around roof poles in the approximate locations given by the diagram in the shelter drawing, Fig. A.2.2. Make these loose loops of rope, or strong wire, or 16-in-wide cloth, such as 50cv polyester bedsheet rolled up to form a "rope". (For making hammocks and hammock shelter, see Chapter 14. These are not essential, although decidedly useful.)

18. Cover the cracks between the logs with cloth, leaves, clay, or any other material that will prevent rain from falling down between the cracks. CAUTION: DO NOT try to rainproof this flat roof, and then cover it with earth. Water will seep through the loose earth cover, puddle on the flat roofing material, and then run down the joints between pieces of roofing material or through small holes.

19. Place 6-ft-long poles, one on top of the other, next to the entrances. This will keep the roof of the entryway trenches from falling into the openings. Secure these poles with wire or rope (see Fig. A-A 1 in Fig. A.2.1.) If wire or rope is not available, make earth-filled "rolls" to hold the trench roof next to each opening. (See the introductory section of this appendix.)

20. Mound earth to a center depth of about 18 in. over the shelter roof (as shown in View A-A 1) to form the surface of the future "buried roof." Be sure to slope both sides of the mound. Then smooth the surface and remove sharp roots and stones that might puncture thin rain proofing materials used under it.

21. Place the waterproofing material on the "buried roof." If small pieces must be used, use them in a patchwork fashion, starting at the lower sides of the mounded earth.

22. Cover the buried roof with another 18 in. of mounded earth, and smooth this final earthen surface.

23. Finish the entrances by placing some shorter poles between the two longer poles next to the entrances. Bank and pack earth at least 6 in. deep around the sides of the entrances, so that rainwater cannot run into the shelter entrances.

24. Dig surface drainage ditches around the outside of the mounded earth and around the entrances.

25. Place a piece of water-shedding material over each of the entrances, forming an open-ended flap.

fallout and rain from the shelter openings. (See Fig. A in the introductory section of Appendix A.2.1. Pole-Covered Trench Shelter. ORNL DWG 74-11755R (nwl67.tif)) fallout would settle on these suspended canopies, rather than falling into shelter openings at their edges and onto the ground like sand.

26. Flang the KAP from the roof of the trench opening into which outdoor air can be felt from the trench opening. (If you have no KAP, make and use a KAP Fan.)

27. Fill all available water containers, including pits which have been dug and lined with available materials. If possible, disinfect all water stored in expedient containers, using a chlorine bleach, such as Clorox, for each 10 gallons of water. Even if only muddy water is available. If you do not have a disinfectant, it may be possible to boil water when needed.

28. Put all of your emergency tools inside your shelter.

29. As time and materials permit, continue to improve your chances of surviving by doing the following things as possible:

(1) Make a homemade fallout meter, as described in Appendix C, and expedient lights. (Prudent planning will have made these extremely useful items well ahead of time.)

(2) Make and hang expedient bedsheet-hammocks and bedsheet-chairs, following the installation shown in Fig. A.2.2.

(3) Install screens or mosquito netting over the two openings, if mosquitoes or flies are a problem. Remember, however, that screen or netting reduces the air flow through a shelter even when air is pumped through with a KAP.

(4) Dig a stand-up hole near the far end of the shelter. Make the hole about 15 in. in diameter enough to permit the tallest of the shelter occupants to stand erect occasionally.

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Fig. A.2.1. Pole-Covered Trench Shelter. ORNL DWG 74-11755R (nwl67.tif)

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Fig.. A.2.2. Pole-Covered Trench Shelter ORNL DWG 74-1 1756R (nwl68.tif)

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### Appendix A.3 Small-Pole Shelter

#### PROTECTION PROVIDED

Against fallout radiation: Protection Factor 1000 (PF 1000), if the shelter is covered with a pole-covered trench shelter. That is, a person in the open outside this shelter will receive a gamma ray dose 1000 times

inside the shelter. See drawings at the end of Appendix A.3.

Against blast: This shelter is excellent for preventing fatalities if it is built with str still quite good if built without them. (These instructions are for fallout shelters. The and other essentials for adequate blast protection are given in Appendix D. Without blast to suffer serious injuries above 7 psi.)

Against fire: Excellent, if sufficiently distant from fires that produce carbon monoxide a

#### WHERE PRACTICAL

In wooded areas with small trees, for builders who have a saw (bow saw, crosscut, or chain tools. Any location is suitable if the necessary poles may be obtained there. Try to avoid

For belowground, semiburied, or aboveground construction. (However, aboveground constructi cavation and movement of so much earth that it is not practical for 2-day construction by tools.)

#### FOR WHOM PRACTICAL

For families or other groups with most members able to work hard 12 hours a day for 2 days realize how hard and long they can work if given a strong incentive.)

#### CAPACITY

The drawings and lists of materials given in these instructions are for a 12-person shelte cupant beyond 12, add 1 ft to the length of the shelter room.

This shelter requires less work and materials per occupant if its room is sized for about entrances are the same regardless of the length of the room. (To make the shelter room twi horizontal, ladder- like braces on the floor and near the ceiling of the room can be made er than one long pole on a side.)

If the room is sized for more than 24 people, management and hygiene problems become more occupied.

For 12 people to live for many days in this shelter without serious hardship, the benches with the dimensions and spacings given in the illustration. Or, materials must be availabl 12 expedient bedsheet hammocks that can be converted each day into 12 bedsheet-chairs.

#### BUILDING INSTRUCTIONS

1 . Study both of the two drawings (Fig. A.3. 1 and A.3.2 at the end of Appendix A.3) and instructions before beginning work. CHECK OFF EACH STEP WHEN COMPLETED.

2. By the time the shelter is finished, plan to have completed ( 1) a ventilating pump (a in. high), essential except in cold weather, and (2) the storage of at least 15 gallons of

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3. Start to assemble the required materials. For building a 12-person Small-Pole Shelter,

° Green poles. No pole should have a small end of less diameter than the minimum diameter for its use by Figs. A.3. 1 and A.3.2. The table below lists the number and sizes of poles build a 12-person Small-Pole Shelter.

Pole Length

Minimum  
Diameter of  
Small End

Number of  
Poles  
Required

Width b

6 ft 2 in.

5 in.

2

-

3 ft 1 in.

5 in.

12

-

2 ft 4 in. a

5 in.

12

-

10 ft 8 in.

5 in.

-

7 ft.

8 ft 8 in.

5 in.

-

7 ft.

10 ft 6 in.

4 in.

4

-

7 ft 2 in.

4 in.

-

47 ft.

5 ft 6 in. a

4 in.

12

-

6 ft 10 in.

4 in.

-

3 ft.

6 ft 3 in.

4 in.

8

-

2 ft 6 in. a

4 in.

16



-

2 ft 3 in.

4 in.

4

-

5 ft 2 in.

3-1/2 in.

-

8 ft.

3 ft 10 in.

3-1/2 in.

-

36 ft.

10ft c

2 in.

12

-

a To be shortened to fit for crossbraces.

b Width equals the distance measured across a single layer of poles when a sufficient number of poles are laid on the ground side by side and touching, to cover a rectangular area.

c For supports during construction.

NOTE: The above list does not include flooring materials, to be placed between the poles and on the earth floor.

° Rainproofing materials: Preferably one 100-ft roll, 12 ft wide, of 6-mil polyethylene. The amount needed is 200 sq. ft. of 4-mil polyethylene, or 200 sq. ft. of other waterproof plastic tablecloths, shower curtains, and or vinyl floor covering. Also include 100 ft. of sticks for drainage ditch drains ( 1/2-in. diameter, any lengths).

° Nails, wire, and; or cord: Ten pounds of 40-penny nails plus 4 pounds of 16-penny nails. However, 7 pounds of 16-penny nails can serve.

° Boards for benches and overhead bunks, if bedsheet-hammocks are not to be used. (Boards desirable, but not essential; small poles can be used instead.) 2 X 4-in. boards 70 feet (or use 3-in. -diameter poles). 1 X 8-in. boards 100 feet (or use 1-to 2-in-diameter poles).

° Materials to build a homemade ventilating pump (a KAP 24 in. wide and 36 in. high see Appendix 8) and to store at least 15 gallons of water per occupant (see Chapter 8).

4. Desirable muscle-powered tools for building a 1 2-person, Small-Pole Shelter are listed below. We have succeeded without having this many tools. A backhoe, chain saws, and other mechanized tools would be helpful, but not essential.)

#### Tools Quantity

Ax, long-handle 2

Bow-saw, 28-in. 2

(or 2-man crosscut saw) 1

Pick 2

Shovel, long-handle 3

Claw hammer 2

File, 10-in. 1

Steel tape, 10-ft 1

(Also useful: a 50-ft steel tape and 2 hatchets)

5. To help drain the floor, locate the shelter so that the original ground level at the entrance is lower than the original ground level at the far end of the shelter unless the location is otherwise dictated.

6. Stake out the trench for the entire shelter. Even in very firm ground, if the illustration is being built, make the excavation at the surface 9 ft 8 in. wide and 18 ft long (3 ft longer than the wooden shelter). The sloping sides of the excavation are necessary, even in very firm ground, to provide adequate space for backfilling and tamping. (The trench illustrated in Fig. A.3.1 is 6 ft wide and 18 ft long work when providing only for excellent fallout protection. For improved blast protection, the trench should be at least 7 ft deep.)

7. Check the squareness of the staked trench outline by making its diagonals equal.

8. Clear all brush, tall grass, and the like from the ground, to a distance of 10 ft all around the shelter so that later you can easily shovel loose earth back onto the roof.

9. If the ground is unstable, excavate with sides that are appropriately less steep.

10. When digging the trench for the shelter, use a measuring stick 7 ft 8 in. long (the mi repeatedly check the excavation width.

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11. When digging with a shovel, pile the earth dug from near ground level about 10 ft. awa excavation. Earth dug from 5 or 6 ft below ground level then can easily be piled on the su from the edge of the excavation.

12. Finish the bottom of the excavation so that it slopes vertically 1/2 in. per foot of l and also slopes toward the central drain ditch. (Eater, sticks covered with porous fabric the ditches, to serve like a crushed-rock drain leading to a sump.)

13. While some persons are excavating, others should be cutting green poles and hauling th poles that have tops no smaller than the specified diameters for each type of pole (not in bark).

14. For ease in handling poles, select wall and roof poles with top diameters no more than specified minimum diameters.

15. Sort the poles by size and lay all poles of the same size together, near the excavatio

16. Before the excavation is completed, start building the ladder- like, horizontal braces Construct these braces on smooth ground near the excavation. Place two straight poles, eac (with small-end diameters of 4 in.), on smooth ground, parallel and 6 ft 2 in. apart. Hold so that their outer sides are exactly 6 ft 2 in. apart, by driving two pairs of stakes int just touch the outsides of the two long poles. Each of the four stakes should be located a the end of a pole. To keep the 10 ft 6 in. poles from being rotated during the next step, small poles across them perpendicularly as temporary' braces, about 4 ft apart.

Then with an ax or hatchet, slightly flatten the inner sides of the two poles at the spots 6 cross-brace poles will be nailed. Next, saw each cross-brace pole to the length required place. Finally, toenail each cross-brace pole in place, preferably with two 40-penny nails

17. Place the lower, ladder-like horizontal brace of the main room on the floor of the com

18. Build the frame of the main room. Near the four corners of the room, secure four of it final vertical positions by nailing, wiring, or tying temporary brace-poles to the inner s and to the inner sides of the two long poles of the ladder-like horizontal brace on the bc To keep the two pairs of vertical wall poles exactly 6 ft 2 in. apart until the upper ladd is secured in its place, nail a temporary horizontal brace across each pair of vertical po their tops.

19. To support the upper ladder-like horizontal brace, nail blocks to the inner sides of t as shown in the lower right- hand comer of the pictorial view, Fig. A.3.2. If you have lar about 3 in. thick and 6 in. long, preferably cut from a green, 4-in.- diameter pole.

20. In the finished shelter, DO NOT leave any vertical support poles under the long poles horizontal brace; to do so would seriously reduce the usable space along the walls for ber occupants.

21. While some workers are building the frame of the main room, other workers should make horizontal braces for the two entrances, then make the complete entrances. To keep the ladder braces square during construction and back-filling, nail a temporary diagonal brace across

22. When the four wall poles and the two ladder-like horizontal braces of the main room are remaining vertical wall poles in place, touching each other, until all walls are completed wall poles, keep them vertical by alternately putting a butt and a top end uppermost. Wall position by backfilling and tamping about a foot of earth against their lower ends, or the position until backfilled.

23. Be sure to use the two 5-in.-diameter poles (6 ft 2 in. long) by placing one next to the bottom of each of the main doorways to the room. Study the drawings. Use braces, earth to hold apart the top and bottom of each doorway thus making sure that a 24-in. -wide air either doorway.

24. To prevent earth from coming through the cracks between wall poles, cover the walls with roofing, or even cardboard. If none of these are available, use sticks, twigs, or grass to cracks.

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25. After all horizontal bracing and vertical wall poles are in place, begin backfilling, walls and trench sides. Pay particular attention to the order of filling. The earth fill be brought up quite evenly, so that the earth fill behind one side is no more than 12 in. than the earth on the opposite side. Fightily tamp the earth fill in 6-in. layers. A pole do not use a mechanical tamper.

26. Next, lay the roof poles side by side, touching each other on top of the wall poles. Cover cracks with plastic, roofing, boards, or sticks to keep earth from falling through. If the the whole roof with some material such as bedsheets or plastic to keep sand from running to CAUTION : Do not try to rainproof this flat roof and simply cover it with earth. If you do straight through the loose earth cover, puddle on the flat roofing material, and leak through pieces of roofing material or through small holes.

27. Mound earth over the shelter, piling it about 15 in. deep along the centerline of the the sides of the roof, so that the earth is only about 2 in. deep over the ends of the roof mounding earth onto the roof, place grade stakes in position so you will be able to know the depths of roof poles as you cover them.) Continue these slopes to two side drainage ditches mounded earth with a rake or stick and remove any sticks or rocks likely to puncture the roofing material to be laid on it.

28. Place rainproofing material on top of the smooth, mounded earth as shown in sections of A.3.1 to make a "buried roof." Plastic film, such as 4-mil polyethylene, is preferable. Roll shower curtains and tablecloths, or canvas can also be used. Be sure to overlap adjoining

29. Place the rest of the earth cover over the shelter, being sure that the corners of the

ft of earth over them. Mound the dirt, smoothing its surface so that water will tend to run down drainage ditches which should be dug all around the edges of the mounded earth..

30. Build the benches and overhead bunks. If boards are available, use them; if not, use saplings. On each side, build a row of benches and bunks 9 ft long, centered in the shelter. In order to save space to the greatest advantage, make the heights and widths of the benches and bunks the thoroughly tested heights (14 in. and 4 ft 5 in.) and widths (16 in. and 24 in.) given by the tests. Space their vertical supports 3 ft apart so two adults can sit between each pair of vertical supports.

31. Narrow the ends of the overhead bunks so that the aisle between them is about 28 in. wide at the ends and 38 in. from each doorway. This allows room for installation and operation of an expedient fan for prevention of dangerous overheating in warm weather.

32. Place a canopy (open on all sides) over each entrance, to minimize the entry of sand-laden air in rain.

33. To improve the floor, lay small poles between the lower brace poles, so that the floor is made of a grid. Or, use sticks covered with scrap boards.

34. Fill all available water containers, including pits which have been dug and lined with available materials. If possible, disinfect all water stored in expedient containers, using a chlorine bleach, such as Clorox, for each 10 gallons of water. Even if only muddy water is available. If you do not have a disinfectant, it may be possible to boil water when needed.

35. Put all of your emergency tools inside your shelter.

36. As time and materials permit, continue to improve your chances of surviving by doing a number of the following things as possible:

(1) Make a homemade fallout meter, as described in Appendix C, and expedient lights. (Prudhomme will have made these extremely useful items well ahead of time.)

(2) Install screens or mosquito netting over the two openings, if mosquitoes or flies are a problem. Remember, however, that screen or netting reduces the air flow through a shelter – even when the air is pumped through with a KAP.

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#### EXPEDIENT VENTILATION AND COOLING

(Those workers who are to work only on the shelter itself, if pushed for time, need not re-arrange things at the beginning their work.)

Install a KAP (one that is 24 in. wide and 36 in. high) near the top of the doorway through which the air is naturally flowing into the shelter room at that time. (If the direction of the natural air flow is from the other opening.) To enable the KAP to efficiently pump fresh air from the outdoors all the time, block the lower half of the doorway in which the KAP is installed with a quickly removable barrier made of a plastic-covered frame made of sticks. Be sure to connect the KAP's pullcord only 1 1/2 in. below the top of the barrier.

prevents excessive arm motions which would cause unnecessary fatigue.

If short of time or materials, make a small Directional Fan.

In windy or cold weather, control the natural flow of air through the shelter by hanging at doorways at both ends, and or by making and using trapdoors on the tops of the vertical end curtains, use pieces of plastic, each with a supporting stick attached to its upper edge. openings in the doorways: (1) an opening under the lower edge of the adjustable curtain at room, and (2) an opening over the top of the curtain at the air-exhaust end of the room. This ment usually will provide adequate chimney-type ventilation for the shelter without using

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Fig. A.3.1. Plan and Elevation of Small-Pole Shelter (ORNL-DWG 71-3428 R4) (nwl74.tif)

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Fig. A.3.2. Pictorial View of Small-Pole Shelter (ORNL-DWG 71-3429 R3) (nwl75.tif)

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#### Appendix A.4

##### Aboveground, Door-Covered Shelter

#### PROTECTION PROVIDED

Against fallout radiation: Protection Factor about 200 (PF 200) if covered with 30 in. of open outside this shelter would receive about 200 times more fallout radiation than if he the end of Appendix A.4 shows the earth cover only 20 in. thick, resulting in a PF of about

Against blast: Better protection than most homes. Blast tests have indicated that this shelter aged at least up to the 5-psi overpressure range from large explosions. Without blast door could be injured at this overpressure range, although probably not fatally.

Against fire: Fair, if the cloth in the entries is covered with mud and if the shelter is producing carbon monoxide and toxic smoke.

#### WHERE PRACTICAL

In a location where at least one hollow-core door per occupant is available, where a dry trench deep can be dug without difficulty, but the water table or rock is too close to the surface to be practical. (A family evacuating in a pickup truck or large station wagon can carry equipment removed. Strong boards at least 6 feet long and at least one full inch thick, or plywood can be used to roof this shelter and to support its overhead earth shielding.

Warning: Some doors with single-thickness panels if loaded with earth will break before they result in protective earth arching.

## FOR WHOM PRACTICAL

For a typical family or other group with two or more members able to work hard for most of building skill is needed. (An urban family of six, with 14- and 12-year-old sons and 13- completed this shelter, sized for six persons, in one long working day: 13 hours and 43 minutes. step, will illustrated instructions at their Florida home 10 miles from the rural building truck to carry them, the interior doors, and other survival items.)

## CAPACITY

The shelter illustrated in Fig. A.4 is the minimum length for 4 persons. It is roofed with

For each additional person, add another door. (If more than about 7 persons are to be sheltered, use separate shelters.)

## BUILDING INSTRUCTIONS

1 . Before beginning work, study the drawing and read ALL of the following instructions. Do some will be digging while others are building an air pump, storing water, etc. CHECK OFF WHEN COMPLETED.

2. By the time the shelter is finished, plan to have completed a ventilating pump (a 16-inch KAP essential except in cold weather) and the storage of 15 gallons of water per occupant (see Appendix B and Chapter 8.)

3. Start to assemble the materials and tools needed. For the illustrated 4-person shelter,

### A. Essential Materials and Tools

- ° Six doors. Boards or plywood at least 5/8-in. thick can be used to replace one or more of

- ° At least 4 double-bed sheets for each of the first four persons, and 3 double-bed sheets

additional person to be sheltered or enough pieces of fabric and or of plastic to cover at least as much area as the sheets would cover. (This material is for making aboveground shelter walls, to make sleeping bags.)

- ° Rainproofing materials (plastic film, shower curtains, plastic tablecloths, mattress protection) 2 square yards for the first 4 persons and 2-1/2 square yards for each additional person.

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- ° A shovel (one shovel for each two workers is desirable). A pick or mattock if the ground is hard.

- ° A knife (the only essential tool for making a small shelter- ventilating KAP) and materials for making it 18 in. wide and 24 in. high. (See Appendix B.)

- ° Containers for storing water. (See Chapter 8.)

### B. Useful Materials and Tools

- ° Two or more buckets, large cans and or large pots with bail handles to carry earth, and water or wastes.
  - ° Saw (or ax or hatchet) to cut a few boards or small poles.
  - ° Hammer and at least 15 small nails (at least 2-1/2 in. long).
  - ° Tape measure, yardstick, or ruler.
  - ° Additional cloth and / or plastic equivalent in size to 2 more double-bed sheets for each person.
  - ° Additional waterproof material 2 more square yards per person.
  - ° Pillowcases, or cloth or plastic bags to serve as earth-filled sand bags. The more, the better.
4. To save time and work, sharpen all tools and keep them sharp.

5. Wear gloves from the start, to help prevent blisters and infections.

6. Select a building site where there is little or no chance of the ground being covered by water. The water table (groundwater level) is not likely to rise closer than 18 inches to the surface.

7. To avoid the extra work of digging among roots, select a site away from trees, if practical.

8. To lessen the dangers of fire and smoke from nearby houses or trees that might catch fire, build the shelter as far as is practical from houses and flammable vegetation.

9. Before staking out your shelter, provide one door per person to roof the main room plus one for each of the two entries. Be sure the door knobs have been removed. Use the two widest entries.

10. To be sure that all the walls will be in the proper positions to be roofed with the available doors on the ground, touching each other and in the same relative positions they will have when they are on the shelter. When all the roof doors are on the ground, side by side, determine the exact size of the room. (Note that Fig. A.4 illustrates a shelter sized for only 4 persons.)

11. Stake out the shelter.

12. Make the earth-filled "rolls" that will form the aboveground walls of your shelter. To make the rolls:

(1) Use doors as vertical forms to hold the earth-filled rolls in place until the walls are finished. Use the same doors that you will use later to roof the shelter.)

(2) Brace the door-forms with 36-in.-long braces (boards or sticks) that press against the ground. See Fig. A.4. Nail only the upper braces, using only very small nails.

(3) After the forms for the two inner sides of the shelter have been finished, put parts of the bedsheets on the ground, as illustrated. (Or use other equally wide, strong cloth or plastic.) A 2-ft width of cloth should be on the ground, and the rest of each sheet should be folded over the earth-filled rolls.



way, over the outsides of the door-forms. Adjacent sheets should overlap about 1 ft when more than one sheet is longer than one sheet.

(4) Shovel earth onto the parts of the sheet on the ground to the height of the rolls you have. Note that the roll to be made on one side is 2 in. higher than the roll on the other side.

(5) Shape the surface of the shoveled-on earth as illustrated, to hold the "hooks" of cloth. The exposed sides of the sheets are folded down.

(6) Fold down the upper side of each sheet while pulling on it to keep it tight and without wrinkles. It should lie on the prepared earth surface, including the small narrow trench, as illustrated in this appendix.

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(7) Pack earth onto the part of the folded-down sheet that is in the narrow, shallow trench. Following the sketches at the bottom of the accompanying drawing, fold back the loose edge over this part of packed earth to form a "hook." (The hook keeps the weight of the earth inside a roll from pulling the cloth out of its proper position.)

(8) Make a roll first on one side of the shelter, then on the other, to keep the heights of the sides of the shelter about equal. This will keep the unequal heights of earth from pushing out of their vertical positions.

(9) Add additional earth on top of the rolls so that the height of the level earth surface outside a roll, is the same as the height of the cloth-covered part of the roll that is against the shelter.

(10) When the roll walls have been raised to their planned heights on both sides of the shelter, remove the braces and the doorforms - being careful to keep the brace nails from damaging the doors.

(11) The door-forms of the side-walls of the shelter can be removed before building the end walls.

13. When smoothing the earth surfaces of the final tops of the roll walls on both sides, care should be taken to have the same slope as the lower sides of the roof doors will have after they are placed on the ground. This slope is necessary so that rainwater reaching the waterproof covering to be placed over the shelter will run to the lower side.) Study Fig. A.4.

14. After the side-walls have been completed (except for their ends that form the sides of the shelter), the door-forms have been removed, use the same doors for forms to build the two 22-in. -wide end walls.

15. Use earth-filled "sand bags" (made of pillowcases or sacks, and / or the tucked-in ends of the sheets) to make the outer ends of each entryway.

16. Make the two doorway frames if lumber, nails, and a saw are available. Make each frame of two 2 by 4-in. boards on each side of it, and slope the top board of each frame so that it will press flat against the earth and be supported. (If materials for a frame are lacking, place a single 2 by 4-in. board or a pole across the top of the entry, in the position shown in Fig. A.4 for the top of the doorway.)

17. After carefully removing all the temporary braces from the door-forms and the doors that were used to support the slopes of the tops of all supporting walls so that the doors will be supported evenly and,

will make contact with the smooth, sloping earth or cloth upon which they will rest.

18. If more than enough waterproof plastic or similar material is available to cover all the tops of the walls on which the roof doors will rest. This will keep the doors from absorbing damp earth.

19. Dig the illustrated 14-in.-deep, 36-in.-wide trench inside the shelter. (If the water is down 14 in., in some locations the walls can be raised to a height of 38 in. by cutting trenches on top of the walls. Another way the wall height can be increased is by making additional trenches.)

20. Place the roof doors in their final positions, and cover them with waterproof material. The waterproof material is folded under the higher edges of the doors to keep the material from sliding downward on the sloping doors as earth is shoveled onto the roof.

21. Extend the waterproof material on top of the doors a couple of feet beyond the lower edges. If enough material is available to cover all of the roof doors.

22. When shoveling the first layer of earth onto the rainproof material protecting the doors, be careful not to possibly puncturing it with rocks or sharp pointed roots in the earth.

23. To make earth arching more effective in supporting most of the earth to be placed on top of the doors, mound earth on and near the ends of the doors.

24. Cover the roof with at least 20 in. of earth. Make sure that there also is a thickness of earth at the corners of both the room and entries.

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25. To prevent surface water from running into the shelter if it rains hard, mound packed earth just inside the two entries. Rain can be kept out by a small canopy or awning that extends beyond the outermost edge of a doorway that roofs an entry.

26. If any waterproof material remains, use it to cover the floor of the shelter.

27. If the weather is warm or hot, install a 16-in. -wide by 24-in. -high air pump (a KAP) board across the roof of the entry into which outside air is moving naturally at the time. (If you have materials for a KAP, make a small Directional Fan.)

28. Cover all exposed combustible material with mud, earth, or other fireproof material, to prevent exposed cloth being ignited from a nuclear explosion or heat from a nearby fire.

29. Fill all available water containers, including pits which have been dug and lined with available materials. If possible, disinfect all water stored in expedient containers, using a chlorine bleach, such as Clorox, for each 10 gallons of water. Even if only muddy water is available. If you do not have a disinfectant, it may be possible to boil water when needed.

30. Put at least your most useful emergency tools inside your shelter.

31. As time and materials permit, continue to improve your chances of surviving by doing the following things as possible:

(1) Make a homemade fallout meter, as described in Appendix C, and expedient lights. (Pruc have made these extremely useful items well ahead of time.)

(2) Install screens or mosquito netting over the two openings, if mosquitoes or flies are Remember, however, that screen or netting reduces the air flow through a shelter even when pumped through with a KAP.

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Fig. A.4. Aboveground, Door-Covered Shelter. (ORNLDWG 74-8 132R) (nwl80.tif)

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## Appendix A.5

### Aboveground, Ridgepole Shelter

#### PROTECTION PROVIDED

Against fallout radiation: Protection Factor 300 (PF 300) if covered with 24 in. of earth. outside this shelter would receive about 300 times more fallout radiation than if he were ing drawing at the end of Appendix A.5.

Against blast: Better protection than most homes. Blast tests have indicated that this she aged at least up to the 5-psi overpressure range from large explosions. Without blast door could be injured at this overpressure range, although probably not fatally.

Against fire: Good, if the shelter is sufficiently distant from fires producing carbon mon

#### WHERE PRACTICAL

In many wooded areas and wherever enough poles are available.

In locations where belowground expedient shelters are impractical because the water table the surface for a covered-trench shelter.

#### FOR WHOM PRACTICAL

For a family or other group with five or more members able to work hard for most of 48 hou member able to saw and fit poles and use the hand tools listed on the following page. (A c lies, with 1 2 of the 1 5 members able to work, completed a shelter like this 23 hours and step-by-step, well illustrated instructions 12 miles from the wooded building site. They u tools, and moved over 50 tons of sandy shielding earth.)

#### CAPACITY

The shelter illustrated in Fig. A.5 is the minimum length for 5 persons. For each additional length of the ridgepole and shelter room. (If more than about 15 persons are to be sheltered, shelters.)

BUILDING INSTRUCTIONS

- 1 . Before beginning work, study Fig. A.5 and read ALL of the following instructions.
- 2. Divide the work. CHECK OFF EACH STEP WHEN COMPLETED.
- 3. By the time the shelter is finished, plan to have completed a ventilating pump (a KAP 2 high, essential for this shelter except in cool weather) and the storage of at least 15 gallons of water for each occupant. (See Appendix B and Chapter 8.)
- 4. Start to assemble the materials. For the illustrated 5-person shelter, these are:
  - A. Essential Materials and Tools
    - ° Poles. (Fresh-cut, green poles are best, sound, untreated poles are satisfactory.) See table for the number of poles required for a 5 -person shelter.

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Use

Pole Length

Minimum  
Diameter of  
Small End

Number of  
Poles  
Required

Width When  
All Are Laid  
on the Ground

For main room:

Ridgepole

4 ft 9 in.

6 in.

Column-posts

4 ft 3 in.

5 in.

2

Footing log

8 ft 0 in.

6 in.

1

Cross braces

6 ft 2 in.

3 in.

2

Roof poles

9 ft 0 in.

4 in.

-

5 ft a

Vertical end-wall poles

5 ft 0 in.

3-1/2 in.

-

14 ft b

Slanting end- wall poles and extras

6 ft 6 in.

3-1/2 in.

-

18 ft b

For outer sections of entry ways:

Horizontal poles

8 ft 0 in.

3-1/2 in.

4

Cross braces (material for 16)

5 ft 0 in.

3 1/2 in.

6

Wall poles

3 ft 4 in.

3 in.

-

32 ft b

Roof poles

2 ft 8 in.

2-1/2 in.

-

12 ft b

For inner sections of entryways:

Long, sloping poles

14 ft 0 in.

4 in.

## Cross braces

1 ft 8 in. 0

4 in.

8

## Vertical support poles

4 ft 0 in. 0

4 in.

8

## Roof poles

3 ft 0 in.

2-1/2 in.

-

13 ft b

a This width equals the distance measured across the tops of a single layer of poles when of poles are laid on the ground side by side with all the same ends in a straight line and poles will be placed butt-ends down to form the walls of the shelter room.)

b This width equals the distance measured across a single layer of poles when a sufficient are laid on the ground side by side and touching, with large ends and small ends alternating rectangular area.

c To be cut into the various lengths needed to close the ends of the main room and also to entryway.

° A saw and an ax or hatchet, to cut green poles. (A bow saw or crosscut saw serves well a more dependable than a chain saw. Having an extra blade for a bow saw may be essential.)

° Two shovels (one shovel for each two workers is desirable). A pick will also be needed, hard.

° Large buckets, cans, or pots with bail handles in which to carry earth, and later to store  
° A knife.

A hammer and at least 80 nails (3 in. or longer). If these are not at hand, rope, wire, or be used to lash poles together. At least 200 ft. of rope or strong wire will be needed, or bedsheets for each person to be sheltered. (Other fabric of equal strength can be used.) T cut or torn into foot-wide strips and twisted slightly to make rope. "

- ° Three double -bed sheets for the illustrated 5 -person shelter or a piece of strong fabr the same size. One additional sheet for each additional 2 occupants. (If sufficient sheets are not available, use many sticks and small poles, placed across the 9-ft side poles.)

- ° At least 2 square yards per person of rainproofing material (shower curtains, plastic ta mattress covers, or the like) essential in rainy, cold weather.

- ° Materials for building a ventilating pump, a KAP 20 in. wide and 26 in. high. (See Appen

- ° Containers for storing 15 gallons of water per occupant. (See Chapter 8.)

## B. Useful Tools and Materials

- ° Additional saws, axes, hatchets, shovels, and large buckets or cans.

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- ° A chain saw if there is a person in the group who is skilled at operating one.

- ° Kerosene, turpentine, or oil to keep a handsaw from sticking in green, gummy wood.

- ° A measuring tape, yardstick, or ruler.

- ° One bed sheet for each person to be sheltered, or a piece of strong fabric or plastic of size.

- ° A total of 40 square yards of rainproofing materials for the illustrated 5-person shelte yards for each additional person. (Even thin plastic will serve for the "buried roof. ")

5. To save time and work, SHARPEN ALL TOOLS AND KEEP THEM SHARP.

6. Wear gloves from the start even tough hands can blister after hours of digging and chop painful and infected.

7. Select a shelter location where there is little chance of the ground being covered with

(If you are sure the water table will not rise to cover the floor of a shallow excavation, first lowering the area of the planned main room by a foot or two. After the shelter is rc earth can be shoveled back to help cover the completed pole roof.) To avoid the extra work when excavating earth, select a site at least as far away from a tree as the tree is tall.

8. For a shelter that is completely aboveground, clear grass, weeds, etc. from the area wh built. (This reduces the possible problem of chiggers, ticks, etc.) Do not remove any eart



9. Stake out the entire shelter. Check the squareness of the shelter room by making its diagonal. Drive two lines of stakes to mark the outside edges of the completed earth covering. Place stakes outside the future positions of the lower ends of the roof poles.

10. Check the squareness of the future floor area inside the two lines marking where the trenches will be dug, to secure the lower ends of the sloping side-poles of the shelter. The parallel lines are 14 ft 6 in. apart. When the two diagonals joining the ends of these two lines are equal in length, the area between them has square corners.

11. While some persons are staking out the shelters, others should be cutting green pole at the site. Cut poles that have tops with diameters (excluding bark) no smaller than the diameters shown in the illustration for each type of pole.

12. To make the hauling and handling of the longer poles easier, select poles with top diameters 50% larger than the specified minimum diameters.

13. Sort the poles by length and diameter and lay all poles of each size together, near the site.

14. AS SOON AS POLES ARE BROUGHT TO THE SITE, SOME WORKERS SHOULD START BUILDING THE FOUR LADDER- LIKE HORIZONTAL BRACES FOR THE ENTRYWAYS TO THE SHELTER. AVOID DELAYS LATER. Study the drawing. Then construct these braces on smooth ground near the excavation. Place two straight poles, each 8 ft long (with small-end diameters of 3- 1/2 in.) parallel and so that their outer sides are 3 ft apart. Hold these poles securely so that they are exactly 3 ft apart, by driving two pairs of stakes into the ground so that they just touch the long poles. Each of the four stakes should be located about one foot from the end of a pole. To prevent the poles from being rotated during the next step, nail two boards or small poles across them as temporary braces, about 4 ft apart.

Then with an ax or hatchet, slightly flatten the inner sides of the two poles at the spots where the 4 cross-brace poles will be nailed. Next, saw each cross-brace pole to the length required for its place. Finally, toenail each cross-brace pole in place, preferably with two large nails in each end.

15. If more than 5 persons are to be sheltered, use 3 column-posts for 6 to 9 persons, and 4 for 10 to 14 persons.

16. For each additional person beyond 5, make the ridgepole and the footing log each 1 foot longer. See Fig.A.5.

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17. After notching the footing log (see drawing), place it in a trench dug deep enough so that the notches are about 4 inches below the surface of the ground.

18. Carefully dig the 4-in. -deep, V-shaped, straight trenches in which the lower ends of the poles will rest. Dig each of these two parallel trenches 7 ft 3 in. from the center line of the footing log.

19. Carefully notch a "V" only about 1/4-in. deep in the top of each of the two outer column-posts. The other ends so that each is 4 ft 3 in. long. (When they are placed on the notched footing log, the ridgepole is placed on them, the upper side of the ridgepole will be about 4 ft 4 in. above the ground.)

20. Place the two outer column-posts in their notches in the footing log, and secure the b against sideways movement by placing two small-diameter, 4-ft horizontal poles just below both sides, as illustrated. Then temporarily place and brace the ridgepole in position.

21. For shelters sized for more than 5 occupants, make and place the inner column-post, cutting a V"-notched column-post too short, first carefully V"-notch each remaining column 1 in. too long, and trim it off to fit in its final position under the ridgepole.

22. If nails at least 4 in. long are available, nail sloping cross-braces to the inner side. If nails are not available, notch slightly bowed cross-braces and the column-post as illustrated; then place them in position. (Strips of ordinary bedsheets, torn about a foot wide and twisted together slightly, serve as lashing rope.") To hold the tops of the column-posts securely against the upper ends of the cross-braces, a tightened rope" loop that encircles the tops of the column-posts can be used.

23. Next put four of the larger-diameter, 9-ft roof-poles in position, with the outside poles each only about 1 in. from an end of the ridge pole.

24. Place the rest of the 9-ft roof-poles in position, making sure that all their small ends are pressed together and overlap on the ridgepole at least as far as illustrated. Pack the lower ends. If the earth is clay, put small spacers of wood between the ends.

25. At each end of the shelter room, build extra shelter space and an entryway. First position the end-wall poles, their upper ends resting on the outermost wall poles. Study Fig. A.5. Place the two 14-ft poles parallel, and equally distant from the centerline of the ridgepole. Nail four 20-in. -long cross-braces to each pair of 14-ft poles, as illustrated. To make sure that the upper ends do not move before the roof is built, holds them in place, tie the upper ends of the 14-ft poles together. Drive a stake against the upper end of each 14-ft pole, to keep it from slipping outward. Under the center of each 14-ft pole, place two vertical posts.

26. Dig 4-in. -deep trenches for the lower ends of the sloping end-wall poles of the main shelter. The poles should be cut to length so that their upper ends will be about 4 in. above the outermost 9-ft roof poles. The poles they lean. Dig narrow, vertical trenches, about 8 in. deep, for all vertical wall poles and for the horizontal brace poles near the ground.

27. Start placing the sloping end-wall poles. First place the longest pole, then the shorter ones.

28. Across the open spaces between the 9-ft roof poles, place limbs and/ or sticks roughly parallel to the ridgepole, in the lower left-hand drawing. Be sure to use limbs or sticks that have diameters of at least 1 in. and place them no farther apart than 6 in. Leave needles and leaves on the limbs. Do not leave sharp points exposed upward. Do not place more than a 6-in. -thick mass of limbs and leaves over the side-poles. The earth cover necessary for excellent fallout protection might be unintentionally reduced if the cover is too thick.

29. Place bedsheets (or 4-mil-thick polyethylene film or equally sturdy material) over the shelter, to keep the earth from falling through the roof.

30. To prevent sand or dry earth from falling between the cracks where the poles are side by side, cover the parts of the roof with cloth, plastic, or paper. If these materials are not available, use dry grass or leaves. (In tick or chigger season, avoid using grass, or leaves from on or near the ground.)

31 After the entryways are completed, begin to cover the shelter with earth. Starting from a full 1-ft thickness of earth cover. First raise its height about a foot on one side and then on the other repeatedly. This is to prevent unequal loading from tipping the shelter.

(Do not excavate any earth closer than 3 ft to the line of stakes marking the final outer completed, 2-ft-thick earth cover.)

32. Fill the spaces between the entryways and the main room only with earth. (An equal thickness of other light material provides much less protection against radiation.)

33. Before placing the rainproofing material for the "buried roof," smooth the surface of the earth. This will prevent sharp rocks or sticks from puncturing the plastic or other rainproofing material. If you have sufficient waterproofing materials to cover the whole roof, use what is available to you. If not, use part, on both sides of the ridgepole.

34. To prevent rainwater on the ground outside from running into the entryways, make a mound of earth about 4 inches high across the entryway floors, about 2 ft from their outer ends. Dig a shallow ditch completely around the earth mounded over the shelter.

35. Unless the weather is cold, install your shelter ventilating KAP in the entry into which the wind blows naturally. (If short of time or materials, make a small Directional Fan.)

36. Complete the storage of water and other essentials.

37. To prevent fallout or rain from falling onto the floor of the outer entryways, place a tarp (as illustrated) over each opening.

38. Fill all available water containers, including pits which have been dug and lined with available materials. If possible, disinfect all water stored in expedient containers, using a chlorine bleach, such as Clorox, for each 10 gallons of water. Even if only muddy water is available. If you do not have a disinfectant, it may be possible to boil water when needed.

39. Put all of your emergency tools inside your shelter.

40. As time and materials permit, continue to improve your chances of surviving by doing the following things as possible:

(1) Make a homemade fallout meter, as described in Appendix C, and expedient lights. (Prudhomo has made these extremely useful items well ahead of time.)

(2) Make and hang expedient bedsheet-hammocks.

(3) Install screens or mosquito netting over the two openings, if mosquitoes or flies are present.

Remember, however, that screen or netting reduces the air flow through a shelter even when it is pumped through with a KAP.

(4) Dig a stand-up hole near the far end of the shelter. Make the hole about 15 in. in diameter enough to permit the tallest of the shelter occupants to stand erect occasionally.

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Fig. 5. A. Aboveground, Ridgepole Shelter, (nwl86.tif)

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## Appendix A.6

### Above ground, Crib-Walled Shelter

#### PROTECTION PROVIDED

Against fallout radiation: Protection Factor 200 (PF 200) if the earth-filled cribs are built 6 ft, as illustrated in Fig. A.6 at the end of these instructions. (A person in the open out about 200 times as much fallout radiation as he would if inside.) If earth is mounded to the depth of the shelter, the protection factor can be raised to PF 500 or better. See the accompanying text of Appendix A.6.

Against blast: Better protection than most homes. Without blast doors, occupants could be probably not fatally at lower overpressure ranges than those that would destroy this shelter.

Against fire: Poor, if the shelter is built as illustrated. The cloth and outer poles would provide a small pulse and other possible sources of intense heat. However, if earth is mounded around the shelter, exposed cloth and wood, good fire protection would be provided.

#### WHERE PRACTICAL

The crib-walled shelter is practical in many wooded areas and whenever enough poles are available in locations where belowground expedient shelters are impractical because the water table or rock is too close for a covered-trench shelter.

#### FOR WHOM PRACTICAL

For a family or group with three or more members able to work very hard for most of 48 hours; a family with an ax or saw and materials found in most American homes can build this shelter. (Groups with the nails, tools, skill, and the number of workers required to build a Ridgepole shelter; a Crib-Walled Shelter requires almost twice the total length of poles and more work than a Ridgepole shelter for the same number of persons.)

#### CAPACITY

The shelter illustrated in Fig. 6. 1 is the minimum length for 5 persons. For each additional person, increase the length of the room. (If more than about 12 persons are to be sheltered, build 2 or more shelters.)

#### BUILDING INSTRUCTIONS

- 1. Before beginning work, study the drawing and read ALL of the following instructions.
- 2. Divide the work. CHECK OFF EACH STEP WHEN COMPLETED.
- 3. By the time the shelter is finished, plan to have completed a ventilating pump (a KAP 2 high, essential for this shelter except in cool weather) and the storage of at least 15 gallons of water for each occupant. (See Appendix B and Chapter 8.)
- 4. Start to assemble materials and tools.

A. Essential Materials and Tools

° Poles. (Fresh-cut, green poles are best; sound, untreated poles are satisfactory.) For the 5-person shelter, the required poles are listed on the following page.

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Use

Pole Length

Minimum  
Diameter of  
Small End

Number of  
Poles  
Required

Width When All  
Are Laid on the  
Ground 3

Sides of longest crib

12-1/2 ft

3 in.

7 ft

Sides of middle-sized crib

10 ft

3 in.

7 ft

Sides of shortest crib

7 ft

3 in.

7 ft

Ends of all cribs

3- 1/2 ft

3 in.

21ft

Vertical poles at the  
comers of all cribs

3- 1/2 ft

2 in.

56

Main roof

9 ft

3-1/2 in.

12 ft

Entry way roofs

5 ft.

2-1/2 in.

22 ft

a This width is the distance measured across a single layer laid on the ground side by side large ends of poles when a sufficient number of them are and small ends alternating so as rectangular area.

- ° A saw (preferably a bow saw with an extra blade, or a crosscut saw) and / or an ax for cutting poles.
- ° Containers for storing 15 gallons of water per occupant. (See Chapter 8.)
- ° A shovel (one for each two workers is desirable).
- ° A pick (if the ground is very hard).
- ° Two to five large cans, buckets, and/ or pots with bail handles, in which to carry earth or wastes later.
- ° A knife.
- ° A minimum of 300 ft of wire at least as strong as clothesline wire. Second choice would be rope, or (third choice) 8 double -bed sheets that could be torn into 1-ft wide strips and then serve as rope. For each additional person beyond 5, supply 20 ft of wire or rope or half a sheet.
- ° Rainproof roofing materials at least 2 square yards per person. Such materials as plastic curtains, plastic tablecloths or plastic mattress covers can be used. These materials are prolonged shelter occupancy in rainy, cold weather.
- ° Fifteen double-bed sheets (or equal square-yardage of other strong cloth or plastic).
- ° Materials for building a ventilating pump, a KAP 20 inches wide and 30 in. high. (See Appendix B.)

#### B. Useful Materials and Tools

- ° Additional saws and shovels, chain saw, pick-mattock, hammer, hatchet.
  - ° Kerosene, turpentine, or oil to keep a hand-saw from sticking in gummy wood.
  - ° A file.
  - ° Two additional double -bed sheets per person, or equivalent square-yardage of other equal material or plastic.
  - ° A measuring tape, yardstick, or miler.
  - ° Old newspapers (about 15 pounds).
  - ° A total of 30 square yards of rainproofing materials for the illustrated 5 -person shelter for each additional person to be sheltered. (Even thin plastic will serve to make a rainproof shelter.)
5. To save time and work, SHARPEN ALL TOOLS AND KEEP THEM SHARP.

6. Wear gloves from the start. Even tough hands can blister and become painful and infected from digging and chopping.

7. Select a shelter location where there is little or no chance of the ground being covered by a heavy rain.

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8. If the building site is near the edge of a woods, pick a site at least 40 ft from the nearest trees.

9. Clear off grass, weeds, etc., from the area where you plan to build the shelter this area should be free of chiggers or ticks. Do not remove any earth.

10. Stake out the entire shelter, locating the 6 required cribs. BE SURE TO MAKE THE INSIDE OF THE MAIN ROOM EQUAL TO THE NUMBER OF PERSONS TO BE SHELTERED MULTIPLIED BY 1-1/2 FT. The illustrated shelter is sized for 5 persons, and the poles listed are those for a 5-person shelter.

11. While some persons are staking out the shelter, others should be cutting green poles at the building site. Cut poles with tops no smaller than the diameters specified. (Note: the specified diameters are for the bark.)

12. Select poles with small-end diameters no more than 50% larger than the specified minimum. This will make handling of the long wall and roof poles easier.

13. Sort the poles by length and diameter and lay all poles of each size together, near the building site.

14. Use larger trees and poles, up to 6 in. in diameter, to make the 3-1/2-ft-long end-poles. Do not use poles with small-end diameters of less than 3 in. for the side-wall poles of the shelter. For the brace-poles, use poles with diameters of at least 2 in., cut off at the height of the upper horizontal poles against which they are tied.

15. Be sure to cut off all limbs so that the poles are quite smooth. Usually it is easier to cut the poles at the building site before cutting them into the required lengths. Pull them by the small, 1-in. end.

16. Determine if there are enough long poles to make the side-poles of the two cribs form the shelter room without splicing two shorter poles together. If the shelter is being built for more than 5 persons, it will require side poles that are longer than 15-1/2 ft. Therefore, if a shelter for more than 5 persons is to be built, it would be best to use 2 cribs placed end-to-end on each side of the shelter room, as illustrated by Fig. A.6.

17. Place the lowermost four poles of each of the cribs in their final positions, so that the crib-walls are in position on the ground. Use the thicker, heavier poles at and near the back of the shelter.

BE SURE THE ROOM IS LONG ENOUGH TO PROVIDE 1-1/2 FT OF ROOM LENGTH FOR EACH PERSON TO BE SHELTERED.

18. To build each crib:

(1) Place two 3- 1/2-ft end-poles on the ground. Put two of the side-poles on top of the two end-poles.



the ends of all four poles extend 3 in. (no more) beyond where they cross. The thicker pole used first to add stability.

(2) Stack additional pairs of end-poles and side-poles to form the crib, keeping each wall until the tops of the uppermost side-poles are at least 42 in. above the ground. To keep the poles of the crib about level while the crib is being raised, alternate the large ends and

(3) Place a pair of small, vertical brace-poles in each of the four corners of the crib. The brace-poles should be no higher above the ground than the upper sides of the crib's uppermost poles.

(4) Tie each pair of vertical brace-poles together tightly at bottom, middle, and top. For of strong wire, rope, or slightly twisted, foot-wide strips of cloth at least as strong as Square knots with back-up overhand knots are best, but three overhand knots one on top of will hold.

(5) If the crib is more than 8 ft long, place an additional pair of vertical brace-poles, outside center of each long crib-wall. Tie this pair of vertical brace-poles together per

above the ground, but not yet in the middle or near the top of the crib. Temporarily tie the center vertical brace-poles to the uppermost side-pole of the wall it touches.

(6) Line the crib with cloth or plastic film, making sure that several inches of the lining uppermost poles. So that the lining will not be pulled down when the crib is being filled, upper edge of the lining to the uppermost wall pole about every 2 ft. First cut a small hole to thread a tie-string or a 2-in. -wide tie-strip of cloth. (If plenty of cloth and / or plastic lining the cribs, secure the lining by simply wrapping a greater width of the upper edge of the uppermost crib wall-pole.)

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(7) Permanently tie together the pair of vertical center brace-poles, using horizontal ties just below the uppermost horizontal wall-poles of the crib. Use the strongest material you horizontal ties across the center of the crib.

(8) Excavate earth 10 ft or so beyond the outer sides of the cribs. To save work, carry it dump it inside the cribs. (Two children can carry a heavy bucket of earth by running a string through the bail or handle of the bucket and tying the bail to the center of the stick before earth closer to the cribs to put on the roof.

(9) Fill the lined crib with earth from which almost all grass, roots, and the like have been placing hard lumps of earth in contact with the lining. Fill the crib so that the surface is about 4 in. above the upper sides of the uppermost horizontal poles.

19. Line the narrow spaces between adjacent cribs with cloth or plastic; then fill these spaces at a time, tamping repeatedly so as to avoid leaving air spaces.

20. Place the 9-ft roof poles over the main room. (If poles are unavailable and boards 1-1 use two thicknesses of boards.) Use the strongest roof poles (or double-thickness boards or

nearest the entryways. Then put shorter, 5- or 6-ft poles or boards over the entryways.

21 To keep earth from falling through the cracks between the roof poles, put sticks in the cover the roof with two or more thicknesses of cloth, plastic, or other material. Newspaper materials are lacking.

22. Put earth on the roof to the depths shown for the illustrated "buried roof." Be sure to smooth this gently mounded earth surface so that the buried roof will shed water.

23. So that the earth cover near the outer edges of the roof will be a full 2 ft thick, make steeply near the edges. Steep earth slopes can be made and kept stable by using large lump steep bank, or by using earth-filled "rolls" of cloth or other material along the edges of

24. Put in place the waterproof material of the buried roof.

25. Pile on the rest of the earth cover, as illustrated, to at least a full 2-ft thickness

26. Smooth the surface of the earth cover, including the sides, so that rain will run off. finished roof.

27. To prevent rainwater on the ground outside from running into the entryways, make mound about 4 in. high across the entryway floors. Make the mounds about 2 ft from the outer end

Dig a shallow drainage ditch completely around the shelter.

28. Unless the weather is cold, install your shelter ventilating KAP in the entry into which naturally. (If short of time or materials, make a small Directional Fan.)

29. To prevent fallout or rain from falling onto the floor of the outer entryways, place s (illustrated) over the openings.

30. If time and energy are available, mound earth all around the shelter. Doing so will reduce covering flammable materials; it also will increase fallout protection.

31 . Fill all available water containers, including pits which have been dug and lined with available materials. If possible, disinfect all water stored in expedient containers, using a chlorine bleach, such as Clorox, for each 10 gallons of water. Even if only muddy water. If you do not have a disinfectant, it may be possible to boil water when needed.

32. Put all of your emergency tools inside your shelter.

33. As time and materials permit, continue to improve your chances of surviving by doing the following things as possible:

(1) Make a homemade fallout meter, as described in Appendix C, and expedient lights. (Prud have made these extremely useful items well ahead of time.)

(2) Make and hang expedient bedsheet-hammocks.

(3) Install screens or mosquito netting over the two openings, if mosquitoes or flies are a problem. Remember, however, that screen or netting reduces the air flow through a shelter even when pumped through with a KAP.

(4) Dig a stand-up hole near the far end of the shelter. Make the hole about 15 in. in diameter enough to permit the tallest of the shelter occupants to stand erect occasionally.

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Fig. A.6. Aboveground, Crib-Walled Shelter (ORNL-DWG 74-81 30R) (nwl9 1 .tif)

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## Appendix B

### How to Make and Use a Homemade Shelter-Ventilating Pump, the KAP

#### I. THE NEED FOR SHELTER AIR PUMPS

In warm weather, large volumes of outside air MUST be pumped through most fallout or blast shelters that are crowded and occupied for a day or more. Otherwise, the shelter occupants' body heat and water vapor will raise the temperature-humidity conditions to DANGEROUSLY high levels. If adequate volumes of outdoor air are pumped through typical belowground shelters in hot weather, many times the number of persons could be accommodated. Otherwise, only a few could survive in these same shelters without adequate forced ventilation. Even if only 3 cubic feet per minute (3 cfm) of outdoor air usually should be pumped through shelters, primary carbon dioxide exhaled by shelter occupants from rising to harmful concentrations.

The KAP (Kearny Air Pump) is a practical, do-it-yourself device for pumping adequate volumes of outdoor air through shelters with minimum work. The following instructions have been improved after repeated use by dozens of small groups to build KAPs including families, pairs of housewives, and children. Although many shelter builders had previously heard of this kind of pump, yet almost all groups succeeded in making one after assembling the materials. Their successes prove that almost anyone, if given these clear instructions, can build a serviceable, large-volume air pump of this simple type, using common materials from most American homes.

If possible, build a KAP large enough to pump through your shelter at least 40 cubic feet of outdoor air for each shelter occupant. If 40 cfm of outdoor air is pumped through a shelter of the size specified below, even under heat-wave conditions the effective temperature of the shelter will be 10°F higher than the effective temperature outdoors. (The effective temperature is a measure of the combined effect due to its heat, humidity, and velocity.) The 36-inch-high by 29-inch-wide KAP described in the instructions as specified, will pump at least 1000 cfm of outside air through a shelter that has the air flow specified in these instructions.

If more than 25 persons might be expected to occupy a shelter during hot weather, then it is necessary to build a larger KAP. The 72-inch-high by 29-inch-wide model described can pump between 4000 and 5000 cfm of outside air.

To maintain tolerable temperature-humidity conditions for people in your shelter during hot weather, pump outside air through your shelter at the rate specified in these instructions.

° Pump enough outdoor air all the way through the shelter (40 cfm for each occupant in very hot weather).

° Distribute the air evenly within the shelter. If the KAP that pumps air through the shelter does not create the air movement that can be felt in all parts of the shelter in hot weather, one or more additional fans may be needed to circulate the air and gently fan the occupants.

° Encourage the shelter occupants to wear as little clothing as practical when they are hot (clothing evaporates and cools best on bare skin.)

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° Supply the occupants with adequate water and salt. For prolonged shelter occupancy under conditions in a hot part of the country, about 4 quarts of drinking water and 1/3 ounce (1/3 ounce) of salt per person are required every 24 hours, including salt in food that is eaten. Normal supply about 1/4 ounce of salt daily. Salt taken in addition to that in food should be dissolved in drinking water.

° Pump outdoor air through your shelter day and night in warm weather, so that both the occupants and the shelter are cooled off at night.

Almost all of the danger from fallout is caused by radiation from visible fallout particle fallout. The air does not become radioactive due to the radiation continuously giving off particles.

The visible fallout particles rapidly "fall out" of slow moving air. The air that a KAP pumps moves at a low speed and could carry into the shelter only a very small fraction of the fallout radiation hazard outside. This fraction, usually not dangerous, can be further reduced if the cautions described in these instructions are followed.

## CAUTION

Before anyone starts to build this unusual type of air pump, ALL WORKERS SHOULD READ THESE INSTRUCTIONS AT LEAST UP TO SECTION V, INSTALLATION. Otherwise mistakes may be made and work may be divided inefficiently.

When getting ready to build this pump, all workers should spend the first half-hour studying the instructions and getting organized. Then, after materials are assembled, two inexperienced persons working together can complete the 3-foot model described in the following pages in less than 4 hours. To speed up the work; for example, one person can start making the flaps while another begins work on the frame.

## II. HOW A KAP WORKS

As can be seen in Figs. 1 and 2, a KAP operates by being swung like a pendulum. It is hinged to a swinging frame. When this air pump is pulled by a cord as illustrated, its flaps are closed and air in front of it and "sucks" air in back of it. Thus a KAP pumps air through the opening in its power stroke. During its power stroke, the pump's flaps are closed against its flap-stop which is fastened across the face of the frame.

Fig. 1. Section through the upper part of a doorway, showing operation of a KAP. (ORNL-DWG (nwl94a.tif)

Fig. 2. KAP in doorway (with flaps open during its return stroke). (ORNL-DWG 66-12319A) (r Book Page: 195

When a KAP swings freely back as a pendulum on its return stroke, all its flaps are opened pumped air stream continues to flow in the direction in which it has been accelerated by the pump itself swings in the opposite direction (see Fig. 2). Thus the flaps are one-way valves that flow in one direction, where desired.

The KAP can be used: (1) to supply outdoor air to a shelter, (2) to distribute air within occupants.

1. To force outdoor air through a shelter, an air-supply KAP usually is operated as an air pump pulling it with a cord (see Fig. 1). (Only rarely is it necessary to operate a KAP as an air pump pushing it with a pole, as described in the last section of these instructions.)

2. To distribute air within a shelter and/or to fan the occupants, air-distribution KAPs may be used overhead and operated as described later.

### III. INSTRUCTIONS FOR BUILDING A KAP

In this section, instructions are given for making a KAP 36 inches high and 29 inches wide when swinging in a typical home basement doorway 30 inches wide. If your doorway or other opening is narrower or wider than 30 inches, you should make your KAP 1 inch narrower than the narrowest opening you plan to install it. Regardless of the size of the KAP you plan to build, first study the 36 X 29-inch model.

In Section VII you will find brief instructions for making a narrower and even simpler KAP for use in the narrow openings of small trench shelters and other small expedient shelters. Section VIII describes large shelters.

#### A. MATERIALS NEEDED FOR A KAP 36 INCHES HIGH BY 29 INCHES WIDE

The preferred material is listed as first (1st) choice, and the less-preferred materials as second (2nd), third (3rd), and fourth (4th) choices. It is best to assemble, spread out, and check all your materials before beginning.

##### 1. The pump frame and its fixed support:

° Boards for the frame:

(1st) 22 ft of 1 X 2-in. boards. (A nominal 1 X 2-in. board actually measures about 3/4 X 1 1/2 in. by 1 1/2 X 2 in. usual, nominal dimensions will be given throughout these instructions.) Also, 6 ft of 1 X 2-in. wood is better.

(2nd) Boards of the same length that have approximately the same dimensions as 1 X 2-in. lumber.

(3rd) Straight sticks or metal strips that can be cut and fitted to make a flat-faced KAP

° Hinges: (1st) Door or cabinet butt-hinges; (2nd) metal strap-hinges; (3rd) improvised hinges of leather, woven straps, cords, or 4 eyescrews which can be joined to make 2 hinges. (Screws attaching hinges. If nails are used, they should go through the board and their ends should be clinched flattened against the surface of the board.)

° A board for the fixed horizontal support: (1st) A 1 X 4-in. board that is at least 1 ft across the opening in which you plan to swing your pump; (2nd) A wider board.

° Small nails (at least 24): (1st) No. 6 box nails, about 1/2 in. longer than the thickness of the board; their pointed ends can be bent over and clinched); (2nd) other small nails.

## 2. The flaps (See Figs. 1, 2, 6, 7, and 8):

° Plastic film or other very light, flexible material – 12 square feet in pieces that can be cut into rectangular strips, each 30 X 5- 1/2 in. : (1 st) polyethylene film 3 or 4 mils thick (3 mils is about 1/64 of an inch); (2nd) 2-mil polyethylene from large trash bags; (3rd) tough paper.

° Pressure-sensitive waterproof tape, enough to make 30 ft of tape 3/4 in. to 1 in. wide, for the hem-tunnels of the flaps: (1st) cloth duct tape (silver tape); (2nd) glass tape; (3rd) scotch tape, freezer or masking tape, or sew the hem tunnels. (Do not use a tape that stretches: it may stretch afterward and cause the flaps to wrinkle.)

## 3. The flap pivot-wires:

(1st) 30 ft of smooth wire at least as heavy and springy as coat hanger wire, that can be cut into straight pieces each 29 in. long (nine all-wire coat hangers will supply enough); (2nd) 35 ft of thinner wire, including light, flexible insulated wire; (3rd) 35 ft of smooth string, preferably about the diameter of coat hanger wire.

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## 4. The pull cord:

° ( 1 st) At least 10 ft of cord; (2nd) strong string; (3rd) flexible, light wire.

## 5. The flap-stops:

° (1st) 150 ft of light string; (2nd) 150 ft of light, smooth wire; (3rd) 150 ft of very strong thread, to provide 4 threads for each stop-flap.

° (1st) 90 tacks (not thumbtacks); (2nd) 90 small nails. (Tacks or nails are desirable but not necessary; the flap-stops can be tied to the frame.)

## B. TOOLS

A hammer, saw, wirecutter pliers, screwdriver, scissors, knife, yardstick, and pencil are a strong, sharp knife is essential for making some models.

### C. BUILDING A KAP 36 INCHES HIGH BY 29 INCHES WIDE

A 36 X 29-in. KAP is most effective if operated in an air-intake or exhaust opening about wide. (If your shelter might have more than 25 occupants in hot weather, read all these instructions and understand how to build a larger pump, briefly described in Section VIII.)

NOTE THAT THE WIDTHS AND THICKNESSES OF ALL FRAME PIECES ARE EXAGGERATED IN ALL ILLUSTRATIONS.

#### 1. The frame

a. Cut two pieces of 1 X 2-in. boards, each 36 in. long, and two pieces of 1 X 2-in. board then nail them together (see Fig. 3). Use nails that do not split the wood, preferably long through the boards and stick out about 1/2 in. on the other side. (To nail in this manner, under the frame so that the nail points will not strike the floor.) Bend over nail points

Next, cut and nail to the frame a piece of 1 X 1-in. lumber 36 in. long, for a center vertical brace (if you lack time to make or to find a 1 X 1-in. board, use a 1 X 2-in. board.) Figure 3 shows the frame; the flap valves will be attached on the front (the opposite) side.

Fig. 3. KAP frame (looking at the back side of the frame). ORNL 71-7003 (nwl96.tif)

b. To make the front side smooth and flat so that the flaps will close tightly, fill in the two pieces of 1 X 2-in. boards long enough to fill in the spaces on top of the 36-in. side between the top and bottom horizontal boards, and nail these filler boards in place. Do this with a 1 X 1-in. board (or a board the size of that used for the center brace) as a filler board brace (see Fig. 4).

If the frame is made of only one thickness of board 3/4 in. to 1 in. thick, it will not be able to swing back far enough on its free-swinging return stroke.

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#### 2. The hinges

Ordinary door butt-hinges are best. So that the pump can swing past the horizontal position, screw them onto the front of the frame, at its top, in the positions shown in Fig. 4. (Pick one at the top.) If you do not have a drill for drilling a screw hole, you can make a hole by using a nail. Screw the screw into the nail hole.

Fig. 4. Completing the frame. ORNL- DWG 66-12322A (nwl97a.tif)

#### 3. The pivot-wires and flaps

a. Make 9 flap pivot-wires. If you have smooth, straight wire as springy and thick as coat hanger wire, make nine 29-in. -long straight lengths of wire. If not, use wire from all-wire coat hanger strings. First cut off all of the hook portion of each coat hanger, including the twisted

only ordinary pliers, use the cutter to "bite" the wire all around; it will break at this

Next, straighten each wire carefully. Straighten all the bends so that each wire is straight as compared to a straight line. Proper straightening takes 1 to 5 minutes per wire. To straighten, repeatedly grasp the bent part of the wire with pliers in slightly different spots, each time pulling a little with your other hand. Then cut each wire to a 29-in. length. Finally, bend no more than 1/2 in. at each end at a right angle and in the same plane that is, in directions so that all parts can lie flat against a smooth surface. The bent ends are for secure attachment later (see Fig. 5).

b. Make 9 polyethylene flaps that will be the hinged valves of the KAP. First cut 9 strips of plastic 30 in. long by 5- 1/2 in. wide (see Fig. 5). To cut plastic flaps quickly and accurately, use a sharp knife. Cut plastic 30 in. wide. Then cut off a flap in this way: (1) draw a cutting guideline on a work surface; (2) place the 30-in. -wide plastic strip so that it lies on this board, with its edges just reaching the edge of the board; (3) place a second board over the plastic on the work surface with a straight edge of this second upper board over the guideline on the lower board; and (4) cut off a flap by running a sharp knife along the straight edge of the upper board.

Fig. 5. ORNL DWG 7 1-7004A (nwl97b.tif)

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To form a hem along one of the 30-in. sides of a 5- 1/2 X 30-in. rectangular strip, fold it in half lengthwise to make the finished flap 4- 1/2 in. wide.

To hold the folded hem while taping it, paper clips or another pair of hands are helpful. Use two pieces of pressure-sensitive tape, each about 1 in. wide and 16 in. long. Or make the hem by taping it very close to the cut edge to form a hem-tunnel (see Fig. 5).

After the hem has been made, cut a notch with scissors in each hemmed corner of the flap (Fig. 8). Avoid cutting the tape holding the hem. Each notch should extend downward about 1/2 in. and extend horizontally from the outer edge of the flap to 1/4 in. inside the inner side of the flap. The flap is positioned on the frames as shown in Fig. 6.

Fig. 6. Sizes of notches in flaps. ORNL-DWG 66-12324 (nwl98a.tif)

Also cut a notch in the center of the flap (along the hem line) extending 1/2 in. downward and horizontally 1/4 in. beyond each of the two sides of the vertical brace (see Fig. 6). The notch should be wider than the brace. [However, if you are building a pump using wire netting for flap-stop, then do NOT cut a notch in the center of each flap.]

c. Take the 9 pieces of straightened wire and insert one of them into and through the hem of each flap, like a curtain rod running through the hem of a curtain. Check to see that each flap is attached to its pivot-wire, as illustrated by Fig. 7. Also see Fig. 8.

Fig. 7. End view. ORNL-DWG 66-12325 (nwl98b.tif)

Fig. 8. Flap. ORNL DWG 71-7005A (nwl98c.tif)

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d. Put aside the flaps and their pivot-wires for use after you have attached the flap-stop to the frame, as described below.

e. Using the ruler printed on the edge of this page, mark the positions of each pivot-wire (numbered 0, 3-5/8, 7-1/4 in.) and the position of each flap-stop (the four marks between each numbered arrowheads on this ruler). All of these positions should be marked both on the vertical and the 36-in. -long boards of the frame and on the vertical brace. Mark the position of the upper pivot-wire (the "0" arrowhead on this ruler) 1/4 in. below the top board to which the hinge is attached (see Figs. 9 and 10).

Fig. 9. (Frame 3/4 View) ORNL DWG 71-7006A (nwl99a.tif)

Fig. 10. (Top of Frame Straight on View) ORNL-DWG 66-12328A (nwl99b.tif)

#### 4. The flap-stops

So that the flaps may swing open on only one side of the frame (on its front, or face), you will use flap-stops made of strings or wires across the face of the frame. (See Figs. 10 and 11.) No flap-stops between the marked points where each pair of the horizontal pivot-wires for the flaps are attached. Do not to connect any flap-stops in such a way that they cross the horizontal open spaces in the flap pivot-wires.

Fig. 11. Positions of pivot-wires and flap-stops. ORNL -DWG 71-7007A (nwl99c.tif)

If you have tacks (NOT thumbtacks) or very small nails, drive three in a horizontal line through the frame: one in each of the two vertical 36-in. sides of the frame and one in the vertical center brace. Push all of these horizontal lines of tacks about three-quarters of the way into the boards. Then take a string or thin wire quickly to a tack, wind the string around the tack and immediately drive the string into the frame to grip the string (see Fig. 11).

If you have no tacks or nails, cut notches or slots where the flap-stops are to be attached into the edges of the vertical sides of the frame and in an edge of the center brace. Next, secure the flap-stops (wires) by tying each one in its notched position. This tying should include wrapping each wire around the vertical center brace. The stops should be in line with (in the same plane as) the flap-stops. Do not stretch flap-stops too tightly, or you may bend the frame.

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#### 5. Final assembly

a. Staple, nail, or tie the 9 flap pivot-wires or pivot-strings (each with its flap attached) to the marked 3-5/8 in. spacings. Start with the lowest flap and work upward (see Fig. 11). Connect each pivot-wire at both ends to the 36-in. vertical sides of the frame. Also connect it to the vertical center brace. BE CAREFUL TO NAIL THE PIVOT-WIRES ONLY TO THE FRAME AND THE BRACE. DO NOT NAIL ANY PLASTIC DIRECTLY TO THE WOOD. All flaps must turn freely on their pivot-wires.

If any flap, when closed, overlaps the flap below it by more than 1 in., trim off the excess.

overlaps by only 1 in.

b. Screw (or nail, if screws are not available) the upper halves of the hinges onto the ho board on which the KAP will swing. (A 1-in. -thick board is best, 3- 1/2 in. wide and at l than the width of the doorway or other opening in which this KAP is to be installed.)

Be careful to attach the hinges in the UNusual. OUT-OF-LINE POSITION shown in Fig. 12.

CAUTIONS: Do NOT attach a KAP's hinges directly to the door frame. If you do, the hinges w tom loose on its return stroke or on its power stroke.

If you are making a KAP to fit into a rectangular opening, make its frame 4 in. SHORTER th height of its opening and 1 in. NARROWER than the width of the opening.

c. For this 3-ft model, tie the pull-cord to the center brace about 12- 1/2 in. below the in Fig. 12. (If you tie it lower, your arm movements will waste energy.) Use small nails c keep the tie end from slipping up or down on the center brace. (For a more durable connect Fig. 22.)

Cut a slot in the flap above the connection of the pull-cord to the vertical brace, deep e flap will close completely when the KAP is being pulled. Tape the end and edges of the slo

Fig. 12. Hinge is attached so pump can swing 180 degrees. ORNL-DWG 66-12330AR (nw200.tif)

#### IV. MORE RAPID CONSTRUCTION

(Skip this section if you cannot easily get chicken wire and 1/4 in.-thick boards.)

If chicken wire and boards about 1/4 in. thick are available, use the chicken wire for fle materials, the time required to build a given KAP can be reduced by about 40%. One-inch w (Hardware cloth has sharp points and is unsatisfactory.)

Figure 13 illustrates how the mesh wire should be stapled to the KAP frame. Next, unless t 3 ft, the front of the whole frame (except for the center brace) should be covered with th in. thick, such as laths. Then the pivot-wires, with their flaps on them, should be staple This construction permits the flaps to turn freely in front of the chicken- wire flap-stop

With this design, the center of each pivot-wire should NOT be connected to the center brac ter of the flap be notched. However, pivot-wires that are attached this way must be made a pivot-wires used with flap-stops made of straight strings or wires.

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Fig. 13. Flaps attached 1/4 inch in front of chicken wire used for flap-stops. ORNL-DWG 66-12333A (nw201a.tif)

Note in Fig. 13 that each pivot-wire is held firm and straight by 2 staples securing each at least as springy as coat hanger wire. If string is used instead of wire, nylon cord abo wire is best for the pivot-strings.

If the KAP is wider than 3 ft, its center vertical brace should also be covered with a 1/4 pivot-wire should be attached to it. Furthermore, the center of each flap should be notched.

## V. INSTALLATION AND ACCESSORIES

### A. MINIMUM OPEN SPACES AROUND A KAP

To pump its maximum volume, an air-supply KAP with good metal hinges should be installed so that it swings only about 1/2 in. above the bottom of the opening and only 1/2 in. to 1 in. from the top.

### B. ADEQUATELY LARGE AIR PASSAGEWAYS

When using a KAP as an air-supply pump to force air through a shelter, it is essential to provide a low-resistance air passageway all the way through the shelter structure from an outdoor air-intake door or air to a separate air-exhaust opening to the outdoors (see Fig. 14).

Fig. 14. (ORNLDWG) 66-12331AR2 (nw201b.tif)

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A low-resistance air passageway is one that is no smaller in cross-sectional area than the opening through which the air is being pumped. For example, a 36 X 29-in. KAP should have a passageway no smaller than a 6 X 6-in. duct. An air-supply KAP of this size will force at least 1000 cubic feet per minute (1000 cfm) through the passageway, if it is installed as illustrated in Fig. 14.

If smaller air passageways or air-exhaust openings are provided, the volume of air pumped will be reduced. For example, if the air-exhaust opening is only 1-3/4 sq. ft (1/4 the size of this opening), only about 500 cfm will be pumped. And if the air-exhaust opening is only a 6 X 6-in. exhaust duct (1/4 the size of the 36 X 29-in. KAP), only about 50 cubic feet per minute will be pumped. This would not provide enough outdoor air for one shelter occupant in a well-insulated shelter under heat-wave conditions in the hottest States. In contrast, when the weather is freezing cold and the shelter itself is still cold, the air is drawn by the shelter occupants, this same 6 X 6-in. exhaust duct and the air-intake doorway will allow a large volume of outdoor air to flow by itself through the shelter without using any pump. The reason: the cold air is drawn in, and the warm air rises if cold air can flow in to replace it. Under these cold conditions, 50 cfm is enough outdoor air for one occupant.

air, and the warm air rises if cold air can flow in to replace it. Under these cold conditions, 50 cfm is enough outdoor air for one occupant.

To provide adequately large air passageways for air-supply KAPs used to ventilate shelters, attention to opening and closing doors and windows, it may be necessary to build large ducts (a large hole in windows, ceilings, or walls is another way to make large, efficient air passageways).

Figure 15 illustrates how a 3-ft KAP can be used as a combined air-supply and air-distribution device to ventilate a small underground shelter that has an exhaust opening too small to provide enough outdoor air. (A similar installation can be used to ventilate a basement room having only one window, by installing a divider in the doorway and entryway, the single entryway is converted into a large air duct and a separate, large air-exhaust duct. To obtain the maximum increased volume of fresh air pumped through the shelter a total of about 1000 cfm for a 36 X 29-in. KAP the divider should be installed horizontally into the shelter room, as shown in Fig. 15. The 6 ft at the end of the divider (the KAP) can be made of plywood, provided it is installed so that it can be taken out and replaced by a larger KAP if needed.)

Fig. 15. Ventilating a shelter when the air-exhaust opening is too small. ORNL DWG 72-663 (nw202.tif)

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Note how the entry of fallout into a shelter can be minimized by covering the entryway with a screen, allowing the slow-moving entering air to rise over an obstruction (the "wall") before it flows in. This way, fallout particles fall to the ground outside the "wall."

#### C. ADEQUATE DISTRIBUTION OF AIR WITHIN THE SHELTER

To make sure that each shelter occupant gets a fair share of the outdoor air pumped through the shelter, air-distribution KAPs should be used inside most large shelters. These KAPs are used within the shelter and in addition to air-supply KAPs (see Fig. 16). Air-distribution KAPs can serve in place of ducts and cooling fans. For these purposes, one or more 3-ft-high KAPs hung overhead from the ceiling are usually most practical. If KAPs cannot readily be hung from the ceiling, they can be supported by boards or metal, somewhat like those used for a small child's swing.

Fig. 16. The use of air distribution KAPs. ORNL DWG 72-7547 (nw203a.tif)

You should make and use enough KAPs to cause air movement that can be felt in all parts of the shelter. Remember that if KAPs are installed near the floor and the shelter is fully occupied, the occupants can block the pumped airflow more than if the same KAPs were suspended overhead.

As a general rule, for shelters having more than about 20 occupants, provide one 3-ft air-distribution KAP for every 25 occupants. In relatively wide shelters, these interior KAPs should be positioned so that air circulates around the shelter, preventing the air that is being pumped into the shelter from being exhausted. Figure 16 illustrates how four KAPs can be used in this way to distribute air to fan the 100 occupants of a 1000-sq.-ft shelter room. Avoid positioning an air-distribution KAP in a direction greater than a right angle turn from the direction of airflow to the location of the occupants.

#### D. OPERATION WITH A PULLEY

A small KAP especially one with improvised hinges or one installed at head-height or higher can be operated efficiently by running its pull-cord over a pulley or over a greased homemade pulley" such as shown in Fig. 17. A pulley should be hung at approximately the same height as the hinges of the KAP. To make

.Fig. 17. Improvised "Pulley" From A Wide-Angled Forked Limb. ORNL DWG 71-7242 (nw203b.tif)

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Fig. 18. IMPROVISED "PULLEY" ORNL DWG 71-7243 (nw204a.tif)

a comfortable hand-hold on which to pull downward, tie two or three overhand knots in a loop at the end of the pull-cord.

(Such a "pulley" can also be used to operate a bail-bucket to remove water or wastes from anyone having to go outside.)

#### E. QUICK-REMOVAL BRACKETS

The air-supply KAP that pumps air through your shelter is best held in its pumping position by homemade quick-removal brackets (see Fig. 19) for the following reasons:

- ° A KAP provided with quick-removal brackets can be taken down easily and kept out of the way of persons passing through its doorway when it is not in use. It can be kept in a place where it is unlikely to damage it.

- ° By installing two sets of quick-removal brackets in opposite shelter openings, you can control the direction in which the KAP pumps air, to take advantage of changes in the direction of natural wind flow through the shelter.

Fig. 19. Quick -removal bracket for KAP. ORNL DWG 72-6365A (nw204b.tif)

- ° If the KAP is installed on quick-removal brackets, in an emergency a person standing beside it could grasp its frame with both hands, lift it upward a few inches to detach it, and carry it away in 3 to 5 seconds. Being able to move the KAP quickly could prevent blast winds from blowing it off a pump, which might also be blown into your shelter possibly injuring occupants. In extensive fallout shelters and their occupants would survive the blast effects of typical large warhead explosions, several seconds would elapse between the time shelter occupants would see the extremely bright light of an explosion and the arrival of a blast wave strong enough to wreck a KAP or other pumps left in place as a ventilation opening.

Note in Fig. 19 that the KAP's "fixed" support-board (a 3-1/2 in. -wide board to which it is attached) is held in a bracket only 2 inches deep. To prevent too tight a fit in the bracket, be sure to use a spacer (the cardboard back of a writing tablet will do) between two boards of the bracket, leaving spaces about 1/16 inches wide between the lower inner corners of the stop-blocks and the sides of the support-board.

To prevent your hands from being cut, you should put tape over the exposed ends of wires and sharp edges of a KAP that you want to be able to remove rapidly.

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In a small expedient shelter, a small KAP can be quickly jerked loose if its "fixed" support-board is attached to the roof with only a few small nails.

### VI. OPERATION AND MAINTENANCE

#### A. PUMPING

Operate your 3-foot KAP by pulling it with an easy, swinging motion of your arm. To pump it out of air, you should pull the KAP toward you until its frame swings out to an almost- horizontal position. Then move your hand so that the pull-cord is kept slack during the entire, free-swinging return motion. Section VIII, LARGE KAPs, illustrates this necessary motion.

Be sure to provide a comfortable hand-hold on the pull-cord (see Fig. 14). Blisters can be prevented by using gloves or padding in conditions.

To pull a KAP via an overhead pulley with minimum effort, sit down and pull as if you were that you should raise your hand quickly with the return stroke and keep it raised long enough

remains slack during the entire return stroke. Or, if the pulley is not overhead, operate extended arm back and forth from the shoulder. -

#### B. PLACEMENT TO TAKE ADVANTAGE OF THE NATURAL DIRECTION OF AIR FLOW

A KAP can pump more air into a shelter if it is installed so that it pumps air through the direction in which the air naturally flows. Since this direction can be reversed by a wind change outside, a way to quickly remove your pump and reposition it so that air can be pumped in the opposite direction is done in several ways, including making one set of quick-removal brackets for one air pump and the other.

#### C. MAINTENANCE

To operate your KAP efficiently, keep the flaps in good repair and make sure that there is an area of open spaces in and around the KAP through which air can flow back around the pump in the pumped direction. So keep at least some extra flap material in your shelter, along with some tools you may need to make repairs.

### VII. NARROW KAPS AND SMALL KAPS

#### A. NARROW KAPS

To swing efficiently in an entrance or emergency exit of an expedient trench shelter that is best made 20 in. wide and 36 in. high. One of less height is not as efficient as a 36-in.-high one, but is uncomfortably fast. So, when ventilation openings can be selected or made at least 38 in. high.

In a narrow trench shelter, it is best to have the pull-cord run the full length of the trench so that occupants will face when sitting. Then each occupant can take a turn pulling the pump and seats.

Good metal hinges on a narrow KAP allow it to swing properly if pulled with the pull-cord attached to the frame. (Pumps with improvised hinges and large pumps must be pulled from a connection to a vertical brace to make them swing properly.) Therefore, if you have small metal hinges and the KAP is less than 20 inches wide, build a rectangular frame without a vertical center brace. Make two pull-cord attachments on each side of the frame and each 9 inches below the top of the frame. (For a small KAP, a pivot point can readily be made by driving two nails so that their heads cross, and wiring change in wind direction outside causes the direction of natural air flow in the trench to change. In the direction in which air is being pumped, you can move your KAP to the opening at the other end and the pull-cord can easily be connected to the other side of the frame, and convenient pumping can be maintained.)

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So that the horizontal support board can be nailed easily to the roofing poles or boards and to use cabinet hinges. Screw them onto an edge of the support board, in the UNusual. OUT-COMMON position shown in Fig. 20. This hinge connection allows the pump to swing a full 180 degrees. To find

tal support board, connect it to the roof with a few small nails, so that it can be pulled

Fig. 20. Roof Poles. Horizontal Support Board. ORNL-DWG 78- 10358 (nw206a.tif)

## B. SMALL KAPS

If the only available opening in which a KAP can be installed is small, build a KAP to fit to make the frame and make the flaps of thinner material, such as the polyethylene of large pumps 24 inches or less in height, make the finished flaps only 3- 1/2 inches wide and space

apart. The flaps should overlap no more than inch. A KAP 24 inches high will pump enough for a few people, except in cold weather.

Small, yet efficient KAPs can be made even if the only materials available are straight sticks of diameter, strips of cloth to tie the frame together and to make the hinges and the pull cords. Use large trash bags for the flaps, freezer or duct tape (or needle and thread) to make the flap-stops, string for the pivot- wire, and string or ordinary thread for the flap-stops. A sharp knife and a few nails and tacks are not available. The flap-stop strings or threads should be secured by wrapping around each stick to which they are attached, so they will be gripped by the out-of-line KAP.

Fig. 21. Sticks ready to be tied together to make a KAP frame. ORNL-DWG 78-2 1897 (nw206b.tif)  
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## VIII. LARGE KAPS

### A. CONSTRUCTION

A 6-ft-high by 29-in. -wide model can be constructed in the same way as a 3-ft model except for both horizontal and vertical center braces (IX 2-in. boards are best). To increase the strength of its double- thickness frame and its vertical center brace should be made of two thickness boards, securely held together with clinched nails. Also, to increase the distance that the pump travels during its return stroke, it is worthwhile to attach a 6-ft piece of 1 X 2-in. board to each side of the frame. Do NOT attach weights to the bottom of the frame; this would slow

This 6-ft-high pump requires 18 flaps, each the same size as those of the 36-in. -high KAP. Each part of a large KAP must withstand hard use. If 1/2-in.-wide strips of tape are attached to the edges of these lower flaps, then even flaps made of ordinary 4-mil polyethylene will remain in use for hours of pumping. However, the lower flaps of large KAPs can advantageously be made of 6-mil polyethylene. The width and spacing of all flaps should be the same as those of the 3 6-in. -high model.

The pull-cord should be attached to the vertical center brace of a 6-ft KAP about 1 6- 1/2 ft from the top. A 3/ 16-in. nylon cord is ideal.

To adequately ventilate and cool very large and crowded shelters in buildings, mines, or caves, a 72 X 29 in. should be used. You can take better advantage of large doorways, elevator shafts, etc., by making" each large air- supply KAP to the size of its opening that is, by making it as large as the opening and brace members should be appropriately strengthened, and one or more "Y" bridles should be scribed in the section below. A 7-ft-high X 5-1/2 ft- wide KAP, with a 1/4 in. -diameter pivot pin at its hinge line, and with two "Y" bridles for its two operators, pumped air at the rate of

through a large basement shelter during tests.

To make a durable connection of the pull-cord to the center vertical brace: (1) Attach a wire 16- 1/2 in. below the hinge line. This loop can be made of coat hanger wire and should go over the center brace. This fixed loop should be kept from slipping on the center brace by bending four 6-3/8 in. nails as illustrated, and two smaller nails in back. (2) Make a free-turning, triple-wire loop of the same size. Cover part of the free-turning loop with tape and tie the pull-cord to this loop. Tie the other end of the pull-cord to the part.

Fig. 22. (Pull Cord to center vertical brace) ORNL DWG 72-8204 (nw207a.tif)

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## B. OPERATION OF LARGER KAPS

A larger KAP can be pulled most easily by providing it with a "Y" bridle (see Fig. 23) attached to the pull-cord.

Fig. 23. Y-bridle for pullcord on KAP. ORNL DWG 71-8069 (nw208a.tif)

A man of average size and strength can operate a 6 ft X 29 in. KAP by himself, pumping over 3000 cfm per minute through a typical large shelter without working hard; tests have shown that he can operate 1/20 of a horsepower. However, most people prefer to work in pairs when pulling a 6-ft KAP with a Y-bridle, when pumping over 3000 cfm.

To pump the maximum volume of air with minimum effort, study Fig. 24 and follow the instructions for operating a large KAP.

Fig. 24. At End Of Its Forward Swing, Pump Should Just Touch Tightened Pull-Cord. ORNL DWG 72-6526 (nw208b.tif)

1 . Gradually start the pump swinging back and forth, moving your arms and body as illustrated. Move mostly with your legs and body.

2. Stand at such a distance from the pump that you can pull the pump toward you until the pump just touches the tightly stretched pull-cord and at such a distance that you can keep the pull-cord slack during the whole of the pump's free backswing.

3. To be sure you do not reduce the amount of air pumped, rapidly move your arms forward as the forward-swinging pump touches the tightened pull-cord. Hold your arms forward until the pump starts to swing toward you.

## IX. SOLUTIONS TO SPECIAL PROBLEMS

### A. INCREASING THE USEFULNESS OF SHELTERS BY SUPPLYING 40 CFM PER



## PLANNED OCCUPANT

If a shelter is fully occupied for days during hot weather and is cooled both day and night and distributing at least 40 cubic feet per minute of outdoor air for each occupant more tolerable temperatures at night these advantages result:

- ° The shelter occupants will be exposed to effective temperatures less than 2° F higher than effective temperatures outdoors, and at night will get relief from extreme heat.

- ° The floors, walls, etc. of a shelter so ventilated will be cooled at night to temperatures. Therefore, during the day a considerable fraction of the occupants' body heat the floors, walls, and other parts of the shelter and less body heat will have to be carried exhaust air during the hottest hours of the day. Thus daytime temperatures will be reduced

- ° Since the shelter occupants will be cooler and will sweat less, especially at night, than they would require if the shelter were ventilated at a rate of less than 40 cfm per occupant outdoor air is very hot and desert-dry, it usually is better to supply less than 40 cfm per occupant the hottest hours of the day.)

- ° If the shelter were to be endangered by the entry of outside smoke, carbon monoxide or other gases, or heavy descending fallout under windy conditions, ventilation of the shelter could be restricted or stopped for a longer period than would be practical if the shelter itself were the beginning of such a crisis period.

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The shelter could be occupied beyond its rated capacity without problems caused by overcrowding becoming as serious as would be the case if smaller- capacity air pumps were to be installed.

## B. PRE-COOLING SHELTERS

If the shelter itself is cooler than the occupants, more of the body heat of occupants can be carried by the ceiling, and floor. Therefore, it would be advantageous to pre-cool a shelter that may soon be used during hot weather. KAPs (or other air pumps or fans) can be used to pre-cool a shelter by forcing outdoor air through the shelter and by distributing it within the shelter. A shelter is used sometimes when the air temperature outdoors is lower than the air temperature inside the shelter. If a shelter is used, the occupants will be kept cooler at a given rate of ventilation than if the shelter were cooled, because the air will not have to carry all of their body heat out of the shelter.

## C. INCREASING THE EFFECTIVENESS OF A KAP

If you want to increase the volume of air that a KAP with good metal hinges can force through side baffles (see Fig. 25). Side baffles should be rigidly fixed to form two stationary "wing" swinging pump frame. They can be made of plywood, boards, doors, table tops, or even well-laminated. A clearance of 1/2 to 1 in. should be maintained between the inner side of each baffle and the swinging frame.

By installing side baffles you may be able to increase the volume of air your KAP will pump.



of small boards. It can be used to discharge the exhaust air far enough away from the KAP airflow outside the one-opening room so that no exhausted air can be "sucked" back into the

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Fig. 27. Use of a "divider" to ventilate a shelter with only one opening. ORNL DWG 75-8203 (nw211a.tif)

#### F. INSTALLING A KAP IN A STEEL-FRAMED DOORWAY

If you need to install a KAP in a steel-framed doorway and it is not feasible to screw or connect it to the doorway, you can attach the KAP by using a few boards and some cord, as shown in Fig. 29. The two horizontal boards shown extending across the doorway are squeezed tightly against the wall in which the doorway is located by tightening two loops of cord, one near each side as is illustrated. A cord is first tightened around the two horizontal boards. Then the loops are bound together in the center with another cord, as illustrated.

Two large "C" clamps serve even better than two looped cords. However, secure support for the KAP still requires the use of a vertical support board on each side of the doorway, as illustrated in Fig. 28.

Fig. 28. Steel Door Frame. ORNL DWG 72-6564 (nw211b.tif)

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Figure 29 shows a quick-removal bracket supported by two horizontal boards tightened across the doorway by looped cords, as described above. Also, study Fig. 19 and its accompanying instructions.

Fig. 29. (Quick Removal Bracket on Doorway) ORNL DWG 72-6617 (nw212a.tif)

#### G. BUILDING MORE DURABLE KAPS

If you are building KAPS in normal times, you may want to use materials that will make your KAPs even though these materials are more difficult to obtain and are more expensive.

Durability tests have shown that the KAP parts that wear out first are the flaps and the pulleys. The lower flaps are subject to hard use. Lower flaps made of 6-oz (per sq. yd), clear, nylon-66 have lasted undamaged for over 1000 hours of full-stroke pumping, without having their edges reinforced. The pulleys made of 6-mil nylon-reinforced polyethylene, without edge reinforcements, have lasted for over 324 hours without damage.

The best pulley tested was a marine pulley such as that used on small sailboats, with a Deere 2-in. -diameter wheel and 3/16-in. stainless steel shaft. This pulley was undamaged after 324 hours. The pulley appeared to be good for hundreds of hours of further operation.

The best pulley-cords tested were of braided dacron or nylon.

#### H. USING AIR FILTERS

To supply shelter occupants with filtered air usually would be of much less importance to health than to provide them with adequate volumes of outdoor air to maintain tolerable temperature. Filtering the entering air could prove worthwhile, provided:

- ° Your shelter is not in an area likely to be subjected to blast, or it is a blast shelter with blast valves protecting everything inside.
- ° Work on filters is started after you have completed more essential work, including the building of a high-protection-factor shelter, making, installing, and testing the necessary number of KAPs, obtaining adequate water, making a homemade fallout meter, etc.
- ° You have enough low-resistance filters (such as fiberglass dust filters used in furnace air conditioners) and other materials for building the necessary large, supported filter in frame.
- ° Your KAP can pump an adequate volume of air through the filter and shelter.
- ° The filter is installed so that it can be easily removed if shelter temperatures rise too high.

To prevent a filter used with a KAP from causing too great a reduction in the volume of air pumped through your shelter, you must use large areas of low-resistance filter material. An actual test, a large basement shelter was used which had two ordinary doorways at its opposite ends, one for its air-intake and its air-exhaust openings. A 72 X 29 in. KAP operating in one doorway pumped 3400 cfm through the shelter. But when a filter frame holding 26 square feet of 1-inch-thick fiberglass dust filter was placed across the air-intake stairwell, the KAP could pump only about 3400 cfm through the shelter.

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## Appendix C

A Homemade Fallout Meter, the KFM  
- How to Make and Use It

FOLLOWING THESE INSTRUCTIONS MAY SAVE YOUR LIFE

Fig. (Nuclear Ground Burst Explosion - Drawing) (nw2 13.tif)

The complete KFM instructions include patterns to be cut out and used to construct the fallout meter. The instructions are extra patterns on 4 unnumbered pages. The reader is urged to use these KFM's in normal peacetime and to keep the complete instructions intact for use during a real emergency.

If Xerox copies of the patterns are used, they should be checked against the originals in order to be sure they are the same size as the originals. Some older copiers make copies with slightly enlarged dimensions. Enlarged copies of all the KFM patterns can be made satisfactory provided: (1) on the PAPER AROUND KFM CAN, the distances between the 4 marks for the HOLES FOR STOP-THREAD are corrected, and (2) the dimensions of the FINISHED-LEAF PATTERN are corrected.

These instructions, including the heading on this page and the illustrative photos, can be all screening and rapidly reproduced by a newspaper or printer. If you keep the KFM instructions during a worsening crisis you will be able to use them to help your friends and thousands of your friends. Make them available for reproduction.

Pg. 1 - (1) LOGO (The reverse sides of these pages are numbered to allow for the creation of an appendix.)

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## I. THE NEED FOR ACCURATE AND DEPENDABLE FALLOUT METERS

If a nuclear war ever strikes the United States, survivors of the blast and fire effects will need some means of knowing when the radiation in the environment around their shelters had dropped enough to go safely outside. Civil defense teams could use broadcasts of surviving radio stations to give information about the fallout radiation in some broadcast areas. However, the fallout radiation can vary widely and the measurements are likely to be made too far from most shelters to make them accurate enough. Therefore, each shelter should have some dependable method of measuring the changing radiation in the area.

During a possible rapidly worsening nuclear crisis, or after a nuclear attack, most unprepared people will not buy or otherwise obtain a fallout meter — an Instrument that would greatly improve their chances of surviving a nuclear war. The fact that the dangers from fallout radiation— best expressed in terms of the radiation dose in rads per hour (R/hr) — quite rapidly decrease during the first few days, and then decrease more slowly, makes it very important to have a fallout meter capable of accurately measuring the unseen fallout dangers. Occupants of a fallout shelter should be able to minimize the radiation dose received. To effectively minimize the radiation doses, a dependable measuring instrument is needed to check the dose received while they are in the shelter and while they are outside for emergency tasks, such as getting food and water. Also, such an instrument would permit them to determine when it is safe to leave the shelter. Untrained families, guided only by these written instructions and using only low cost materials, have been able to make a KFM by working 3 or 4 hours. By studying the operating sections for about 1-1/2 hours, average untrained families have been able to successfully use the KFM to measure dose rates and to calculate radiation doses received, permissible times of exposure, and to determine when it is safe to leave the shelter.

The KFM (Kearny Fallout Meter) was developed at Oak Ridge National Laboratory. It is under development, and as accurate as most civil defense fallout meters. In the United States in 1950, there was no commercially available dose-rate meter that is accurate and dependable and that measures both indoor and outdoor dose rates.

The KFM is a simple, accurate, and dependable instrument that measures both indoor and outdoor dose rates. Its wartime use is a British instrument that retails for \$375. Comparable American instruments are not available.

## II. SURVIVAL WORK PRIORITIES DURING A CRISIS

Before a nuclear attack occurs is the best time to build, test and learn how to use a KFM. The KFM is so simple that it could be made even after fallout arrives provided that all the materials listed in Sections V, VI, and VII) and a copy of these instructions have been carried into the shelter.

Before building a KFM, persons expecting a nuclear attack within a few hours or days and who are not in a shelter they intend to await attack should work with the following priorities : ( 1 ) build or improve the shelter (if possible, a shelter covered with 2 or 3 feet of earth and separate from flammable materials) and make and install a KAP (a homemade shelter- ventilating pump) - if instructions and materials are available.

available, at least make a Directional Fan. Also store at least 15 gallons of water for each container is available. (2) Assemble all materials for one or two KFM s. (3) Make and storing wallboard gypsum, as later described) for both the KFM and its dry-bucket. (4) Complete

### III. HOW TO USE THESE INSTRUCTIONS TO BEST ADVANTAGE

1. Read ALOUD all of these instructions through section VII, "Tools Needed," before doing
2. Next assemble all of the needed materials and tools.
3. Then read ALOUD ALL of each section following Section VII before beginning to make the in that section.

A FAMILY THAT FAILS TO READ ALOUD ALL OF EACH SECTION DESCRIBING HOW TO MAKE A PART, BEFORE BEGINNING TO MAKE THAT PART, WILL MAKE AVOIDABLE MISTAKES AND WILL WASTE TIME.

4. Have different workers, or pairs of workers, make the parts they are best qualified to less skilled worker should start making the drying agent (as described in Section VIII) before start making other parts. The most skilled worker should make and install the aluminum-foil X and XI).
5. Give workers the sections of the instructions covering the parts they are to build- so step-by-step instructions, checking off with a pencil each step as it is completed.
6. Discuss the problems that arise. The head of the family often can give better answers in different possible interpretations of some instructions with other family members, especially
7. After completing one KFM and learning to use it, if time permits make a second KFM- the better instrument.

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### IV. WHAT A KFM IS AND HOW IT WORKS

Fig. 215a. Charged KFM (nw215a.tif)

A KFM is a simple electroscope-ionization chamber fallout meter with which fallout radiation is measured. To use a KFM, an electrostatic charge must first be placed on its two separate aluminum leaves are insulated by being suspended separately on clean, dry insulating threads.

To take accurate readings, the air inside a KFM must be kept very dry by means of drying agent (easily made by heating gypsum wallboard, "sheetrock") or silica-gel. (Do not use calcium salt.) Pieces of drying agent are placed on the bottom of the ionization chamber (the house)

An electrostatic charge is transferred from a homemade electrostatic charging device to the leaves of a KFM by means of its charging-wire. The charging-wire extends out through the transparent KFM.

When the two KFM leaves are charged electrostatically, their like charges (both positive or both negative) force them to be forced apart. When fallout gamma radiation (that is similar to X rays but more penetrating) enters the ionization chamber of a KFM, it produces charged ions in this enclosed air. These ions neutralize or all of the electrostatic charge on the aluminum- foil leaves to be discharged. As a result, the KFM leaves move closer together.

A KFM-maker who wants visual proof that his instrument can be partially or wholly discharged should persuade his dentist to place the charged KFM about 20 inches directly below a typical dental X-ray machine. For example, when a typical 90 kvp machine was set at 15 milliamps and for a 1/20th second exposure, the X-ray beam partially discharged the KFM's separated aluminum-foil leaves, promptly reducing the separation from 19 mm to 9 mm. Other types of machines will require different settings. Many dental X-ray machines are not calibrated, nor do they produce gamma rays, so such tests should not be used in an attempt to verify the accuracy of a KFM.

Fig. 215b. (How to position yourself to take a reading.) (nw215b.tif)

To read the separation of the lower edges of the two KFM leaves with one eye, look straight down at the scale on the clear plastic cover. Keep the reading eye 12 inches above the SEAT. The KFM must be on a horizontal surface. To be sure the reading eye is always at this exact distance, place the ruler on the SEAT, while the upper end of the ruler touches the eyebrow above the reading eye. Hold the KFM can with one hand and the ruler with the other. Using a flashlight makes the reading more accurate.

If a KFM is made and maintained with the specified dimensions and of the specified materials, it will give a reading that is AUTOMATICALLY AND PERMANENTLY ESTABLISHED BY UNCHANGING LAWS OF NATURE. Unlike factory made radiation measuring instruments, A KFM NEVER NEEDS TO BE CALIBRATED OR TESTED AGAINST A RADIATION SOURCE. A KFM is used with a watch and the following table that is based on the calibrations made at Oak Ridge National Laboratory.

The millimeter scale is cut out and attached (see photo illustrations on the following pages) to the KFM so that its zero mark is directly above the two leaves in their discharged position on a horizontal surface. A reading of the separation of the leaves is taken by noting the position of the lower edge of one leaf appears to be on, on one side of the zero mark on the scale, and adding the number of millimeters the lower edge of the other leaf appears to be on, on the other side of the zero mark. The sum of these two apparent positions of the lower edges of the two leaves is called a KFM reading. The photo illustrations shows the lower edges of the leaves of a KFM appearing right of zero and 10 on the left, giving a KFM reading of 19 mm. (Usually the lower edges of the leaves are the same distance from the zero mark.)

As will be fully explained later, the radiation dose rate is determined by:

- 1 . charging and reading the KFM before exposure;
2. exposing it to radiation for a specified time in the location where measurement of the dose rate is desired; when outdoors, positioning the KFM about 3 ft. above the ground;
3. reading the KFM after its exposure;
4. calculating, by subtraction, the difference between the reading taken before exposure and the reading taken after exposure;

5. using this table to find what the dose rate was during the exposure – as will be descri

DIFFIN

READ

INGS

TIME INTERVAL OF AN EXPOSURE

Difference between the reading before exposure and  
the reading after exposure (8 ply standard foil leaves)

15 SEC.

R/HR

1 MIN.

R/HR

4 MIN.

R/HR

16 MIN.

R/HR

1 HR.

R/HR

2mm

6.2

1.6

0.4

0.1

0.03

4mm

12.

3.1

0.8

0.2

0.06



6mm

19.

4.6

1.2

0.3

0.08

8mm

26.

6.2

1.6

0.4

0.10

10 mm

31.

7.7

2.0

0.6

0.13

12 mm

37.

9.2

2.3

0.6

0.15

14 mm

43.

11.0

2.7

0.7

0.18

Instructions on how to use a KFM are given after those detailing how to make and charge the KFM.  
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To get a clearer idea of the construction and use of a KFM, look carefully at the following captions.

A. An Uncharged KFM. The charging wire has been pulled to one side by its adjustment-thread. The photo was taken looking straight down at the upper edges of the two flat, 8-ply aluminum leaves. The leaves are barely visible, hanging vertically side by side directly under the zero mark, together with their ends even. Their suspension-threads insulate the leaves. These threads are almost in contact (but do not cross) each other where they extend over the top of the rim of the can.

Fig. 216a (Top Down view of KFM) (nw216a.jpg)

B. Charging a KFM by a Spark-Gap Discharge from a Tape That Has Been Electrostatically Charged. The Unwound Quickly. Note that the charged tape is moved so that its surface is perpendicular to the charging-wire.

The high-voltage electrostatic charge on the unwound tape (that is an insulator) jumps the gap between the tape and the upper end of the charging-wire, and then flows down the charging-wire to the insulated aluminum-foil leaves of the KFM. (Since the upper edges of the two leaves are 3 inches apart on the scale and this is a photo taken at an angle, both leaves appear to be under the right side of the scale.)

Fig. 216b. (Unwound tape by KFM to produce static charge) (nw216b.jpg)

C. A Charged KFM. Note the separation of the upper edges of its two leaves. The charging-wire is raised to an almost horizontal position so that its lower end is too far above the aluminum leaves to cause electrical leakage from the leaves back up the charging-wire and into the outside air.

Also note the SEAT, a piece of pencil taped to the right side of the can, opposite the charging-wire.

Fig. 216c. A Charged KFM (nw216c.jpg)

D. Reading a KFM. A 12-inch ruler rests on the SEAT and is held vertical, while the reader reads the upper end of the ruler. The lower edge of the right leaf is under 8 on the scale and the left leaf is under 6 on the scale, giving a KFM reading of 14.

For accurate radiation measurements, a KFM should be placed on an approximately horizontal surface so that the charges on its two leaves and their displacements do not have to be equal.

Fig. 216d. Reading a KFM (nw216d.jpg)

NOTE: In these photos, the paper scale is taped to the top of the transparent plastic-film the scale to the under side of the cover, where it is less likely to be damaged.

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KFM Diagrams (Inside, Top, and Cutaway views) (nw217.tif)

## V. MATERIALS NEEDED

A. FOR THE KFM: (IN THE FOLLOWING LIST, WHEN MORE THAN ONE ALTERNATIVE

MATERIAL IS GIVEN, THE BEST MATERIAL IS LISTED FIRST.)

1 . Any type metal can. approximately 2-9/16 inches in diameter inside and 2-7/8 inches high. Clean with soap. (This is the size of a standard 8-ounce can. Since most soup cans, pop cans, etc. are also about 2-9/16 inches in diameter inside, the required size of can also can be made of height of more widely available cans as described in Section IX of these instructions.)

2. Standard aluminum foil - 2 square feet. (In 1987, 2 square feet of widely sold U.S. brand weighed between 8.0 and 8.5 grams. One gram equals 0.035 ounce.) (If only "Heavy Duty" or "Extra Duty" aluminum foil is available, make 5-ply leaves rather than 8-ply leaves of standard foil. The fallout meter will be almost as accurate.)

3. Doorbell- wire, or other light insulated wire (preferably, but not necessarily) a single strand of insulation - 6 inches.

4. Any type of clean, fine thread that has not been anti-static treated will serve to suspend the KFM leaves. (Almost all kinds of sewing thread and fly-tying thread manufactured in 1987 are anti-static treated, and are unsatisfactory. ) In 1987 the best widely available excellent insulating thread was dental floss; floss is not anti-static treated. Most unwaxed dental floss is too thick and stiff to use for suspending KFM leaves, but, since dental floss is not a twisted thread, you can make flexible threads from it. Make each no more than one-quarter as thick as the floss, and about 12 inches long. Cut several strands at the end of the floss outside its dispenser. Then separate strands while they are still at the end of the strand-thread that you want and the other way on the unwanted strands. Use a needle to touch and separate the strands in the middle 6 inches of the 12-inch-long piece of floss.

A widely sold dental floss. Johnson and Johnson's Extra Fine Unwaxed, drawn out of its dispenser by splitting it, makes quite satisfactory leaf-supporting threads. However, better leaf-supporting threads can be made by first separating any dental floss into thinner, more flexible threads.

Very thin monofilament fishing line or leader is an excellent insulator. The 2-pound- test Ponton's "Stren" monofilament fishing line, is best. "Trilene" 2-pound "nylon leader." a monofilament manufactured by Berkley and Company, also is excellent. (A 4-pound monofilament line will be disadvantageously stiff.) Some modern monofilament lines or leaders such as "Trilene" contain plasticizers that makes them pliant, but also makes them poorer insulators for the first several hours

of their dispenser and used to suspend the leaves of a KFM. However, in about 6 hours the anhydrite drying agent in a KFM removes this additive and the monofilament becomes as good even strands of unwaxed dental floss.

To minimize the chance of using a piece of monofilament or other thread that has been soiled or changed into a poor insulator, always first remove and discard the outermost layer of thread. Thread that has not been kept clean in a plastic bag or other packaging after being initially unwrapped should be discarded.

During a worsening crisis or after an attack, neither thread that has not been anti-static dental floss, nor clean 2-pound or 4-pound monofilament line may be available. However, most homes have an excellent insulator, very thin polyethylene film - especially clean dry clear insulating strip cut only 1/16 inch wide can be used to suspend each KFM leaf, instead of dental floss. (Installed leaves suspended on strips of thin plastic film must be handled with care.)

To cut 1/16-inch-wide strips from very thin polyethylene film, first cut a piece about 6x12 inches. Tape the two 6-inch-wide ends to a piece of paper (such as a brown grocery bag), so that the film is held flat and smooth on the paper. Make 10 marks, 1/16-inch apart, on each of the two tapes that are holding the film.

Place a light so that its reflection on the film enables you to see the edge of the film through the tape. Then use a very sharp, clean knife or clean razor blade, guided by the edge of a finger, to cut the film into 9 strips, of which you will select the best two. When cutting, hold the knife almost horizontally, with its blade perpendicular to the taped-down film. Throughout this procedure avoid touching the film with your fingers. Parts of the strips.

5. A piece of clear plastic film - a 6 x 6 inch square. Clear vinyl (4 mils thick) used for food wrap is best, but any reasonably stout and clear plastic will serve. The strong clear plastic used for cheese, if washed with hot water and soap, is good. Do not use weak plastic or cellophane. Paper from cellulose (such as Flex-O-Pane) and roasting bags are too permeable to water vapor.

6. Cloth duct tape ("silver tape"), or masking tape, or freezer tape, or Scotch-type tape. (A roll of Scotch Magic Transparent Tape, if available, should be saved for use in charging the KFM.)

7. Band-Aid tape, or masking tape, or freezer tape, or Scotch transparent tape, or other tapes - about 2 square inches.

8. Gypsum wallboard (sheetrock) - about 1/2 square foot, best about 1/2 inch thick, for a drying agent. (Silica gel with dark blue color indicator is an even better drying agent, but not available in most communities. Available from chemical supply firms that supply high school chemistry equipment. A small amount of blue silica gel in the bottom of a KFM, white typing correction fluid or white ink is needed to make the edges of a KFM's aluminum leaves easier to see.)

9. Glue - not essential, but useful to replace Band-Aid and other thin tapes. "One hour" epoxy glue or airplane cement is satisfactory.

10. An ordinary wooden pencil and a small toothpick (or split a small sliver of wood. )

11. Two strong rubber bands, or string.

12. Several small, transparent plastic bags, such as sandwich bags, to cover the KFM when fallout particles may get on it and contaminate it. Or pieces of thin, transparent plastic film.

bread bags. Also small rubber bands or string.

#### B. FOR THE CHARGING DEVICES:

1 . Most hard plastic rubbed on dry paper. This is the best method.

a. Plexiglas and most other hard plastics, such as are used in draftsmen's triangles, commercial plastic rulers, etc. at least 6 inches long.

b. Dry paper — Tough paper, such as clean, strong grocery bag or typing paper. Tissue paper, newspaper, or facial tissue such as Kleenex, or toilet paper are satisfactory for charging if durable.

2. Scotch Magic Transparent tape (3/4 inch width is best), or Scotch Transparent Tape, or (polyethylene or polyvinyl chloride) insulating electrical tapes, or a few of the other common brands of Scotch type tapes do not develop sufficiently high voltage electrostatic charges when unrolled quickly cannot be used for charging a KFM inside a dry-bucket, needed for charging when the air is

#### C. FOR DETERMINING DOSE RATES AND RECORDING DOSES RECEIVED:

1. A watch with a second hand.

2. A flashlight or other light, for reading the KFM in a dark shelter or at night.

3. Pencil and paper — preferably a notebook.

D. FOR THE DRY-BUCKET: A KFM MUST BE CHARGED INSIDE A DRY BUCKET IF THE AIR IS VERY HUMID, AS IT OFTEN IS INSIDE A CROWDED, LONG-OCCUPIED SHELTER LACKING ADEQUATE FORCED VENTILATION.)

1. A large bucket, pot, or can, preferably with a top diameter of at least 11 inches.

2. Clear plastic (best is 4-mil-thick clear plastic used for storm windows). A square piece larger in side than the diameter of the bucket to be used.

3. Cloth duct tape, one inch wide and 8 feet long (or 4 ft., if 2 inches wide). Or 16 ft. wide.

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4. Two plastic bags 14 to 16 inches in circumference, such as ordinary plastic bread bags. these bags should be at least 5 inches greater than the height of the bucket.

5. About one square foot of wall board (sheetrock), to make anhydrite drying agent.

6. Two 1-quart Mason jars or other airtight containers, one in which to store anhydrite and to keep dry the KFM charging devices.

7. Strong rubber bands — enough to make a loop around the bucket. Or string.

## VI. USEFUL BUT NOT ESSENTIAL MATERIALS

—Which Could be Obtained Before a Crisis—

1. An airtight container (such as a large peanut butter jar) with a mouth at least 4 inches in diameter, along with some drying agent, when it is not being used. Keeping a KFM very dry greatly extends the time during which the drying agent inside the KFM remains effective.

2. Commercial anhydrite with a color indicator, such as the drying agent Drierite. This granular anhydrite remains light blue as long as it is effective as a drying agent; it turns pink when it becomes ineffective. Or use silica gel with color indicator, that is dark blue when effective and white when it becomes ineffective. Heating in a hot oven or in a can over a fire reactivates the agent and restores their blue color. Obtainable from laboratory supply sources. Use enough to cover the bottom of the KFM's can no more than 1/2 inch deep.

3. Four square feet of aluminum foil, to make a moisture-proof cover for the dry-bucket.

## VII. TOOLS NEEDED

Small nail - sharpened

Stick, or a wooden tool handle (best 2-2 1/2 inch diameter and at least 12 inches long)

Hammer

Pliers

Scissors

Needle - quite a large sewing needle, but less than 2 1/4 inches long  
Knife with a small blade (about 2 inches)

Desirable but not essential tools: a file and a fine-toothed hacksaw blade.

## VIII. HOW TO MAKE THE DRYING AGENT

— The Easiest Part to Make, but Time Consuming —

1. For a KFM to measure radiation accurately, the air inside its ionization chamber must be kept very dry. An excellent drying agent (anhydrite) can be made by heating the gypsum in ordinary gypsum wallboard (sheetrock). Do NOT use calcium chloride.

2. Take a piece of gypsum wallboard approximately 12 inches by 6 inches, and preferably with a 3/8 inch thick. Cut off the paper and glue, easiest done by first wetting the paper. [Since normal air penetrates the plastic cover of a KFM and can dampen the anhydrite and make it ineffective in a short time as two days, fresh batches of anhydrite must be made before the attack and kept in a shelter for replacement. The useful life of the drying agent inside a KFM can be greatly lengthened by keeping the KFM inside an airtight container (such as a peanut butter jar with a 4-inch-diameter mouth) along with some drying agent, when the KFM is not being used.]

3. Break the white gypsum filling into small pieces and make the largest no more than 1/2 of pieces larger than this may be too close to the aluminum foil leaves.) If the gypsum is pliers makes breaking it easier. Make the largest side of the largest pieces no bigger than

Illustration - Simple 1/2 In cube of gypsum. (nw2 19.tif)

4. Dry gypsum is not a drying agent. To drive the water out of the gypsum molecules and produce anhydrite, heat the gypsum in an oven at its highest temperature (which should be 250 F) for one hour. Heat the gypsum after placing the small pieces no more than two pieces deep. Heat the pieces over a fire for 20 minutes or more in a pan or can heated to a dull red.

5. If sufficient aluminum foil and time are available, it is best to heat the gypsum and seal it as follows:

a. So that the right amount of anhydrite can be taken quickly out of its storage jar, put gypsum in a can with the same diameter as the KFM, measuring out a batch of gypsum that almost covers the bottom of the can with a single layer.

b. Cut a piece of aluminum foil about 8 in. x 8 in. square, and fold up its edges to form a container in which to heat one batch of gypsum pieces.

c. Measure out 10 or 12 such batches, and put each batch in its aluminum foil "bowl."

d. Heat all of these filled "bowls" of gypsum in hottest oven for one hour.

e. As soon as the aluminum foil is cool enough to touch, fold and crumple the edges of each foil "bowl" together, to make a rough aluminum-covered "ball" of each batch of anhydrite.

f. Promptly seal the batches in airtight jars or other airtight containers, and keep containers sealed except when taking out an aluminum-covered "ball. "

6. Since anhydrite absorbs water from the air very rapidly, quickly put it in a dry airtight container quite hot. A Mason jar is excellent.

7. To place anhydrite in a KFM, drop in the pieces one by one, being careful not to hit the stop-threads. The pieces should almost cover the bottom of the can, with no piece on top of another.

8. To remove anhydrite from a KFM, use a pair of scissors or tweezers as forceps, holding the can in position and not touching the leaves.

## IX. MAKE THE IONIZATION CHAMBER OF THE KFM

(To Avoid Mistakes and Save Time, Read All of This Section ALOUD Before Beginning Work.)

1 . Remove the paper label (if any) from an ordinary 8-ounce can from which the top has been removed. Wash the can with soap and water and dry it. (An 8-ounce can has an inside diameter of about 2 1/8 inches and an inside height of about 2-7/8 inches.)

2. Skip to step 3 if an 8-ounce can is available. If an 8-ounce can is not available, reduce the size of the can to 4-ounce.

can having an inside diameter of about 2-9/16 inches (such as most soup cans, most pop can cans). To cut off the top part of a can, first measure and mark the line on which to cut. bending the can while cutting, wrap newspaper tightly around a stick or a round wooden tool. the wood is covered with 20 to 30 thicknesses of paper and the diameter (ideally) is only diameter of the can.

One person should hold the can over the paper-covered stick while a second person cuts the along the marked cutting line. If leather gloves are available, wear them. To cut the can file, or use a hacksaw drawn backwards along the cutting line. Or cut the can with a sharp pocketknife by: (1) repeatedly stabbing downward vertically through the can into the paper repeatedly making a cut about 1/4 inch long by moving the knife into a sloping position, w still pressed into the paper covering the stick.

Next, smooth the cut edge, and cover it with small pieces of freezer tape or other flexibl

3. Cut out the PAPER PATTERN TO WRAP AROUND KFM CAN. (Cut one pattern out of Pattern Page A.) Glue (or tape) this pattern to the can, starting with one of the two short sides of th starting short side directly over the side seam of the can. Wrap the pattern snugly around taping it securely as it is being wrapped. (If the pattern is too wide to fit flat between a little off its lower edge.)

4. Sharpen a small nail, by filing or rubbing on concrete, for use as a punch to make the install the stop-threads in the ionization chamber (the can). (The stop-threads are insula charged aluminum leaves from touching the can and being discharged.)

5. Have one person hold the can over a horizontal stick or a round wooden tool-handle, the diameter about as large as the diameter of the can. Then a second person can use the sharp hammer to punch four very small holes through the sides of the can at the points shown by on the pattern. Make these holes just large enough to run a needle through them, and then the holes so as to bend back the obstructing points of metal.

(Illustration) Single thread threaded through 4 holes to make 2 stop threads, (nw220.tif)

6. The stop-threads can be installed by using a needle to thread a single thread through a very clean thread, preferably nylon, and do not touch the parts of this thread that will b will serve as the insulating stop-threads. Soiled threads are poor insulators.

(See illustrations.)

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Pattern to wrap around can (Table to find Dose Rates (R/Hr) (nw221.tif)

Before threading the thread through the four holes, tie a small toggle (see the preceding thread. (This toggle can easily be made of a very small sliver of wood cut about 3/8 in. l been pulled through the four holes, attach a second toggle to the thread, about 1/2 inch f comes out of the fourth hole. Then the thread can be pulled tightly down the side of the c toggle can be taped securely in place to the side of the can. (If the thread is taped down move under the tape.)

The first toggle and all of the four holes also should be covered with tape, to prevent air ter it has been covered and is being used as an ionization chamber.



X. MAKE TWO SEPARATE 8-PLY LEAVES OF STANDARD [NOT HEAVY  
DUTY\*] ALUMINUM FOIL

Proceed as follows to make each leaf:

Fig. 222a (How to fold aluminum foil - 1 ) (nw222a.tif)

A. FOR THE KFM: (IN THE FOLLOWING LIST, WHEN )

- 1 . Cut out a piece of standard aluminum foil approximately 4 inches by 8 inches.
2. Fold the aluminum foil to make a 2-ply (= 2 thicknesses) sheet approximately 4 inches by 8 inches.
3. Fold this 2-ply sheet to make a 4-ply sheet approximately 2 inches by 4 inches.
4. Fold this 4-ply sheet to make an 8-ply sheet (8 sheets thick) approximately 2 inches by 4 inches, so that the two halves of the second-fold edge are exactly together. This third folding makes an 8-ply foil sheet with one corner exactly square.
5. Cut out the FINISHED-LEAF PATTERN, found on the following Pattern Page B . Note that the pattern is NOT a square and that it is smaller than the 8-ply sheet. Flatten the 8 thicknesses of aluminum foil with your fingers until they appear to be a single thin, flat sheet.
6. Hold the FINISHED-LEAF PATTERN on top of the 8-ply aluminum foil sheet, with the pattern's THIRD-FOLD EDGE on top of the third-fold edge of the 8-ply aluminum sheet. Be sure that one corner of the FINISHED-LEAF PATTERN is on top of the exactly square corner of the 8-ply aluminum sheet.

Fig. 222b (How to fold aluminum foil -2) (nw222b.tif)

Fig. 222c (How to fold aluminum foil -3) (nw223.tif)

7. While holding a straight edge along the THREAD LINE of the pattern, press with a sharp object to make a shallow groove for the THREAD LINE on the 8-ply aluminum sheet. Also using a sharp object, trace around the top and side of the pattern, so as to indent (groove) the 8-ply foil.
8. Remove the pattern and cut out the 8-ply aluminum foil leaf. Then, in order to prevent electrical discharge from overly sharp points on the lower corners of the leaf, cut about 1/8 inch off each of its two lower corners.
9. While holding a straight edge along the indented THREAD LINE, lift up the OPEN EDGE of the leaf (keeping all 8 plies together) until this edge is vertical, as illustrated. Remove the straight edge from the 8-ply aluminum along the THREAD LINE so as to make a flat-folded hem.
10. Open the flat-folded hem of the finished leaf until the 8-ply leaf is almost flat again, from which the FINISHED-LEAF PATTERN has already been cut.

1 1 . Prepare to attach the aluminum- foil leaf to the thread that will suspend it inside

\*If only heavy duty aluminum foil (sometimes called "extra heavy duty") is available, make the same size, and use the table for the 8-ply KFM to determine radiation dose rates. To make the 5-ply leaf, start by cutting out a piece of foil approximately 4 inches by 4 inches. Fold it to make a square approximately 2 inches by 2 inches, with one corner exactly square. Next, fold a single thick square approximately 2 inches by 2 inches. Slip this square into a 4-ply sheet, thus making the 5-ply leaf, using the FINISHED-LEAF PATTERN, etc. as described for making an 8-ply

If no epoxy glue\* is available to hold down the hem and prevent the thread from slipping in pieces of tape (Band-Aid tape is best; next best is masking or freezer tape; next best, Scotch tape). After first peeling off the paper backing of Band-Aid tape, cut each piece of tape 1/8 inch by 1 inch. Attach these two pieces of tape to the finished 8-ply aluminum leaf with the sticky sides up, exactly as shown by the pattern on the following pattern page, secure 1/8 inch of one end of a tape strip to the 8-ply aluminum foil leaf by first turning under this 1/8-inch end; that is, with the sticky side up. Then turn under the other 1/8-inch-long end, and attach this end below the THREAD LINE. See the tape strip as illustrated on Pattern (C).

Be sure you have read through step 18 before you do anything else.

12. Cut an 8-1/2 inch piece of fine, unwaxed, very clean thread that has not been anti-static treated. See INSTRUCTIONS, Page 6 for excellent insulating threads and substitutes. In 1986 most sewing threads were anti-static treated and are too poor insulators for use in a KFM.

Cut out Pattern (C), the guide sheet used when attaching a leaf to its suspending thread. Tape Pattern (C) to the top of a work table. Cover the two "TAPE HERE" rectangles on Pattern (C) with pieces of tape, each piece the size of the rectangle. Then cut two other pieces of tape each the same size as the rectangles. Tape ONTO the thread ONTO the guide sheet, on top of the "TAPE HERE" rectangles.

Be very careful not to touch the two 1-inch parts of the thread next to the outline of the leaf. Oil and dirt even on clean fingers will reduce the electrical insulating value of the thread between the top rim of the can.

13. With the thread still taped to the paper pattern and while slightly lifting the thread at the center of the thread, slip the finished leaf under the thread and into position exactly as outlined on the pattern page. Hold the leaf in this position with two fingers.

14. While keeping the thread straight between its two taped-down ends, lower the thread slowly until it is between two plastic strips. Then press the thread against the plastic strips.

15. With the point of the knife, hold down the center of the thread against the center of the leaf. Then, with two fingers, carefully fold over the hem and press it almost flat. Be careful not to pull the hem out of the corners of the hem. Remove the knife, and press the hem down completely flat against the rest of the leaf.

16. Make small marks on the thread at the two points shown on the pattern page. Use a ballpoint pen or sharp pencil, if available.

17. Loosen the second two small pieces of tape from the pattern paper, but leave these tapes

1 8. Cut 5 pieces of Band-Aid tape, each approximately 1/8 inch by 1/4 inch, this small.

Fig. Pg. 224a (Shows size of Band -Aid tape) (nw224a.tif)

Use 2 of these pieces of tape to secure the centers of the side edges of the leaf. Place t illustrated in the SIDE VIEW sketch below. Or use tiny droplets of epoxy, applied with a r the side edges and to hold down the hem.

Fig. pg. 224b View 8-oz. can showing the two leaves charged (when not charged, the leaves perpendicular and touching.) ORNL-DWG 76-6542 (nw224b.tif)

19. To prevent possible partial discharge from overly sharp lower comers of the leaves, us 1/16 inch ( L) off each lower comer of the two leaves. (Partial discharge from an overly s prevent a KFM's leaves from being adequately charged and adequately separated.)

20. To make it easier to take accurate readings:

a. Make a black stripe 1/8-inch wide on the hem side of the lower edge of each leaf, if th to be used is white anhydrite made from gypsum, or light blue Drierite. It is best to use marker, such as black Marko by Flair.

b. Make a white stripe, if the drying agent to be used is dark blue silica gel. Fiquid Pap fluid, or white ink, serves well.

\*If using epoxy or other glue, use only a very little to hold down the hem, to attach the and to glue together any open edges of the plied foil. Most convenient is "one hour" epoxy Model airplane cement requires hours to harden when applied between sheets of aluminum foi stiffens the free thread beyond the upper comers of the finished leaf, put no glue within thread will go out from the folded hem of the leaf.

The instructions in step 1 1 are for persons lacking "one hour" epoxy or the time required Persons using glue instead of tape to attach the leaf to its thread should make appropriat following page and of some of the procedures detailed in steps 12 through 18.

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COVER THE TWO "TAPE HERE" RECTANGLES WITH SAME-SIZED PIECES OF TAPE, IN ORDER TO KEEP FROM TEARING THIS PAPER WHEN REMOVING TWO ADDITIONAL PIECES OF TAPE. THEN, BY PUTTING TWO OTHER PIECES OF TAPE THIS SAME SIZE ON TOP OF THE FIRST TWO PIECES, TAPE THE THREAD ONTO THIS GUIDE SHEET, AND LATER ATTACH A LEAF TO THE TAPED-DOWN THREAD.

Fig. 225 Pattern (C) how to tape aluminum leaves to thread line, (nw225.tif)

(Cut out this guide along its border lines and tape to the top of a work table.)

WARNING: The parts of the thread that will be inside the Can and on which the leaf will be to insulate the high-voltage electrical charges to be placed on the leaf Therefore, the su

must be kept very clean.

#### XI. INSTALL THE ALUMINUM-FOIL LEAVES

1 . In preparation for suspending the leaves inside the can, make two shallow notches in the can. Make one notch above each of the two lines ("FASTEN THREADS HOLDING ALUMINUM LEAVES HERE") on the paper Pattern attached to the outside of the can. Make flat-bottomed first filing a V-shaped notch, and then using a fine-toothed hacksaw blade to make the notch. If a file and/or a hacksaw blade are not available, the leaf- suspending threads can be taped to the rim of the can.)

2. Use the two small pieces of tape stuck to the ends of a leaf- suspending thread to attach the thread to the outside of the can. Attach the tapes on opposite sides of the can, so as to suspend the leaf- suspending thread as shown in the END VIEW sketch. Each of the two marks on the attached thread MUST rest exactly in a notch in the top of the rim of the can, if you are unable to make notches). Be sure that the hem-side of the leaves will face outward. See END VIEW sketch.

3. Position and secure the second leaf, being sure that:

a. The smooth sides of the two leaves are not wrinkled or bent and face each other, and are flat against each other ("The leaves are flat against each other") when not charged. See END VIEW sketch and study the first photo illustration, "Uncharged KFM".

b. The upper edges of the two leaves are suspended side by side and at the same distance from the rim of the can.

c. The leaf-suspending threads are in their notches in the top of the rim of the can (or are attached to the rim of the can with Band-Aid to the top of the rim of the can) so that putting the cover on will not move the threads.

d. No parts of the leaf-suspending threads inside the can are taped down to the can or otherwise restricted.

e. The leaf-suspending parts of the threads inside the can do not cross over, entangle or otherwise interfere with each other.

f. The threads come together where they go over the rim of the can, and the leaves are flat against each other as shown in the first photo illustration, "An Uncharged KFM"

g. The leaves look like these photographed leaves. If not, make new, better leaves and install them.

4. Cover with tape the parts of the threads that extend down the outside of the can, and also tape the small pieces of tape near the ends of the threads on the outside of the can. Or use waterproof glue to attach the parts of the threads on the outside of the can securely to the can.

5. To make the SEAT, cut a piece of a wooden pencil, or a stick, about one inch long and tuck it into the side of the can along the center line marked SEAT on the pattern. Be sure the upper end of the pencil is at the same position as the top of the location for the SEAT outlined on the pattern. The SEAT is 3/4 inch below the top of the can. Be sure not to cover or make illegible any part of the pattern on the paper pattern.

6. Cut out one of the "Reminders for Operators" and glue and/or tape it to the unused side is best to cover all the sides of the finished KFM with clear plastic tape or varnish. This on the end of an adjustment thread or moisture from damaging the "Reminders" or the table.

## XII - MAKE THE PLASTIC COVER

1. Cut out the paper pattern for the cover from the Pattern Page (B ).

2. From a piece of clear, strong plastic, cut a circle approximately the same size as the (Storm-window vinyl film, 4 mils thick, is best.)

3. Stretch the center of this circular piece of clear plastic over the open end of the can the sides of the can, making small tucks in the "skirt," so that there are no wrinkles in lower part of the "skirt" in place with a strong rubber band or piece of string. (If another diameter as the KFM can is available, use it to make the cover - to avoid the possibility leaf-suspending threads.)

Fig. 226 (Shows positioning of tape and rubber band to secure plastic cover on KFM) (nw226)

4. Make the cover so it fit snugly, but can be taken off and replaced readily.

Just below the top of the rim of the can, bind the covering plastic in place with a 1 :4-strong tape. (Cloth duct tape is best. Use two thicknesses. If only freezer or masking tape three or four thicknesses.)

Keep vertical the small part of the tape that presses against the rim of the can while pull tape horizontally around the can so as to bind the top of the plastic cover snugly to the the tape is kept vertical, the lower edge of the tape will not squeeze the plastic below to such a small circumference as to prevent the cover from being removed quite easily.

## REMINDERS FOR OPERATORS

### REMINDERS FOR OPERATORS

r h e DFMNG agent inside a KFM is O.K. if,  
when (he charged KFM is not exposed

nm or less in 3 hours.

HEADING: With the reading eye 12 inches  
'ertically above the seat, note on the

edges of the leaves. If the right leaf is  
t 10 mm and the left leaf is at 3 mm.

he KFM reads 10 mm. Never take a  
reading while a leaf is touching a stop-  
hread. Never use a reading that is less

h a n 5 mm.

Ending A DC6E RATE: I f before exposure a KFM  
eads 17 mm and if after a 1-minute ex-  
losure it reads 3 mm. the difference in  
eadings is 12 mm the attached table  
hows the dose rate was 9.6 R hr dur-  
ng the exposure.

ENDING A DOSE: I f a person works outside  
or 3 hours where the dose rate is 2 R  
i r . w h a t i s h i s r a d i a t i o n d o s e ? An-  
wer: 3 hr x 2 R hr = 6 R.

FtNQNS HCW LCN3 IT TAKES TO GET A CERTAIN R DOSE: I f the  
dose rate is 1.6 R hr outside and a per-  
6 R H.6 R/hr 3.75 hr = 3 hours and 45

Fallcut RADIATION GUIDES for a healthy person

atio P n dose of more than 100 R during a  
2-week period:

6Rper day can be tolerated for up to  
two months without losing the ability

100 R in a week or less is not likely to

350R i n a few days results in a 50-50  
chance of dying, under post-attack  
conditions.

600Rin a week or less is almost certain  
to cause death within a few weeks.

The DFNNGAGENTi n s i d e a KFM is O.K. if,  
when the charged KFM is not exposed

mm or less in 3 hours.

READING: With the reading eye 12 inches  
vertically above the seat, note on the  
tom scale the separation of the lower  
edges of the leaves. If the right leaf is  
at 10 mm and the left leaf is at 3 mm.  
the KFM reads 10 mm. Never take a  
reading while a leaf is touching a stop-  
thread. Never use a reading that is less  
than 5 mm.

FINDING A DC6E RATE: If before exposure a KFM

reads 17 mm and if after a 1-minute exposure it reads 5 mm. the difference in readings is 12 mm the attached table shows the dose rate was 9.6 R hr during the exposure.

FINDING A DOSE: If a person works outside for 3 hours where the dose rate is 2 R/hr. what is his radiation dose? Answer:  $3 \text{ hr} \times 2 \text{ R/hr} = 6 \text{ R}$ .

FMXCHCW LCN3 IT TAKES TO GET A CERTAIN R DOSE: If the

dose rate is 1.6 R/hr outside and a person is willing to take a 6 R dose, how

$6 \text{ R} / 1.6 \text{ R/hr} = 3.75 \text{ hr} = 3 \text{ hours and } 45$

Fallout RADIATION GUIDES for a healthy person

a total dose of more than 100 R during a 2-week period:

two months without losing the ability to work.

100R in a week or less is not likely to seriously sicken.

350R in a few days results in a 50-50 conditions/

600R in a week or less is almost certain to cause death within a few weeks.

5. With scissors, cut off the "skirt" of the plastic cover until it extends only about one rim of the can.

6. Make a notch in the "skirt," about one inch wide, where it fits over the pencil SEAT at "skirt" in this notched area should be only about 5/8 of an inch long, measured down from the can.

7. Remove the plastic cover, and then tape the lower edges of the "skirt," inside and out, 1/4-inch-wide tape. Before securing each short piece of tape, slightly open the tucks that on their edges, so that the "skirt" flares slightly outward and the cover can be readily removed.

8. Make the charging-wire by using the full-size, exact-size pattern on the right.

Doorbell wire with an outside diameter of about 1/16 inch is best, but any lightweight ins part of a lightweight two-wire extension cord split in half, will serve. The illustrated wire. To stop tape from possibly slipping up or down the wire, use a very little glue

If a very thin plastic has been used for the cover, a sticky piece of tape may need to be attached to the bare-ended adjustment thread, so both threads can be used to hold the charging wire in place.

The best tape to attach to an end of one of the adjustment-threads is cloth duct tape. A strip about 3/4 inch wide is the sticky base. To keep this tape sticky (free of paper fibers), the paper side of the tape should be covered with transparent tape or varnish. A piece about 1/8 inch wide by 3 1/4 inch long serves to stick the sticky base, to hold the adjustment-thread. A 3/4 inch by 1- 1/4 inch rectangular piece of tape makes the finger hold – important for making adjustments inside a dry-bucket.

With a needle or pin, make a hole in the plastic cover 1/2 inch from the rim of the can at the upper end of the CENTER LINE between the two leaves. The CENTER LINE is marked on the pattern and is wrapped around the can. Carefully push the CHARGING-WIRE through this hole (thus stretching the wire) until all of the CHARGING- WIRE below its Band- Aid-tape stop is inside the can.

9 .

From the Pattern Page (B ) cut out the SCALE. Then tape the SCALE to the under side of the cover in the position shown on the pattern for the cover, and also by the drawings. Preferably use clear tape. Be careful not to cover with tape any of the division lines on the SCALE between 20 on the left and the left of 0.

10. Put the plastic cover on the KFM can.

Fig. 228 (Charging Wire. Sticky Ended Adjustment Thread.) (nw228.tif)

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### XIII. TWO WAYS TO CHARGE A KFM

When preparing to charge a KFM, be sure its anhydrite is fresh. (Under humid conditions, some anhydrite will go through the plastic cover to make the drying agent ineffective. If the anhydrite is on top of another piece. Re-read VIII 7 and VIII 8.

1 . Charging a KFM with Hard Plastic Rubbed on Dry Paper.

a. Adjust the charging- wire so that its lower end is about 1/16 inch above the upper edge of the aluminum-foil leaves. Use the sticky-tape at the end of one adjustment-thread to hold the charging- wire in this position. Stick this tape approximately in line with the threads supporting the leaves, either on the side of the can or on top of the plastic cover. (If the charging-wire is held loosely by the cover, it may be necessary to put a piece of sticky-tape on the end of each adjustment-thread in order to adjust the charging- wire securely. If a charging- wire is not held in place, its lower end may be forced up by the like charge on the leaves before the leaves can be fully charged.)

Fig. 229a (Charging a KFM) (nw229a.tif)

b. Select a piece of Plexiglas, a draftsman's plastic triangle, a smooth plastic ruler, or other hard, smooth plastic. (Unfortunately, not all types of hard plastic can be used to generate a uniform charge.)



electrostatic charge. ) Be sure the plastic is dry.

For charging a KFM (especially inside a dry-bucket), cut a rectangular piece of hard plastic Plexiglas about 1-1/2 by 6 inches. Sharp corners and edges should be smoothed. To avoid contaminating the charging end with sweaty, oily fingers, it is best to mark the other end of tape, and to hold it only by its taped end.

c. Fold DRY paper (a piece of clean paper bag, or other smooth, clean paper) to make an approximate square about 5 inches on a side and 15 to 20 sheets thick. (This many sheets of paper less to the fingers of the electrostatic high-voltage charges to be generated on the hard plastic rubbed paper.)

d. Fold the square of paper in the middle, and move the hard plastic rapidly back and forth. The middle of this folded square is rubbed vigorously on the paper while the outside of the square of paper is squeezed firmly between thumb and little finger on one side, and the other fingers on the other. To avoid discharging the charge on

the plastic to the fingers, keep them away from the edges of the paper. See sketch..

Fig. 229b (Using plastic and paper to generate a static charge) (nw229b.tif)

e. Move the electrostatically charged part of the rubbed plastic rather slowly past the upper charging-wire, while looking straight down on the KFM. Keep the hard plastic approximately perpendicular to the charging-wire and about 1/4 to 1/2 inch away from its upper end. The plastic jumps the spark gaps and charges the leaves of the KFM. Charge the leaves sufficiently to give a reading of at least 15 mm.

f. Pull down on an insulating adjustment-thread to raise the lower end of the charging-wire. The charging-wire has been held in its charging position by its sticky ended adjustment-thread. To move it to the top of the clear plastic cover, to avoid possibly damaging the threads: (1) pull down on the bare-ended adjustment-thread; and (2) detach, pull down on, and secure the sticky-ended adjustment-thread to the side of the can, so as to raise and keep the lower end of the charging-wire close to the underside of the clear plastic cover.) Do not touch the charging-wire, because the insulation usually is not good enough to prevent the charge from bleeding off into the fingers.

g. To get the most accurate readings possible, lightly bump or shake the charged KFM (to remove the unstable part of the charge) before taking the initial reading.

h. If the initial reading is more than 20 mm, to get the most accurate reading possible, completely discharge the leaves (by touching them with the charging-wire while guiding the wire with the fingers on its insulation), to reduce to 20 mm or slightly less the initial reading that you get. Then completely discharge, and recharge to 20 mm or slightly less.

i. To keep a KFM in excellent condition and to enable its drying agent to last much longer before becoming ineffective, put the whole KFM in an airtight container, such as a large peanut butter jar, with drying agent about an inch deep on its bottom. Or at least keep the charging paper and the plastic charging strip dry in a sealable container, such as a Mason jar, with some drying agent.

2. Charging a KFM from a Quickly Unwound Roll of Tape. (Quick unwinding produces a harmless static charge of several thousand volts on the tape.)

a. Adjust the charging- wire so that its lower end is about 1/16 inch above the upper edge aluminum-foil leaves. Use the sticky-tape at the end of one adjustment-thread to hold the charging- wire in this position. Stick this tape approximately in line with the leaves, either on the can or on the plastic cover. (If the plastic cover is weak, it may be necessary to stick the sticky-tape on the end of each adjustment-thread, in order to hold the charging-wire secure. If the charging- wire is not secure, its lower end may be forced up by the like charge on the leaves and the leaves can be fully charged.)

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b. The sketch shows the "GET SET" position, preparatory to unrolling the Scotch Magic Tape, PVC, electrical tape, or other tape. Be sure to first remove the roll from its dispenser. The other kinds of tape will not produce a high enough voltage.

(Illustration) "Get Set" position (nw230a.tif)

c. QUICKLY unroll 10 to 12 inches of tape by pulling its end with the left hand, while the right hand allows the roll to unwind while remaining in about the same "GET SET" position only an inch or so away from the KFM.

d. While holding the unwound tape tight, about perpendicular to the charging- wire, and about 8 inches away from the end of the charging- wire, promptly move both hands and the tape to the right, slowly taking about 2 seconds to move about 8 inches. The electrostatic charge on the unwound tape "jumps" the spark gaps from the tape to the upper end of the charging-wire and from the end of the charging-wire to the aluminum leaves, and charges the aluminum leaves.

Be sure neither leaf is touching a stop-thread.

Try to charge the leaves enough to spread them far enough apart to give a reading of at least 20 mm but no more than 20 mm after the KFM has been gently bumped or shaken to remove any unstable part of the charge.

e. Pull down on an insulating adjustment-thread to raise the lower end of the charging-wire. If the charging- wire has been held in charging position by its sticky-ended adjustment-thread between the top of the clear plastic cover, it is best first to pull down a little on the bare-end adjustment-thread, and then to move, pull down on, and secure the sticky-ended adjustment-thread to the side of the can so that the lower part of the charging-wire is close to the underside of the plastic cover.

Do not touch the charging-wire.

(Illustration) Transferring charge (nw230b.tif)

f. Rewind the tape tight on its roll, for future use when other tape may not be available.

TESTING YOUR KIM TO LEARN IF IT CAN  
ACCURATELY MEASURE LOW DOSE RATES

Put fresh drying agent in your KFM and then charge and test the KFM in a location where it

mal radiation. Take an initial reading. If after 3 hours its reading has decreased by 1 mR, leaf-suspending threads are good insulators and that your KFM can reliably measure dose rates of 30 milliroentgens per hour (30 mR/hr). By post-attack standards, 30 mR/hr is a low dose rate of continuous exposure (an impossibility, because fallout decays). 30 mR/hr would result in a dose not enough to incapacitate. Warning: In heavy fallout areas, for the first few days after the attack, rates inside even most good shelters will be higher than 0.03 R/hr.

#### TROUBLE SHOOTING

If charging does not separate the two leaves sufficiently, take these corrective actions:

1. Be sure the pieces of anhydrite in the bottom of the ionization chamber (the can) are stacked one piece on top of another and the top of no piece more than 1/2 inch above the bottom of the chamber.
2. Check to be sure that the threads suspending the leaves are not crossed; then try to charge.
3. If the KFM still cannot be charged, replace the used anhydrite with fresh anhydrite.
4. If you cannot charge a KFM when the air is very humid, charge it inside its dry-bucket.
5. If you cannot charge the KFM while in an area of heavy fallout, take it to the place of shelter for protection against radiation, and try to charge it there. (A dose rate of several hundred mR/hr charges on both the charging device and the instrument so rapidly that a KFM cannot be charged.)

If a KFM or other radiation measuring instrument gives unexpectedly high readings inside a shelter, take it off the outside of the instrument and repeat the radiation-measurements. Especially when outdoors where there is fresh fallout, keep the instrument in a lidded pot, plastic bag, or other container to avoid possibility of having it contaminated with fallout particles and afterwards getting erroneous measurements.

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#### XIV. MAKE AND USE A DRY-BUCKET

By charging a KFM while it is inside a dry-bucket with a transparent plastic cover (see illustration) it can be charged and used even if the relative humidity is 100% outside the dry-bucket. The bucket is kept very dry by a drying agent placed on its bottom.

About a cupful of anhydrite serves very well. The pieces of this dehydrated gypsum need not be large; but is best for use inside a KFM, but do not use powdered anhydrite.

Fig. 23 1a (Bucket with Anhydrite) (nw23 1a.jpg)

A dry-bucket can be readily made in about an hour by proceeding as follows:

1. Remove the handle of a large bucket, pot, or can preferably with a top diameter of at least 14 inches. A 4-gallon bucket having a top diameter of about 14 inches and a depth of about 9 inches is approximately this size is satisfactory. If the handle-supports interfere with stretching plastic film across the top of the bucket, remove them, being sure no sharp points remain.

2. Cut out a circular piece of clear plastic with a diameter about 5 inches larger than the bucket. Clear vinyl 4 mils thick, used for storm windows, etc., is best. Stretch the plastic over the top of the bucket, and tie it in place, preferably with strong rubber bands looped together.
3. Make a plastic top that fits snugly but is easily removable, by taping over and around the top of the bucket. One-inch-wide cloth duct tape, or one-inch-wide glass-reinforced strapping tape, will do well. When taping, do not permit the lower edge of the tape to be pulled inward below the top of the bucket.

Fig. 23 1a (Bucket with Anhydrite - Enlarged View) (nw23 1b.jpg)

(Illustration) Cutting the clear plastic cover. (nw23 1c.tif)

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4. Cut two small holes (about 1 inch by 2 inches) in the plastic cover, as illustrated. The holes are (shown by dotted lines) outward from the small holes, out to the solid-line outlines of the hand-holes, so as to form small flaps.
5. Fold the small flaps upward, so they are vertical. Then tape them on their outer sides, forming a "wall" about 3/4 inch high around each hand-hole.
6. Reduce the length of two ordinary plastic bread bags (or similar plastic bags) to a length greater than the height of the bucket. (Do not use rubber gloves in place of bags; gloves allow much more humid outside air being unintentionally pumped into a dry-bucket when it is being charged with a KFM inside it.)
7. Insert a plastic bag into each hand-hole, and fold the edge of the plastic bag about 1 inch up to form a vertical "wall" around each hand-hole.
8. Strengthen the upper parts of the plastic bags by folding 2-inch pieces of tape over the top edge around each hand-hole.
9. Make about a quart of anhydrite by heating small pieces of wall-board gypsum, and keep it in a Mason jar or other airtight container with a rubber or plastic sealer.
10. Make a circular aluminum-foil cover to place over the plastic cover when the dry-bucket is used for minutes to hours. Make this cover with a diameter about 4 inches greater than the diameter of the bucket, and make it fit more snugly with an encircling loop of rubber bands, or with string if essential, an aluminum-foil cover reduces the amount of water vapor that can reach and pass through the plastic cover, thus extending the life of the drying agent.

1 1 . Charge a KFM inside a dry-bucket by:

a. Taking off wrist watch and sharp-pointed rings that might tear the plastic bags.

b. Placing inside the dry-bucket:

(1) About a cup of anhydrite or silica gel;

(2) the KFM, with its charging-wire adjusted in its charging position; and

(3) dry, folded paper and the electrostatic charging device, best a 5-inch-long piece of E

smoothed edges, to be rubbed between dry paper folded about 4 inches square and about 20 s  
(Unrolling a roll of tape inside a dry-bucket is an impractical charging method.)

c. Replacing the plastic cover, that is best held in place with a loop of rubber bands.

d. Charging the KFM with your hands inside the plastic bags, operating the charging device  
another person illuminate the KFM with a flashlight. When adjusting the charging- wire, m  
hands very slowly. See the dry-bucket photos.

12. Expose the KFM to fallout radiation either by:

a. Leaving the KFM inside the dry-bucket while exposing it to fallout radiation for one of  
intervals, and reading the KFM before and after the exposure while it remains inside the d  
(The reading eye should be a measured 12 inches above the SEAT of the KFM, and a flashligh  
other light should be used.)

b. Taking the charged KFM out of the dry-bucket to read it, expose it, and read it after t

(If this is done repeatedly, especially in a humid shelter, the drying agent will not be e  
many KFM chargings, and will have to be replaced.)

## XV. HOW TO USE A KFM ALTER A NUCLEAR ATTACK

### A. BACKGROUND INFORMATION

If during a rapidly worsening crisis threatening nuclear war you are in the place where yo  
pone studying the instructions following this sentence until after you have:

(1) built or improved a high-protection-factor shelter (if possible, a shelter covered wit  
and separate from flammable buildings), and

(2) made a KAP (homemade shelter- ventilating pump) if you have the instructions and mater

(3) stored at least 15 gallons of water for each shelter occupant if you can obtain contai

Having a KFM or any other dependable fallout meter and knowing how to operate it will enab  
ation injuries and possible fatalities, especially by skillfully using a high-protection-f  
limit exposures to radiation. By studying this section you first will learn how to measure  
(roentgens per hour = R/hr), how to calculate doses [It] received in different time interv  
time intervals (hours and/or minutes) in which specified doses would be received. Then thi  
doses (number of R) that the average person can tolerate without being sickened, that he i  
he is likely to be killed by.

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Most fortunately for the future of all living things, the decay of radioactivity causes th  
come less and less dangerous with the passage of time. Each fallout particle acts much lik  
would if it were made so that its rays, shooting out from it like invisible light, became  
Contrary to exaggerated accounts of fallout dangers, the radiation dose rate from fallout  
ground in the areas of the heaviest fallout will decrease quite rapidly. For example, cons

a relatively nearby, large surface burst, at a place where the fallout particles are deposited after the explosion. At this time one hour after the explosion, assume that the radiation dose rate (at a particular time) measures 2,000 roentgens per hour (2,000 R/hr) outside. The dose rate is reduced to 200 R/hr by normal radioactive decay. Two days after the explosion the dose rate is reduced by radioactive decay to 20 R/hr. After two weeks, the dose rate is less than 2 R/hr. After two weeks, the dose rate is less than 2 R/hr, people can go out of a good shelter and work outdoors for 3 hours a day, receiving a dose without being sickened.

In places where fallout arrives several hours after the explosion, the radioactivity of the fallout is in its time period of most rapid decay while the fallout particles were still airborne. If you are in a place where fallout arrives 8 hours after the explosion, two days must pass before the dose rate measured at your location will decay to 1/10 its initial intensity.

## B. FINDING THE DOSE RATE

1. Reread Section IV, "What a KFM Is and How it Works." Also reread Section XIII, "Two Way Charge a KFM," and actually do each step immediately after reading it.

2. Charge the KFM so that it reads at least 15 mm. Next raise the lower end of the charging rod and gently bump or shake the KFM to remove any unstable part of the charge. Read the apparent separation of the lower edges of the leaves while the KFM rests on an approximately horizontal surface. If the reading is larger than 20 mm, bleed off enough charge to reduce the initial reading to 20 mm or slightly less, for maximum accuracy. Never take a reading while a leaf is touching the stop-thread.

3. To prevent possible contamination of a KFM (or of any other fallout meter) with fallout, keep it inside a plastic bag or other covering when there is risk of fallout particles being blown onto it. An instrument contaminated with fallout particles can give too high readings for the low dose-rate measurements made inside a good shelter.

4. Expose the KFM to fallout radiation for one of the time intervals shown in the vertical table attached to the KFM - (Study the following table.) If the dose rate is not known even approximately, first expose the fully charged KFM for one minute. For dependable measurements outdoors, expose the charged KFM about 3 feet above the ground. The longer outdoor exposures usually are best made by attaching the KFM with 2 strong rubber bands to a stick or pole, careful never to tilt the KFM too much.

5. Read the KFM after the exposure, while the KFM rests on an approximately horizontal surface.

6. Find the time interval that gives a dependable reading by exposing the fully charged KFM for one or more of the listed time intervals until the reading after the exposure is;

(a) Not less than 5 mm.

(b) At least 2 mm less than the reading before the exposure.

7. Calculate by simple subtraction the difference in the apparent separation of the lower leaves before the exposure and after the exposure. An example: If the reading before the exposure is 18 mm and the reading after the exposure is 6 mm, the difference in readings is 18 mm - 6 mm = 12 mm.

8. If an exposure results in a difference in readings of less than 2 mm, recharge the KFM again for one of the longer time intervals listed. (If there appears to be no difference in readings taken before and after an exposure for one minute, this does not prove there is absolutely no danger. Take a longer reading.)

9. If an exposure results in the reading after the exposure being less than 5 mm, recharge the KFM and expose it again for one of the shorter time intervals listed.

10. Use the table attached to the KFM to find the dose rate (R/hr) during the time of exposure. The dose rate (R/hr) is found at the intersection of the vertical column of numbers under the time interval and of the horizontal line of numbers that lists the calculated difference in readings at the end of the exposure.

An example: If the time interval of the exposure was 1 MIN. and the difference in the readings was 12 mm, the table shows that the dose rate during the time interval of the exposure was 9.2 roentgens per hour).

Another example: If the time interval of the exposure was 15 SEC. and the difference in readings was 11 mm, the table shows that the dose rate during the exposure was halfway between 31 and 37 R/hr; that is, the dose rate was 34 R/hr.

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11. Note in the table that if an exposure for one of the listed time intervals causes the difference in readings to be 2 mm or 3 mm, then an exposure 4 times as long reveals the same dose rate. example: If a 1-min. exposure results in a difference in readings of 2 mm, the table shows that the dose rate was 1.6 R/hr; then if the KFM is exposed for 4 minutes at this same dose rate of 1.6 R/hr, the table shows that the resultant difference in readings is 8 mm.

The longer exposure results in a more accurate determination of the dose rate.

12. If the dose rate is found to be greater than 0.2 R/hr and time is available, recharge the KFM and repeat the dose-rate measurement - to avoid possible mistakes.

#### C. CALCULATING THE DOSE RECEIVED

The dose of fallout radiation - that is, the amount of fallout radiation received - determines the danger to men and animals. Being exposed to a high dose rate is not always dangerous - provided the exposure is short enough to result in only a small dose being received. For example, if the dose rate outside an exposure shelter is 1200 R/hr and a shelter occupant goes outside for 30 seconds, he would be exposed for 1/2 of 1 hour, which equals 1/120 hour. Therefore, since the dose he would receive if he stayed outside for 1 hour would be 1200 R, in 30 seconds he would receive 1/120 of 1200, which equals 10 R (1200 R divided by 120 = 10 R). A daily dose of 10 R (10 roentgens) will not cause any symptoms if it is not repeated day after day. In contrast, if the average dose rate of an area were found to be 12 R/hr and if a person stayed in that particular area for 24 hours, he would receive a dose of 288 R (12 R/hr x 24 hr = 288 R). If a person had been exposed previously to very little radiation, there would still be a serious danger. A dose of 288 R would be fatal under the difficult conditions that would follow a heavy nuclear attack.

Another example: Assume that three days after an attack the occupants of a dry, hot cave c

protection against fallout are in desperate need of water. The dose rate outside is found water from a source 3 miles away is estimated to take 2-1/2 hours. The cave occupants estimate backpackers will receive a dose in 2- 1/2 hours of 50 R ( $2.5 \text{ hr} \times 20 \text{ R/hr} = 50 \text{ R}$ ). A dose symptoms (nausea in about 10% of persons receiving a 50 R dose) for persons who previously very small doses. Therefore, one of the cave occupants makes a rapid radiation survey for proposed route, stopping to charge and read a KFM about every quarter of a mile. He finds er than 20 R/hr.

So, the cave occupants decide the risk is small enough to justify some of them leaving she get water.

#### D. ESTIMATING THE DANGERS FROM DIFFERENT RADIATION DOSES

Fortunately, the human body - if given enough time - can repair most of the damage caused ic example: A healthy man accidentally received a daily dose of 9.3 R (or somewhat more) c each day for a period of 106 days. His total accumulated dose was at least 1000 R. A dose gens, if received in a few days, is almost three times the dose likely to kill the average dose in a few days and after a nuclear attack cannot get medical treatment, adequate rest, symptom this man noted was serious fatigue.

The occupants of a high-protection-factor shelter (such as a trench shelter covered with 2 ing crawlway entrances) would receive less than 1/200 of the radiation dose they would rec areas of very heavy fallout, persons who remain continuously in such a shelter would recei dose of less than 25 R in the first day after the attack, and less than 100 R in the first first two weeks, such shelter occupants could start working outside for an increasing leng ing a dally dose of no more than 6 R for up to two months without being sickened.

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To control radiation exposure in this way, each shelter must have a fallout meter, and a c the approximate total dose received each day by every shelter occupant - both while inside

The long- term penalty which would result from a dose of 100 R received within a few weeks Americans fear. If 100 average persons received an external dose of 100 R during and short the studies of the Japanese A-bomb survivors indicate that no more than one of them is lik ing 30 years as a result of this 100 R radiation dose. These delayed radiation deaths would other cancers. In the desperate crisis period following a major nuclear attack, such a rel expectancy during the following 30 years should not keep people from starting recovery wor their fellow citizens from death due to lack of food and other essentials.

A healthy person who previously has received a total accumulated dose of no more than 100 2-week period should realize that:

100 R, even if all received in a day or less, is unlikely to require medical care - provid total additional dose of no more than a few R is received.

350 R received in a few days or less results in a 50-50 chance of being fatal after a larg survivors could get medical care, sanitary surroundings, a well-balanced diet, or adequate

600 R received in a few days or less is almost certain to cause death within a few days.



#### E. FINDING THE PROTECTION FACTOR OF A SHELTER

To avoid the necessity of repeatedly going outside a shelter to determine the changing dose, shelter's protection factor (PF) by measuring the dose rate inside the shelter as soon as reliably measured. Then promptly measure the dose rate outside. The uncontaminated shelter side/Dose Rate Inside.

An example: If the dose rate inside is found to be 0.2 R/hr and the dose rate outside is 30 R/hr / 0.2 R/hr = 155.

Then at future times the approximate dose rate outside can be found by measuring the dose rate inside and multiplying it by 155. Approximate Dose Rate Outside = Dose Rate Inside x PF.

#### F. USING A KIM TO REDUCE RADIATION DOSES RECEIVED

If a charged KFM is discharged and reads zero within a second or two after being taken out means that the dose rate outside is hundreds of roentgens per hour. Get back inside! Also, a short reading is not as accurate as are readings made in longer specified exposure times.

Inside most shelters, the dose received by an occupant varies considerably, depending on the location. For example, inside an expedient covered-trench shelter the dose rate is higher near the entrance than in the trench. In a typical basement shelter the best protection is found in one corner. Especially during the first hours after the arrival of fallout, when the dose rates and doses received are highest, shelter occupants should use their fallout meters to determine where to place themselves to minimize the doses they receive.

They should use available tools and materials to reduce the doses they receive, especially by staying deeper (if practical) and reducing the size of openings by partially blocking them with heavy objects, etc. - while maintaining adequate ventilation. To greatly reduce the slight risk of fallout entering through nose or mouth, shelter occupants should cover nose and mouth with a towel or other cloth. If fallout is being deposited outside their shelter, if at the same time ventilating air is being blown into the shelter, the air inside an occupied shelter often becomes very humid. If a good flow of outdoor air is being pumped into the shelter especially if pumped by briefly operating a KAP or other ventilating pump - a KFM usually cannot be used for the intake of the shelter room without putting it inside a dry-bucket. However, if the air to be pumped has a relative humidity of 90% or higher, the instrument cannot be charged, even by quickly uncharging it. In the most extensive areas of heavy fallout, the occupants of most home basements, that provide in the best protection against heavy fallout radiation, would be in deadly danger. By using a dependable fallout meter, occupants can determine where persons lying on the floor in certain locations would receive the smallest doses, and that by using such shielding in these locations, the doses received could be greatly reduced. Additional shielding can be obtained by putting a very small shelter inside the basement where the dose rate is found to be lowest. Further instructions are given below.

Instructions for using tools and materials for the roof, and water containers, books, and other heavy objects for shielding the roof. Or, if tools are available, breaking through the basement floor and digging a shallow trench to increase available protection against radiation. If a second expedient ventilating pump, a KAP (Kilowatt Air Pump Fan), is made and used as a fan, such an extremely cramped shelter inside a shelter usually can protect many times as many persons as can occupy it without forced ventilation.

END OF INSTRUCTIONS

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PATTERN (A) (nw236.tif)

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PATTERN (B ) (nw237.tif)

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PATTERN (C) (nw238.tif)

(Cut out this guide along its border lines and tape to the top of a work table.)

#### WARNING:

The parts of the thread that will be inside the can and on which the leaf will be suspended high-voltage electrical charges to be placed on the leaf Therefore, the suspended parts Of very clean.

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#### j REMINDERS FOR OPERATORS

##### REMINDERS FOR OPERATORS |

The drying agent inside a KFM is D.K. if, when the charged KFM is tot exposed to radiation, its readings iecrease by 1 mm or less in 3 lours.

heading: With the reading eye 12 nches vertically above the seat, note in the tom scale the separation of he lower edges of the leaves. If the ight leaf is at 10 mm and the left eaf is at 3 mm. the KFM reads 10 nm. Never take a reading while a eaf is touching a stop-thread. Never lse a reading that is less than 5 n m .

Ending a dose rate: If before exposure i KFM reads 17 mm and if after a -minute exposure it reads 5 mm. he difference in readings is 12 mm he attached table shows the dose ate was 9.6 R hr during the ixposure.

Ending a dose: If a person works outside for 3 hours where the dose rate is 2 R/hr. what is his radiation dose? Answer:  $3 \text{ hr} \times 2 \text{ R/hr} = 6 \text{ R}$ .

Finding how long it takes to get a certain R dose: If the dose rate is 1.6 R/hr outside and a person is willing to take a 6 R dose, how long can he remain outside? Answer:  $6 \text{ R} / 1.6 \text{ R/hr} = 3.75 \text{ hr} = 3 \text{ hours and } 45 \text{ minutes}$ .

Fallout radiation guides for a healthy person not previously exposed to a total radiation dose of more than 100 R during a 2-week period:

6 R per day can be tolerated for up to two months without losing the ability to work.

100 R in a week or less is not likely to seriously sicken.

350 R in a few days results in a 50-50 chance of dying, under post-attack conditions.

600 R in a week or less is almost certain to cause death within a few weeks.

The drying agent inside a KFM is O.K. if, when the charged KFM is not exposed to radiation, its readings decrease by 1 mm or less in 3 hours.

Reading: With the reading eye 12 inches vertically above the seat, note on the tom scale the separation of the lower edges of the leaves. If the right leaf is at 10 mm and the left leaf is at 3 mm. the KFM reads 10 mm. Never take a reading while a leaf is touching a stop-thread. Never use a reading that is less than 5 mm.

Finding a dose rate: If before exposure a KFM reads 17 mm and if after a

1-minute exposure it reads 5 mm.  
the difference in readings is 12 mm  
the attached table shows the dose  
rate was 9.6 R hr during the  
exposure.

Finding a dose: If a person works  
outside for 3 hours where the dose  
rate is 2 R hr. what is his radiation  
dose? Answer:  $3 \text{ hr} \times 2 \text{ R hr} = 6 \text{ R}$ .

Finding how long it takes to get a certain  
R dose: If the dose rate is 1.6 R hr  
outside and a person is willing to  
take a 6 R dose, how long can he  
remain outside? Answer:  $6 \text{ R} / 1.6$   
R/hr 3.75 hr = 3 hours and 45  
minutes.

Fallout radiation guides for a healthy  
person not previously exposed to a  
total radiation dose of more than  
100 R during a 2-week period:

6 R per day can be tolerated for up to  
two months without losing the  
ability to work.

100 R in a week or less is not likely  
to seriously sicken.

350 R in a few days results in a  
50-50 chance of dying, under  
post-attack conditions.

600 R in a week or less is almost  
certain to cause death within a few  
weeks.

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Instructions

(A)

Instructions For Persons Concerned With Reproducing The

## K.F.M. Instructions

The KFM instruction pages are printed so that they can be readily cut out and pasted up (under "12-PAGE TABLOID" given on page 242) to expedite rapid reproduction preparatory to mass distribution is required to reproduce this survival information.

All of the paste-ups should be photo-reduced to fit your size newspaper, EXCEPT four cut-outs [(21) and (24)] and one drawing [paste-up (26)] SHOULD REMAIN AT 100%.

To make the instruction pages fully camera-ready for paste-up and photographing, it is necessary to follow the following instructions: (1) To use a camera-invisible copy the numbers on the back of each page onto the front of that page, writing them in a black ink in the approximate original position of the numbers; (3) To cut out each of the 40 paste-ups.

On the back of each paste-up are the number of the tabloid page to which the paste-up is to be attached (in parentheses) the number of the paste-up itself. For example, on the back of "INSTRUCTIONS, Page 1" are the following: "Pg 1- (2)" and "Pg 1- (3)." Thus, this page contains two paste-ups, both of which are to be attached to page 1 of the tabloid paste-up. The positions in which they should be attached to page 1 are indicated by a sketch on page 242.

Timed field tests by two newspapers have shown that less than 40 minutes is required to be able to produce a tabloid. Each test began when the newspaper was given only written instructions like this page, along with KFM instructions like those in this book except that the index numbers were written in camera-invisible blue on each half page of the instructions.

The camera-ready copy is for use with a straight lens (100%, horizontal and 100% vertical).

Fig. Pg. 241 TABLOID LAYOUT SHEET and CENTER FOLD OF A 12-PAGE TABLOID, INDICATING TABLOID Page 6 AND Page 7. (nw241.tif)

All photographs are 85-line screen.

The following layout sketch for a 12-page tabloid indicates where each of the numbered paste-ups should be pasted-up and what spaces should be left blank. This positioning of the paste-ups is for the KFM-maker to cut out the patterns without destroying any instructions printed on opposite pages.

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Fig. Pg. 242 INSTRUCTIONS (B) FOR PHOTOGRAPHER - PRINTER (nw242.tif)

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(Electronic Edition Note: Because the format of the pages has changed these instructions were followed taking this into account, -aj)

## Appendix D

### Expedient Blast Shelters

## INCREASING IMPORTANCE

The majority of urban and suburban Americans would need blast shelters to avoid death or injury and evacuate before an all-out nuclear attack. As nuclear arsenals continue to grow, an increase in the protection of blast shelters. In an attack on militarily relevant targets, as much as 10 states could be subjected to blast damage severe enough to destroy or damage homes depending on the number of warheads assigned to each hard target, weapon reliability, etc. If blast shelters affording 15-pounds-per-square-inch (15 psi) overpressure range were available to everybody and were used in an attack, the great majority of the occupants would survive all blast, fire, and radiation effects subjected to less than 15-psi blast effects.

Fifteen-psi blast shelters will survive as close as about 1.5 miles from ground zero of a 1-megaton air burst and about 2.3 miles from ground zero or a 1-megaton air burst. Except in high-density urban areas, the openings and exits of shelters are all too likely to be covered with blast-hurled debris. Even with good earth-covered 15-psi blast shelters would be killed would be only about 1/6th as many as the most people sheltered in typical American homes probably would die from blast and fire effects.

Blast tests have indicated that the SmallPole Shelter (the most blast-resistant of the shelters described in Appendix A) should enable its occupants to survive up to the 50-psi overpressure range. The blast-resistant and radiation-protective features described in following sections, an area. Calculations show that this earth-covered expedient blast shelter also would give adequate protection against 50-psi blast overpressure range against the intense initial nuclear radiation that is emitted by a 1-megaton explosion. However, to make this shelter (see page 258) provide adequate protection against more intense initial nuclear radiation that would reach the 50-psi overpressure range from a 100-kiloton or smaller explosion, it should have at least 6 feet of earth cover and additional cans of earth should be placed in the horizontal parts of the entryways promptly after the shelter is occupied.

The life-saving potential of well designed, well built blast shelters is a demonstrated fact. People living in high-risk areas would be able to build expedient blast shelters within only a few days. Field-tested instructions, had made some preparations before the crisis arose, had a few days of warning, and during the crisis were motivated by the President. The following information is given to encourage more Americans to make preparations for blast protection. Also, it may serve to convince those who realize the need for permanent blast shelters in high-risk blast areas.

Some informed citizens particularly those who live near large cities or in their outer suburbs would be able to build earth-covered expedient blast shelters in their backyards, rather than to evacuate. Going to build or find good shelter and other essentials of life would entail risks that many people would not take particularly if they live outside the probable areas of severe blast damage. For such citizens the best way to stay at home, build earth-covered expedient blast shelters, supply them with the essential items, and keep their main with their possessions.

The following descriptions of the characteristics and components of expedient blast shelters are given to readers to use locally available materials to provide at least 15-psi blast protection. Practical, as well as the ability to work very hard for two to four days. (Field-tested instructions date only workers who were supervised have built expedient blast shelters. 5 )

## PRACTICALITY OF EXPEDIENT BLAST SHELTERS

At Hiroshima and Nagasaki, simple wood-framed shelters with about 3 feet of earth over wood were not damaged by blast effects in areas where substantial buildings were demolished. 4

Figure D. 1 shows a Fliroshima shelter that people with hand tools could build in a day, i able. This shelter withstood blast and fire at an overpressure range of about 65 psi. Its earth cover brought about effective earth arching; this kept its yielding wooden frame fro

Fig. D. 1. A small, earth-covered backyard shelter with a crude wooden frame-undamaged, although only 300 yards from ground zero at Hiroshima, (nw244a.jpg)

Although the shelter itself was undamaged, its occupants would have been fatally injured b no blast door. The combined effect of blast waves, excessive pressure, blast wind, and bur blown into the shelter (the popcoming effect) and from the heated air would have killed th survive in areas of severe blast, their shelters must have strong blast doors.

In nuclear weapons tests in the Nevada desert, box-like shelters built of lumber and cover were structurally undamaged by 10- to 15-psi blast effects. However, none had blast doors, open shelters would have been injured by blast effects and burned as a result of the popco blast winds blew away much of the dry, sandy earth mounded over the shelters for shielding quate protection against fallout radiation.

Twelve different types of expedient shelters were blast-tested by Oak Ridge National Labor Defense Nuclear Agency's blast tests. 5 Two of these tests each involved the detonation of of conventional explosive; air-blast effects equivalent to those from a 1 -kiloton nuclear by these chemical explosions.

Several of these shelters had expedient blast doors which were closed during the tests. Fi damaged interior of the best expedient blast shelter tested prior to 1978, an improved ver described in Appendix A. Its two heavy plywood blast doors excluded practically all blast rose only to 1.5 psi an overpressure not nearly high enough to break eardrums. The only da ent shelter- ventilating pump (a KAP) in the stoop-in entryway. Two men worked about 5 min flap-valves that were blown loose.

Fig. D.2. Undamaged interior of a Small-Pole Shelter after blast testing at the 53-psi ove range. Large buildings would have been completely demolished, (nw244b.jpg)

When blast-tested at 5-psi overpressure, not even the weakest covered-trench shelters with (described in Appendix A) were damaged structurally. However, if the covering earth were s were exposed to the blast winds of a megaton explosion at the 5-psi overpressure range, so blown away that the shelter would give insufficient protection against fallout radiation. earth mounded over some of the above- ground shelters was, in fact, removed by the blast w small test explosions, even at the lower overpressure ranges at which homes would be wreck tests where the steeply mounded earth was damp, little blast-wind erosion resulted. (The r even if shelters without blast doors are undamaged, the occupants are likely to suffer inj

Millions of Americans if given good instructions, strong motivation, and several days to build blast shelters with materials found in many rural areas and suburban neighborhoods. could be cut down for poles and sticks, and a garage or part of a house could be torn down. Citizens could build expedient blast shelters if they learn to:

- ° Utilize earth arching by making a yielding shelter. The remarkable protection that earth provides in those parts of a shelter designed to use it is illustrated by Fig. D.3.

Fig. D.3. Effective earth arching in the earth covering of this 4-ft-wide Pole-Covered Trench Shelter prevented a single pole from being broken by blast forces that exerted a downward force of 10 tons per square foot) on the overlying earth, (nw245a.jpg)

This picture shows the unbroken roof of a 4-foot-wide Pole-Covered Trench Shelter that was covered with rock-like soil and blast tested where the blast pressure outside was 53 psi. Its strong blast resistance prevented the blast wave from entering. Without the protection of earth arching that developed in the 5-foot-thick earth over the yielding roof poles, the poles would have been broken like straws. In contrast, the earth pressure produced by 1-kiloton blast effects almost completely collapsed the unsupported earth walls.

Fig. D.4. Post-blast interior of an Above-ground, Door-Covered Shelter that survived 1-kiloton blast effects at the 5.8-psi overpressure range. The shelter walls were made of bedsheets as described in Appendix A. (nw245b.jpg)

Figure D.4 also indicates the effectiveness of earth arching. This photo shows the roof of an earth-covered fallout shelter, as it appeared after surviving blast effects severe enough to destroy homes. The roof was made of light, hollow-core, interior doors and looks as though it had broken. In fact, only the lower sheets of 1/8-inch-thick veneer of the hollow-core interior doors were broken. (These breaks were caused by a faulty construction procedure a front-end loader had dumped earth onto the uncovered doors.) The upper 1/8-inch-thick sheets of veneer were bowed down but unbroken, until an earth arch formed in the 2-foot-thick earth covering and prevented the veneer from being broken. Earth arching also prevented this roof from being smashed in by blast overpressure of 5.8 psi (835 pounds per square foot) on the surface of the earth mounded over the shelter. (See Appendix A for details of construction.)

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- ° Make shelters with the minimum practical ceiling height and width. Most of the narrow covered-trench shelters used by tens of thousands of Londoners during the World War II blitz had only 4- 1/2 foot ceilings, to maximize blast protection and minimize high water-table problems. They were found to be among the safest for protection against nearby explosions. The Chinese also had a good understanding of this 'design principle and skillfully utilize the protection provided by the earth. The civil defense handbook states: '... the height and width of tunnel shelters should be kept to the minimum required to accommodate the sheltering requirements,' and 'The thicker the protective layer, the greater the ability to resist blast waves.' 2 1

- ° Shore earth walls to prevent their caving in as a result of ground shock and earth pressure.



unshored (that is, unsupported) earth walls are partially collapsed by ground shock at much overpressures than those at which a flexible roof protected by earth arching is damaged. Figure D.5 is a picture of a seated dummy taken by a high-speed movie camera mounted inside an unshored, Pole-Covered Trench Shelter of the Russian type tested at the 20-psi range. (A second dummy was obscured by the blanket curtains made of blankets.) The shelter had an open stairway entryway, positioned at right angles to the stand-up-height trench and facing away from the targeted "city" so as to minimize the entrance of air blast wave and blast wind.

Fig. D.5. A dummy in an unshored Pole-Covered Trench Shelter as it is struck by collapsing rock-like earth walls. The photo also shows the shelter's blanket-curtains as they are torn by the 180-mph blast wind. (Immediately after this photo was taken, the dummies were killed by the air blast wave and blast wind. Outside, the blast wind peaked at about 490 mph.) (nw246a.jpg)

Figure D.6 is a post-blast view of the essentially undamaged earth-covered roof poles and collapsed, unshored shelter walls of the Russian shelter tested at 20 psi. This shows that unshored fallout shelters do not survive closer to the blast than the 7-psi overpressure limitation was confirmed by an identical shelter tested at 7 psi; parts of its unshored walls collapsed by the ground shock from an explosion producing merely 1-kiloton blast effects.

Fig. D.6. Dummies after ground shock from 1-kiloton blast effects at the 20-psi range had rock-like walls of a hardened desert soil called caliche. The dummies' steel "bones" and limbs were not from being knocked down and buried. The fallen caliche all around them kept them from being killed by the air blast wave and 180-mph blast wind that followed, (nw246b.jpg)

Unsupported earth walls should be sloped as much as practical. The length and strength of the shoring material should be considered and, in order to attain effective earth arching, the thickness of the shoring should be at least half as great as the distance between the edges of the trench.

The stability of the earth determines the proper method for shoring the walls of a trench,

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Methods for shoring both loose, unstable earth and firm, stable earth are described below:

\* In loose, unstable earth such as sand, the walls of all underground shelters must be shored. An oversized trench must be dug with gently sloping sides. Next, the shoring is built, often a roofless structure. Then earth must be backfilled around the shoring to a level a few inches above the uppermost parts of the shoring, as in Fig. D.7. Finally, the roof poles or planks must be supported only by the backfilled earth. Blast tests have indicated that a Pole-Covered Trench Shelter thus proportioned and lightly shored should protect its occupants against disastrous earth walls at overpressure ranges up to 15 psi.

\* In firm, stable earth, it is best first to dig a trench a few inches wider than 7 feet (the width of the roof poles) and 1 foot deep. Next, dig the part to be shored, down the center of this shallow trench to the dimensions given for the shoring in Fig. D.7. The trench walls should be sloped and smoothed accurately, so that the shoring can be tightened against the earth. If the shoring does not fit snugly against the trench walls, large wedges of earth may be jarred loose, hit the shoring, and cause it to collapse.

A different, comparatively simple way to tighten shoring is indicated by Fig. D.8. This sketch shows a 4-pole frame designed to be installed every 2-1/2 feet along a trench in stable earth and against trench-wall shoring with the same dimensions as those shown in Fig. D.7. Note that the horizontal brace poles have shallow "V" notches sawed in both ends.

Fig. D.7. An illustration of several ways to shore a trench in unstable earth, using various materials. A 4-piece frame (consisting of 4 poles, or 4 boards, installed as shown above) every 2- 1/2 feet along the length of the trench, including the horizontal parts of the trench shoring should be at least 2 inches below the roof poles, so that the downward forces are on the earth. (ORNL-DWG 78-14430R) (nw247.tif)

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Fig. D.8. A 4-pole frame designed so that it can be tightened against the shoring material must press firmly against the walls of a trench dug in stable earth. (In this sketch, the three poles have been removed, so that the upper brace pole may be seen more clearly.) (ORNL-DWG 78-17246R) (nw248a.tif)

If these brace poles are driven downward when positioned as shown, the two wall poles are pushed outward against the shoring materials placed between them and the earth walls. An upper brace pole should be cut to the length needed to make it approximately the same height as the roof pole side of it (no higher) after the shoring is tightened. Finally, each "V"-notched end should be against the wall pole.

Light, yielding poles can serve simultaneously both to roof and to shore a shelter. A good example is the Chinese "Man" Shelter illustrated in Fig. D.9, requiring comparatively few poles to build. It is too cramped for long occupancy, and its unshored, lower earth walls can be squeezed in by blast pressure. Therefore, it is not recommended if sufficient materials are available for building a covered-trench shelter. It is described here primarily to help the reader understand the construction of similarly designed entryways, outlined later in this appendix. The room and the horizontal entryways of the model tested were made of 6- 1/2-foot poles averaging only 3 inches in diameter. It had two triangular entries of ORNL design. Each was protected by an expedient triangular blast door made of thin poles. In Fig. D.9, note the two small horizontal poles at the top of the triangle, one tied inside and the other tied outside the triangle, to hold the wall poles together. Before covering this shelter, a 6-inch-thick covering of small limbs was placed horizontally across the approximately 3 -inch gap between the 6-1/2 foot wall poles; the limbs were then covered with bedsheets.

Fig. D.9. Chinese "Man" Shelter tested at 20 psi, and undamaged because the thin poles yielded and were protected by earth arching. This drawing was taken from a Chinese civil defense manual. Dimensions are in centimeters, (nw248b.tif)

When blast-tested in loose, unstable soil, the unsupported earth walls of the trench below the roof were squeezed in. The 12-inch width of the foot trench was reduced to as little as 4 inches after the duration forces produced by 0.2-kiloton blast effects at 50 psi. The much longer duration forces of a megaton explosion would be far more damaging to the shelter at lower overpressure ranges, destabilizing and squeezing-in unshored earth at depths many feet below ground level.

Calculations based on blast-test findings indicate that the unsupported earth walls of a shelter will fail if the aboveground maximum overpressure is greater than 5 to 7 psi and this overpressure

by an explosion that is a megaton or more.

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(Most homes would be severely damaged by the 3-psi blast effects from a 1 -megaton or large. This damage would result one mile closer to ground zero of a 1 -megaton surface burst than at which the unshored earth walls of some shelters would be collapsed. For a 20-megaton surface burst, the corresponding reduction in distance would be about 2.7 miles.)

0 Build sufficiently long and strong entryways. Blast shelters need longer horizontal and vertical entryways, and thicker earth cover than do most fallout shelters; these are needed for increased protection against high levels of initial nuclear radiation. The entryways of the shelters described in Appendix A. 3 (with the improvements for increased blast protection outlined in the last section of this appendix) afford protection against both blast and radiation up to the 50-psi level. However, these entryways require straight poles 14 feet long; these may be difficult to find.

In contrast, both the horizontal and the vertical parts of the triangular entry pictured in Fig. D. 12 require only small-diameter, short poles. Triangular entries of this type were undamaged by blast effects at the 20-psi overpressure range and by 0.2-kiloton blast effects at 50 psi. Its blast door (also triangular and made of short poles) can be used with a wide variety of shelters and should withstand megaton blast effects at 25 psi. Therefore, their construction requires considerable detail.

\* The horizontal part of a triangular entry: if the Chinese "Man" Shelter shown in Fig. D. 9 is used without excavating the unshored lower trench that forms its earth seats, it will serve as an unshored crawlway-entry affording blast protection up to at least the 25-psi overpressure range. For horizontal entries, one at each end of the shelter, should be provided. Each entry should be 14 feet long. This length is needed to reduce the amount of initial nuclear radiation reaching the blast shelter while assuring adequate through-ventilation. The outer part of such a horizontal entry is shown in the background of Fig. D. 10.

\* The vertical part of a triangular entry: The lower section of the vertical part is made in a manner similar to the horizontal shelter shown in Fig. D.9. Figure D. 10 shows 4- 1/2 foot horizontal poles forming a "V", with one end of each pole laid on top of the adjacent lower pole. The other ends of the poles (1) are pressed against the two pairs of vertical posts (2). (After this photo was taken, these two pairs of vertical posts were sawed off as shown in Fig. D. 11 .) The 4- 1/2-foot poles (1) were kept level by the short spacer-poles (3) that were wired or nailed in place.

Fig. D. 10. Uncompleted lower section of a vertical triangular entry. (Photo 2954-77R) (nw250.tif)  
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Fig. D. 11. Lower part of a vertical triangular entry, showing its connection to the horizontal part of the shelter entry. (ORNL-DWG 78-14675) (nw250.tif)

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Fig. D. 12. Completed frame of Chinese "Man" Shelter showing its two ORNL-designed entryways (one at each end) and triangular blast doors made of poles. Before covering the vertical entries with earth, tree branches were placed vertically over the sides; the branches were covered with bedsheets. Horizontal branches, also covered with bedsheets, were laid over the rest of the shelter.

After being covered with earth, this shelter was subjected to 1-kiloton blast effects. Mul and around this yielding structure prevented both the small poles and the bedsheets from b (nw251.jpg)

Each pair of vertical posts (2) was securely wired together at top and bottom. The two pair top and bottom by two horizontal brace- poles toenailed in place to frame the rectangular "doorway" between the vertical entry and the horizontal entry. Only the upper pole (4) of zontal brace-poles is shown.

The two pairs of vertical posts (2) were positioned so that they pressed against two 7-1/2 only the uppermost is shown. These in turn pressed against the outermost two poles (6) of against the earth in two slot-trenches dug in the sidewalls of the excavation. These two 7 at least 6 inches in diameter.

Additional details of the lower section of this vertical triangular entry are given in Fig considerably larger in diameter than those illustrated are used, fewer poles are required. However, the space inside the entry is decreased unless the larger-diameter horizontal pole made longer than 4-1/2 ft.

As shown on the left in Fig. D.10, a small, vertical pole (7) was placed in the small "V" the horizontal poles that form the lower section of the vertical entry. After this photo w tical pole was positioned in the adjacent large "V", inside the entry. These two poles (7) gether so as to make a strong, somewhat yielding, outer- comer connection of the horizontal way that the tops of the side-wall poles of the Chinese "Man" Shelter are bound together.

The upper section of the vertical part of this entry (the section above the tops of the tw shown in Fig. D. 10 and Fig. D. 1 1) is made by overlapping the ends of its nearly horizon poles [marked with a (1) in Fig. D. 10] were each 4 feet 6 inches long and varied uniforml 2- 1/2 inches just above the two pairs of wired-together posts, to 4-inch diameters just b of poles. The triangular- shaped blast door was hinged to and closed against this door fra cut from worn auto tires, to be described shortly.

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The upper section is formed by laying poles in a triangular pattern, ends crossing at the small ends placed so that the poles are as nearly horizontal as is practical. Each of its strong wires that tightly bind an outside and an inside small vertical pole, in the same m

"Man" Shelter (shown in Fig. D.9) is secured. (Instead of No. 9 soft steel wire, rope or t could be used.)

Before starting to install the upper section of a vertical triangular entry, the three out cal poles that will hold the three comers together should be connected temporarily with th Connect them at the height of the door frame planned for the triangular blast door, and sp same size as this door frame.

Next, all the horizontal poles should be laid out on the ground in the order of their incr gular entry then should be started with the smallest poles at the base, with increasingly ward the top so that the three pairs of small vertical poles will press securely against a the entry.

To prevent the negative overpressure ("suction") phase of the blast from yanking out and closing the door and the upper part of the vertical entry to which it is hinged, the uppermost 4 or 5 three sides of the vertical entry should be wired or tied securely together. Rope or strip of strong wire is not available.

Before placing earth around this lightly constructed blast-protective entry, the vertical thickness of about 6 inches with a yielding, compressible covering of limbs, brush, or inner space brush should be placed in three layers, with the innermost layer at right angles to the vertical thickness then is covered with strong cloth, such as 50% dacron bedsheets, or two thicknesses of film. This outermost covering keeps loose earth or sand from filling spaces inside the yielding entry. Thus protected, this vertical entry should be undamaged by 25-psi blast effects of

A vertical blast-protective entry can also be made like a strong box, using 2-inch-thick boards for blast protection up to 50 psi if made as small as shown here and protected with a yielding material and a 6-inch-thick layer of brush covered with strong cloth.

Fig. Pg. 252 A vertical blast -protective entry. 22 in. x 28 in. (nw252.tif)

Install blast doors to keep out airborne blast waves, blast wind, overpressure, blast-born burning-hot dust and air, and fallout.

A fast-rising overpressure of as little as 5 psi will break some people's eardrums. At over 50% of the people who are exposed will have their eardrums broken. However, persons near a blast have their eardrums broken by somewhat less than half of these unreflected overpressures. (Any blast waves and greatly increase overpressures near it.) Broken eardrums are not serious in normal clear attack this injury is likely to be far more dangerous to persons in crowded shelters and medical treatment. Lung damage, that can result from overpressures as low as 10 to 12 psi, would be a post-attack condition.

A blast door must withstand blast waves and overpressure. Not only must the door itself be able to withstand forces at least as great as those which the shelter will survive, but in addition the transverse walls must be equally as strong. The expedient blast door pictured in Fig. D. 13 is made of 1 1/2 inch car tires each a full 2 inches thick. It had a continuous row of hinges made of 1 1/2 inch-long strips of car tires.

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Fig. D.13. Blast door surrounded by 4 blast-protector logs that were notched and nailed to the

The wet, mounded soil had been compacted by the blast but not blown away, (nw253.jpg)

The strips were nailed to the vertical poles on one side of the vertical entry. These and the other strips are shown in Fig. D. 14. Although the two center boards were badly cracked by the shock pressure at the 17-psi range, the door pictured in Fig. D.13 afforded good protection against the surface explosion of a million pounds of TNT. In Fig. D. 14, note the essential, strong tie wires at the bottom of the vertical entry, to prevent the blast door from being yanked open ("suction") that follows the overpressure.

Blast doors must be protected against reflected pressures from blast waves that could strike the door and tear it off its hinges. Note the blast-protector logs installed around the door. When the door was closed, the tops of these four logs were about 2 inches higher than the edges on all sides.

The closed door must be prevented from rebounding like a spring and opening a fraction of an inch. It must be bowed down by overpressure, or from being opened and perhaps torn off its hinges by the pressure that follows the overpressure phase. Figure D. 14 gives the details of such a hold-down system. Near the bottom of the vertical entry the 6 strong wires must encircle a horizontal pole and be nailed to the vertical wall poles with at least a dozen 6-inch (60-penny) nails. Blast tests in the 1950s have proved that this hold-down system works. 5

Figure D. 15 shows a blast door made of 5 thicknesses of 3/4-inch exterior plywood, well secured with 4-1/2-in. nails at 4-in. spacings. This door was protected by 4 blast-protector logs, 6 inches in diameter. The logs were notched, nailed together, and surrounded with earth. For protection by the thermal pulse from an explosion, exposed wood and rubber should be coated with thick lime) or mud, or covered with aluminum foil.

An equally strong blast door and the door base upon which it closes can be made of poles. They are easy to work with ax and saw. Figure D.16 shows the best blast-tested design. This door is made of a row of hinges made from worn auto tire treads. The pole to which the hinges were attached was 12 inches in diameter after peeling and had been flattened on its top and outer sides. The two other poles were 8 inches in diameter and had been flattened with an ax on the bottom, top, and inner sides. Each pole was 55 inches long. They were notched and spiked together with 60-penny nails so that the door could be closed on its similarly constructed base made of three stout poles. Other poles, at least 7 inches in diameter and hewn so that they would fit together snugly, were nailed side-by-side on top of the three

Many Americans have axes and would be able to cut poles, but not many know how to use an axe to make square sides on a pole or log. This easily acquired skill is illustrated by Fig. D. 17. The worker flattens the pole down by nailing two small poles to it and to other logs on the ground. Figure D. 17 shows the worker. When hewing a flat side, the worker stands with his legs spread far apart, and repeatedly looks almost straight down at where his ax head strikes. First, vertical cuts with a sharp axe, 12 inches apart and at angles of about 45 degrees to the surface of the pole, for the length of the pole. Then cuts should be made almost as deep as is needed to produce a flat side of the desired width. Beginning at the starting end, the worker should cut off long strips, producing a flat side.

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Fig. D. 14. Expedient blast door that can be closed and secured in 4 seconds. Four seconds is the time if the shelter is at the 15-psi overpressure range from a 550-kiloton or smaller warhead. This is the Soviet ICBM arsenal. (See the last paragraph on page 255.) However, this door closure is the best blast-tested expedient means to secure a closed blast door. (ORNL-DWG 73-1063) (nw254.tif)

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Fig. D.15. Tire-strip hinges nailed to an expedient, 4-inch-thick blast door made of plywood. The door is designed to withstand 50-psi blast effects of very large weapons and undamaged by blast at 1000 yards range, (nw255a.jpg)

Fig. D.16. Blast-tested triangular blast door made of hand-hewn pine poles, notched and nailed together. This door closed on a triangular pole base that is concealed in this photo by two

blast-protector logs that also withstood 53-psi blast effects, (nw255b.jpg)

Fig. D.17. Hewing flat sides on a pole with a sharp ax. (nw255c.jpg)

To hew a second flat side at right angles to the first side, rotate the pole 90°, secure it as shown in Fig. D.17.

° Provide blast closures for an adequate ventilation system. The following two expedient closures will permit adequate volumes of ventilating air to be pumped through a shelter:

1 . Install two blast doors, one on each end of the shelter, designed to be left open until a bright light from a large blast is seen. Figure D. 14 shows a door held open by a prop-stick yanked away by the attached pull-cord. While propped open, one blast door serves as an external low-resistance air-intake opening, and the other serves as an air-exhaust opening. A large pump can pump air at the rate of several thousand cubic feet per minute through such open doors.

When an attack is expected, each pull-cord should be held by a shelter occupant who stays ready to yank out the prop-stick as soon as he sees the light of an explosion. After the door handle loop at the end of its wire bridle is close to the upper hook of the load-binder and at the

D. 14). The person who closes the door should quickly hook the upper hook of the load-binder loop and pull down on the handle of the load-binder. The door will then be tightly shut. (The emergency would be the millions of load-binders owned by truckers and farmers.)

At distances from a large explosion where blast wave and overpressure effects are not destructive, smash most good expedient blast shelters, there is enough time between the instant the light is seen and the arrival of its blast wave for an alert person to shut and securely fasten a blast door. The smaller the explosion and the greater the overpressure range, the shorter the warning. At the 15-psi overpressure range from a 1 -megaton surface burst (1.5 miles), the blast wave arrives about 4.5 seconds after the light; whereas at the 10-psi overpressure range from a 1- megaton surface burst, the blast wave arrives about 4.5 seconds after the light. For a 20- megaton surface burst, the 15-psi range is about 8 seconds, and at the 30-psi overpressure range, about 4 seconds. Experiments have shown that even people who react quite slowly can close and secure this door within 4 seconds of seeing a spotlight shine on the door without warning.

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2. Build a vertical air shaft next to the outermost side of each vertical entry, with an

Overlapping-Flaps Blast Valve (see Fig. D. 18) connecting each entry to its air shaft, as shown in Fig. D. 19. These air shafts and blast valves permit forced ventilation to be maintained when the shelter is closed. Figure D. 18 illustrates the construction of a fast-closing expedient blast valve that is undamaged by the 65-psi shock wave and other effects produced by the explosion of a milligram of TNT. When blast-tested in a shock-tube at 100-psi, the flaps were undamaged; they closed in 0.006 second (0.006 sec.). This is as fast as the best factory-made blast valves close.

Fig. D.18. Overlapping-Flaps Blast Valve, made of boards, plywood, and strips cut from the treads of worn car tires. (ORNL DWG 73-2229A) (nw256a.tif)

To withstand 50 psi, the load-bearing "2-inch" boards (actually 1-1/2 inches thick) of the inches wide, if the 1 -in. -high air openings are each made 12 in. wide, measured between ter entry. See Fig. D. 19, that gives the dimensions of a valve that has been blast tested of solid wood at each end of each 1 -in. -high air opening. If there are 5 such air openin stalled KAP (Appendix B) can pump air at about 1 25 cubic feet per minute (125 cfm) throug such valves. This ventilation rate is ample for at least 40 people in cold weather. Except constant air supply of about 10 cfm per shelter occupant is enough to maintain tolerable c shelter occupancy for many days.

Fig. D.19. Installation of a 50-psi Overlapping- Flaps Blast Valve in such a way that it w blown into a shelter by the blast overpressure, nor pulled out by the following negative p ("suction") phase. (ORNL-DWG 78-14429R) (nw256b.tif)

If a factory-made blower capable of pumping more than 100 cfm is available, use it. Such a er can pump against much higher air flow resistances than a KAP can. It can pump its full- air through a shelter equipped with two Overlapping-Flaps Blast Valves, one at each end of only 2 air openings-providing a total of 24 square inches of openings per valve. Equally c makeable Plywood Double- Action Piston Pump, made and operated as described in Appendix E.

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Remember that a pressure of 7200 pounds pushes against each square foot of the exposed fac when it is subjected to a 50-psi blast overpressure. Also keep in mind that the "suction" wardly directed force of up to 700 pounds per square foot on the valve face and can yank i securely installed. Figure D. 19 shows how to securely install a blast valve. (Merely nail will not enable it to withstand severe blast forces.)

Note in Figure D. 19 that an opening is shown between the back edge of the uppermost board Flaps Blast Valve and the adjacent horizontal pole of the vertical entry. Both this openin to the lowermost board of the Blast Valve should be closed off with a stout board, to prev these openings and on into a vertical entry and the shelter room.

The top of an air shaft should be a few inches higher than the earth piled around it, as a entries of the Small-Pole Shelter illustrated in Figure A.3 . 1 on page 1 74. To minimize into an air shaft, a shed-like, open-sided miniature roof should be placed over it, a few can be lightly constructed, since it will be blown away by a severe blast.

° Minimize aboveground construction and the mounding of shielding earth. At high overpress ranges, the shock wave and the blast-wind drag can wreck an aboveground shelter entry. For 5-ft-high earth mound over a shelter built with its pole roof at ground level was moved en air-blast effects at the 53-psi overpressure range to break one of the poles of a blast-do of a 1 -megaton explosion at the same overpressure range would have operated 10 times as l would have smashed the vertical entryways of this shelter. Whenever practical, a blast she far enough belowground so that the top of its shielding earth cover is at ground level. Av construction and earth mounds also greatly reduces the chances of damage from blast-hurled such as tree trunks and pieces of buildings.

Dry earth, steeply mounded over a shelter which is subjected to blast winds from a big exp blown away. However, blast- wind "scouring" of wet earth is negligible. The blast winds fr the 31-psi overpressure range scoured away 17 inches of dry, sandy soil mounded at a slope



If it is impractical to build a blast shelter with its roof belowground, good protection can be achieved by covering the shelter with even dry earth at slopes not steeper than 10 degrees.

- ° Provide adequate shielding against initial nuclear radiation. Good expedient blast shelters require a greater thickness of earth cover than is needed on good fallout shelters, for these reasons.

- \* Blast shelters should also protect against initial nuclear radiation emitted by the fire. The radiation is reduced by half when it penetrates about 5 inches of packed earth (as compared to a halving of the radiation from fallout only about 3- 1/2 inches of earth against radiation from fallout).

- \* The initial radiation, in some areas where good blast shelters will survive, can be much greater than the fallout radiation is likely to be.

- \* Initial nuclear radiation that comes through entryways is more difficult to attenuate (reduce) than radiation. Therefore, longer entryways or additional right-angle turns must be provided.

For these reasons, good blast shelters should be covered with at least 4 ft of well-packed earth or 5 ft of unpacked or light earth. (A 3-ft thickness gives excellent protection against radiation from fallout.)

#### A 50-PSI SMALL-POLE SHELTER

This expedient blast shelter is described in detail to enable the reader to build this model shelter and to better understand the design principles of other expedient blast shelters that are capable of withstanding blast effects severe enough to destroy all ordinary buildings and kill the occupants. Blast effects at overpressure ranges up to 50 psi that are produced by an explosion of 1 megaton of TNT. The shelter indicated that the Small-Pole Shelter described and illustrated in Appendix A.3 will afford protection against blast effects at overpressure ranges up to 50 psi that are produced by an explosion of 1 megaton of TNT. The shelter is:

- ° Made with horizontal entryways each with ceilings no higher than 7 ft, 2 in., no wider than 4 ft, and at least 10 ft long to lessen the radiation coming through the entries (see Fig. D.20). Lower ceilings and longer entryways would give better protection but would increase the time required for entry.

- ° Constructed with a floor of poles that are 4 in. or more in diameter, laid side-by-side, resting on the floor poles. The ground shock and earth pressures at a depth of 10 ft or more will counteract the overpressure on the surface of more than about 35 psi, if caused by a large explosion, may squeeze earth upward into the shelter through an unprotected earth floor. The Small-Pole Shelter described in Appendix A.3 has an earth floor.

- ° Installed in an excavation about 13 feet deep, with the shelter's vertical entrances approximately 1 foot above the height so that the blast doors are only about one foot above the original ground level.

To prevent possibly life-endangering cave-in of the 13-foot-deep trench that was dug for the model shelter, the trench walls were sloped about 45 degrees. The shelter was built as a bunker, and then covered. During a crisis it would be impractical to safely excavate a deep trench and then safely build a shelter in it. A 13-foot-deep trench is usually too deep to dig it with safely sloping walls requires the removal of a large amount of earth.

Fig. D.20. Entryway of Small-Pole Blast Shelter shielded against initial nuclear radiation is a simplified vertical section through the centerline of one end of the shelter. ORNL-DW (nw258.tif)

- ° Made with 4 rectangular horizontal braces in each vertical entry, in addition to the end ladder-like braces. The detailed drawings in Appendix A show such braces. The lowest rectangle should be positioned 3- 1/2 feet above the flooring at the bottom of the vertical entry (see Fig. D. 14).

- ° Equipped with blast doors each made of 5 sheets of 3/4-inch exterior plywood (see Fig. D. 14). The sheets are glued together with resin glue and nailed together with 4- 1/2 in. nails. The nails should be driven on 4-in. centers. The protruding ends should be clinched (bent over). The blast doors must be secured against being blown open by negative pressure ("suction") by securing them with a strong wire bridle (see Fig. D. 14). The bridle, a lower, fixed wire strongly connected near the bottom of the entry to all of the vertical poles, is shown in Fig. D. 14.

- ° Provided with an adequate ventilation pump and with ventilation openings protected against blast by blast valves (Fig. D. 14) installed in the vertical entries as shown in Fig. D. 19, to provide for air-exhaust openings. (Ventilation openings should be as far as practical from buildings and other flammable materials. Manually closed ventilation openings are NOT effective at the 50-psi overpressure range of nuclear weapons, because there is insufficient time to close them between the arrival of the warning and the arrival of the blast wave.)

- ° Made with the roof poles covered by a yielding layer of brush or limbs about 6 inches thick. This yielding layer in turn should be covered with bedsheets or other strong covering material to increase the effectiveness of protective earth arching. Brush or limbs should be laid in 3 layers with the middle layer perpendicular to those of the other two layers.

- ° Covered with 5 feet of earth, sloped no steeper than 10 degrees.

- ° Provided with additional shielding materials in the entryways, as shown in Fig. D.20. Such shielding is needed to prevent occupants from receiving possibly incapacitating or fatal doses of initial nuclear radiation through the entryways at the 50-psi overpressure range, if the shelter is subjected to the blast of a nuclear weapon that is one megaton, or larger, in explosive yield.

Damp earth serves better for neutron shielding material than dry earth and can be substituted for concrete if sufficient water containers are not available. (At the 50-psi overpressure range, smaller than one megaton, the entry and shielding shown in Fig. D.20 may not provide adequate shielding against initial nuclear radiation.)

When the shelter is readied for rapid occupancy, the shelter-ventilating KAP is secured and the bags of earth in the doorway (under the KAP) are removed. Persons entering the shelter

the platform adjacent to the vertical entry. This platform is attached to vertical wall-panels supporting shielding water and earth. When all except the person who will shut and secure the shelter room, occupants should quickly begin to place bags of earth in the doorway. When the whole doorway can be closed with bags of earth or other dense objects until ventilation is

The entries of other types of blast shelters can be shielded in similar ways.

° Protected against fire by being built sufficiently distant from buildings and flammable materials. Its exposed wood covered. For maximum expedient protection against ignition by the thermal radiation from a large explosion, all exposed wood should be free of bark, coated with wet mud or damp slaked lime (whitewash), and covered with aluminum sheet metal or foil to reflect heat. (Most of the thermal radiation from an explosion that was 1 megaton or larger would reach the 50-psi over-pressure range before the blast wave had arrived and had torn the expedient protective coverings from the wood. However, as observed in megaton nuclear weapon tests, the dust cloud first produced by the popcoming end of the blast winds would screen solid wood near the ground so effectively against thermal radiation that it would not be ignited, provided it had been initially protected as described above.)

#### PRECAUTIONS FOR OCCUPANTS OF BLAST SHELTERS

Although a well constructed blast shelter may be undamaged at quite high overpressure ranges, occupants may be injured or killed as a result of rapid ground motions that move the whole shelter several feet in a second. Rapid ground motions are not likely to cause serious injuries unless the shelter is subjected to 30-psi or greater blast effects. To prevent possible injury, when the occupants of high-pressure blast shelters expecting attack they should avoid:

° Having their heads close to the ceiling. The "air slap" of the air-blast wave may push down on an undamaged shelter much more rapidly than a person can fall. If one's head were to be only a few inches from the ceiling, a fractured skull could result.

° Leaning against a wall, because it may move very rapidly, horizontally as well as vertically.

° Sitting or standing on the floor, because ground shock may cause the whole shelter (including the floor) to move very fast and injure persons sitting or standing on the floor. The safest thing to do is to lie down on a suspended, strong hammock or chair, or on thick foam rubber such as that of a mattress, or on a bed of branches.

In dry areas or in a dry expedient shelter, ground shock may produce choking dust. Therefore, occupants should be prepared to cover their faces with towels or other cloth, or put on a mask. If a mask is not available, they should keep such protective items within easy reach.

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#### Appendix E

How to Make and Use a Homemade Plywood Double-Action

Piston Pump and Filter

## THE NEED

Ventilating pumps—mostly centrifugal blowers capable of operating against quite high resistances—used to force outdoor air through most high-protection-factor fallout shelters and through mobile shelters. Low-pressure ventilating devices, including ordinary bladed fans and homemade air pumps and Directional Fans, cannot force enough air through a permanent shelter's usual air-supply pipes, or of pipes with a blast valve, a filter, and the valves needed to maintain a positive pressure.

Manually cranked centrifugal blowers, or blowers that can either be powered by an electric motor or manually cranked, are the preferred means of ventilating permanent shelters from Switzerland to China. The advantages of efficient centrifugal blowers are:

- 1 . They are quite expensive. For example, in 1985 a good American hand-cranked blower, that could pump about 50 cubic feet per minute (50 cfm) through a shelter's pipes, blast valve and filter, cost about \$250. An excellent foreign blower that enables one man to pump somewhat larger volumes sells for about twice as much.

2. Not enough centrifugal blowers could be manufactured quickly enough to equip all shelters during a recognized crisis threatening nuclear attack, and lasting for weeks to several months.

Therefore, there is need for an efficient, manually operated, low- cost ventilating pump that:

- 1 . Can pump adequate volumes of outdoor air through shelter- ventilating systems that have high resistances—up to several inches water gauge pressure differential.

2. Will be serviceable after at least several weeks of continuous use.

3. Can be built at low cost in home workshops by many Americans, using only materials available in the U.S.

4. Could be made by the millions in thousands of shops all over the U.S., for mass production during a recognized prolonged crisis, using only plywood and other widely available materials.

To produce such a shelter ventilating pump, during the past 20 years I have worked intermittently, building several types of homemade air pumps. However, until I was traveling in China as early as 1982 and saw a wooden double- action piston pump being used, I did not conceive or come able to develop into a shelter- ventilating pump that meets all of the requirements outlined above and tested a simple homemade Plywood Double-Action Piston Pump, described below, that satisfied all the requirements. Three other persons have used successively improved versions of these instructions and several others have contributed improvements.

## HOW A PLYWOOD DOUBLE-ACTION PISTON PUMP WORKS

Fig. 1 pictures the box-like test model described in these instructions.

Fig. 1 . Plywood Double- Action Piston Pump, with manometer attached for tests, (nw261a.jpg)

Fig. 2 illustrates a vertical section through a slightly improved model, and shows the piston being pushed from right to left, causing air from the outdoors to be "sucked" down the open end of the pump, then down to the right through the open valve in the airtight frame (that is above the PARTITION), and on down into the lower- pressure area behind the leftward-moving piston.

Because the air to the right of the leftward-moving piston is at a lower pressure than the the exhaust valves in the front end (the handle end) of the pump are held closed.

During this half of the pumping cycle, the higher-pressure air in the part of the pump's s the leftward-moving piston opens the air-exhaust valves in the back end of the pump, and f the shelter room, The higher-pressure air to the left of the valve in the airtight frame ( PARTITION) keeps this valve closed, while the lower-pressure air to the right of this valv When the piston is pulled to the right, all of the valves shown closed are quickly opened, ly closed. Then fresh air is forced into the shelter room through the opened exhaust valve pump.

Fig. 2. Vertical Section of the Double-Action Piston Pump showing its square piston being (nw261b.tif)

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#### PERFORMANCE TESTS

The volumetric and durability tests summarized below are proof that this homemade Plywood ton Pump is better than most hand-cranked centrifugal blowers for supplying a shelter with air-intake and exhaust pipes- especially when the ventilation system contains a filter and that give the best protection, Chemical Biological Radiological (CBR) Filters, have quite do commercial blast valves that close quickly enough to protect filters.

##### 1. Volumetric tests.

Because the rapidly pulsating airflows into and out of a piston pump are very hard to meas velocity meter, I made an inflatable cylindrical bag of 2-mil (0.002 inch) polyethylene fi this bag was 256 cubic feet. The bag was suspended on a horizontal strong cord running thr tube 62 inches in circumference connected the back end of the pump (that is opposite the c pendend bag. Bag and pump were in a below ground shelter that normally has essentially moti

Since this type of pump exhausts equal volumes of air from each of its two ends, the total that it pumps equals twice the cfm that it exhausts into the shelter from one of its ends. pump attached with "C" clamps to a small steel table and being used to pump air into the 2 bag.

I measured the pressure differences against which the pump was operated. In a shelter thes caused by the resistance to airflow in pipes, valves, and a filter. I measured pressure di gauge (1 in. w.g. 0.036 psi) with the small-tube manometer attached to the side of the pur pressure differences for several tests, I nailed a piece of plywood over the top of the ai different sized openings: in most tests I placed different layers of filter materials in a over the 6 x 6-in. air-supply duct on the top of the pump. See Fig. 3. (This low -resistar fallout particles of wartime concern, and also most infective aerosols that may be used in "Making and Using a Homemade Filter Box and Filter", by Cresson H. Kearny, October 1985.

Fig. 3. Pump with Homemade Filter (20 x 20 x 8-inches inside dimensions) connected airtigh pump's 6 x 6-inch air-intake duct, (nw262.jpg)

The best centrifugal blowers that I have seen or heard about are those manufactured by a F Oy. (I cranked a Temet Oy blower in an Israeli shelter used for testing ventilation equipment. The blower was better than Swiss, German and captured Russian blowers also undergoing tests.) A few of the volumetric tests of my best model Plywood Double-Action Piston Pump (powered by

are compared with performance data furnished by Temet Oy for its centrifugal blower when I have converted Temet Oy's metric units into the common American units.

In Table 1 the pressure difference of 4.3 inches water gauge is the resistance to airflow realistically gives as typical of a well designed shelter ventilation system of pipes, valves and Biological Radiological (CBR) filter. Temet Oy gives 2.0 inches water gauge as typical of a system with only a low resistance dust filter. The much larger volume pumped by the Double-Action Piston Pump with CBR filter is used (as compared to the cfm pumped by this very good centrifugal blower) is a measure of the effectiveness of even the best centrifugal blowers at high pressure differences.

In areas devastated by a nuclear explosion, the typical very dusty conditions are likely to be very dirty and higher in resistance to airflow. Then the greater effectiveness of a piston pump with a high-resistance air-supply system will be even more important than when its filter

TYPE OF  
PUMP

PRES.  
DIFF  
(in. w.g.)

CUBIC FEET PER MINUTE

HORSE-

POWER

Double-Action  
Piston Pump

one man

4.9

134

?

two men

4.3

182

?

Temet Oy  
Centrifugal Blower

two men

4.3

90

0.15

Double-Action  
Piston Pump

one man

2.3

172

?

two men

2.3

208

?

Temet Oy  
Centrifugal Blower

two men

2

300

0.18

Table 1 . Comparison of Plywood Double- Action Piston Pump with Temet Oy Centrifugal Blower

The horsepower requirements of my pump have not yet been measured. However, based on the pressure on the 12 x 12-in. piston of 22.3 lbs. when the pressure difference was 4.3 in. w.g. the pumpers were making 52 strokes (cycles) per minute while pumping 182 cfm, the horsepower delivered without allowing for friction and the losses of power due to reversals in the directions of flow is that the actual horsepower delivered by the two pumpers (I, a 69 year old with a stiff back

boy) was somewhat less than 0.2 HP. A man in good condition can work for hours delivering

When comparing machines powered by human muscles, what muscles are used and how they are used are as important as are the horsepower requirements. Leg muscles are more efficient and are much stronger than arm muscles. Arm muscles are used much more in cranking a blower than in pushing and pulling a reciprocating piston pump back and forth horizontally. See Fig. 3. If this double-acting pump is mounted at a height above the floor so that its handle is approximately at the height of a standing operator can do most of the work with his legs. See Fig. 3. He efficiently moves his body weight, while moving his hands and forearms horizontally for slightly less than a foot relative to the pump handle. A man who uses the same horsepower by cranking a blower uses less efficient muscles inefficiently, and is much

As shown in Table 2, the volumetric efficiency of my best model is good for a shelter-ventilator. The volumetric efficiency of a piston pump (a positive displacement pump) is found by dividing the actual cfm by the theoretical maximum cfm at the same pumping rate and the same pressure difference, assuming

full length, that all valves open and close instantaneously, and that there is no leakage. As the pressure difference increases, the pressure difference, the lower the efficiency- as one would expect, because of increased

PRES.

DIFF

(in. w.g.)

STROKES PER

MINUTE

dm

EFFICIENCY

4

36

122

84.0%

2.6

45

160

89.0%

0.7

51



188

92.0%

0.4

54

202

94.0%

0.2

55

208

94.5%

Table 2. Volumetric Efficiencies of Double- Action Piston Pump Operated by One Man.

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## 2. Durability tests.

Finding a homemadeable method to seal the moving piston so as to assure at least one month pumping was the most difficult problem. Various rubber seals attached to the edges of the and aluminum sheetmetal strips (shaped and attached like the galvanized steel sheetmetal s wore out in less than a week, even when oiled every 24 hours.

To save money during weeks of continuous durability testing, the pump was operated by an e powered a pulley drive that turned a 2-foot-diameter pulley having an attached 40-in. -lon connection to a horizontally- sliding bar connected to the handle of the pump's wooden pis

After pumping for 380 hours (15.83 days) at 44 strokes per minute against a pressure diffe the worst worn spot on any of the 30-gauge steel sheetmetal sealing strips on the piston w from its original 0.0155 in. to 0.0 145 in. This worst-spot wear of 0.001 in. is only about The flap valves functioned as well as when new, and appeared unworn.

I conclude that this pump would be serviceable after several months of continuous use-prov ter every 24 hours of actual use, as in this durability test. In this test I lubricated th and its rod with Lubriplate Ne. 105, "the original whito grease". This non-sticky "grease- sively, especially to lubricate internal combustion engines before first starting up. Anot found Siloo White Lube, an all-purpose lithium grease, the best of the lubricants that he durability' tests, a very' light oil applied daily serves reasonably well. Ordinary bearin

## MATERIALS

The following materials (that cost about \$65. retail in 1985) are needed to make and opera

pump:

Plywood, 3/4-in. exterior: one 4 x 8-ft. sheet (finished on one side, unwarped).

Plywood, 3/8-in. exterior: 1/4th of a 4 x 8-ft. sheet (finished on one side, unwarped). (See exterior plywood) .

Oak board, 3/4 x 1-3/4 in., straight, well seasoned, 4 ft. long, to make the piston rod. (If strong wood is not available, use a straight fir or pine board.)

Fir or pine board, about 3/4 x 1 -3/4 in., 8 ft. long, to make the piston-rod handle, etc.

2 8 -gauge or lighter galvanized- steel flashing ( sold by lumber yards for roofers) , no galvanized steel or flashing no thinner than 0.0 1 2 in. Or 30 gauge galvanized steel sheet from some sheetmetal shops. (Sheet- metal thicker than 0.0 1 6 is not springy enough for making near-equivalent of piston rings. ) Best to go to a sheet- metal shop and have 3 strips cut and about 30 in. long.

Screws, round-headed, zinc-plated wood screws:

22 each of No. 12 (2-in. long, 12/32 in. dia.), with flat washers 10 each of No. 10 (1-1/2 in. long, 1/2 in. dia. washers)

15 each of No. 6 (3/4 in. long, with flat washers)

Nails, 4-penny (1-1/2 in.), best cement-coated: 1/4 lb. Nails, 3-penny (1-1/4 in.), galvanized

Staples (if an oak board for the piston rod is not available), No. 17, 3/4-in., galvanized

Tacks, No. 6 upholstery, (1/2-in. long): a small container. Tacks, No. 3 upholstery (3/8-in. long): a small container. Felt, weather stripping, 5/8-in. wide: 10 ft.

Tape, silver duct tape, 2-in. wide: a small roll.

Tape, masking tape, 3/4-in. wide: a small roll.

Adhesive, waterproof: "Liquid Nails", or other all purpose construction adhesive: one approx. 10 oz. tube for use in caulking gun).

Epoxy, 5-minute: 2 tubes.

Rubber cement: a small tube.

Sealer (such as polyurethane clear finish, to reduce absorption of oil or other lubricant): 1 pint.

Plastic film, transparent storm-window type (such as 4-mil Flexo-Glass, by Warp Bros.): 3

Grease-type lubricant, an all-purpose motor-breakin lithium grease such as "Siloo White Lithium Grease" or "No. 5 Space Age Lubricant": two approx. 10 oz. tubes.

Inner tube rubber, heavy truck or auto (cut from an old tube): 1 sq. ft.

#### FUNCTIONAL RELATIONS OF PARTS

Look at Figs. 2, 5, and 6. In Figs. 5 and 6, the lower, fixed part of the front end is pictured. The piston rod slides back and forth on the center of the fixed part of the front end (as in Fig. 7), and in the notch in the removable part of the front end.

Fig. 4. Mechanized Drive Used in Weeks-Long Durability Tests, (nw263a.jpg)

Fig. 5. Front End of the Durability Test Pump, showing the lower fixed part (below piston rod) and the removable part, that is held by 6 screws with flat washers. Felt weather-stripping makes the front end airtight, (nw263b.jpg)

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Fig. 6. Pump Built by Dale Huber, of Lake City, Florida in his home workshop, while guided by the second draft of these repeatedly improved instructions. The removable part of the front end is shown inserted into the 12 x 12-in. "cylinder" under the PARTITION. The plastic flaps covering the valve holes are black: transparent plastic film is preferred, (nw264a.jpg)

Note that a single plastic-film flap covers each pair of 2 x 4-in. valve holes, and that, in a side view of all six flaps), all flaps open away from the vertical center-plane of the pump.

In Fig. 5 the removable (upper) part of the front end is shown in place, secured by six screws with flat washers. In Fig. 7 note the pair of 2 x 4-in. flap- valve holes above the piston rod.

In Fig. 6. the removable part of the front end has been removed, exposing the 26-in. -long PARTITION that serves as the top of the 12 x 12-in. "cylinder", in which the piston can make a 24-in. stroke.

Also see Figs. 2 and 7. Fig. 6 also shows the piston while it is being removed and one of the O-rings (made of inner tube rubber) on its piston rod.

The back end of the "box" is made of one piece of plywood, as shown in Fig. 8. The two pleated exhaust valves each cover two 2 x 4-in. valve holes, that are positioned the same as the four valve holes at the front end.

Fig. 7. Front End (Operator's End) of Plywood Double-Action Piston Pump. The two 4 x 12-in. valve holes are shown by dashed lines, as is the 12 x 26-in. PARTITION, (tinted the same as the four valve holes at the front end. (nw264b.tif)

Fig. 8. Back End. Only the plywood is shown, (nw264c.tif)

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#### CUTTING OUT THE PLYWOOD PARTS

1. The four parts of the "cylinder" (its bottom, two sides, and the PARTITION: see Fig. 7) the wood grain of the plywood running in the same direction as the lengths of these parts. friction.

2. Outline on a sheet of exterior 3/4-in. plywood all of the plywood parts-except for the the two 12 x 12-in. construction forms, which are made of 3/8-in. exterior plywood. (If 3/4-in. plywood is available, use 1/4-in.) Do not assume that the comers of a sheet of plywood are truly square. Allow the width of the sawcut of the saw to be used, and allow for this width when drawing adjacent lines on the plywood. Be sure to make all comers square.

3. If you do not have a table saw that saws accurately, or a heavy-duty' saber saw, you will need a professional carpenter or cabinet maker to saw out the plywood parts-and also the piston ring making it out of an oak board. A professional can accurately saw out all of the plywood parts and valve holes in about 2 hours, provided you have accurately outlined all saw lines.

4. Make the following plywood rectangles with tolerances of + or - 1/32 in. :

PARTITION, 12 x 26-in.

Two sides, each 16-3/4 x 32-in. (If your "3/4-in. plywood" actually is less than 1 1/16-in. thick, allow for the thickness of each of your sides 16-3/4-in. less the difference between 3/4-in. and the actual thickness of the plywood. See Fig. 7.)

Bottom, 17-1/2 x 32-in.

Top, 13-1/2 x 32-in.

Two valve frames, each 4 x 12-in.

Piston, 12 x 12-in. (of 3/8-in. plywood).

Two construction forms, each 12 x 12-in. (of 3/8-in. plywood).

5. Make the following plywood rectangles with tolerances of + or - 1/16 in.:

Backend, 13-1/2 x 17-1/4-in. (See Fig. 8.)

Removable (upper) part of front end, 13-1/2 x 10-7/8 in. (See Figs. 7 and 9.)

Fixed (lower) part of front end, 13-1/2 x 6-3/8-in. (See Figs. 7 and 10.) The four parts of the front end are: two each 6-1/2 x 6-in.; two each 6-1/2 x 7-1/2-in.

Two spacers (to be nailed to the bottom) each 3/4 x 3/4 x 32-in.

6. Saw out the 10 valve holes; a tolerance of + or - 1/8 in. is good enough. (See Figs. 7, 8, and 9.)

7. Saw a square 6 x 6-in. hole in the center of the top, as shown in Fig. 2 - if you are going to use a homemade filter (described in separate instructions) directly on top of your pump. (To connect to a round air-intake pipe, cut an appropriate round hole in the top.)

8. Sandpaper the finished sides of the PARTITION, the two sides, and the bottom, to reduce friction on the reciprocating piston. Use fine sandpaper.

9. Make and attach the 6 valve flaps, to complete the flap valves, that are the lowest res type tested.

a. Make a 3-3/4 x 5-3/4-in. cardboard TEMPFATE, using carbon paper to transfer lines of Fi cardboard. (See Fig. 1 1 on page 7, and note that this TEMPFATE outlines the right half of 1 1 - 1 /2-in. plastic-film flap.) Also transfer the dashed tack- line and mark the ends c stop-string lines. Drill 8 small holes through the cardboard at the ends of the 4 stop-str can use a pencil to mark these points on plywood.

Fig. 9. Removable Part of Front End, Unfinished. Only the plywood is shown, (nw265a.tif)

Fig. 10. Fixed Part of Front End, Unfinished. Only the plywood is shown, (nw265b.tif)

b. Use your TEMPLATE to mark around the 2 x 4-in. valve holes in plywood parts: (1) the po ends of each hole's H stop- strings, (2) the right side-edge and the bottom-edge of each f attached, and (3) the tack-lines.

c. Drill a 1/1 6-in. diameter hole through the plywood at each point marked for an end of

d. With nylon kite string (or other nylon string about 1/1 6-in. in diameter, such as 50-1 and a big enough needle, string the "four" stop-strings across each 2 x 4-in. hole. (Use a enough to make "four" uncut stop-strings.) Start on the unfinished, back side of the plywo opposite side from the future valve flap. To secure the starting end, wrap the string arou tack, and then drive it in. Keep pulling the string tight as you thread it through the hol its finishing end around a half-driven tack. Finally epoxy the string in all of the holes, the plywood. (An equally strong nylon string can be made by twisting together 4 pieces of dental floss.)

(Stop-strings also can be positioned by using No. 3 upholstery tacks in place of the 1/16- holes. Drive a tack partly' in, wind the string around it while pulling the string tight, completely in, to hold the string securely. Finally, coat the tack heads and the adjacent smooth covering of adhesive, to provide a smooth seat for the valve flap.)

e. Cut out 6 plastic flaps of transparent 4-mil plastic film (each 3-3/4 x 11-1/2 in.). Th accurately' cut a flap of thin plastic film is to make a cardboard template 3-3/4 x 1 1-1/ film, and cut around it with a very sharp knife.

f. In preparation for attaching a flap over each pair of 2 x 4-in. valve holes, cover the pair of holes with masking tape, up to the straight "tack line" that you already have draw each hole. Use your cardboard TEMPLATE. The masking tape will prevent the adhesive (that w used to attach each valve flap) from being applied too near the 2 x 4-in. holes, where adh keep a flap from opening fully.

g. Position each of the 6 flaps properly' in its closed position, with its lower edge on t have used the TEMPLATE to draw 3/4 in. below each flap's pair of 2 x 4-in. holes. Position side-edge on the line already drawn 1 in. from the right side of the right hole of each pa holes. Then put masking tape over the lower edge of each flap and the adjacent plywood, to temporarily' in its closed position.

h. Gently fold down the upper part of each flap, so that the plywood above its pair of 2 x 4-in. valve holes is uncovered (except where you have placed the protective tape), and place small pieces of masking tape as to hold each flap temporarily in this folded-down position.

i. Quickly apply a thin coat of all-purpose construction adhesive (such as Liquid Nails) to the plywood area above the protective masking tape that covers the plywood up to the "tack line" above each pair of 2 x 4-in. valve holes. Then promptly detach the small pieces of masking tape from the flap in its folded-down position, and turn the flap (the lower part of which is still in its proper closed position by masking tape) into the whole flap's closed position. Press the upper part of the flap against the approximately 1/2-in.-wide coating of adhesive, to secure the valve in its proper position. Allow several hours for the adhesive to harden before removing the tape and using the pump.

j. Drive small tacks (No. 3: 3/8 in.) on the "tack line" (see TEMPLATE), to make sure the pump is securely attached after long use. (Very small tacks are easily driven if held with tweezers or needle-nosed pliers.)

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#### PUTTING THE PUMP 'BOX' TOGETHER

1. The following procedure is the best tested construction method for persons who lack experience in putting parts together so that all corners are exactly square, or who do not have the big clamps and other equipment used by cabinet makers. This procedure is best carried out by two persons working together.

2. On the finished side of the top, draw two parallel lines exactly 12 in. apart and parallel to the edges. Each of these lines will be 3/4-in. from an edge. Also draw a line 6 in. from and parallel to the top, to mark the positions of the two valve frames. See Fig. 2.

3. Build the pump's "box" upside down; start by placing its top on the floor, as indicated in Fig. 12.

Fig. 12. Parts of the Pump "Box", with Dimensions in Inches. The Roman numbers give the best method for attaching these parts to each other, (nw266.jpg)

4. Attach the two valve frames II and III to the top I with construction adhesive, positioning them from an end of the top I. Make sure that each frame's flap valve is upside down and facing the center of the pump. Remove any adhesive that is on the top beyond the ends of the valve frames.

(When using construction adhesive to make this pump, it is best to apply a rather thin coat of adhesive to both plywood surfaces to be joined. Then promptly move one plywood part slightly back and forth, while pressing them together—thus making sure that both surfaces are coated and in contact until the adhesive sets and bonds adequately before attaching more parts.)

5. Draw two parallel lines on the unfinished side of the PARTITION, each 3 inches from one edge.

Adhere the two 12-in.-long unattached edges of the valve frames to the PARTITION on these lines, as illustrated by Figs. 2, 7 and 12. Allow time for the adhesive to set.

6. Before permanently attaching side V, position it vertically with a long edge resting on the side-edge of the PARTITION and ends of the two valve frames I and II in contact with the finished side V. See Fig, 7. On the unfinished (outer) side of side V draw lines showing the position of the PARTITION and of the two valve frames in contact with the finished side of side V.

7. Preparatory to attaching side V to the PARTITION and to the two valve frames, drill 4 screw holes (for your 2-in. roundhead screws) through side V. Drill these holes so that an end of each valve frame about 1 in. from its adhered edge, and the other 2 screws will go into the PARTITION, at points above the valve frames. Next, with side V temporarily in its position with a smaller diameter drill through the 4 holes in side V, into the PARTITION and into the two valve frames. Then with the 4 screws temporarily connect side V, the PARTITION, and the two valve frames. and, while checking with a carpenter's square the squareness of the angle between the PARTITION and side V, adjust the two pairs of screws to attain squareness, Remove side V,

8. Apply adhesive to the 3/4-in. -wide area along the long edge of the top, and if necessary use a heavier adhesive than normal to unattached edges of the PARTITION and of the two valve frames. The position side V, and by again screwing in and adjusting the 4 screws, make the angle between the PARTITION and side V square. Allow the adhesive to set.

9. Use short pieces of duct tape to temporarily attach the two 12 x 12-in. construction forms to the PARTITION and to side V, (Before using these forms, drive 4 small nails into each form, near the corners to serve as handles for removing them from the completed "cylinder".) Attach a construction form to each end of the PARTITION.

10. Adhere the finished side of side VI to the top, to the unattached side-edge of the PARTITION and to the end-edges of the valve frames, while keeping side VI pressed against the two square construction forms. keep side VI pressed against the construction forms until the adhesive sets, use small nails to secure the two small boards horizontally across the ends of the sides, at each end of the "box".

11. On the finished side of the bottom IX, draw two parallel lines 13- 1/2-in. apart, making them parallel to the center line of the bottom, as shown in Fig, 7, Nail the two 3/4 x 3/4 x 32-in. screws into the bottom, VIII to the bottom, 13-1/2-in. apart.

12. To attach the bottom, first place it (with its finished side down) on the exposed long edge of the construction forms. find that the bottom rests on the construction forms and is not in contact with the long-edge of the sides. effect increase the heights of the sides by coating with adhesive both the edges of the sides and the

3/4-in. -wide area of the bottom to which the sides will be adhered, Then adhere the bottom to the sides. Before the adhesive hardens, remove any that has been squeezed into the corner of the

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13. Permanently attach the fixed part of the front end X (see Fig. 7, 10 and 12) with adhesive to the sides and to the bottom. Be sure that its flap valve is upside down and is facing away from the pump, and that a long edge of this part is level with the outer side of the bottom. Remove the construction forms.

14. Paint the interior of the "cylinder" with sealer-after removing all adhesive that may remain.

15. After the sealer dries, sandpaper the interior of the "cylinder" with fine sandpaper,

final coat of sealer.

16. To attach the removable part of the front end XI, stand the "box" on its completely open end. Drill the 7 oversized screw holes (for your 2-in. screws) clear through the removable part of the front end. Fig. 9. With the flap valve facing outward, temporarily attach this part with a few small screws to the top and to ends of the two sides. Then with a smaller-diameter bit, drill the screw holes through the top and the sides so that the 7 screws will hold securely.

17. So that it will be unnecessary to tightly screw on the removable part of the front end, make the repeated temporary attachments airtight, tack felt weather-stripping (best 1/8-in. thick strips made of two thicknesses of flannel, to the contact edges of the top and the sides. The tacks serve well. Then with a razor blade carefully cut the felt covering the screw holes and remove these small pieces of covering felt.

18. Attach with screws the removable part of the front end.

19. To prevent damage to the front-end valve flaps when you stand the pump on its front end, attach a piece of 3/8-in. plywood to the front end, near each of its four corners, as pictured in Fig. 10. When standing the pump on its front end, use small pieces of masking tape to temporarily secure the flaps in their closed positions.

20. Attach the back end XII, using only screws. See Fig. 8. (For repairs, the back end may be removed.) To make the attachment of the back end airtight, coat its attachment "crack" only with rubber cement.

#### MAKING THE PISTON, THE PISTON ROD, AND ITS HANDLE

1. Have a sheetmetal shop cut three 3-ft.-long, 3-in. -wide strips of galvanized steel sheetmetal thicker than 0.016-in., thick and no less than 0.012-in. thick. (Most galvanized steel valley flaps and sold by many lumber yards is less than 0.016-in. thick; 30-gauge galvanized sheet metal from a sheetmetal shops is about 0.015-in. thick.) Steel sheetmetal thicker than about 0.016 in. is satisfactory and is unsatisfactory.

2. With a tolerance of + or - 1/32-in., cut from these strips two strips each 11-13/16-in. long and 1 1-3/4-in. long. (These four strips first must be bent and then tacked to the four sides of the plywood "cylinder". These piston-sealing strips serve rather like piston rings, by making close, sliding, low-friction seals on the sides of the plywood "cylinder". Steel strips resist wear and if properly lubricated make for months of continuous use.)

3 . Preparing the four sheetmetal sealing strips :

a. Since the strips to be tacked to the top and the bottom of the piston must be bent differently, mark "T or B" on each of the two strips that are 11-13/16 in. long. Mark "8" on each of the two strips that are 11-3/4 in. long.

b. On each of the two strips marked "T or B ", draw an ink line along which to make the approximately 90 degree bend, and another line for the approximately 90 degree bend. (See the left half of Fig. 13 for the distances from the edges of these two "T or B" strips to their bends.) Also draw two ink lines on each of the two strips marked "8", to drive tacks, spaced as shown in the left half of Fig. 13.

Likewise draw four lines on each of the two strips marked "8", as specified in the right half of Fig. 13.



noting that some of these lines are spaced differently than corresponding lines on the strip B"

c. Using a small sharpened nail for a punch and placing one strip of sheetmetal at a time punch 2 rows of tack holes in each strip. The tack holes should be about 1-1/2 in. apart.

d. From a nominal 1 x 2-in. straight board, make two boards each about 3/4 x 7/8 x 12-1/4 bending the sealing strips.

e. Securely sandwich a "T or B" strip of sheetmetal between the two 12- 1/4-in. -long boards on top of each other, by tightening two "C" clamps on the ends of the two boards, so that 3/8 -in, from one side of the strip is just visible along the straight edge of a board. Then clamp the boards in a vise so that the 3/8-in. -wide part of the sheetmetal strip is uppermost.

f. Bend the exposed part of the strip about 30 degrees off the vertical, away from the side where the holes have been indented by the punch. To bend evenly, hammer gently and repeatedly with a 3/4 x 3/4 x 18-in. board held against the exposed 3/8-in.-wide part of the strip.

Fig. 13. Piston Sealing Strips, each made of a springy sheetmetal strip 3 in. wide, (nw267

g. With the sheetmetal strip held sandwiched between the two 12- 1/4-in.-long boards by the clamps and the vise, so that the bending line for the almost 90-degree bend is barely visible, bend the part of the strip 90 degrees, in the same direction that the 3/8-in.-wide part was bent. See Fig. 13.

h. Bend the other "T or B" strip, and similarly bend each of the two "8" strips.

4. Attach the four sheetmetal sealing strips to the plywood piston with No. 6 tacks (1/2-in. long). On a solid metal surface the part of the plywood piston opposite the spot to which part of a strip is attached that when a tack is hammered in its point is clinched (bent over) on the far side of the plywood piston, by being hammered against the solid metal surface.

a. First tack a "T or B" sheetmetal strip to the top of the piston, and a "T or B" strip to the bottom.

b. Then tack the two "8" strips to its sides. The strips should fit together so as to make adjacent ends of two strips do not fit neatly together, cut bit by bit a very little off the ends so that the two adjacent ends fit together neatly at their corners.

c. To prevent air leakage between the ends of the sealing strips, put rubber cement in the "cracks" between strips. (This was not done on the test pump's piston.)

Fig. 14. Plywood Piston with Sheetmetal Sealing Strips Attached, (nw268.jpg)

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5. For the piston rod, saw from a straight, well-seasoned oak board a 3/4 x 1-3/4 x 36-1/2 in. rod. Sandpaper it smooth. (A piston rod made of well-seasoned oak is less likely to break if a crack necessitates using screws, in place of nails and staples, for attachments. Piston rods made of

fir boards were undamaged in the tests.)

6. To complete the piston rod:

- a. For the handle, use 4 pieces of a nominal 1 x 2-in. board cut to the lengths shown in Fig. 16. Round all edges and corners, to minimize the chances of the operators' blistering their hands.
- b. Paint the piston rod and its handle with sealer. When dry, sandpaper. Then apply a fine coat of paint.
- c. Use adhesive, screws, and nails (or adhesive and nails if your piston rod is of soft wood) to attach the handle to the piston rod. The handle illustrated by Fig. 15

Fig. 15. Piston Rod Handle Made of 3/4 x 1-3/4-in. Boards, (nw269.jpg)

Fig. 16. The Pump Handle of the Durability-Test Pump, showing how one man best holds it when pumping, (nw269a.tif)

- d. To reduce friction on the piston rod and resultant enlargement of the piston-rod hole with epoxy all four sides of the piston-rod hole. See Figs. 7 and 16. Be sure that the piston rod fits snugly yet freely in its hole when the removable part of the front end is screwed in place.
- e. From a piece of thick truck-tire inner-tube rubber, cut a 2-in.-wide strip 12-in. long. Attach a 2-in.-wide rubber bumper (see Figs. 15 and 16), connect one end of this rubber strip to the 3/4-in.-wide side of the piston rod. Do not place any screw or staple in the strip closer than 1 in. to the strip's forward edge, that may repeatedly bump into the front end. Wrap and attach the strip around the piston rod next to the handle. (If you have only a piece of passenger-car inner-tube rubber, then to make a 2-in.-wide bumper use a 4-in.-wide strip of this thinner rubber folded double.)

7 . Attaching the piston rod to the piston :

- a. On the back of the 12x1 2-in. plywood piston, mark lines to enable you to attach the piston rod as pictured in Fig. 14. Note that the lower side of the piston rod is exactly 5-1/2 in. above the plywood of the piston, and that the center line of the piston rod intersects the vertical center line of the plywood of the piston.
- b. To the end of the piston rod (see Fig. 14) adhere and screw (or adhere and nail if your piston rod is of oak) two pieces of nominal 1 x 2-in. boards each 3 in. long. Each of these two small boards at the end of the piston rod are in contact with and securely connected to the plywood piston, and form the front end of the piston rod.
- c. Connect the piston rod to the piston, best with epoxy (or adhesive) and small screws. Make sure the four piston sealing strips overlap the piston's plywood in the direction of the piston rod. The 1-3/4-in.-wide sides of the piston rod are parallel to the top and bottom of the piston, and the 3/4-in.-wide ends of the piston rod are perpendicular to the piston. See Figs. 2 and 14.
- d. Make and attach to the piston rod a 3-in.-long rubber bumper, positioned close to the piston rod as shown in Fig. 2.

1 . Check to see that the four sheetmetal strips on the four sides of the piston all make walls of the "cylinder" when the piston is moved back and forth. If the piston does not slide quite easily even when not lubricated, carefully bend a strip or strips so that they press "cylinder" walls. If while someone is shining a flashlight through a valve opening in the you observe that parts of a sheetmetal strip do not make close contact with a "cylinder" wall outward that part of the strip.

2. Lubricate all four walls of the "cylinder", the sheetmetal strips that slide against them. Use a very thin motor-breakin white lithium grease (not an ordinary bearing grease, that is a thin oil. The pump should be lubricated after no more than each 24 hours of use, and before again after days of disuse.

3. Install the pump at a height above the floor so that most of the persons who are going to pull with their hands moving at about the same height that their elbows are when they are pulling. For an example of a pump- supporting table raised to an efficient height for operators who use the pumper pictured.

4. To save work and to minimize wear on the pump, usually operate it with a length of stroke equal to the distance between its two rubber bumpers. To save energy especially when pumping air through a resistance ventilation system, move the piston back and forth by using mostly your leg and

#### PROLONGED STORAGE

Wipe off all grease and other lubricants if you do not plan to use this pump for months. Air on wood-tend to become gummy with time.

Keep your supply of pump lubricants taped to your pump.

#### REQUEST

Suggestions for improving this pump and/or these instructions will be appreciated, and may be likely to save lives.

Cresson H. Kearny

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#### FILTER BOX AND FILTER

#### PURPOSES

The primary shelter ventilation requirement is to supply enough outdoor air to maintain endurance conditions.

To keep the concentration of respiratory carbon dioxide low enough for survival, very little is required. Even for an infant or an infirm person remaining in a crowded shelter for days, 3 cfm is adequate. For a healthy adult or child 1.5 cfm is enough. Too much carbon dioxide, not too little, cause of unendurable conditions in inadequately ventilated shelters in which the air does not move.

In contrast, up to 25 cfm of outdoor air per occupant may be needed to maintain endurance inside a crowded shelter occupied for days during a heat wave in a hot, humid part of the country. A large-volume ventilating pump, best with a low-resistance filter.

If outdoor air flows into a shelter through a hood, gooseneck pipe, or other air-supply opening, fallout particles to fall out before the air reaches shelter occupants, breathing this unfiltered air may cause term radiation casualties. However, a very small fraction of the occupants of a shelter survive an area of heavy fallout may contract cancer years later as a result of breathing shelter air containing particles, that a properly designed filter could have removed.

Air that has been in contact with fallout particles before being filtered is not radioactive.

The homemade filter illustrated below, if used with an efficient "suction" pump such as the Piston Pump described separately, will remove practically all fallout particles likely to be encountered later. This filter also will remove most infective aerosols, the air-borne tiny particles likely type of attack on the United States. It will not remove poisonous gasses, an even less likely cause of death if all-out war befalls us.

## CONSTRUCTION

### Filter Box

If 20 x 20-inch furnace filters are available, use plywood or boards to build the filter box. To make permanent connections airtight, first use waterproof construction adhesive or glue, and if smaller filters are available, reduce the horizontal dimensions of the box accordingly, except for the air openings.) Check to be sure that your filters will fit snugly in the box of the size you plan to use.

The square frame on the bottom of the filter box should fit snugly over the square air-intake opening of the Piston Pump. Tape the cracks to make the connection airtight and to support the filter box.

Make the illustrated 4 supports of the hardware cloth no thicker than 3/4 inch, thus providing a low-resistance filter for low-resistance airflow. (Hardware cloth is a stiff, square-mesh, molten-dipped wire.)

Make the square top of the filter box so that it covers the upper edges of the box's sides. Then cut in its center a round hole slightly smaller than 4 inches in diameter. File the hole until a 4-inch-diameter can (such as a coffee can with its top and bottom cut out) fits snugly in the hole. To make the top airtight, first use waterproof construction adhesive or epoxy, and then tape. If epoxy is not available, cut a 2-1/2-inch-diameter hole in the center of the bottom of the top. Then make radial cuts spaced about one-half inch apart, out to the full diameter of the can. Bend the top 90 degrees, preparatory to tacking them with small tacks to the bottom of the filter box top.

So that the top of the filter box can be easily removed, tape it onto its box. A roll of cellophane tape is recommended.

filter box and pump at all times.

To connect the filter box to the shelter's air-intake pipe, the best widely available air 4-inch-diameter flexible duct used with clothes dryers.

(Figure p.270) Homemade Filter To Fit On Plywood Double-Action Piston Pump, and To Be Connected To 4-Inch-Diameter Air-Intake Pipe, (nw270.tif)

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#### Filter Materials

Furnace or air-conditioner dust filters, those made of oiled fiberglass fibers, will remove the smallest fallout particles. Filters that are sold in box-like housings can easily be installed to pass through them, by taping them to the inner sides of the filter box. The illustration shows the filter material, each taped around its edges. (If commercial dust filters are not available,

However, in very dusty areas a cloth filter may become overloaded, thus seriously reducing airflow sooner than if an oiled fiber filter is used as a prefilter.)

To filter out most of the tiny particles that may pass through one or more furnace filters, place a towel on top of the filter-support made of hardware cloth, and tape them around their edges. See the illustration.

Tests by U.S. Army specialists have shown that filtering air through two thicknesses of battery-operated HEPA filters will remove 99.97 percent of even microscopic aerosols as small as 1 to 5 microns in diameter. (See "Emergence of Biological Warfare Against Radiological and Biological Aerosols", by H. G. Guyton et al., A.M.A. Archives of Internal Medicine, July through Dec. 1959.) This is the size of most infective aerosols used in biological warfare. If 85 percent of this size-range of infective aerosols are removed, then most persons breathing this filtered air will not receive enough infective aerosols to harm them.

Persons who are especially desirous of protecting their shelter's occupants against biological warfare should use HEPA filters who can not afford or obtain expensive High Efficiency Particulate Air filters (HEPA filters). Disposable pleated air filters that meet official ASHRAE standards. One 2-in. pleated air filter, 19-1/2 in., will remove over 90 percent of particles in the 1.0-5.0 micron range, yet when the flow of 200 cfm is only about 0.2 in. water gauge (about 0.007 psi). Its cost is about twice that of a furnace filter of the same size. However, it has approximately three times the life of a standard furnace filter becoming overloaded. Disposable pleated air filters are available in larger cities.

#### USE

The illustrated homemade filter has such low resistance to airflow that, when used with a Plywood Double-Action Piston Pump, the air volume is decreased by only about 10 percent compared to the volume pumped with no filter in the ventilation system. With a homemade Plywood Double-Action Piston Pump, up to approximately 200 cfm can be pumped through this filter even when the total pressure (caused by the ventilation pipes, a dirty filter, etc. that restrict airflow) is 0.18 psi).

Even if the United States suffers an all-out Soviet attack, only a small part of its area

severe enough to injure the occupants of fallout shelters. (Fallout shelters are not designed especially typical earth-covered ones afford consequential blast protection. ) In contrast, protected by an efficient blast valve, will be wrecked by a quite low-pressure blast wave the air-intake pipe -even if the small part of the blast wave that would enter the shelter room pipes is not nearly powerful enough to injure the shelter occupants. Thus unprotected installations in an area several times as large as the area in which occupants of fallout shelters will

To be sure of having a filter in good condition, you can:

1 . Make and keep in your shelter an extra complete filter, ready to replace your installed one or if it becomes overloaded with dust and its resistance to airflow becomes too high. Further, if a filter is installed in your shelter room and becomes so radioactive with retained fallout as to be delivering a consequential radiation dose to shelter occupants, it is advantageous to be able to remove it out, and install a replacement filter. (To be able to supply your shelter with unfiltered air after the end of consequential fallout danger, you should make and keep ready a duct with which to connect your pump directly to its air-intake pipe.)

2. If you have only one filter, do not install it before you need to filter the air supply entering the shelter to the air-intake pipe, using an appropriate duct and fittings. Then before the attack and fallout (revealed by your fallout-monitoring instrument), keep your shelter well ventilated. Whether or not your filter is installed, stop ventilating your shelter for a few hours while fallout is deposited outside - unless heat-humidity conditions become unbearable. If before shelter ventilation the shelter air does not contain an abnormally high concentration of carbon dioxide, then the shelter can be supplied for about 5 hours to prevent building up too high a concentration of respiratory gases, provided there is about 70 cubic feet of shelter-room volume for each occupant.

#### AN ENCOURAGING REMINDER

Persons making preparations to improve their chances of surviving an all-out attack should be aware that if the United States is hit with warheads the sizes of those in the 1987 Soviet intercontinental arsenal, the fallout concern will be much larger than the extremely small particles (1 to 5 microns in diameter) removed by this filter. Fallout particles this small produced by large nuclear explosions will persist for many days to months after the nuclear explosions, by which time they have become much less dangerous. All of the larger particles can be removed merely by filtering the air through a few thick

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#### Appendix F

Means for Providing Improved Natural Ventilation and  
Daylight to a Shelter with an Emergency Exit

#### THE NEED

Survivors in areas of heavy fallout can greatly reduce the radiation doses that they will receive by taking their risks of contracting cancer, if they sleep and spend many of their non-outdoor-working

during the first several months after an attack. (See Minimizing Excess Radiogenic Cancer Attack, by Kathy S. Gant and Conrad V. Chester, Health Physics, September 1981.)

A permanent family shelter can serve quite well for months as a post-attack temporary home provide adequate natural ventilation most of the time, to have adequate and easy forced ventilation is needed, and to have daylight illumination, A shelter dependent on ventilation through pipes and on artificial lights even during daytime is much less practical for use

The following instructions should enable a family having an earth-covered shelter with an it much more livable for months-long occupancy. The means described below for providing in daylight illumination also will supply guidance to survivors who will build shelters post-radiation exposures, especially to children and pregnant women,

#### BUILDING AND USING A MULTI-PURPOSE EMERGENCY EXIT HOUSING

Build a multi-use emergency exit housing of the design pictured in Fig. F, 1 and detailed housing to fit snugly over the top of your completed vertical exit shaft. This exit housing plywood, four 2 x 2 x 36-inch boards, and four 16 x 16-inch window panes of 1/8-inch Plexiglas waterproof adhesives are used to assure sturdiness and durability.

Fig. F. 1, Multi-Use Emergency Exit Housing Installed Over the Square Emergency Exit Description 17,1, 17.2, and 17.3. (Photograph) (nw273.jpg)

The adjustable top of this exit housing measures 4x4x1 feet, and can be tilted to make different openings in any of four directions. The top also can be raised straight up to make various openings around, or it can be completely closed - as explained by Fig. F.2 and the following description

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Fig. F.2. Plan and Side View of Multi-Purpose Emergency Exit Housing, on a Square Emergency Exit 34 x 34-Inch Cross-Sectional Outside Dimensions. (Diagram) (nw274.tif)

Fig. F.3. The Top and Four Walls of the Multi-Purpose Emergency Exit Housing, Nested Together to Form a Storage Space. (Photograph) (nw274a.jpg)

In Figs. F.2 and F.3, note the eight beveled plywood guides, two on the inside of each side wall. These guides are needed so that the top can be tilted in the position desired, merely by using a screwdriver

To hold the top in a tilted or raised position, spacer boards are placed between the raised top and the side wall or walls, as illustrated by Fig. F.4.

Fig. F.4. View from Below the Exit, Looking Up the Multi-Purpose Emergency Exit Housing. The housing is shown supported in a tilted position by two 6-inch-wide boards placed between a wall and the housing top. (nw274b.jpg)

The illustrated housing over a vertical exit provides:

\* A means to regulate shelter ventilation, and to increase natural ventilation when the wind is from the side. For example, the shelter's opened exit is to the north of its opened entry and a north wind is blowing. The natural airflow will blow in through the exit and out through the entry. This natural ventilating

inadequate, is increased if the adjustable top of the exit housing is not simply raised 6 but is tilted as shown in Fig. F. 1 , with its south side closed and its north side tilted 26-inch ventilation opening between the upper edge of the entry housing's north wall and i wind striking the north wall produces increased air pressure over and above this wall, for exit and on through the shelter. In contrast, if a south wind is blowing, natural airflow shelter's entry and out through its exit. And if the adjustable top still is tilted open t then reduced air pressure over and above the downwind north wall will "suck" an increased exit and through the shelter.

The measured increases in airflows through a small shelter resulting from the top of this being tilted were only 40-50 cfm when an 8-10 mph breeze was blowing. These rather small i airflow, however, often would make it unnecessary to supply forced ventilation to a family intermittently operating a KAP.

- \* Exclusion of rain, snow, and larger dust and fallout particles. The four 12 x 48-inch ve adjustable top overhang the exit housing's walls by 6 to 12 inches. Thus the top serves as hood over the exit, preventing rain, snow, and larger dust and fallout particles from ente continuing. (To prevent entry of flies and mosquitoes, an insect screen panel, made to fit the emergency exit, should be kept stored in the shelter until needed. A screen door for t doorway also should be stored. Remember that installing screens greatly reduces natural ve

- \* A reliable source of daylight. The four 12x1 2-inch' windows of this exit housing let er exit shaft, that is painted white, to permit a person on the shelter floor below to read, after sunset. See Fig. F.4.

- \* A way to observe what is going on all around the shelter, without having to go outside, exposure to fallout radiation.

- \* Quick installation post-attack, after fallout decays sufficiently. In an installation te expose the upper 12 inches of the emergency exit shaft. Then in just 8 minutes the author the 5 parts of this exit housing 80 feet, positioned its four walls around the already exp the reinforced concrete emergency exit, nailed its walls together, and placed its adjustab position pictured in Fig. F. 1 .

#### BUILDING AND USING AN ENTRYWAY COVER THAT PROVIDES A LARGE, PROTECTED VENTILATION OPENING

Build a shelter entryway cover that keeps out rain, snow, and the bigger dust and fallout a large, protected ventilation opening both for natural ventilation and for easy forced ve needed.

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For an example of one type of entryway cover, see Fig. F.5. This photo shows a 4-piece cov less than 5 minutes carried out of this shelter and installed over the 4 x 6-foot opening stairway doors.

Fig. F.5. A Quickly Installable, 4-Piece Entryway Cover That Provides Easy Access and a La Ventilation Opening, (nw275.jpg)

This cover is made of 4 pieces of 1/4-inch chipboard, each 5 feet wide, and short lengths



boards. These 4 pieces can be tied quickly with their attached nylon cords to inner parts entryway doors, which are pictured in their opened, upright positions.

The lowermost of the 4 chipboard pieces has a groove near each end. The grooves are each made of lengths of 1 x 2 lumber spaced apart to fit the lower ends of the doors and hold them in place apart. The upper edge of this lowermost piece is 8 inches below the lower raised corners of the 48-inch ventilation opening is assured when the lower of the two large covering pieces (pieces) rests on the doors. (This step-over piece of chipboard illustrates a way to reduce the quantity of dust that will be blown into many types of shelters, because most sandlike particles and coarse dust settle to the ground. They are not blown upward and over a vertical obstruction by most winds. If an ordinary doorway, even more fallout particles can be kept out of the shelter room if an 1 x 2 is cut in the door near its top.

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Then air entering the shelter room will have to rise at least 4 feet above the entryway floor. (Fallout particles will be deposited on the entryway floor.)

The chipboard piece attached to the upper ends of the doors also has two 1x2 boards nailed to it forming grooves into which the upper ends of the doors fit. The doors are thus held in the open position etc. is kept from falling or being blown through the upper end into the entryway.

The uppermost of the two large covering pieces of chipboard (or exterior plywood) rests on the doors and is kept from slipping down by a 1 x 2-inch board nailed 4 inches from its upper end. This board is attached to the upper edge of the piece of chipboard (or plywood) attached to the upper ends of the steel columns (on the side of this column.) This large piece of chipboard is securely tied to the doors.

To keep the two large pieces from moving sideways, one 1 x 2-inch board is nailed near each door spaced so as to lie against the outside of each opened, upright steel door. To strengthen each large covering piece, a 1 x 2-inch board is nailed along its lower edge.

Fig. Pg. 276. (nw276.tif)

The lower of the two large covering pieces also has a reinforcing 1x2 nailed near its hinge.

The most practical hinge that the author has devised is illustrated by the drawing. This is much less likely to be broken than are conventional hinges, and makes it easier to build the two large doors and the opened doors. Note that the upper edge of the lower large piece goes under the rainproof flap, which is nailed only along the lower edge of the upper large covering piece. Then the two pieces are hinged together by first stretching each of 2 strong, 2-inch-wide rubber bands (or nylon cords) to the upper large covering piece, and then hooking its attached bent-wire hook onto the lower large covering piece. Each strong rubber band (cut from a truck inner tube) and nylon cords is 5 inches from an opened door. Thus hinged, the lower large piece can be opened by a person to step out of or into the stairway entry. When this hinged lower large piece is closed, a 10-square foot protected ventilation opening with a 10-inch overhang results.

#### OTHER ENTRYWAY COVERS TO PROVIDE LARGE PROTECTED OPENINGS FOR NATURAL VENTILATION AND KAP VENTILATION

The owner of a permanent shelter with an emergency exit may be able to improvise coverings

exit after fallout decays sufficiently to permit work outdoors - provided that he understands low-pressure forced ventilation requirements, and has the boards, nails, pieces of chipboard tools, etc. needed. But if you own a permanent shelter your pre-crisis preparations surely storing ready-to-install entry way and exit coverings of whatever designs you decide will needs for high-protection factor sleeping and living quarters during weeks or months following

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